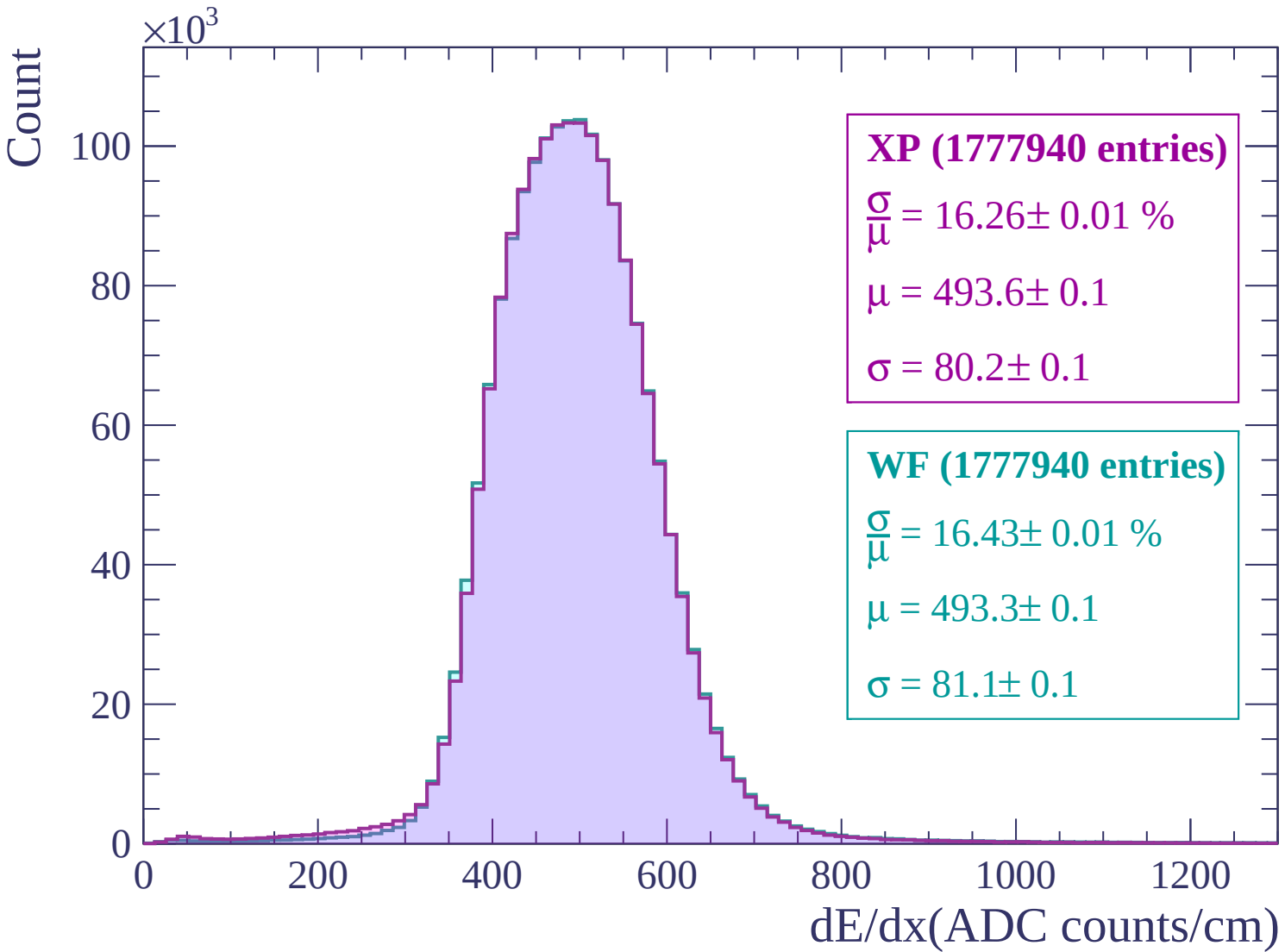
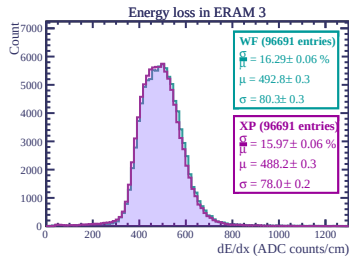
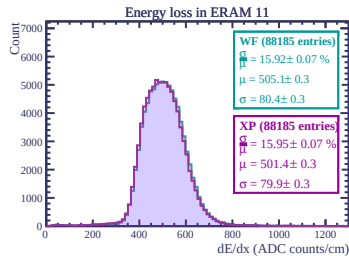
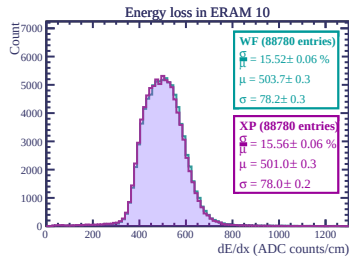
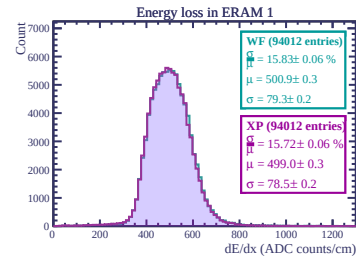
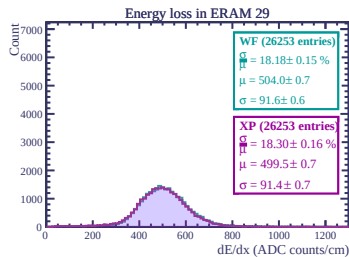
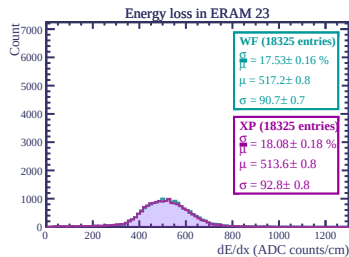
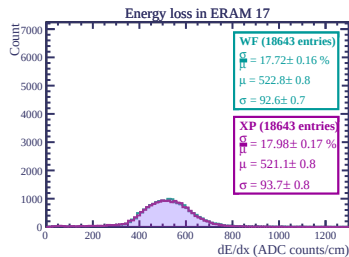
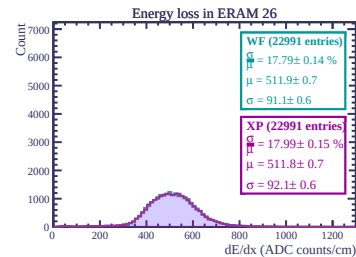
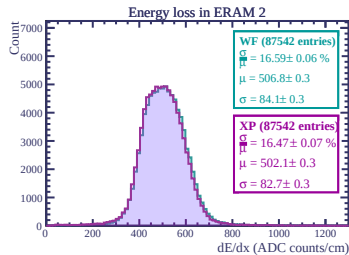
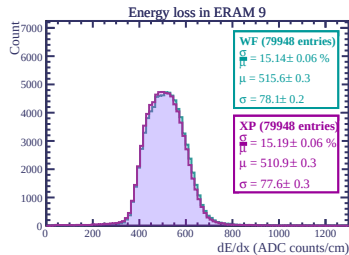
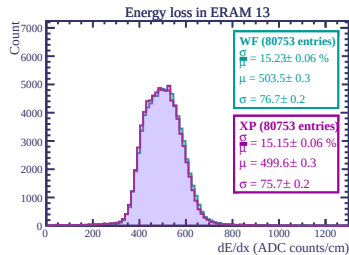
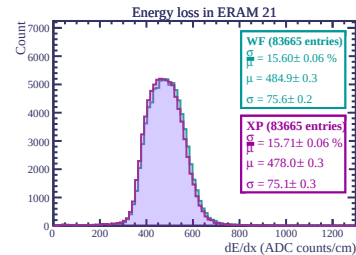
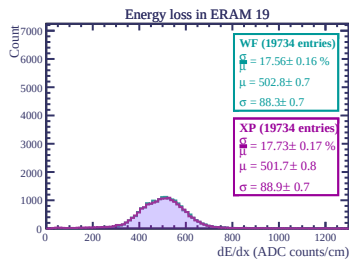
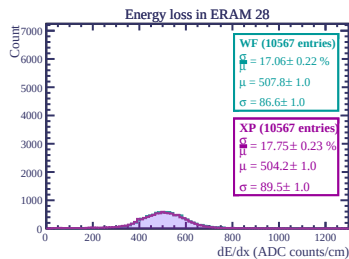
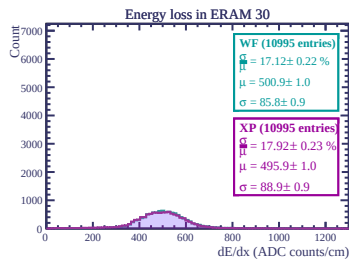
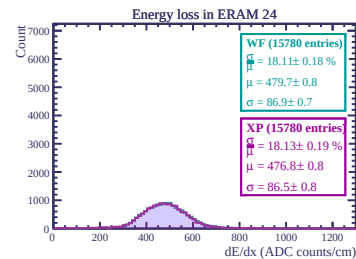
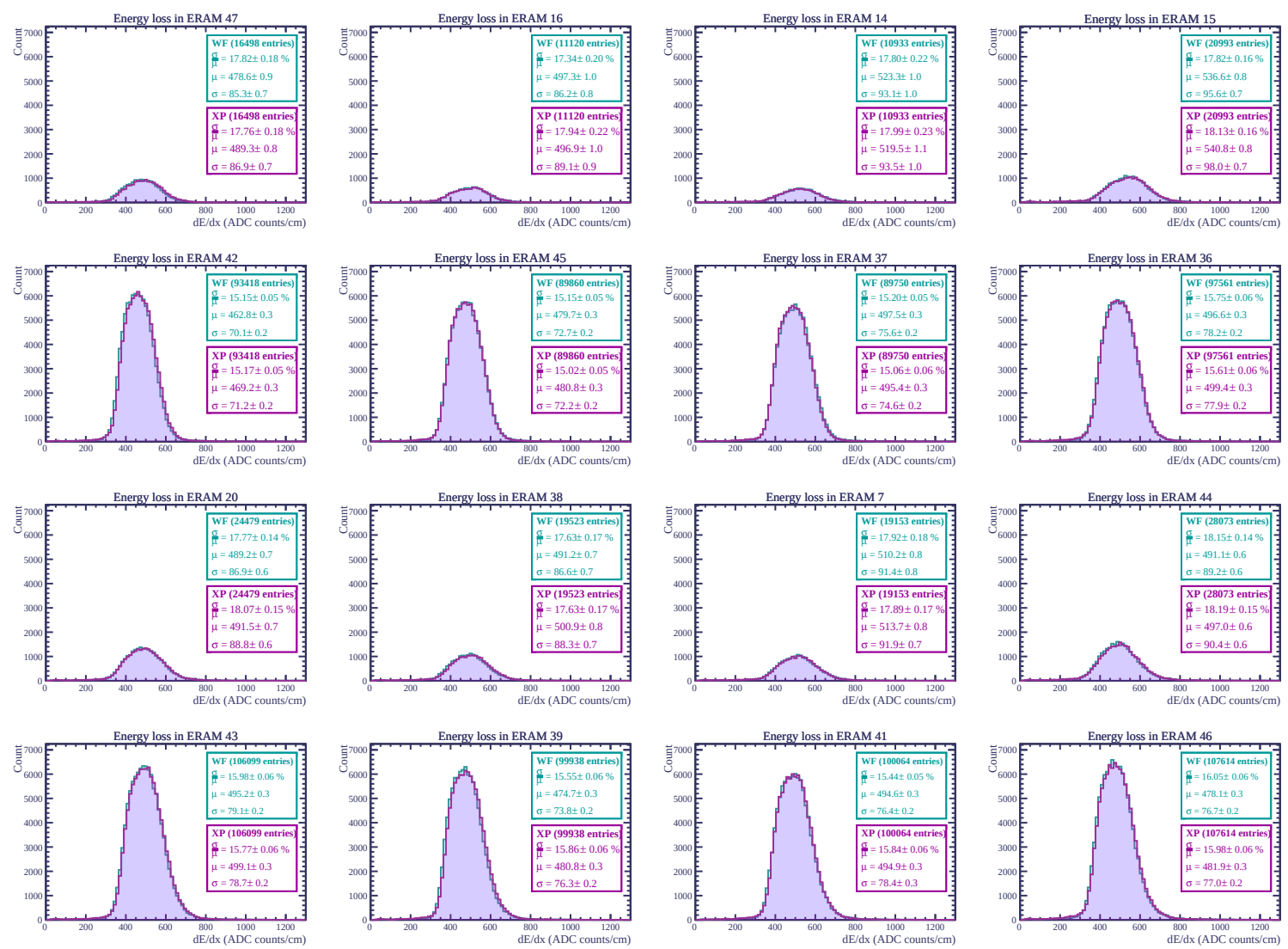
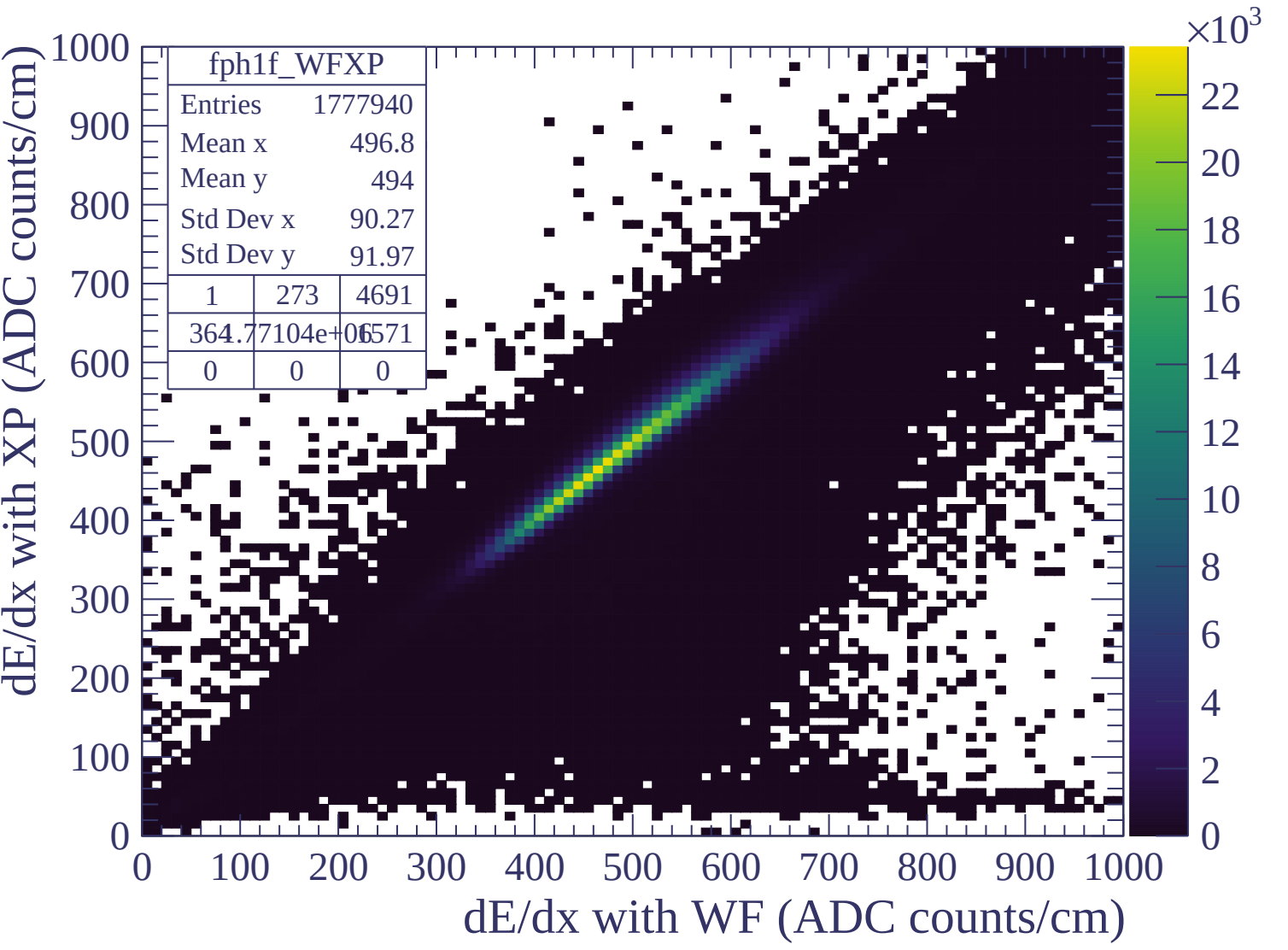


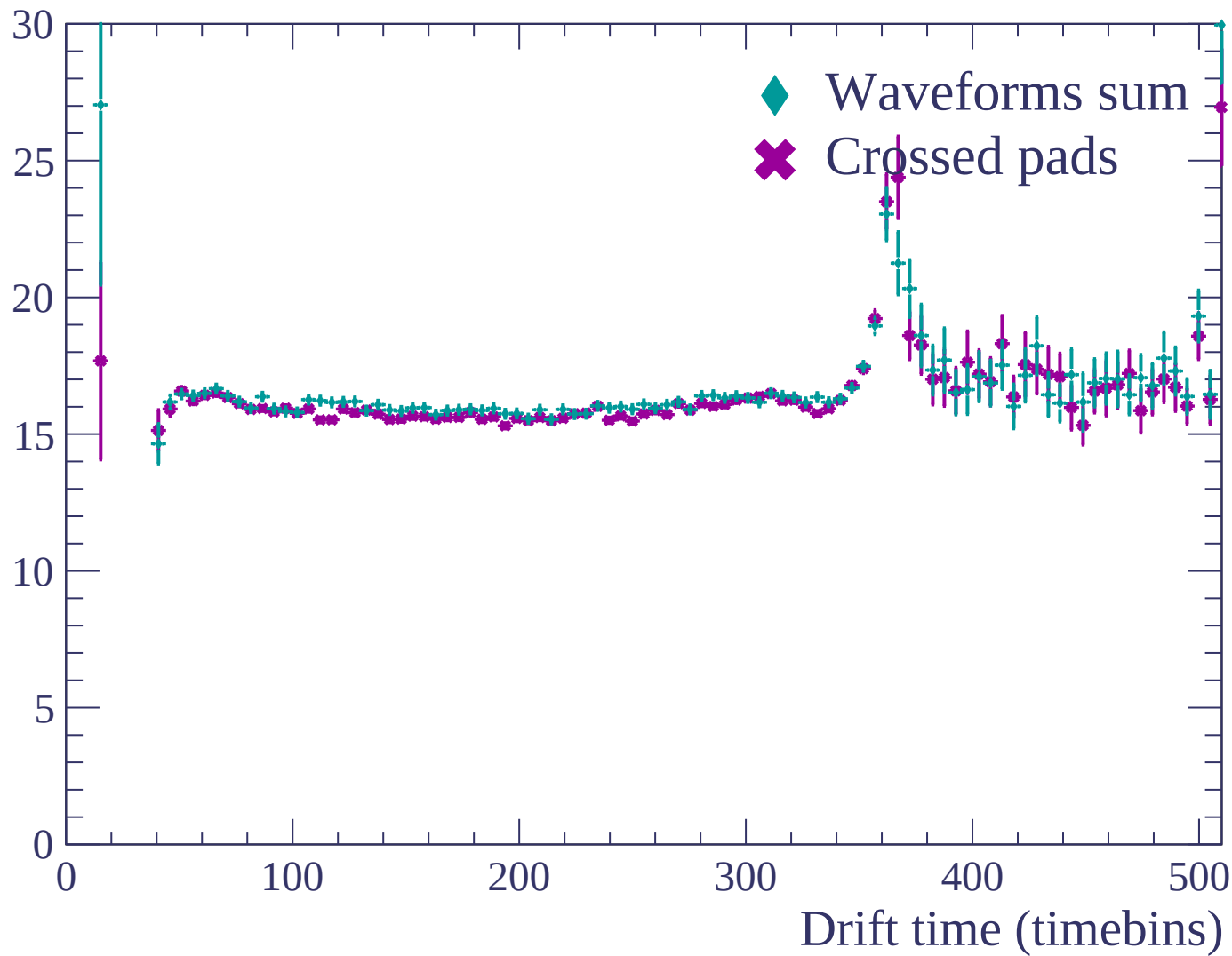


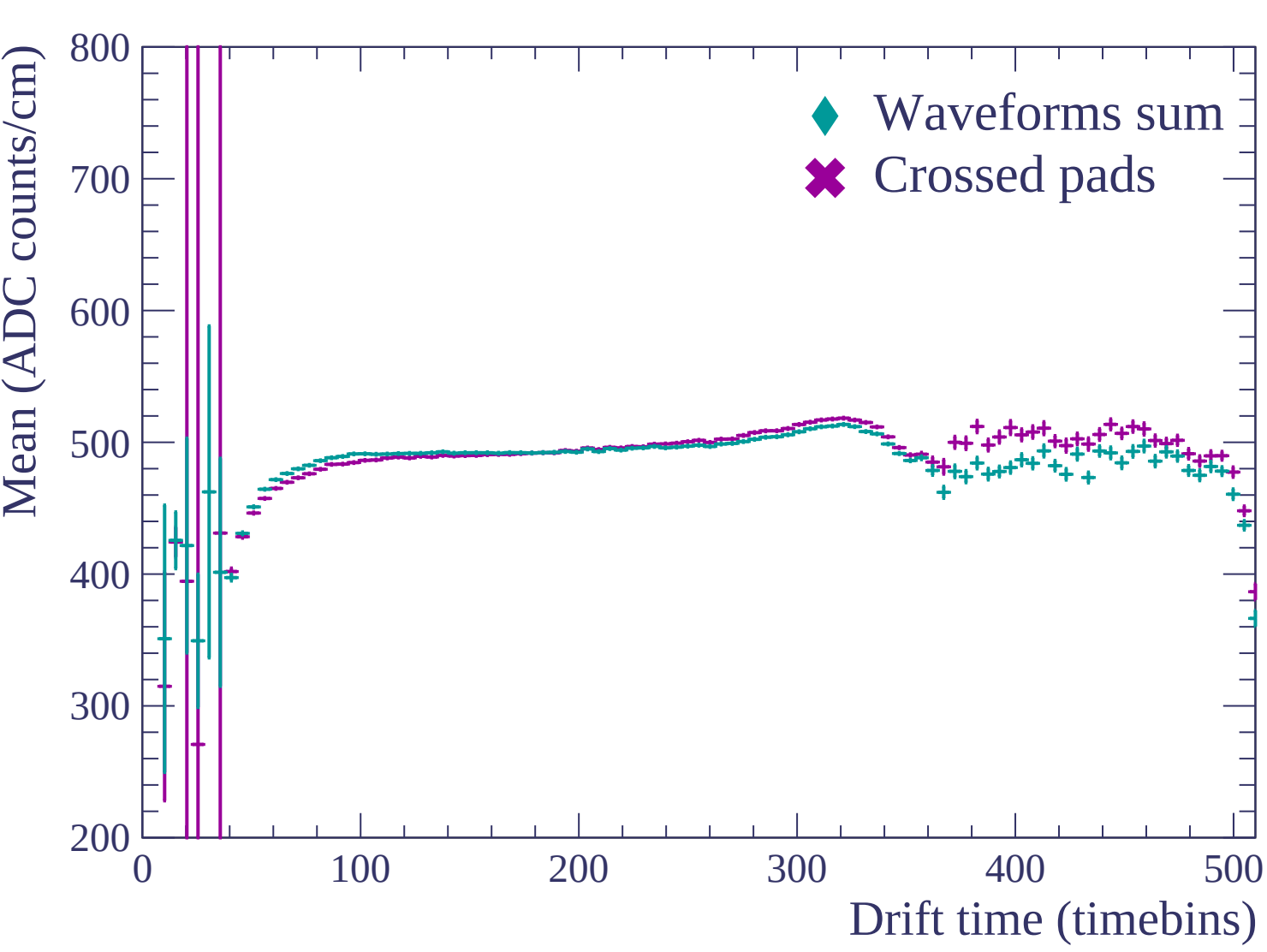


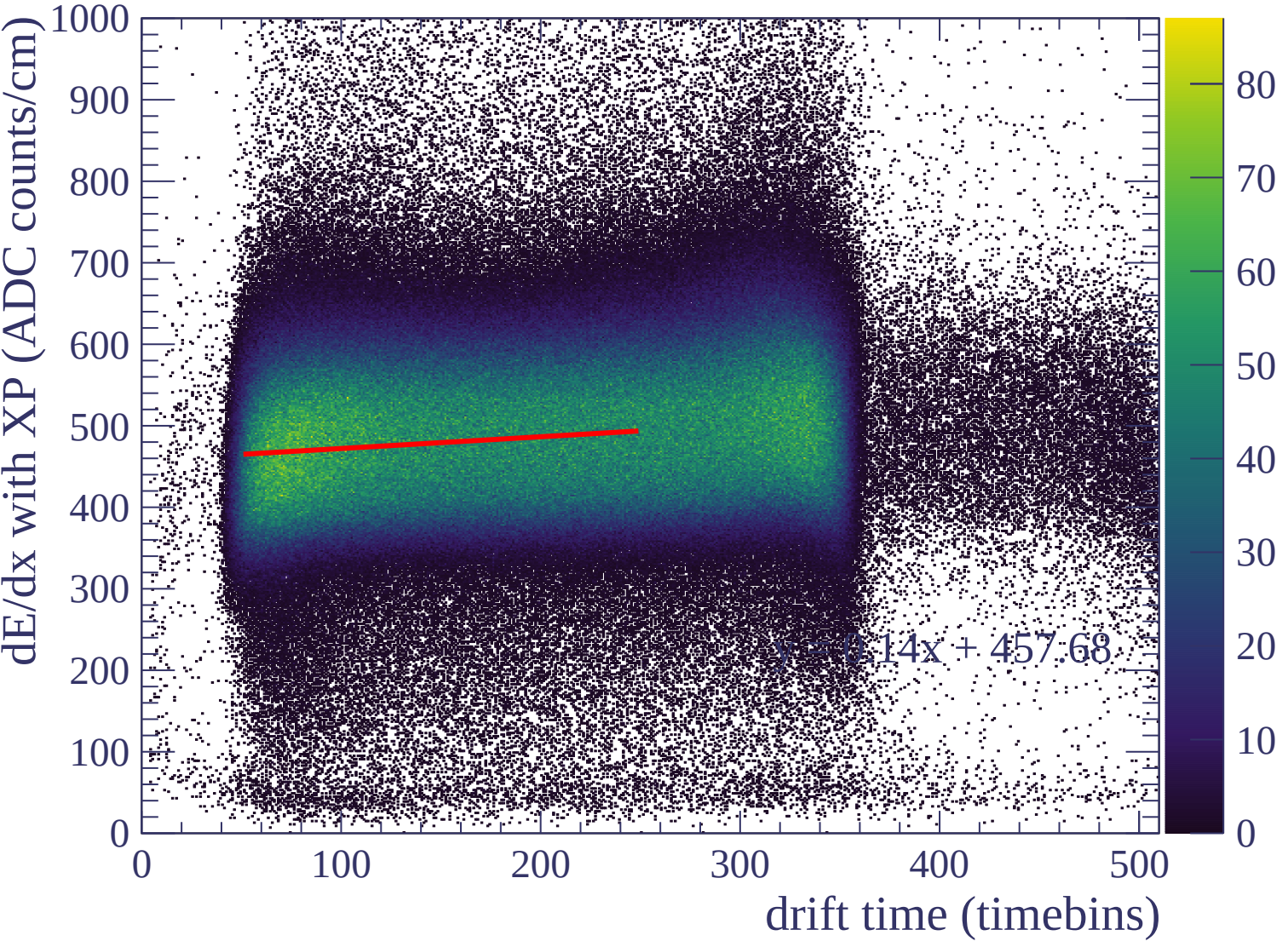


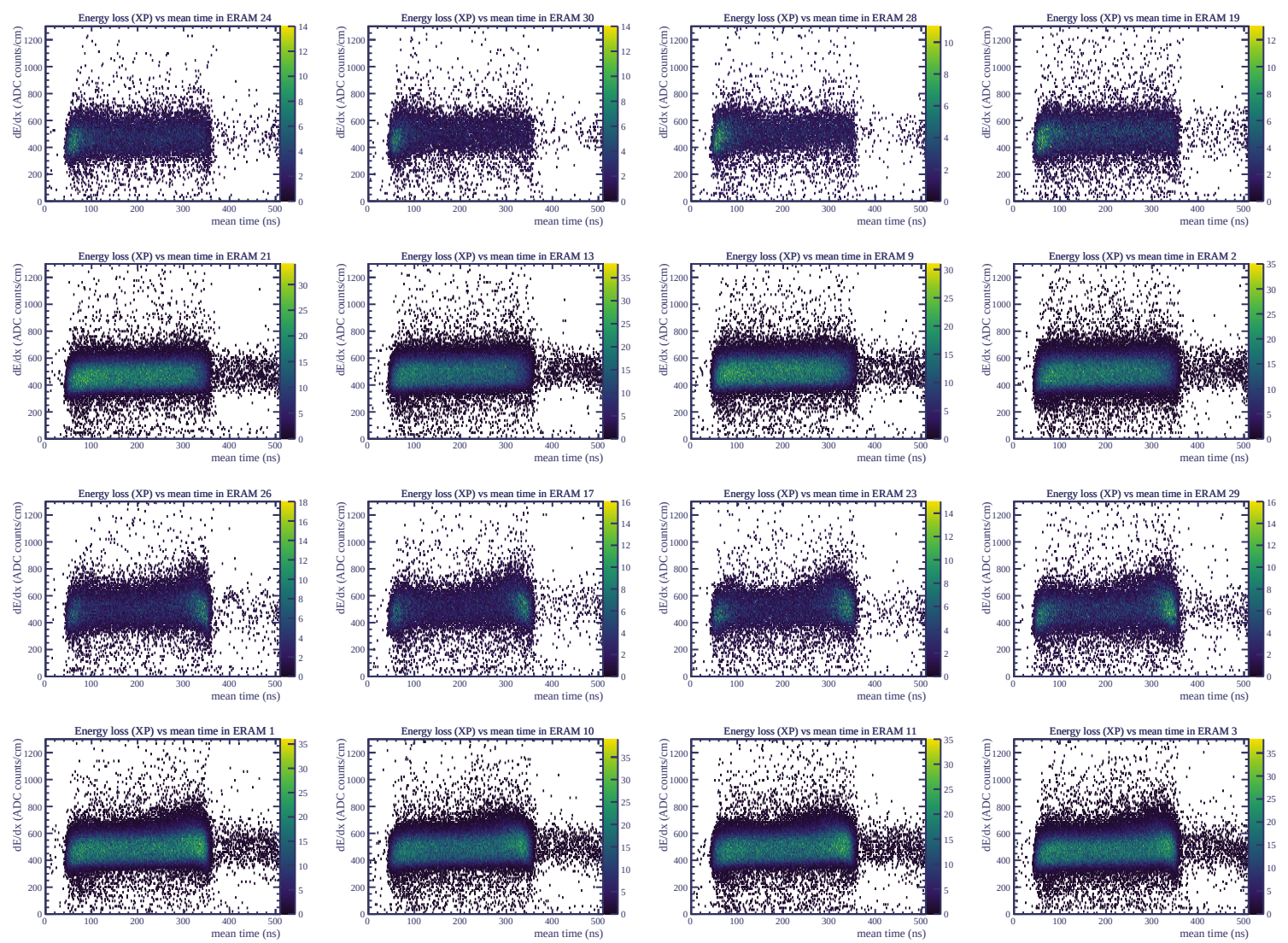


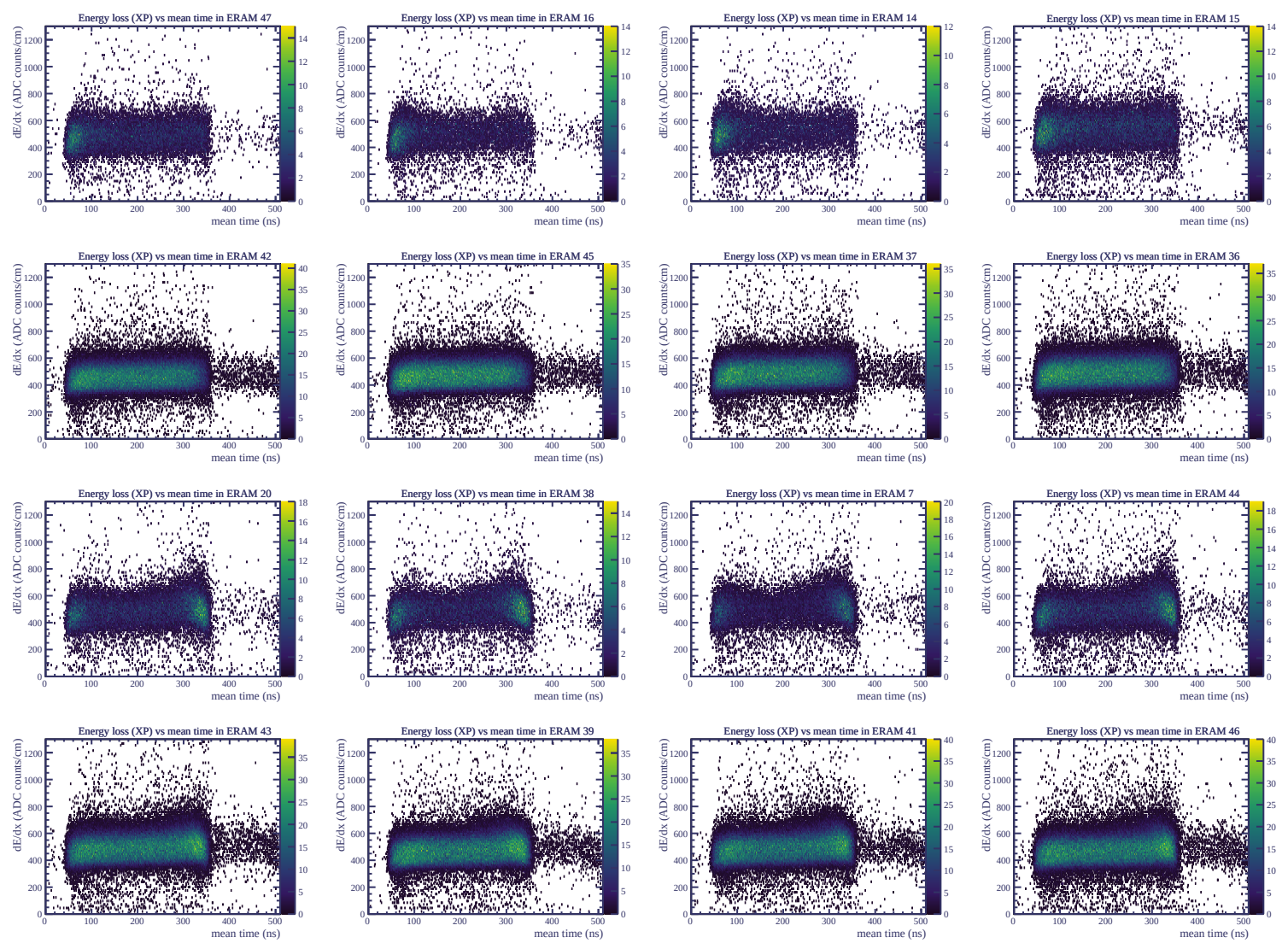
Resolution (%)

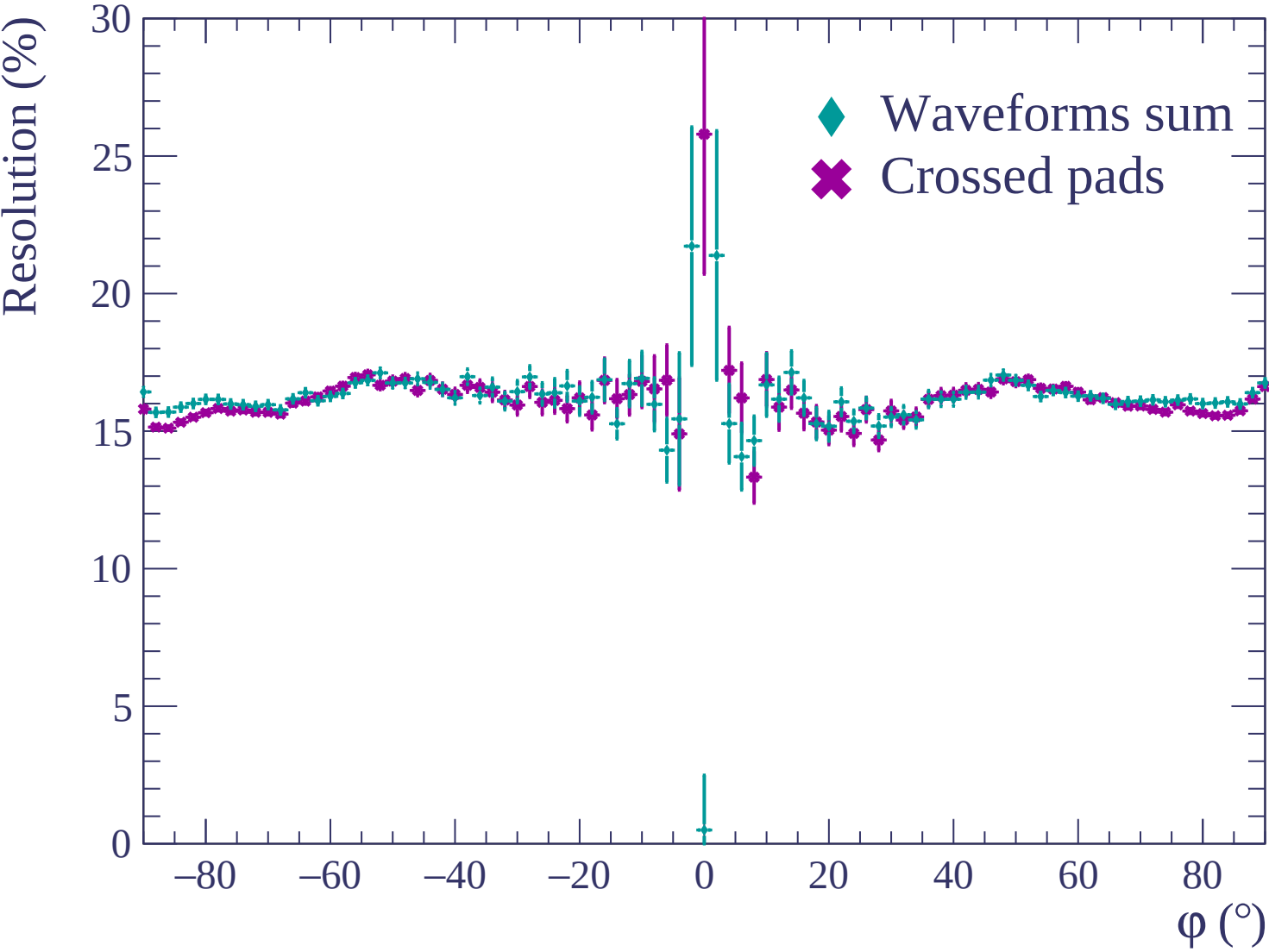


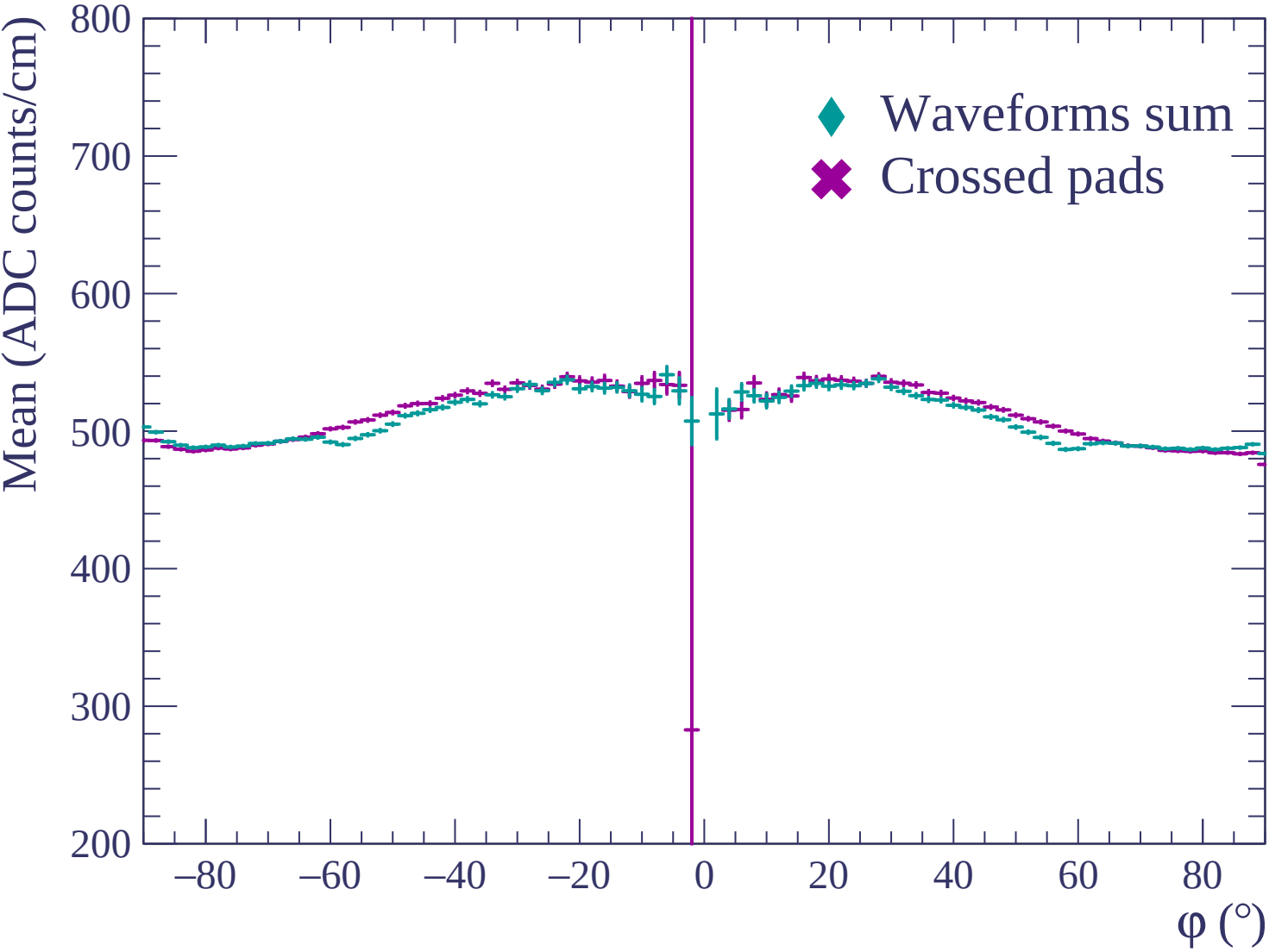


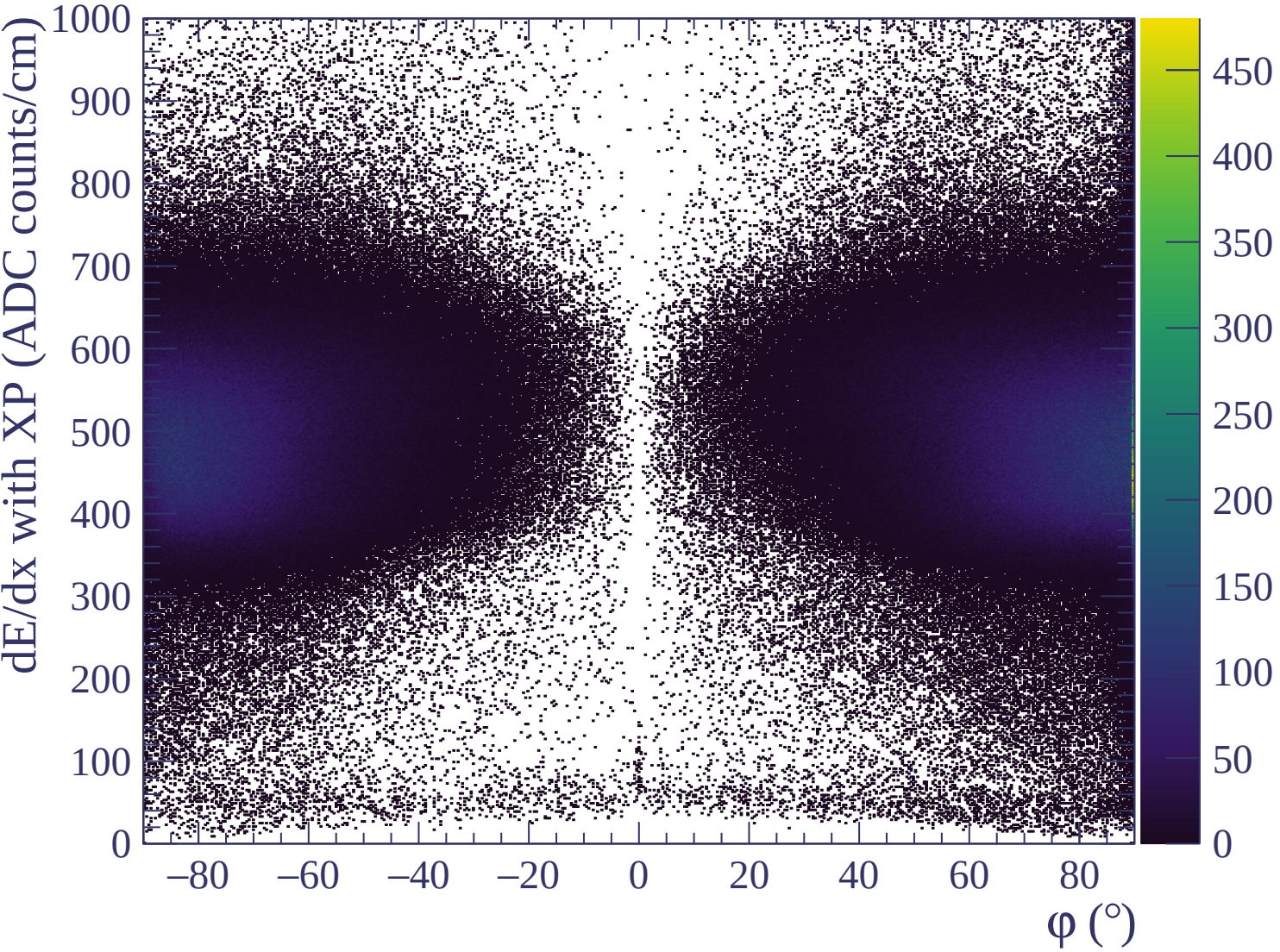


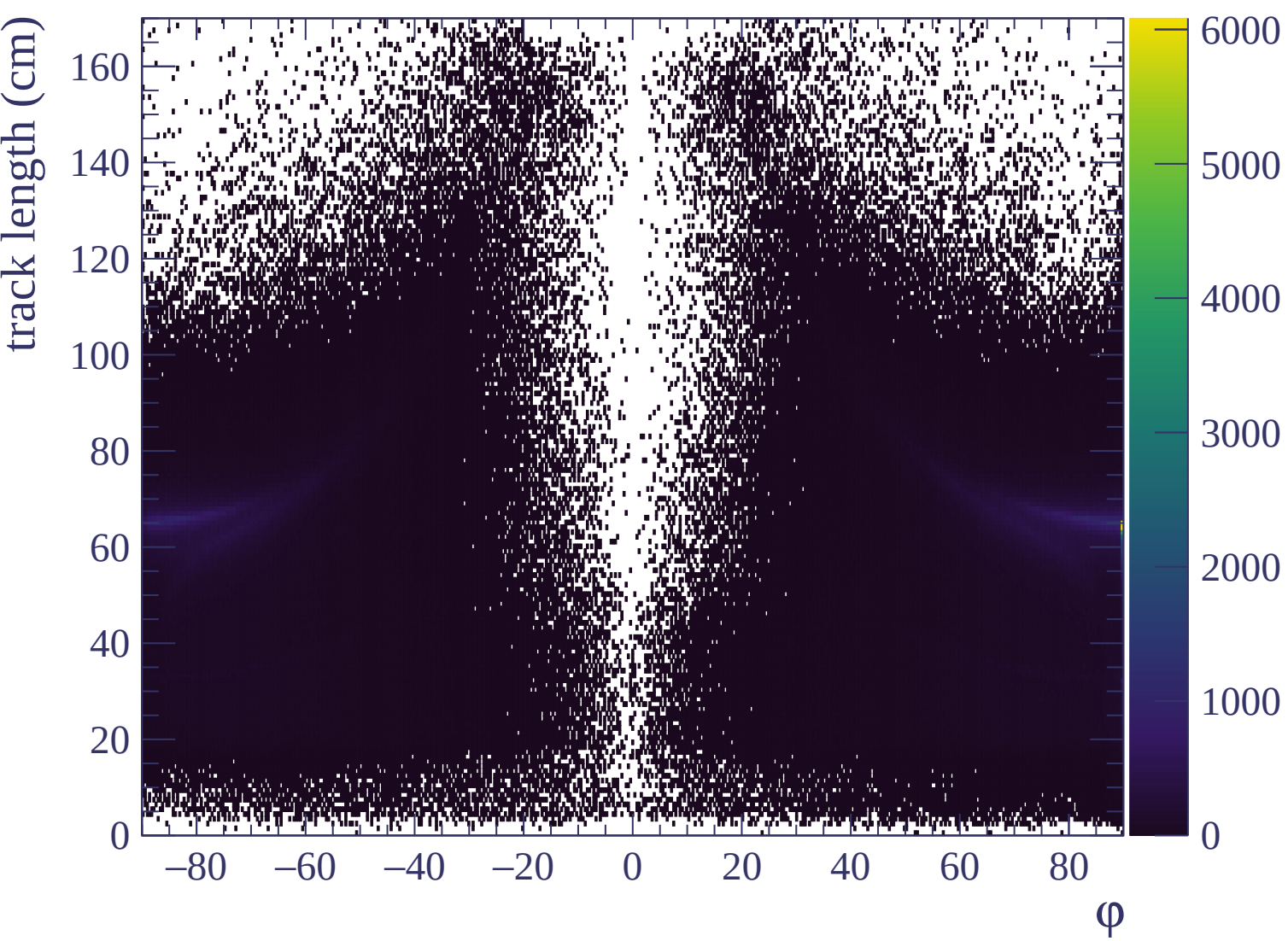


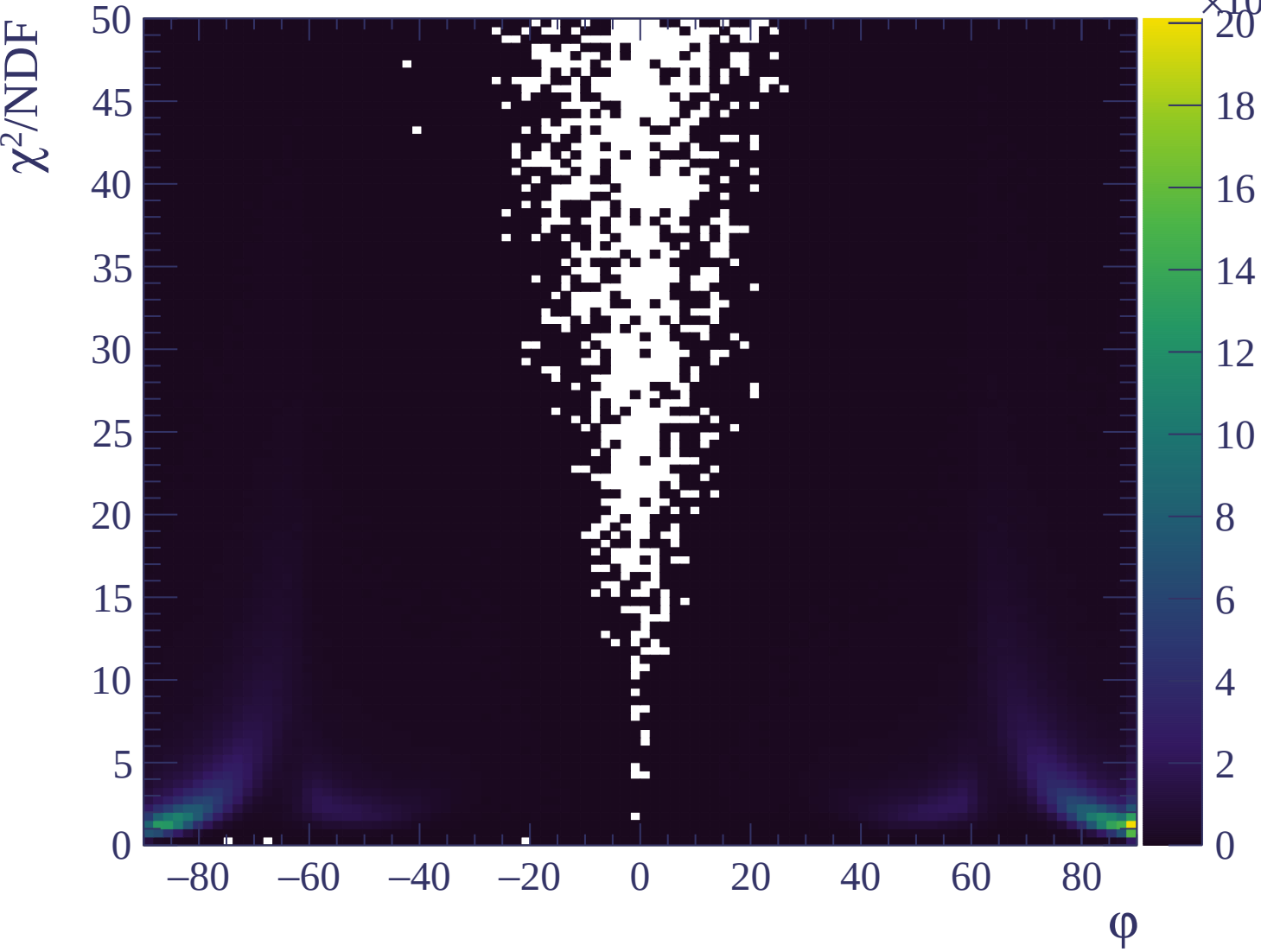


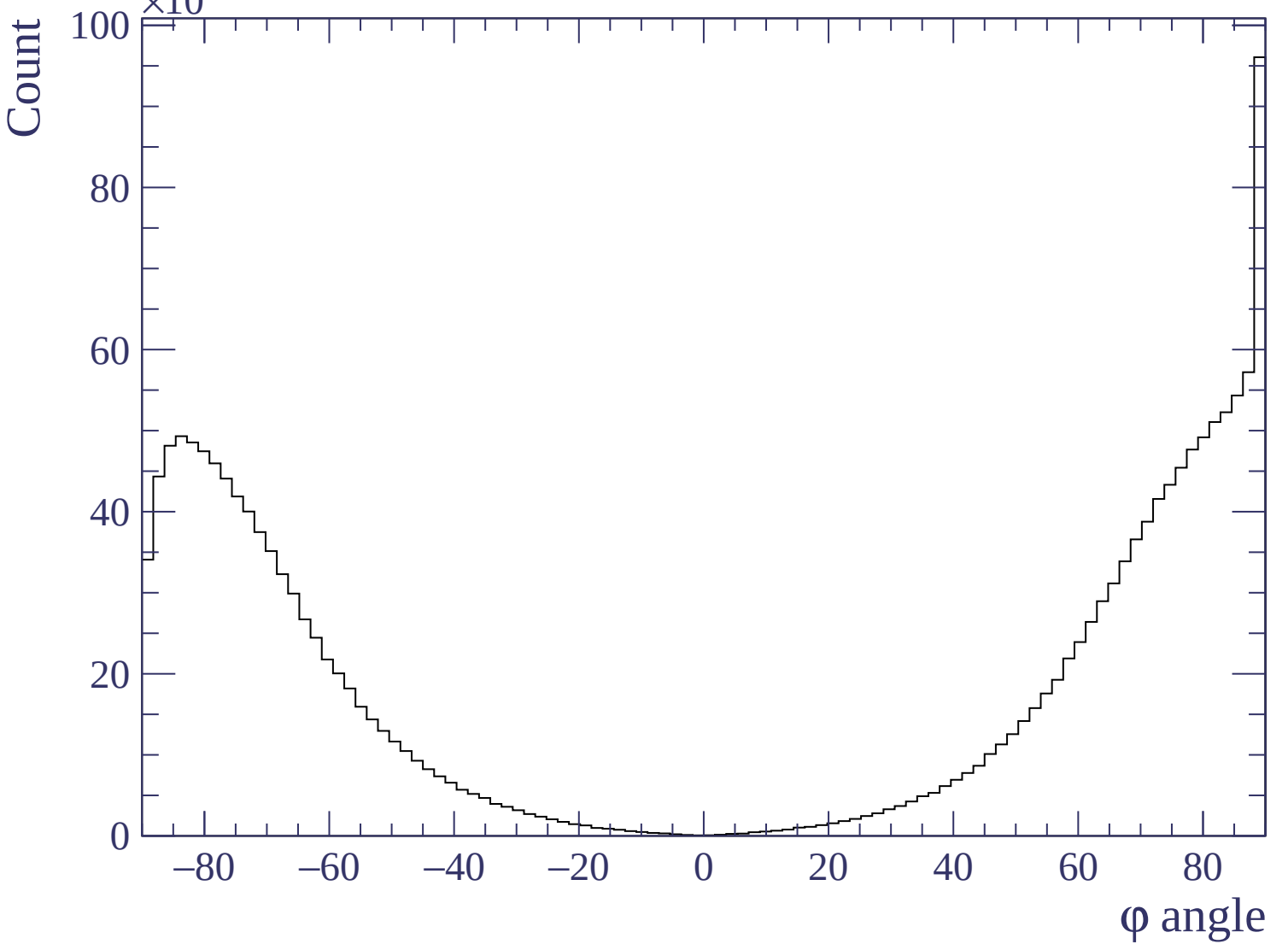


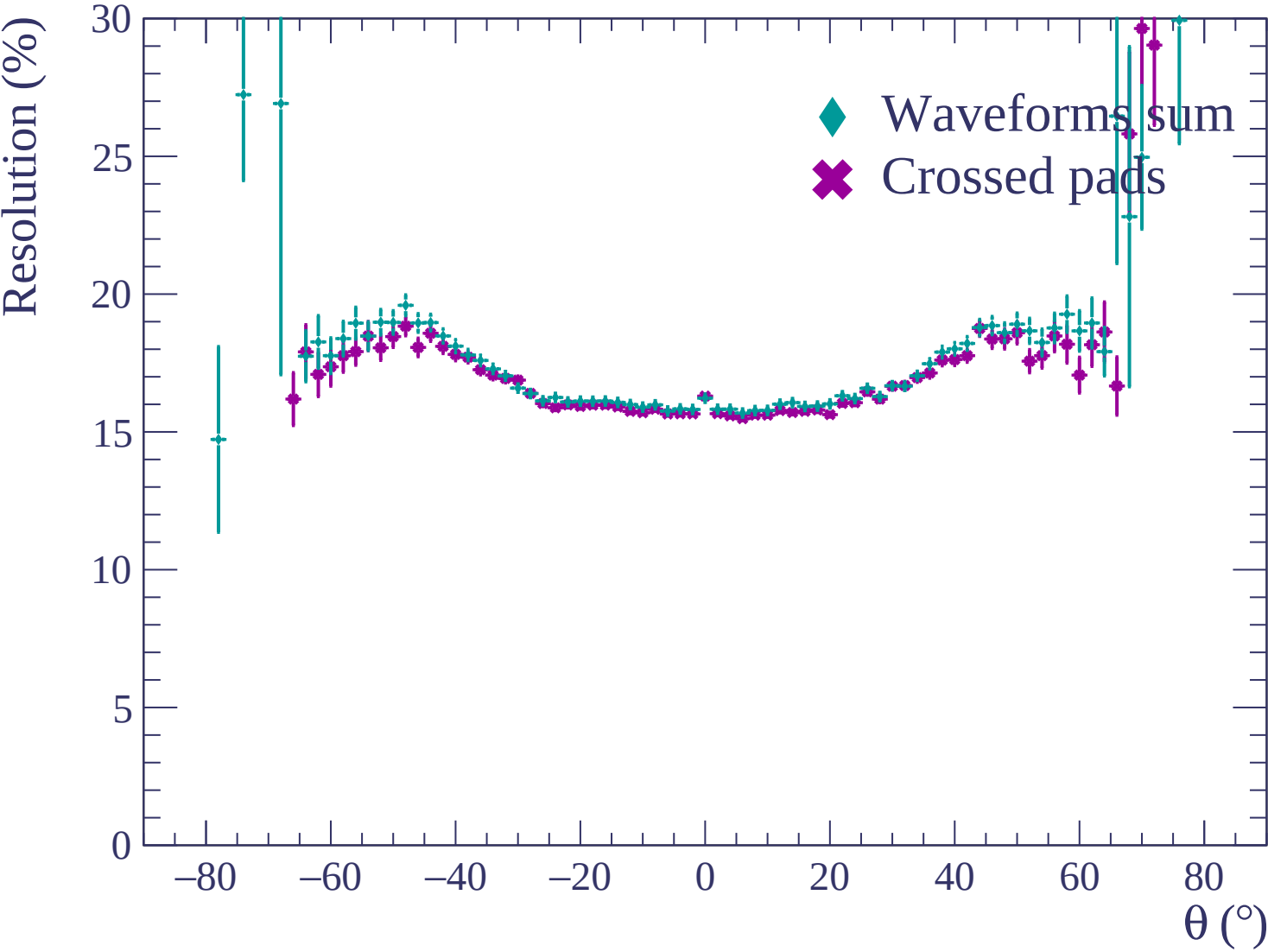


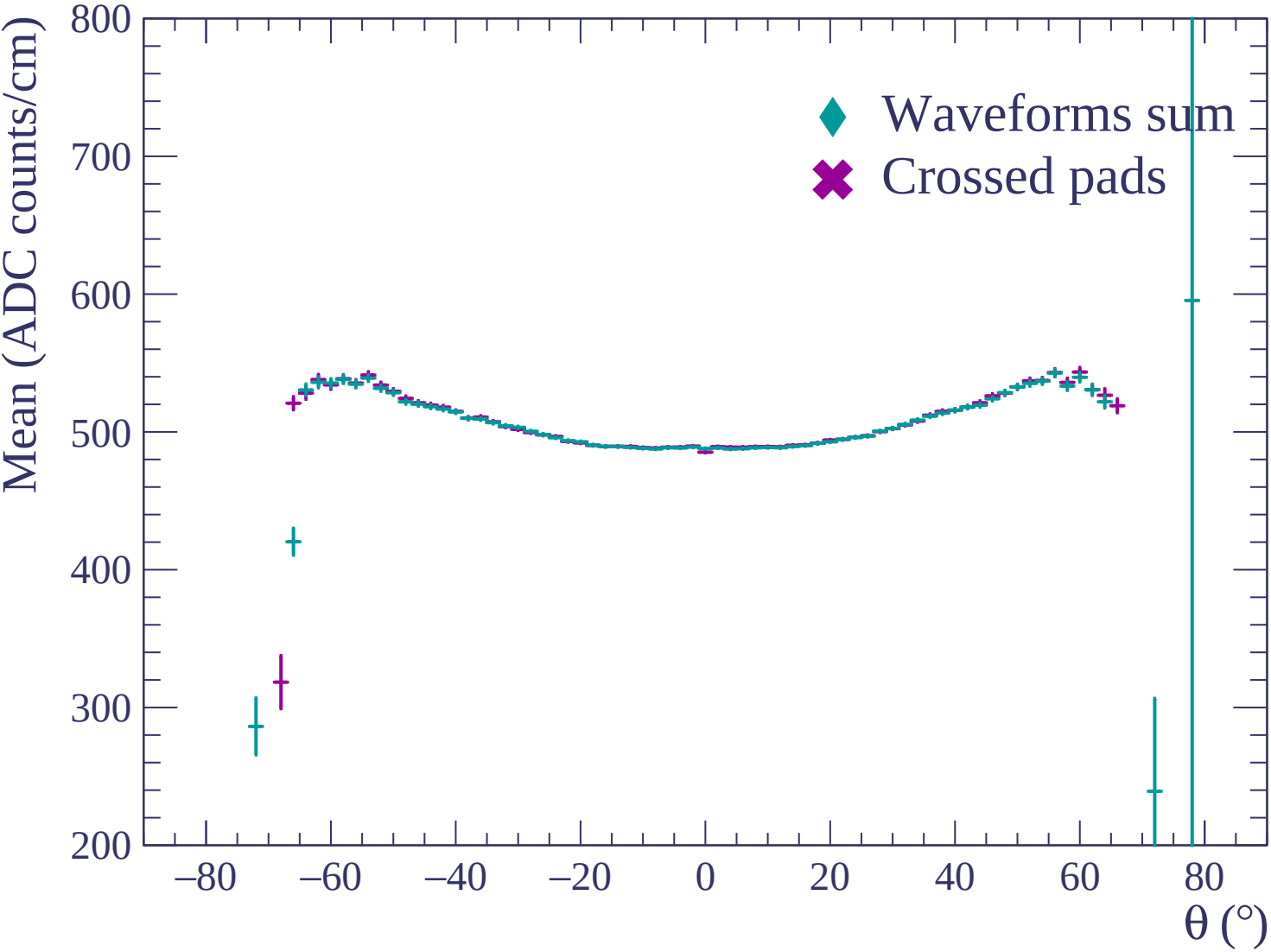


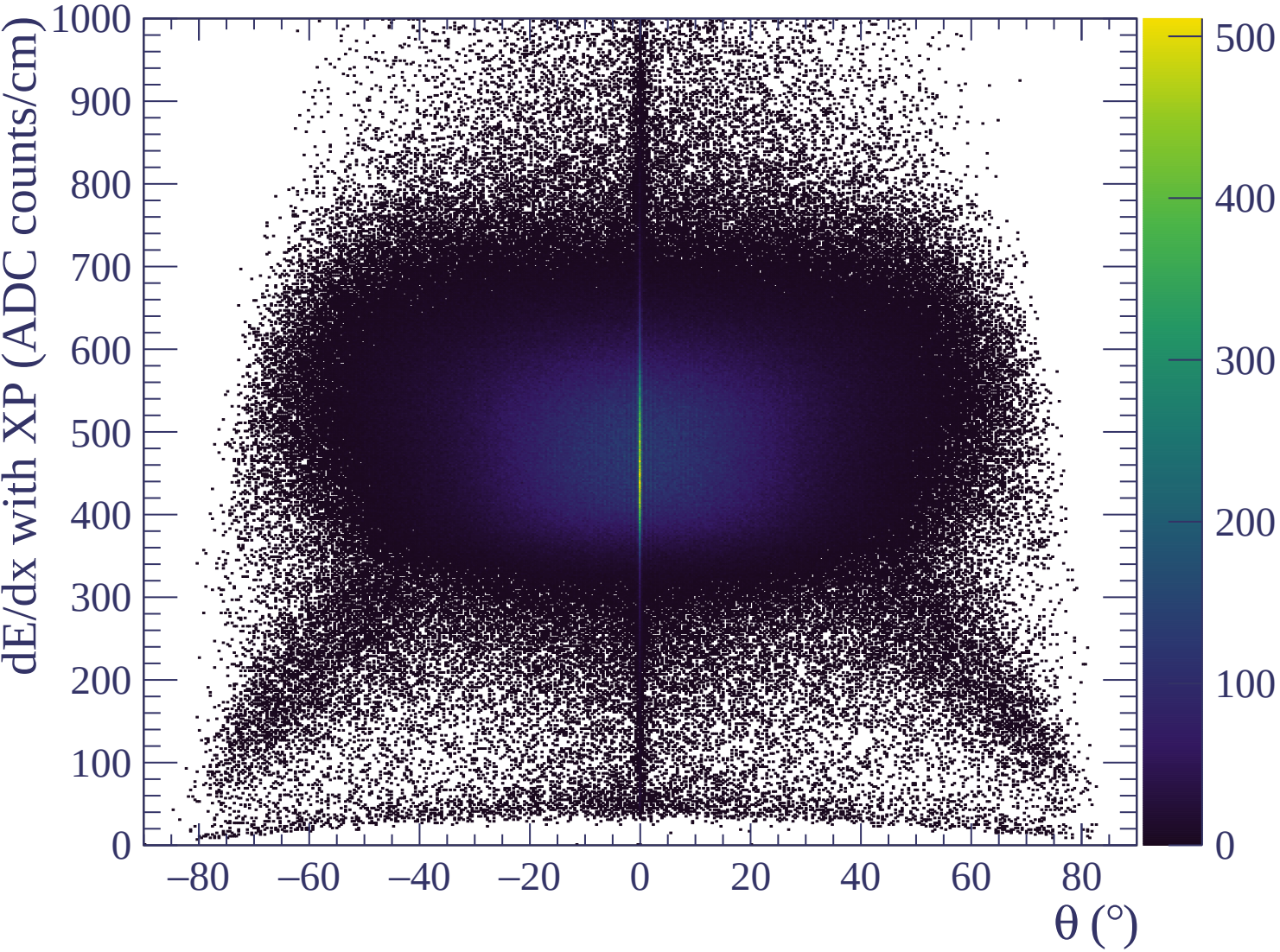


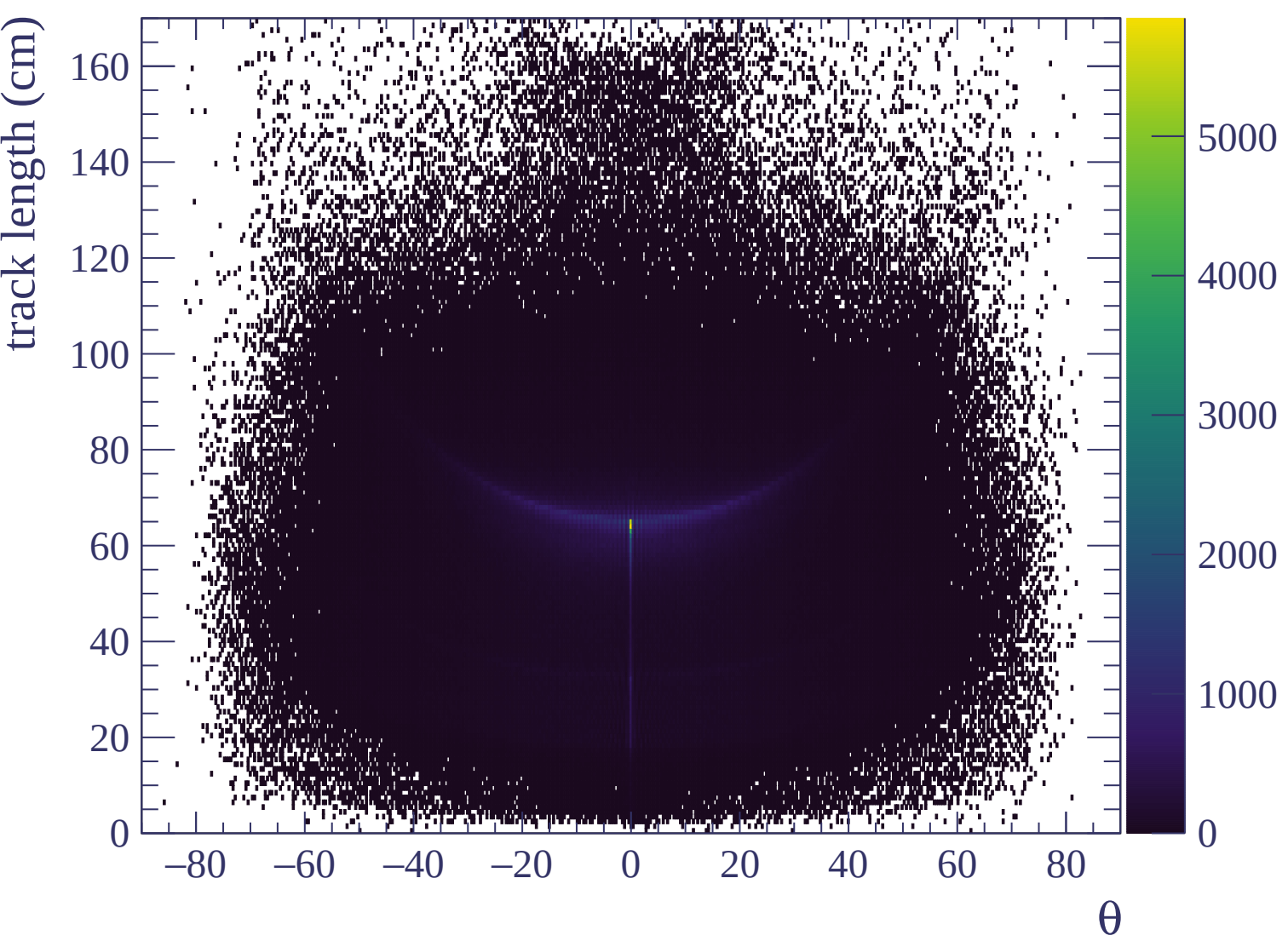


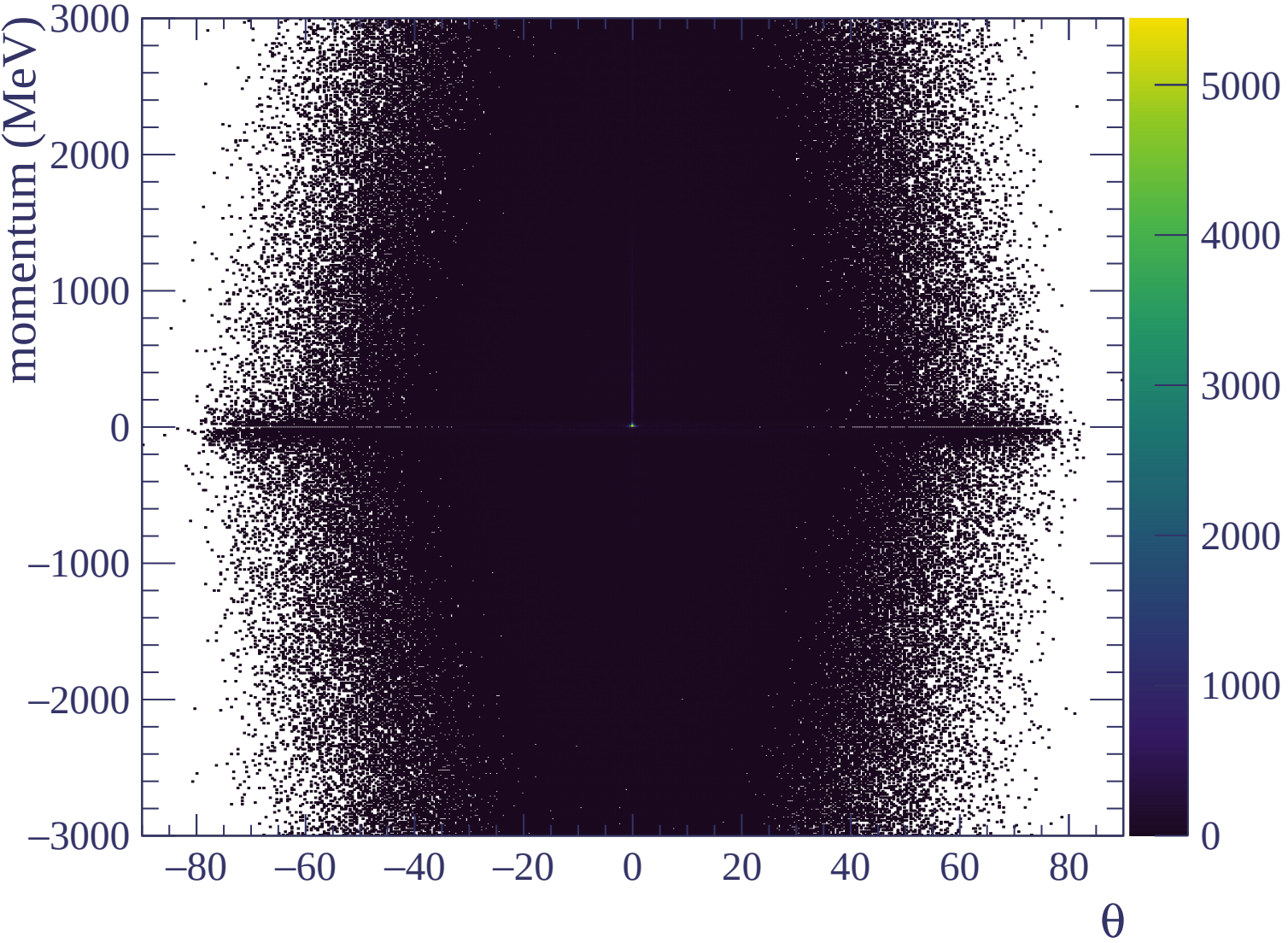


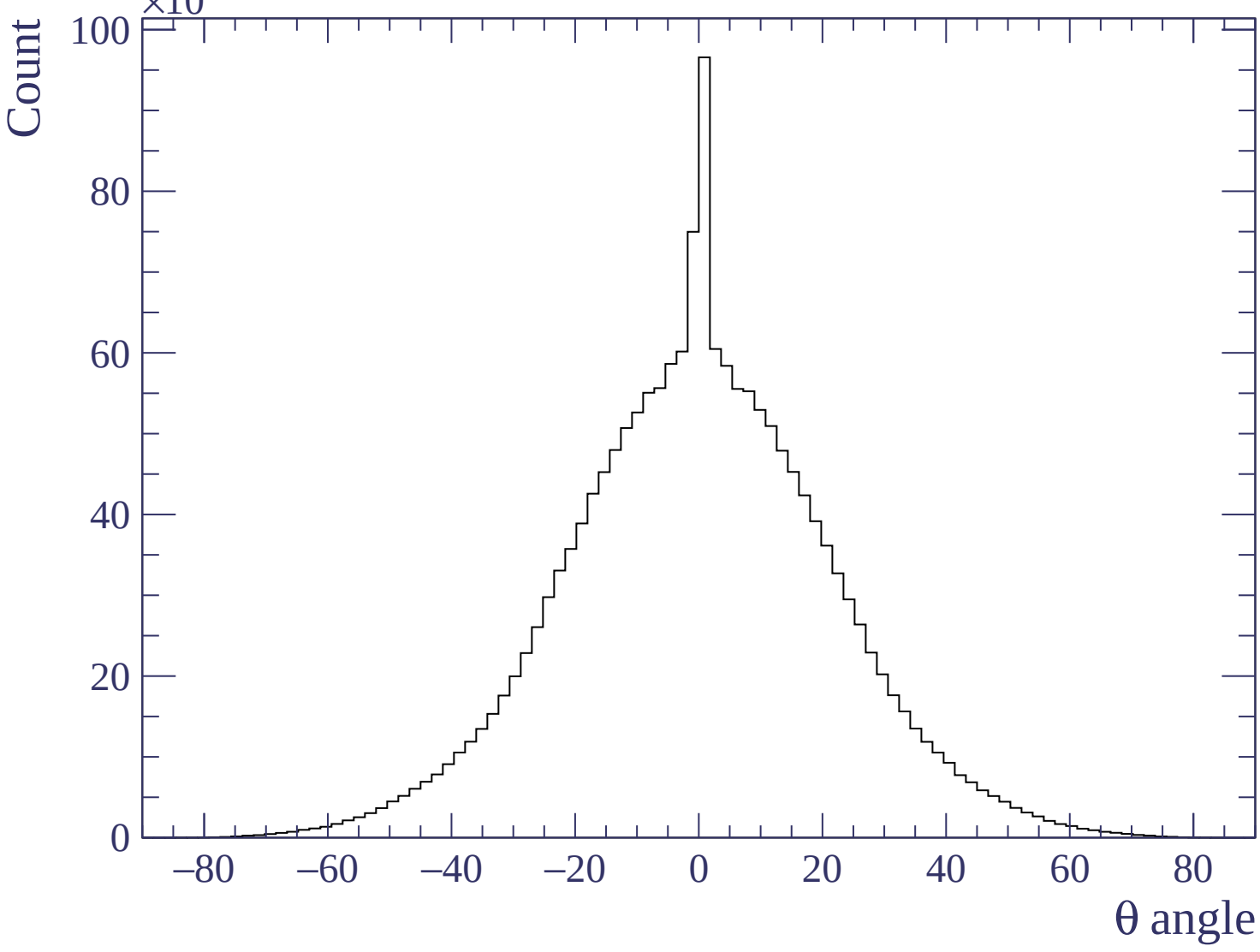


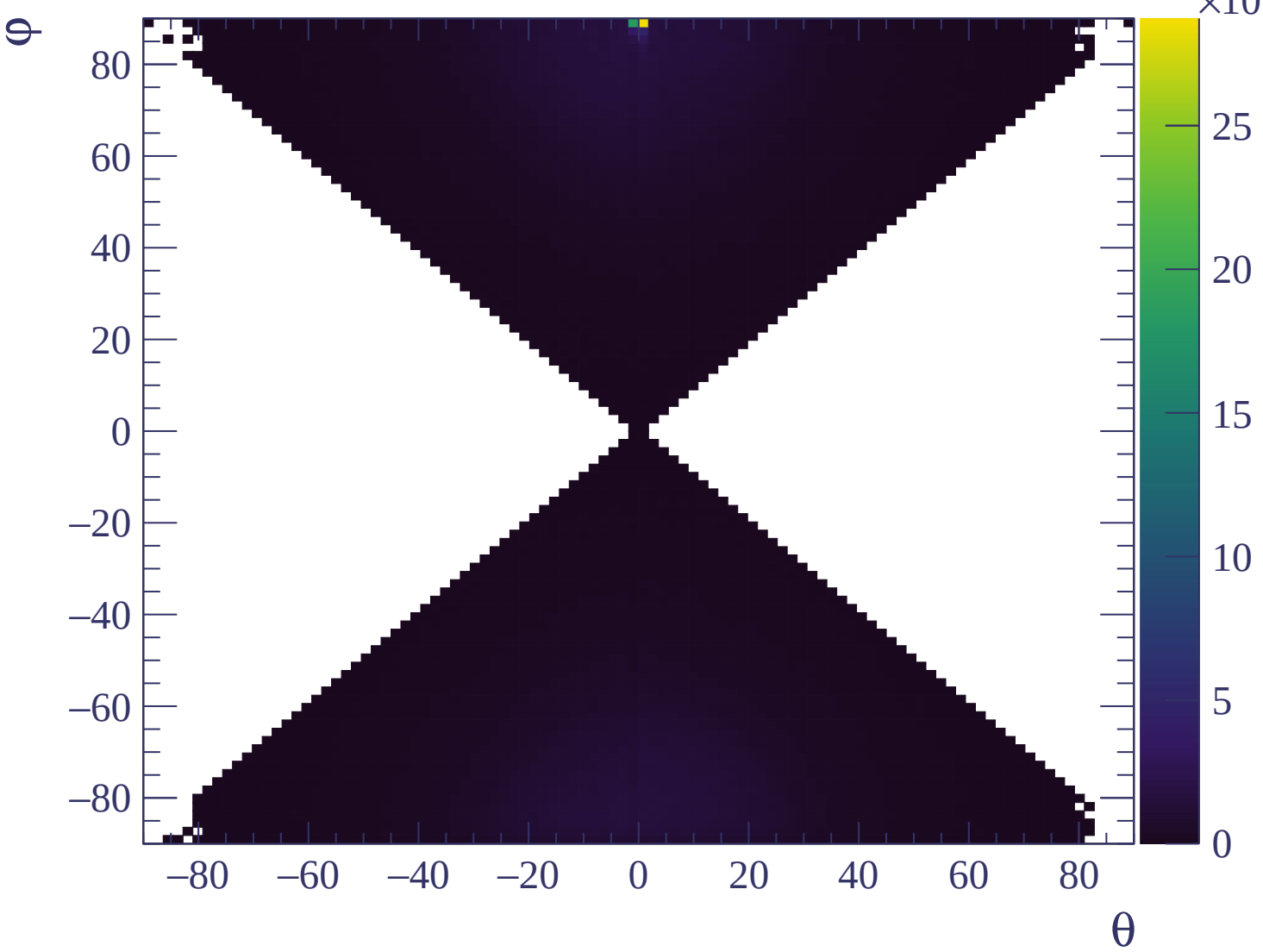


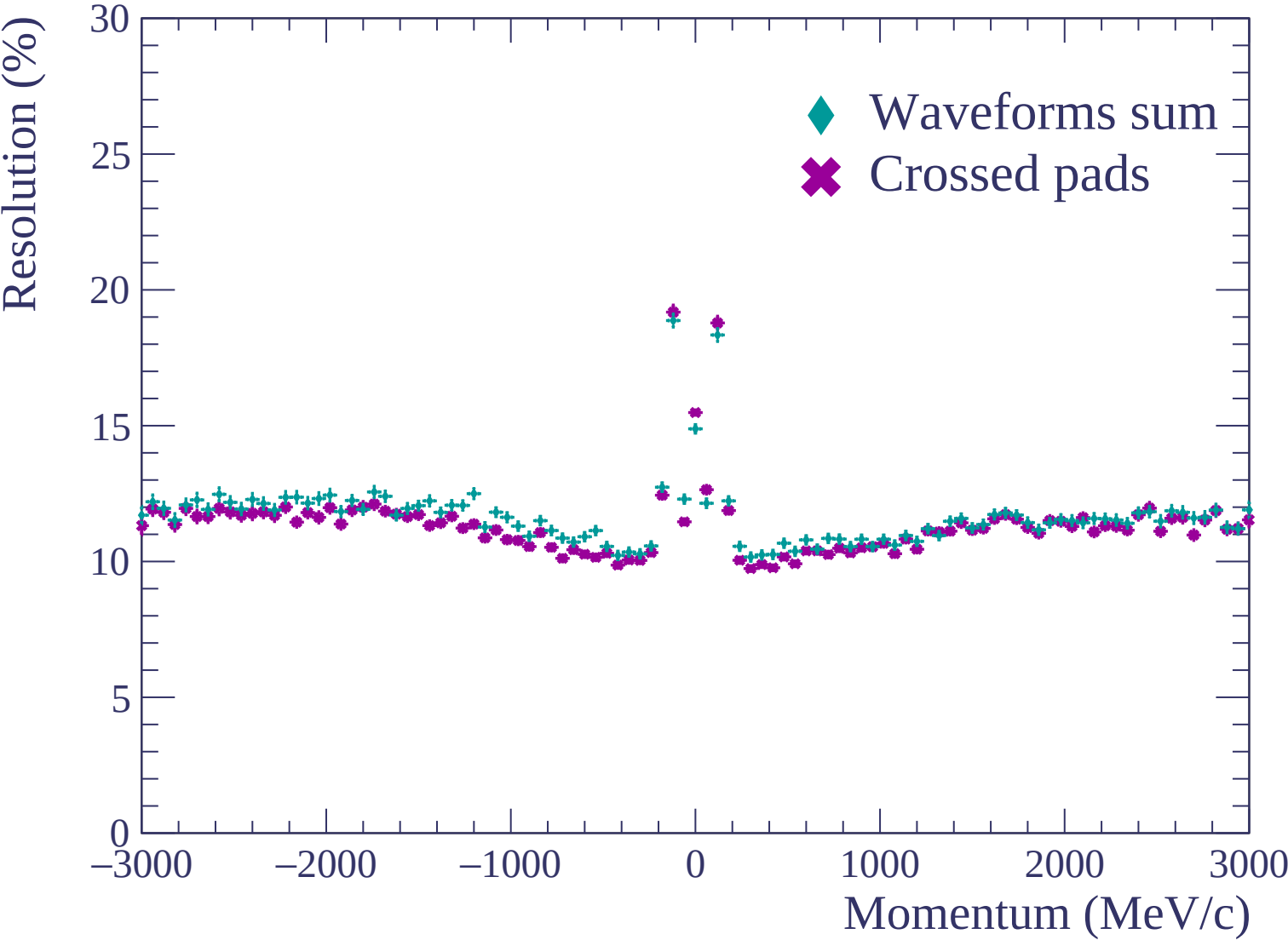


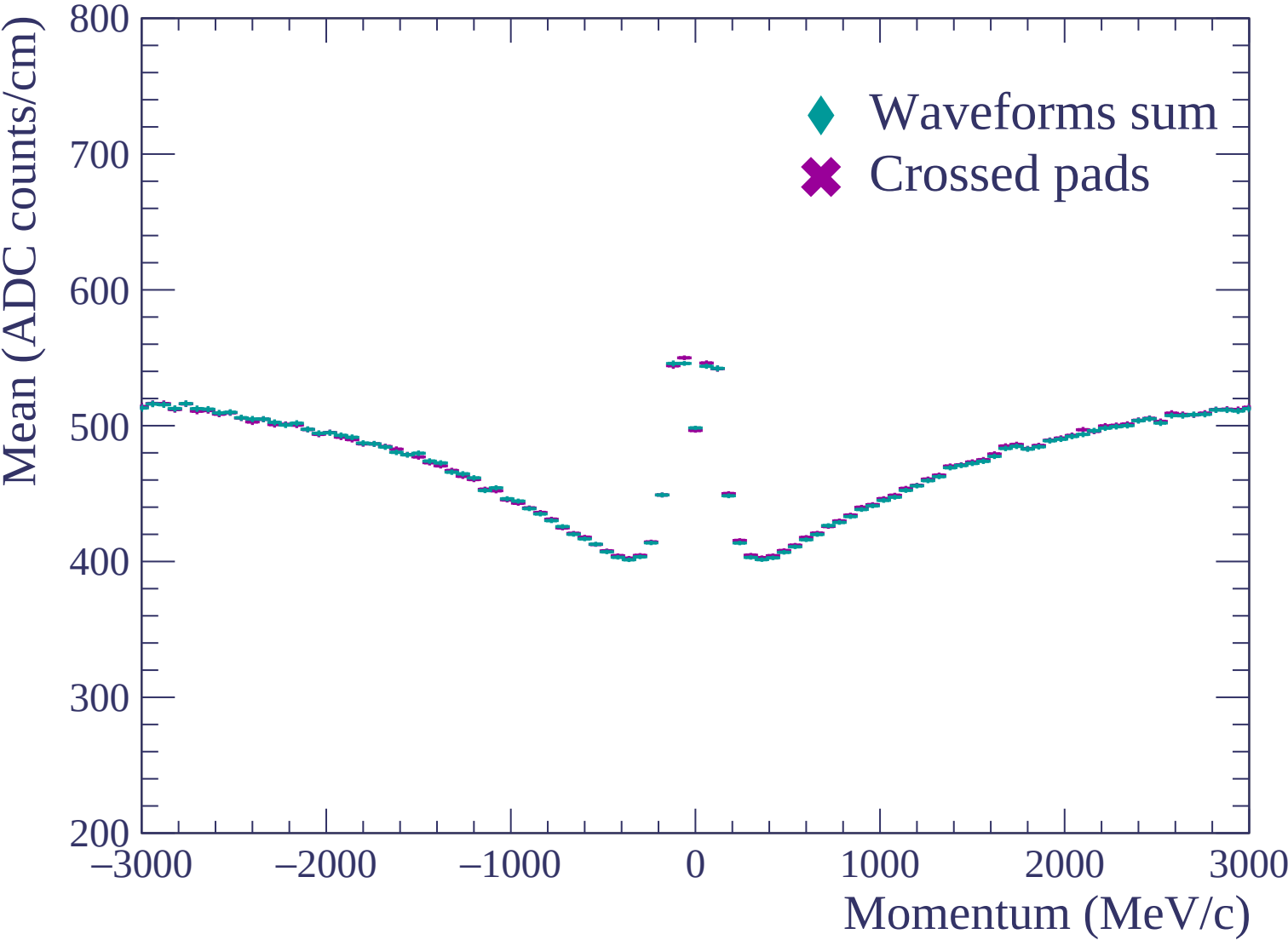


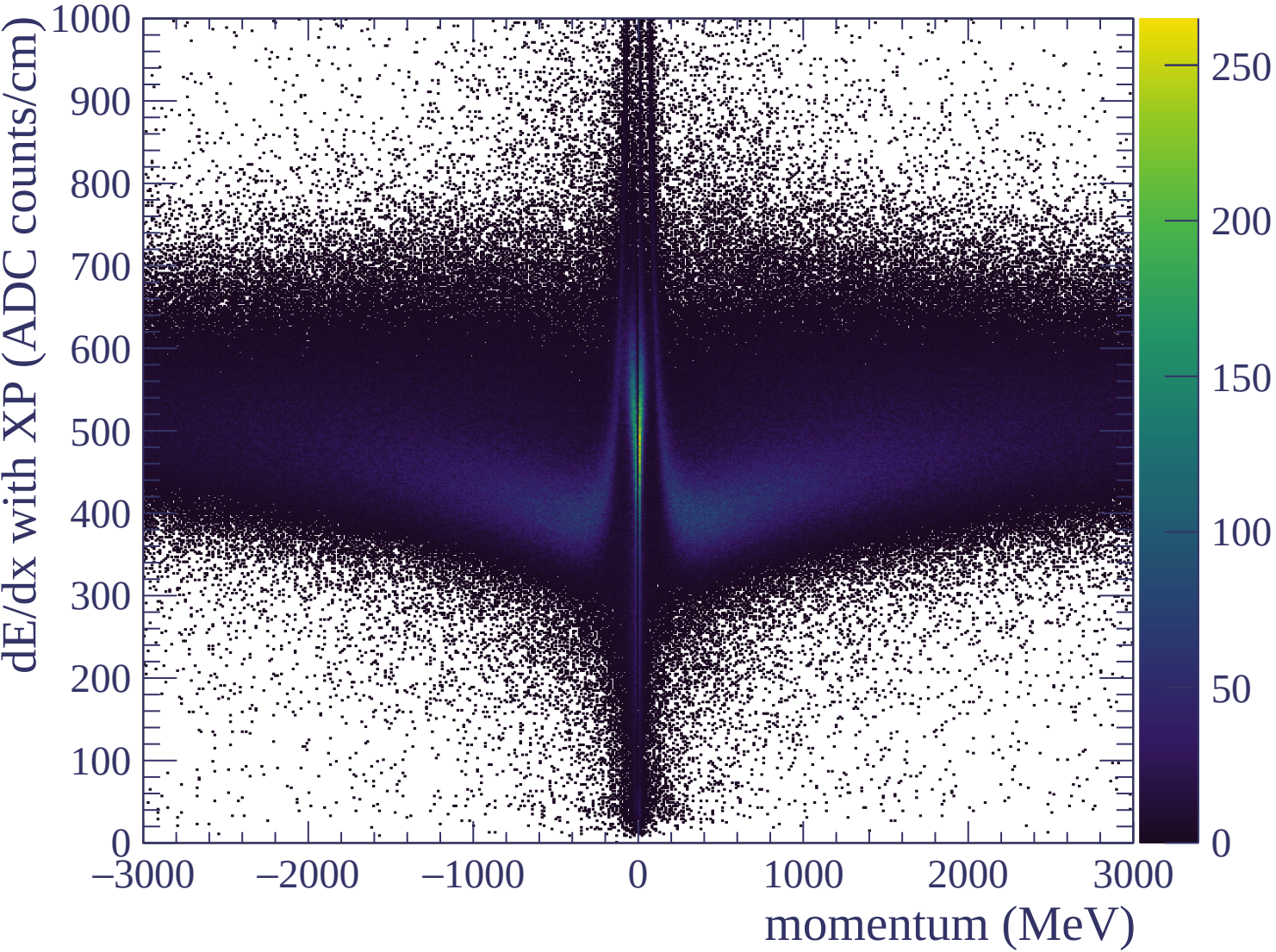


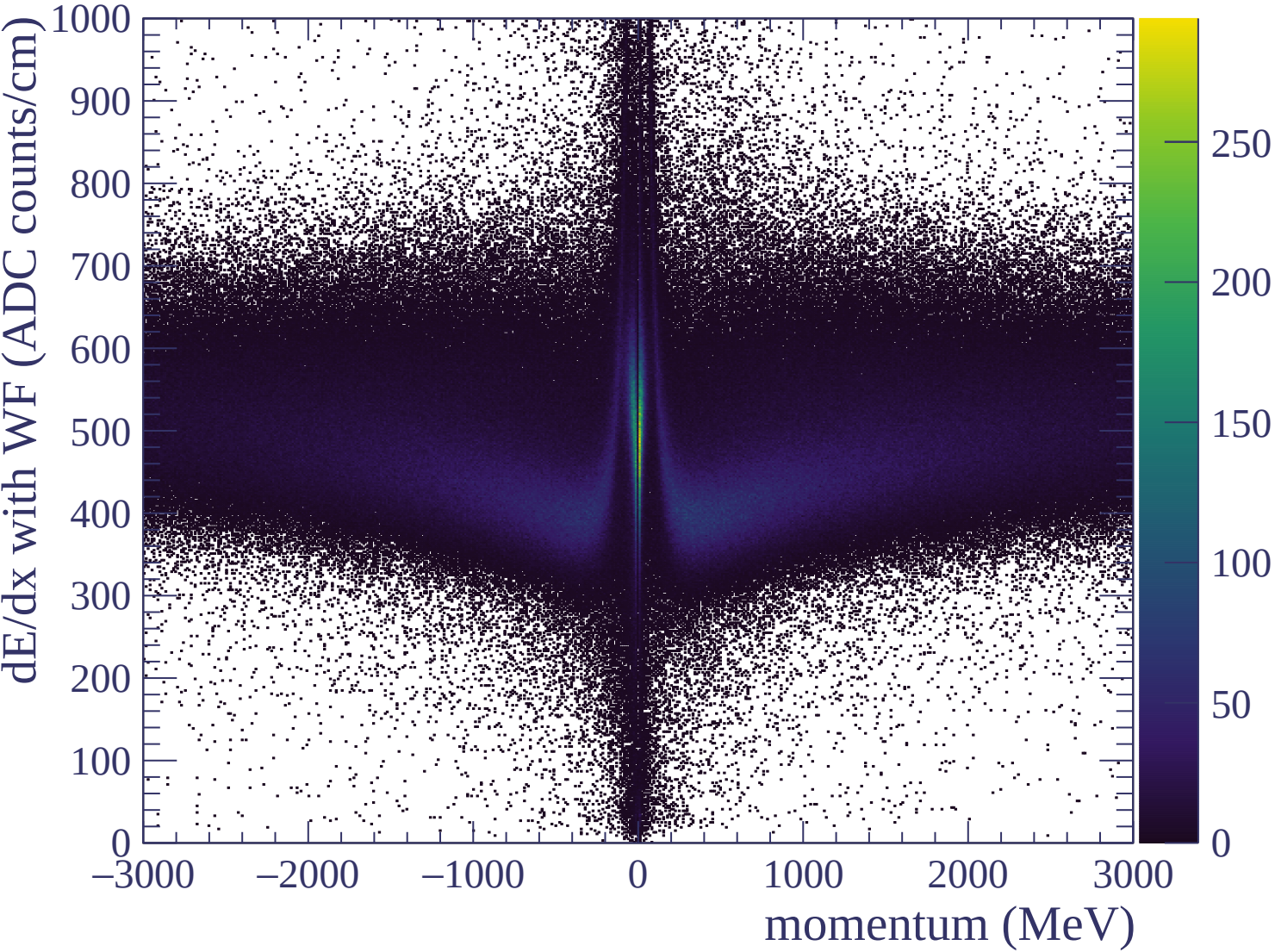


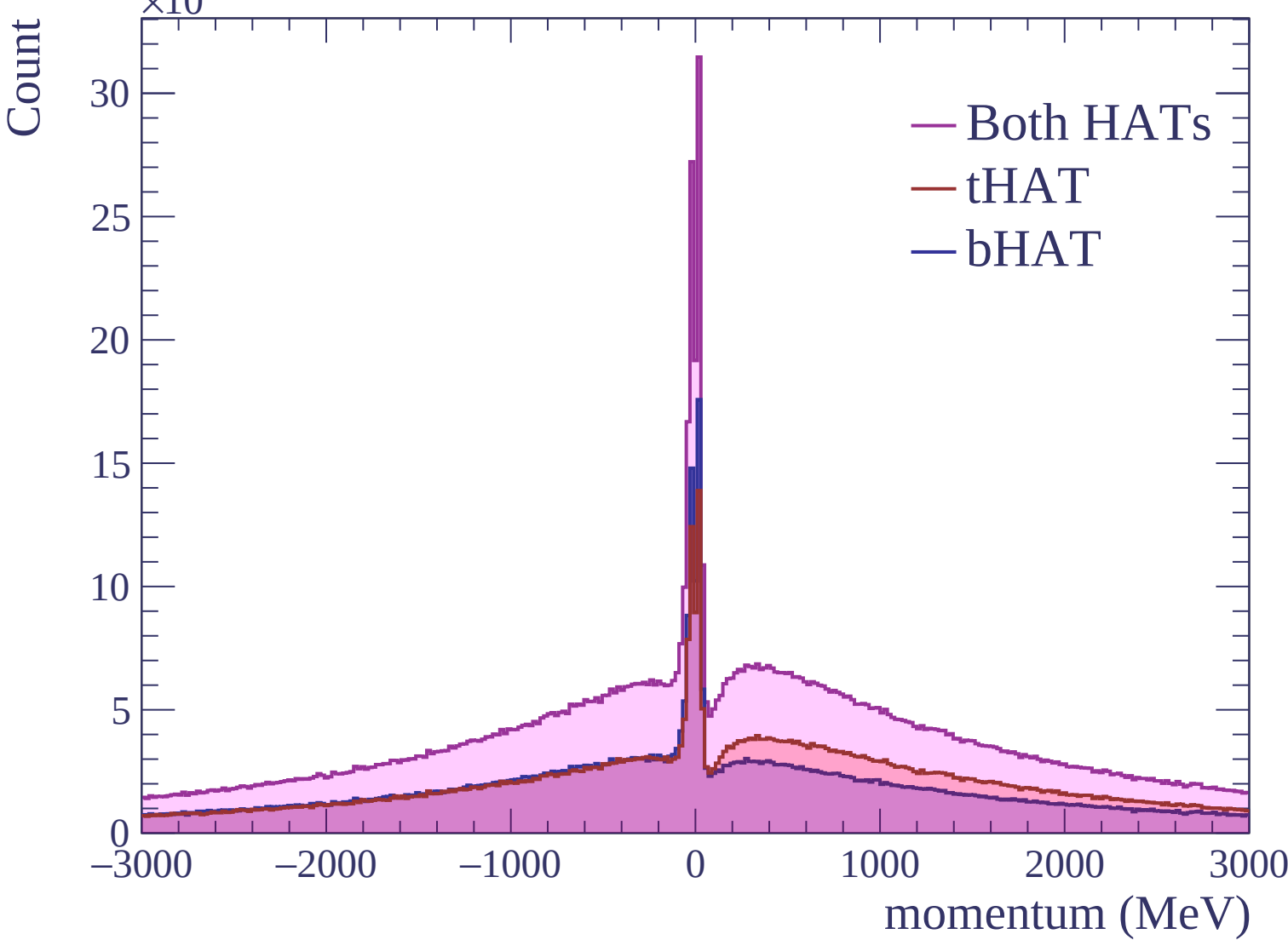


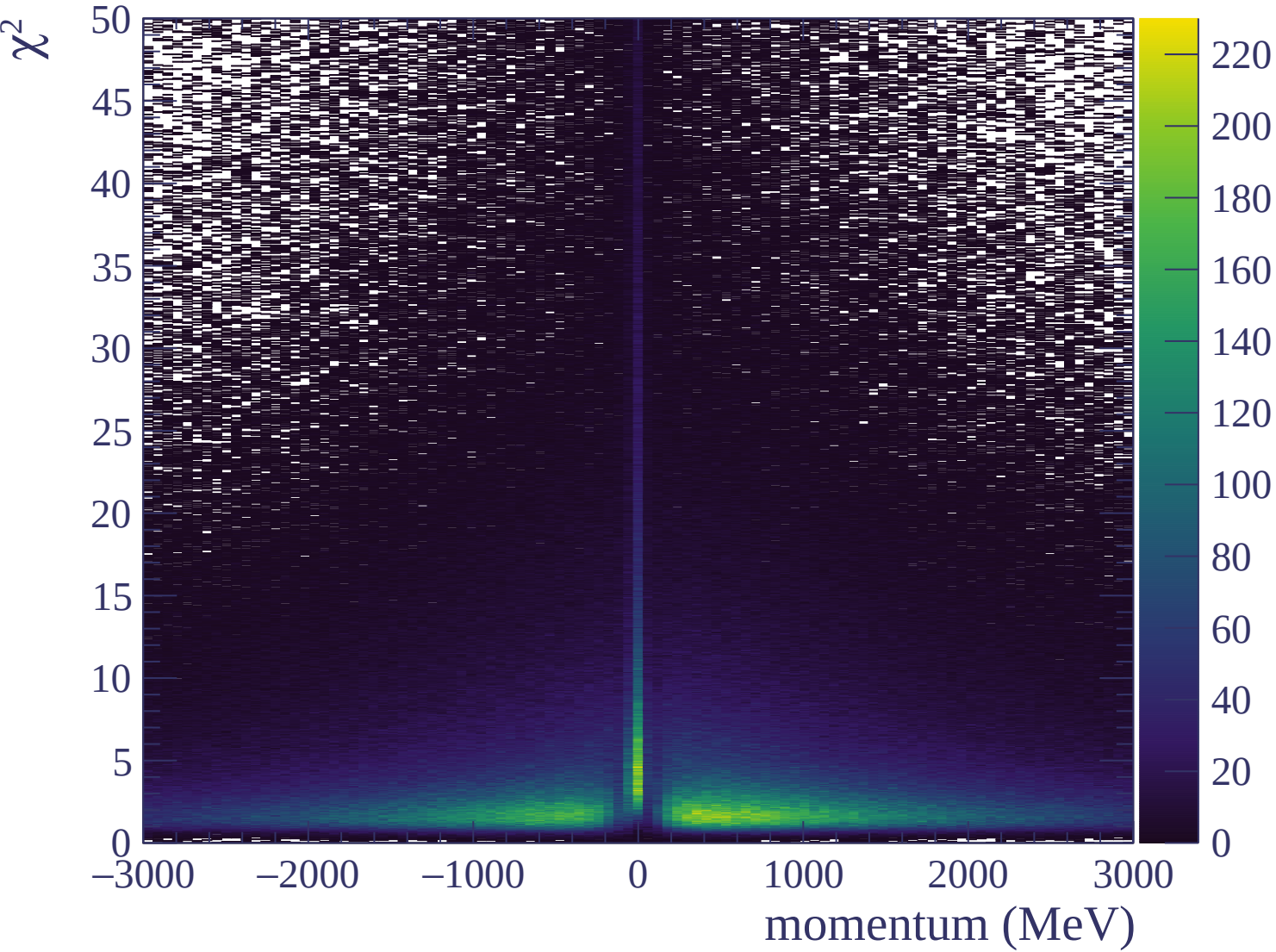


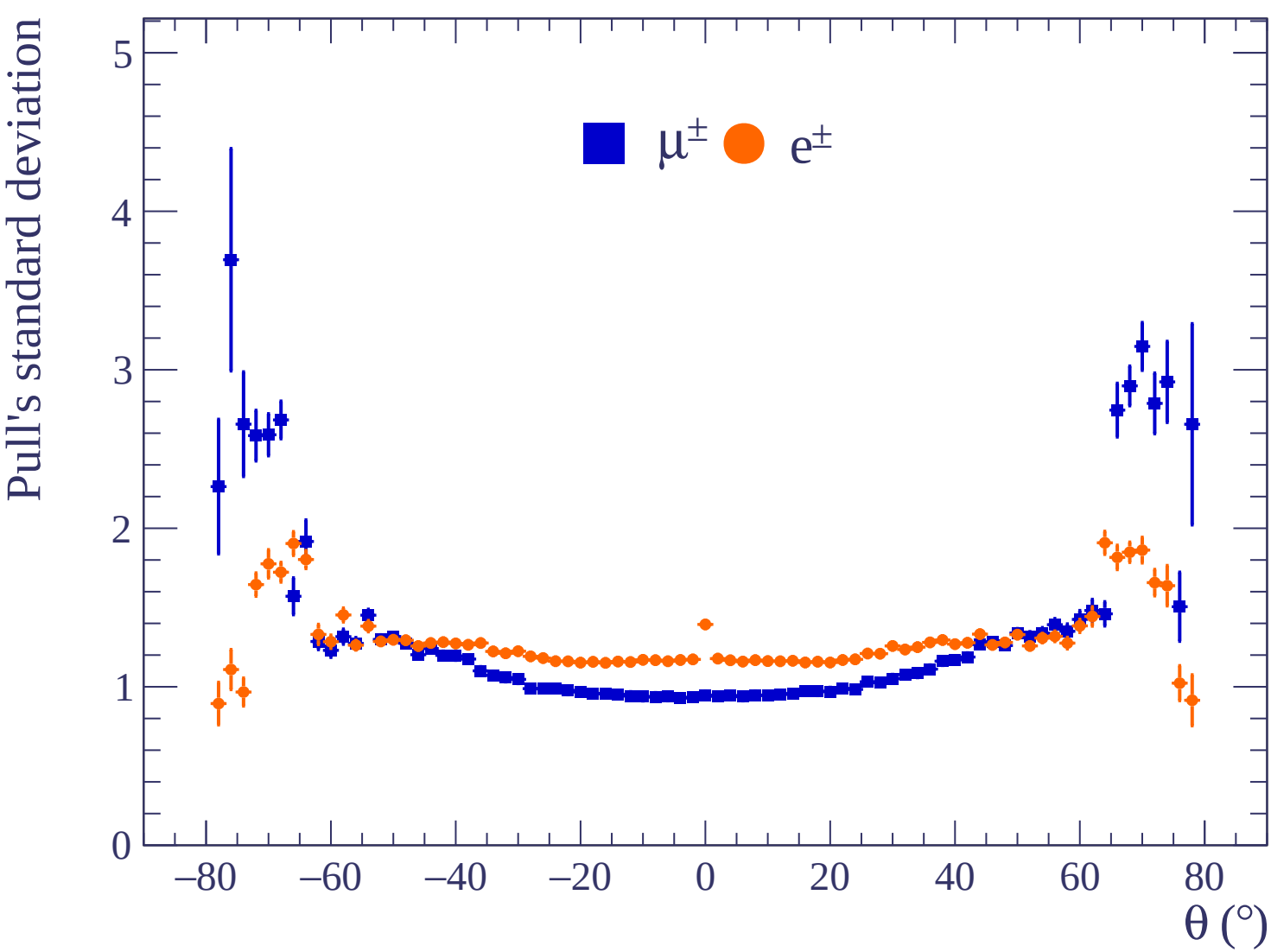


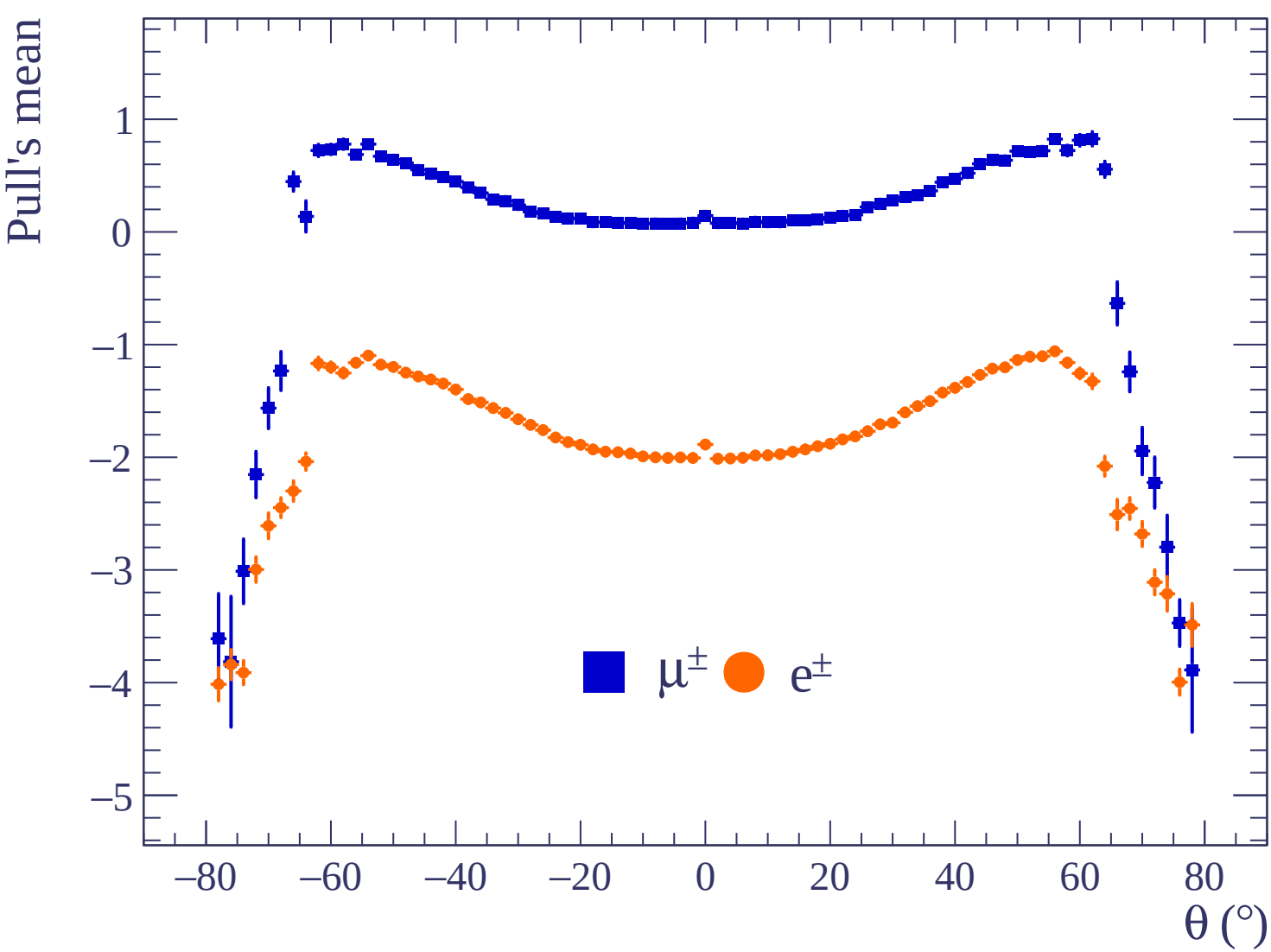


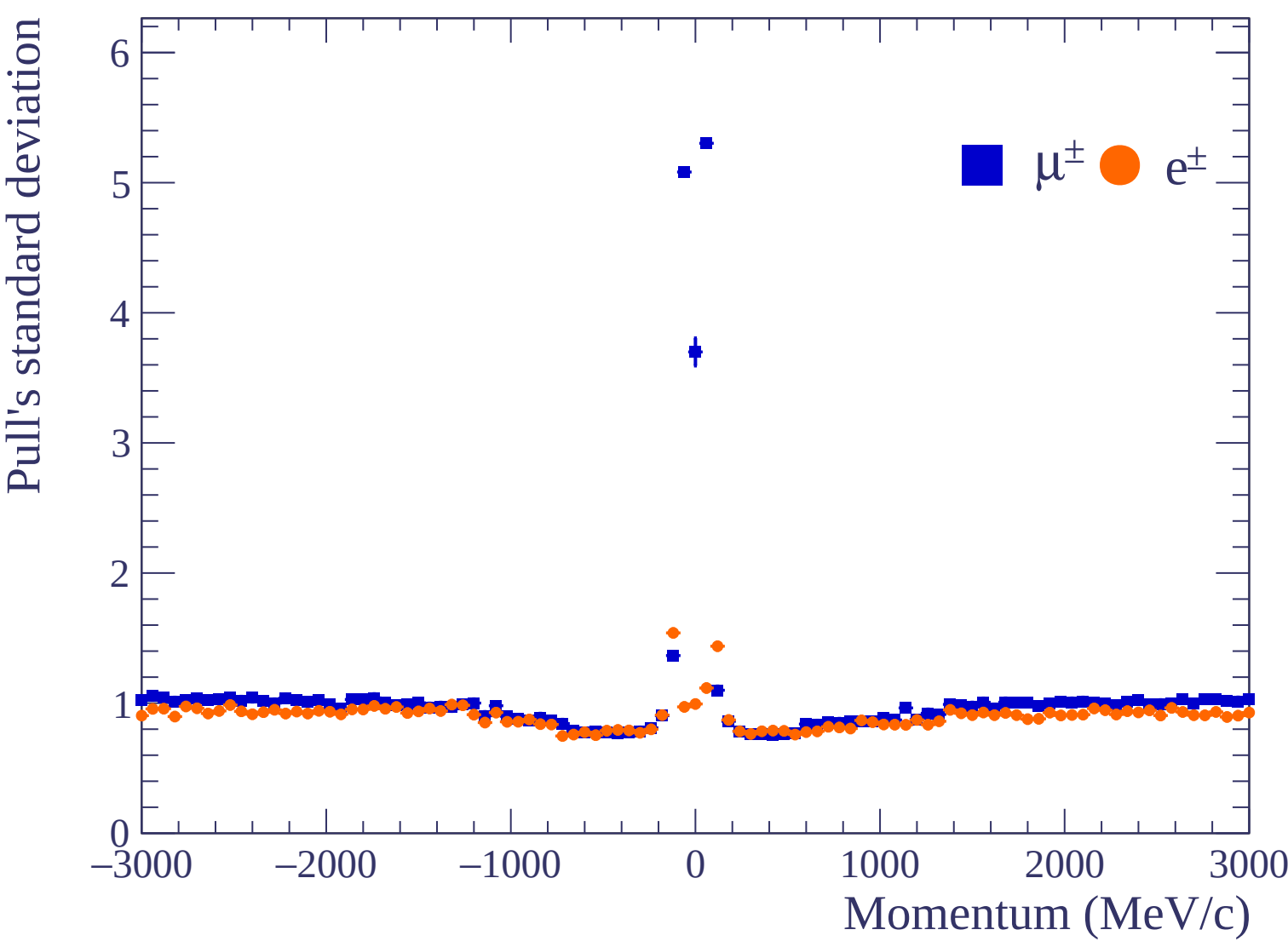


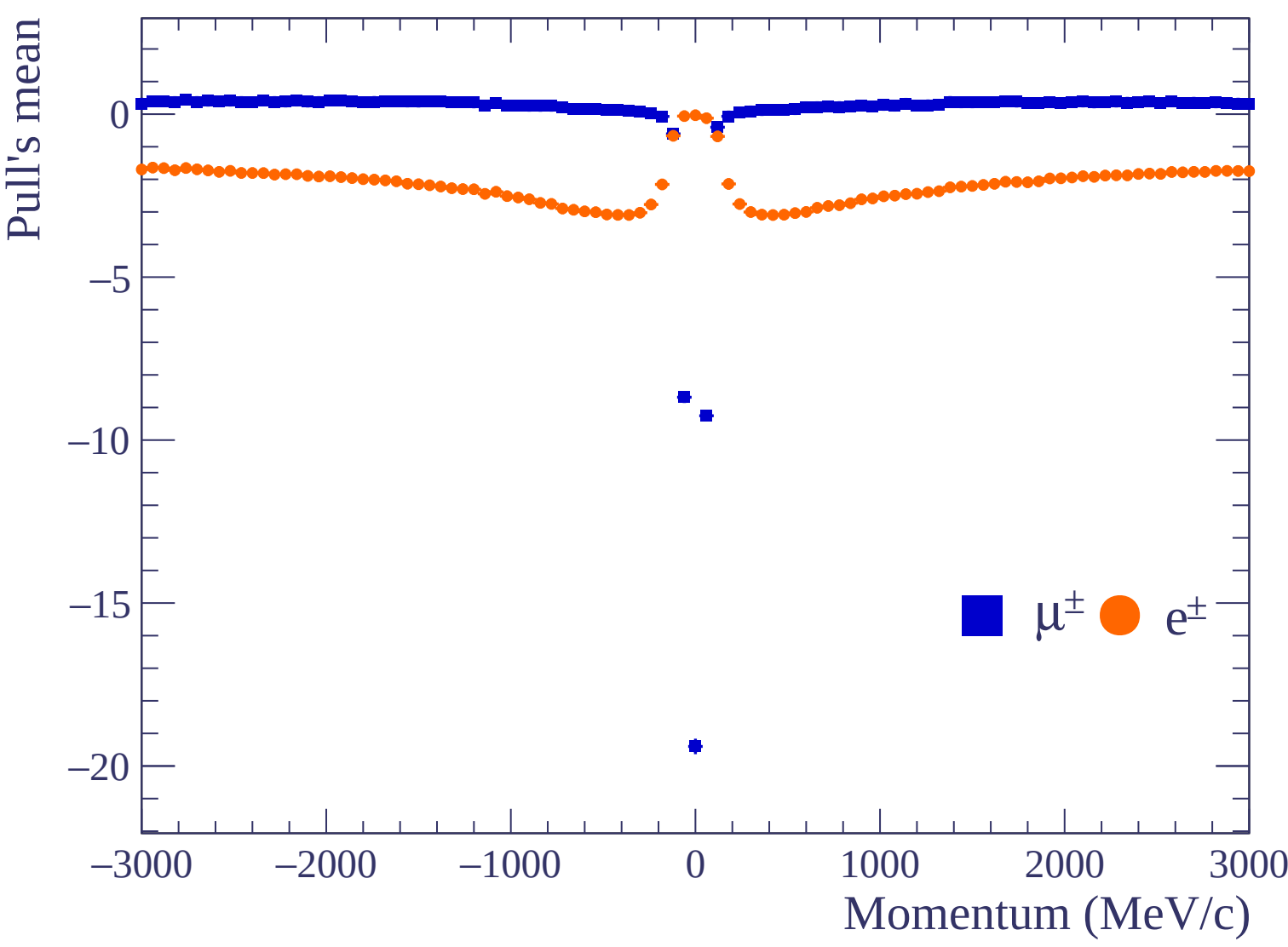


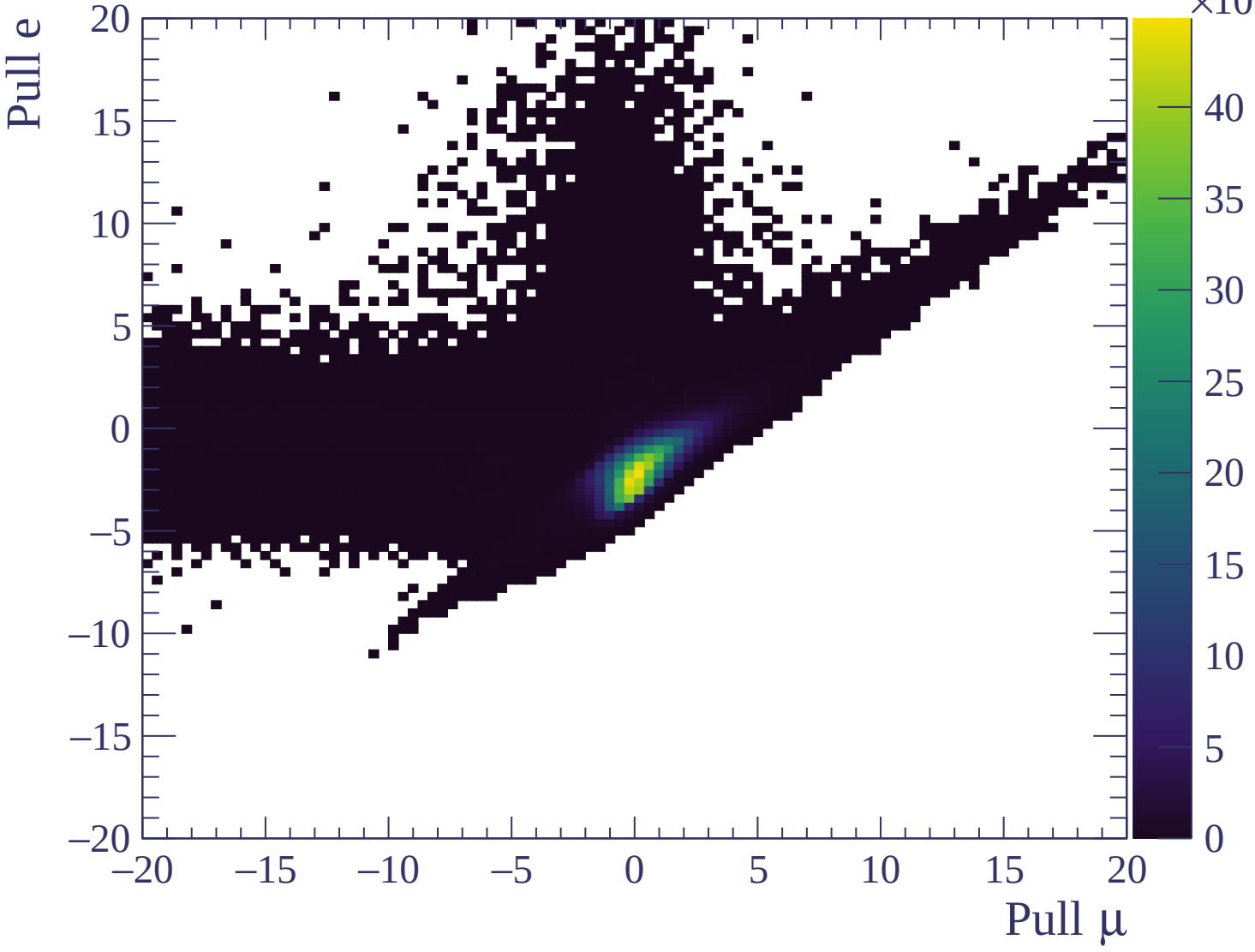


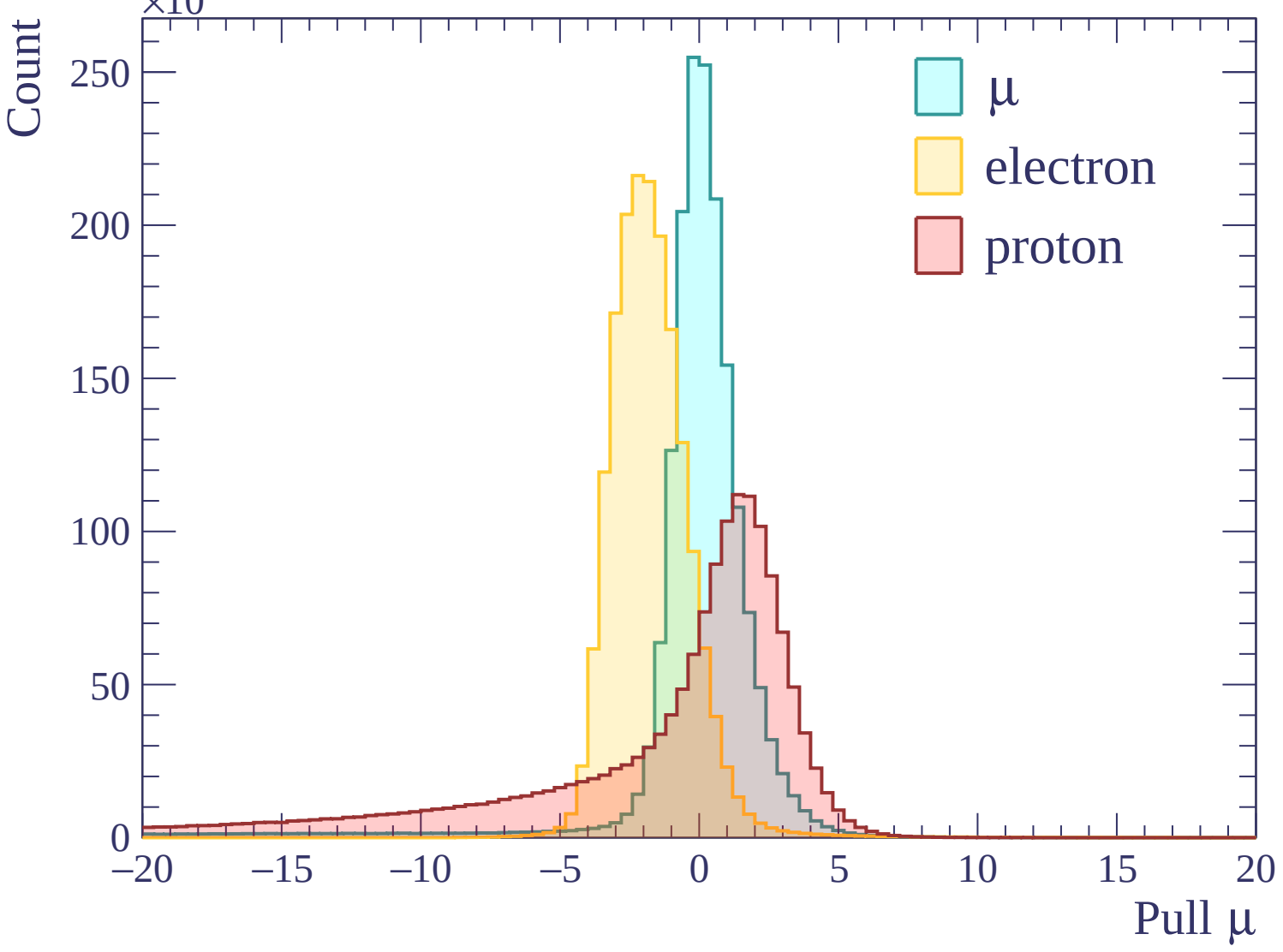


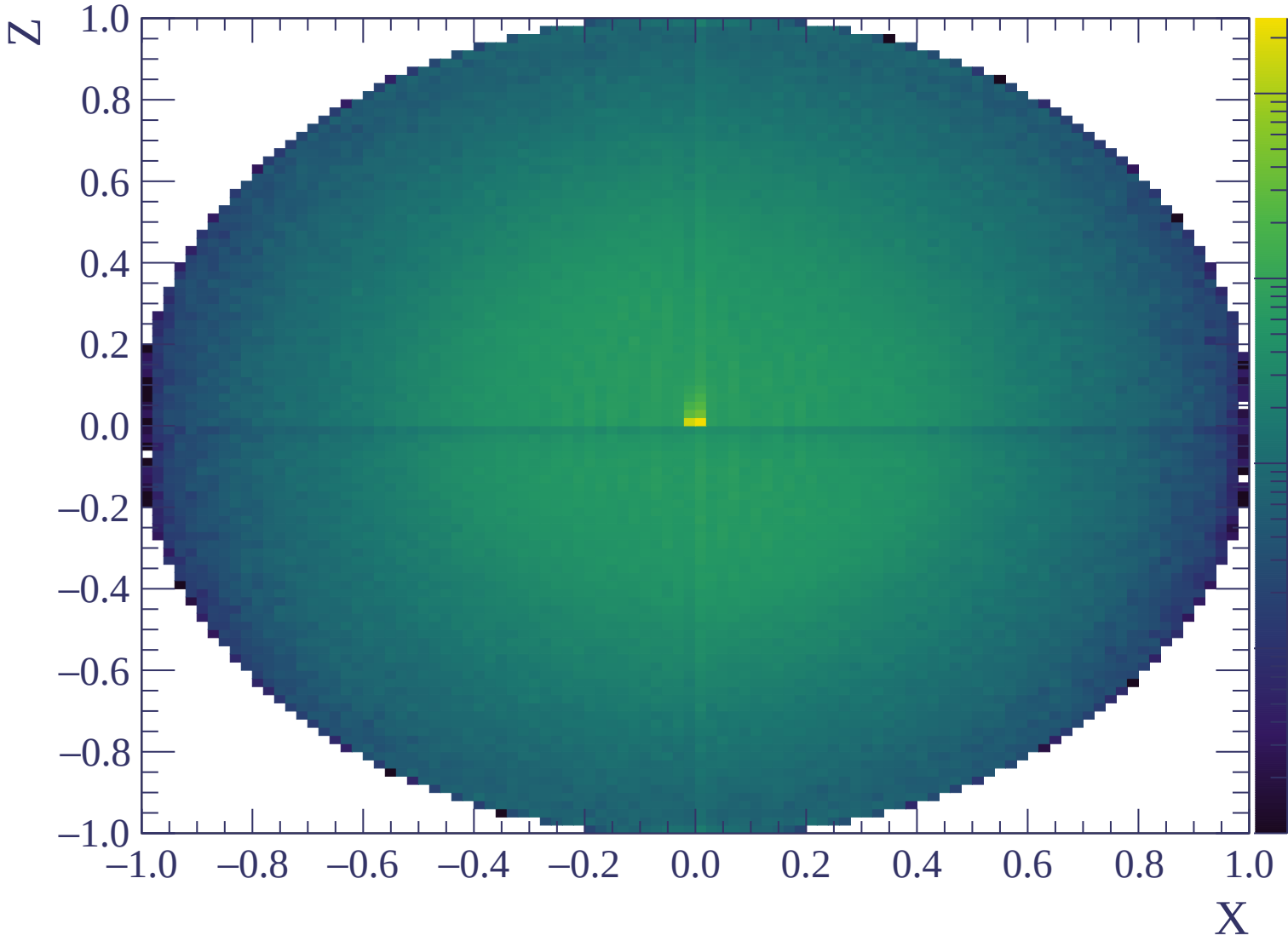


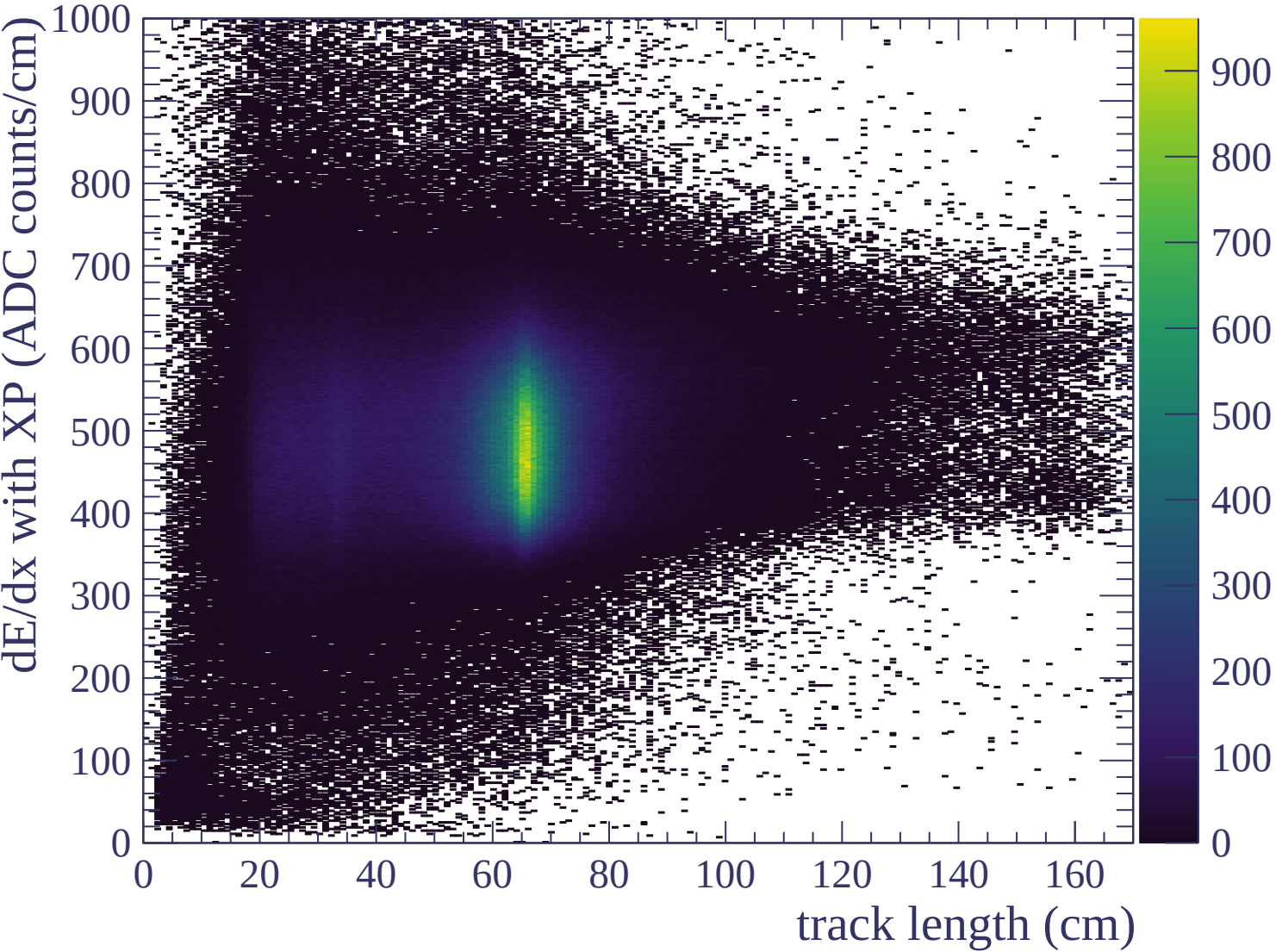


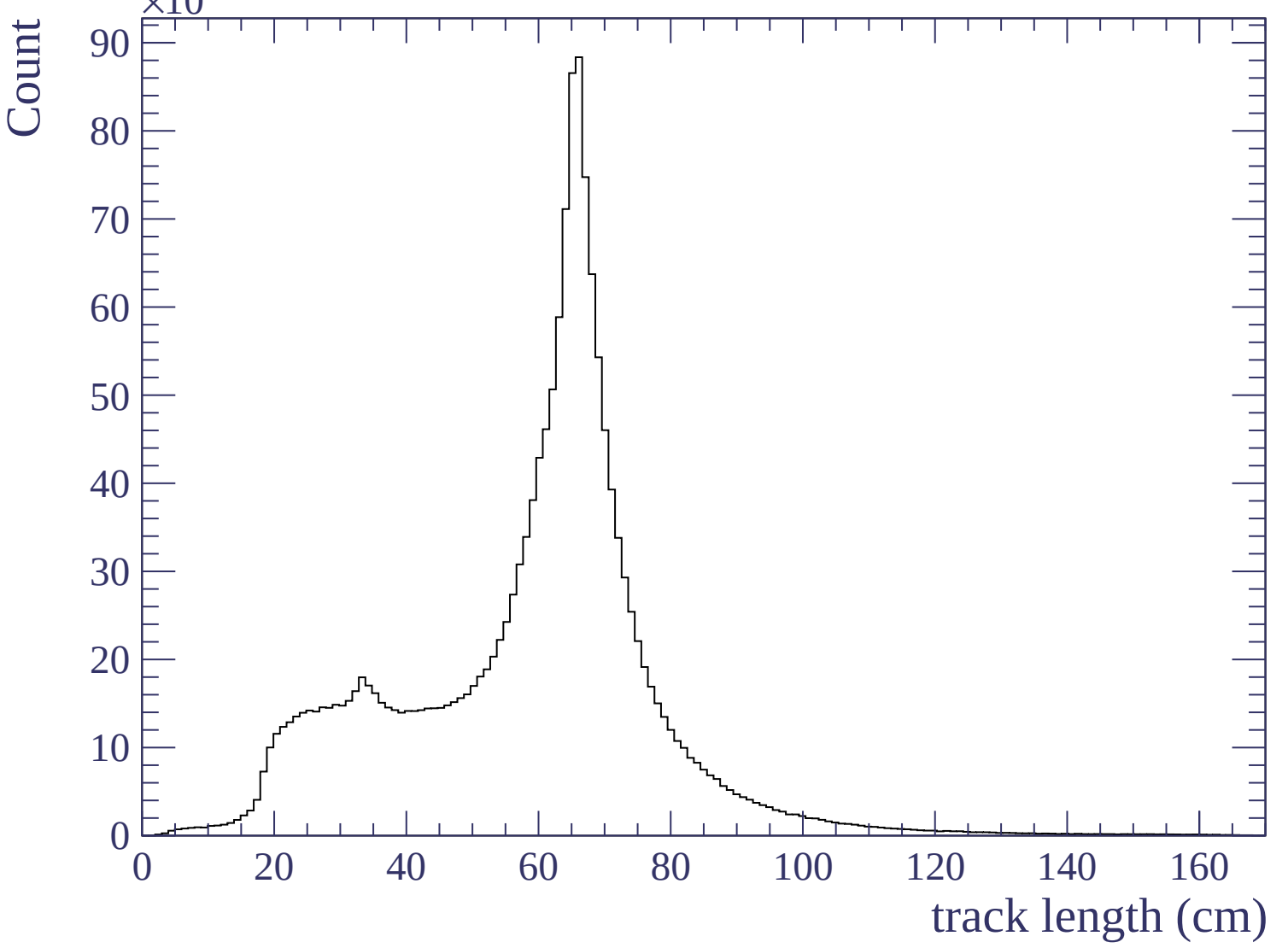


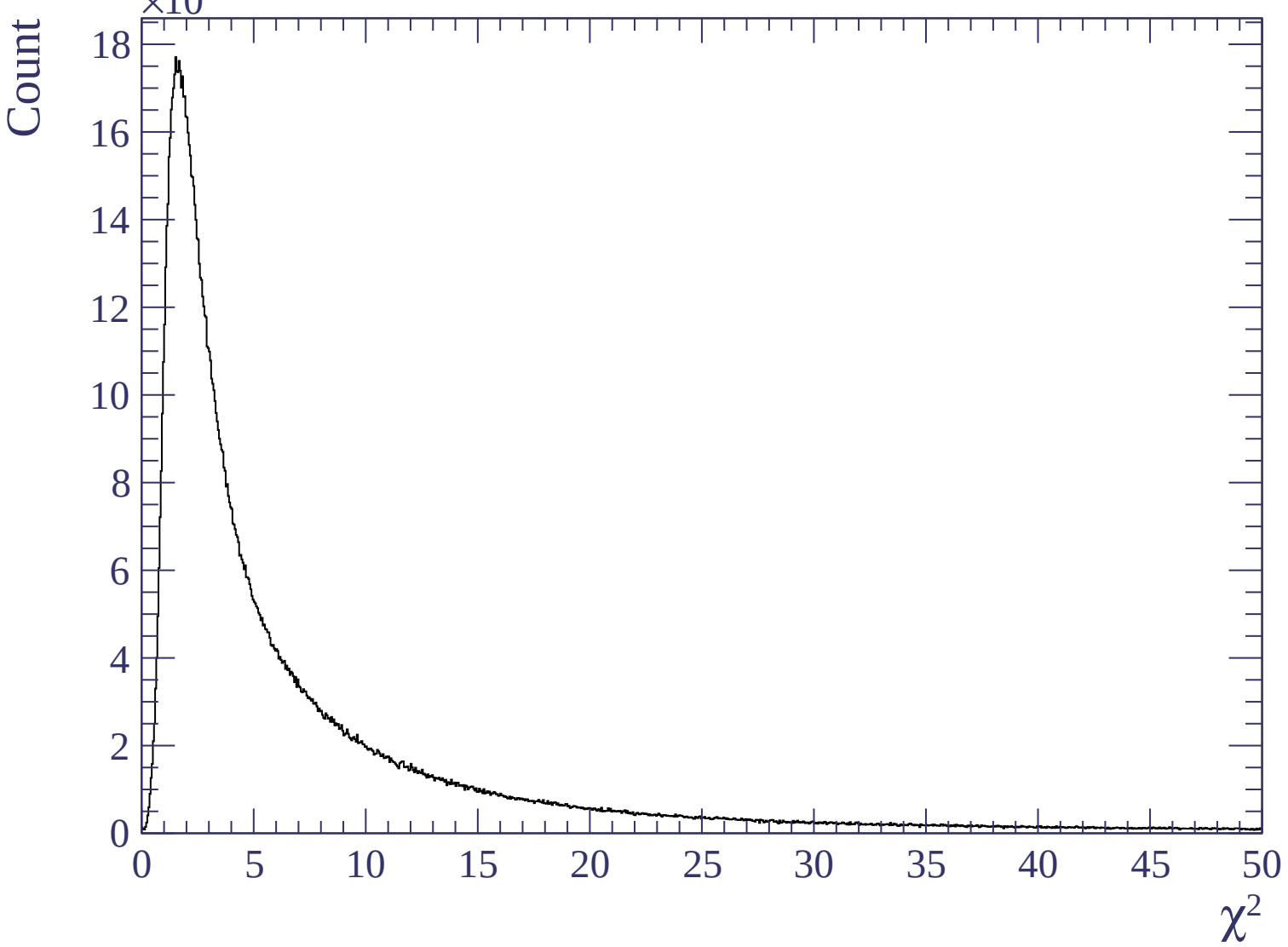


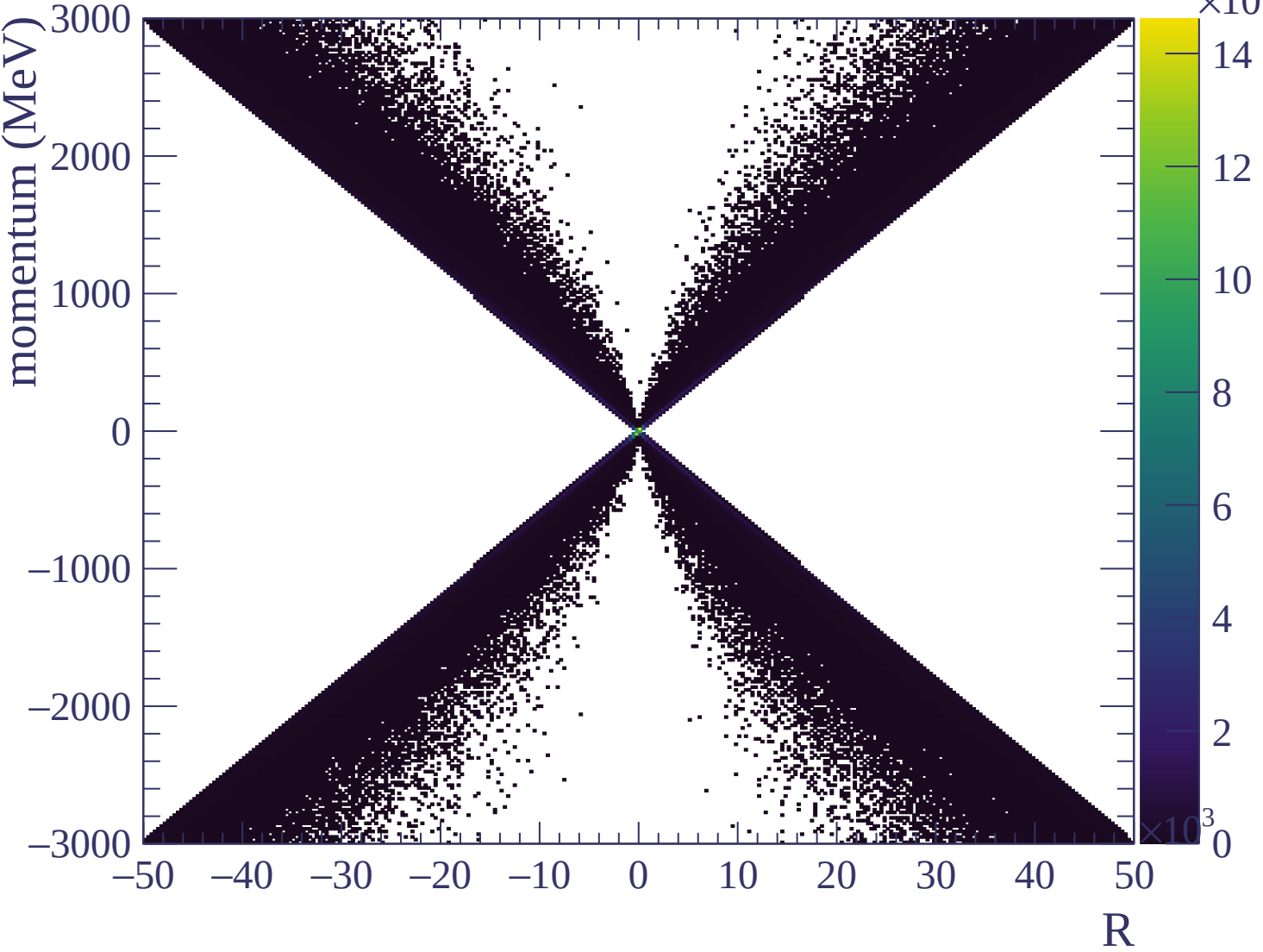


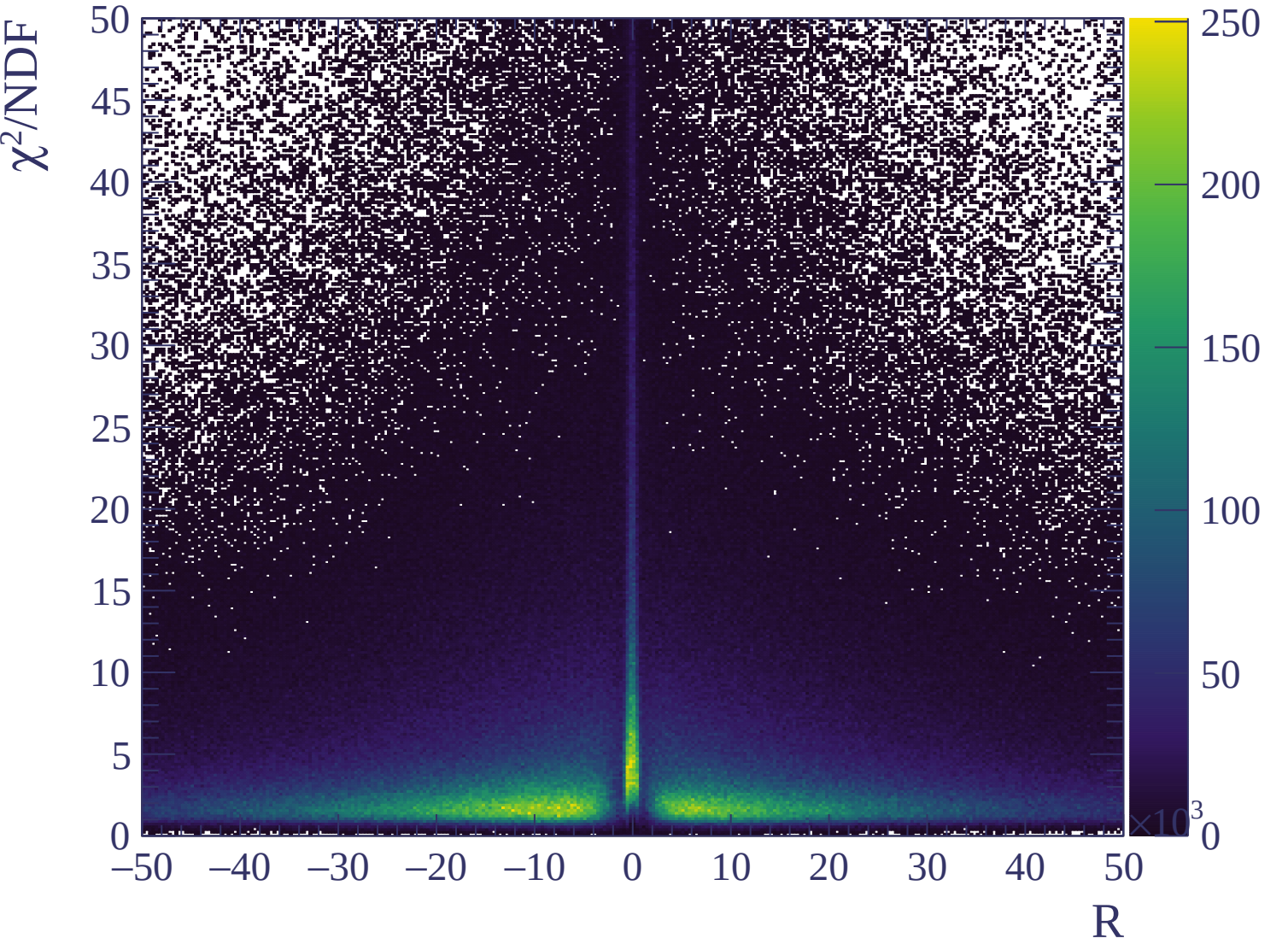






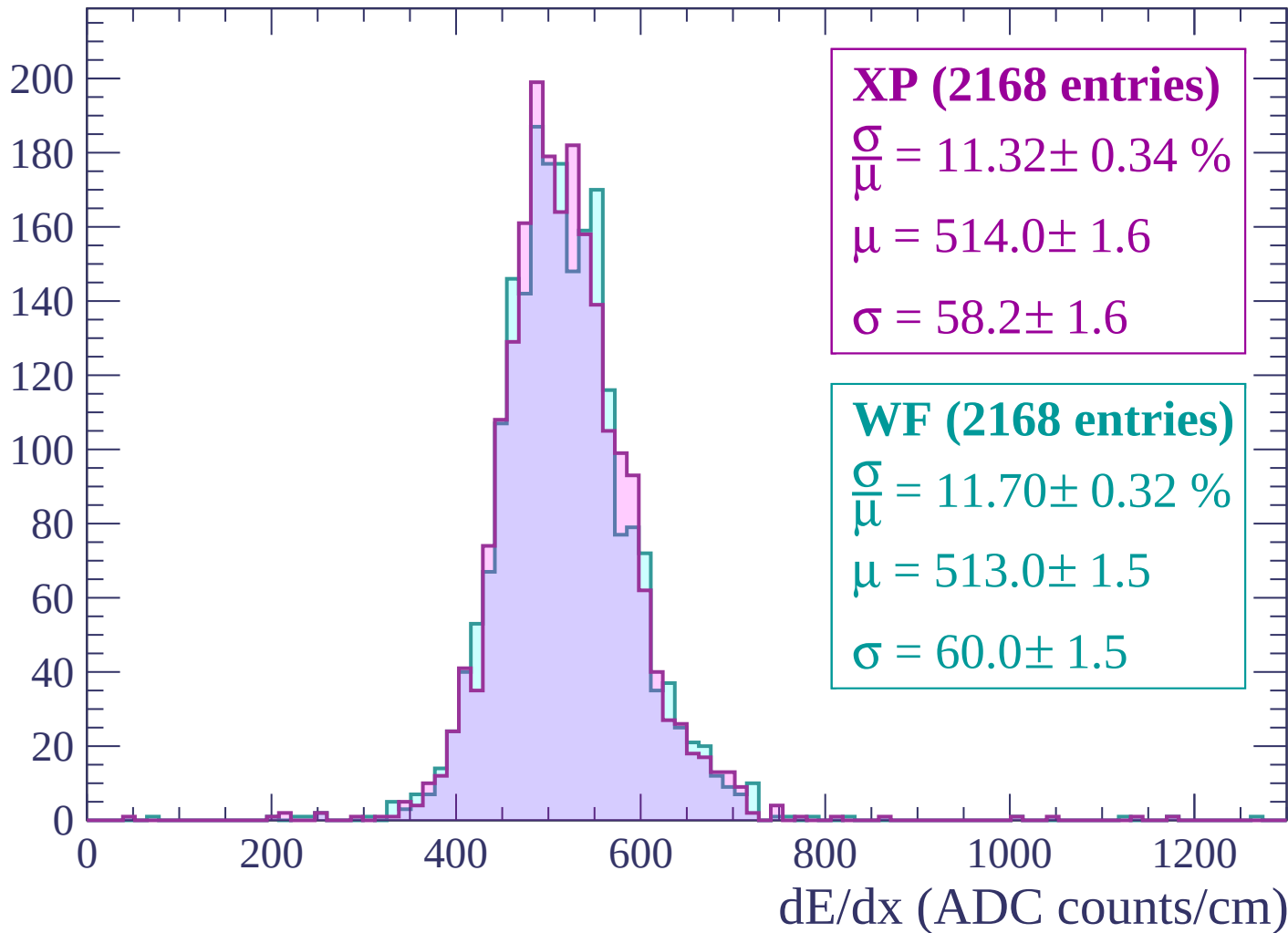






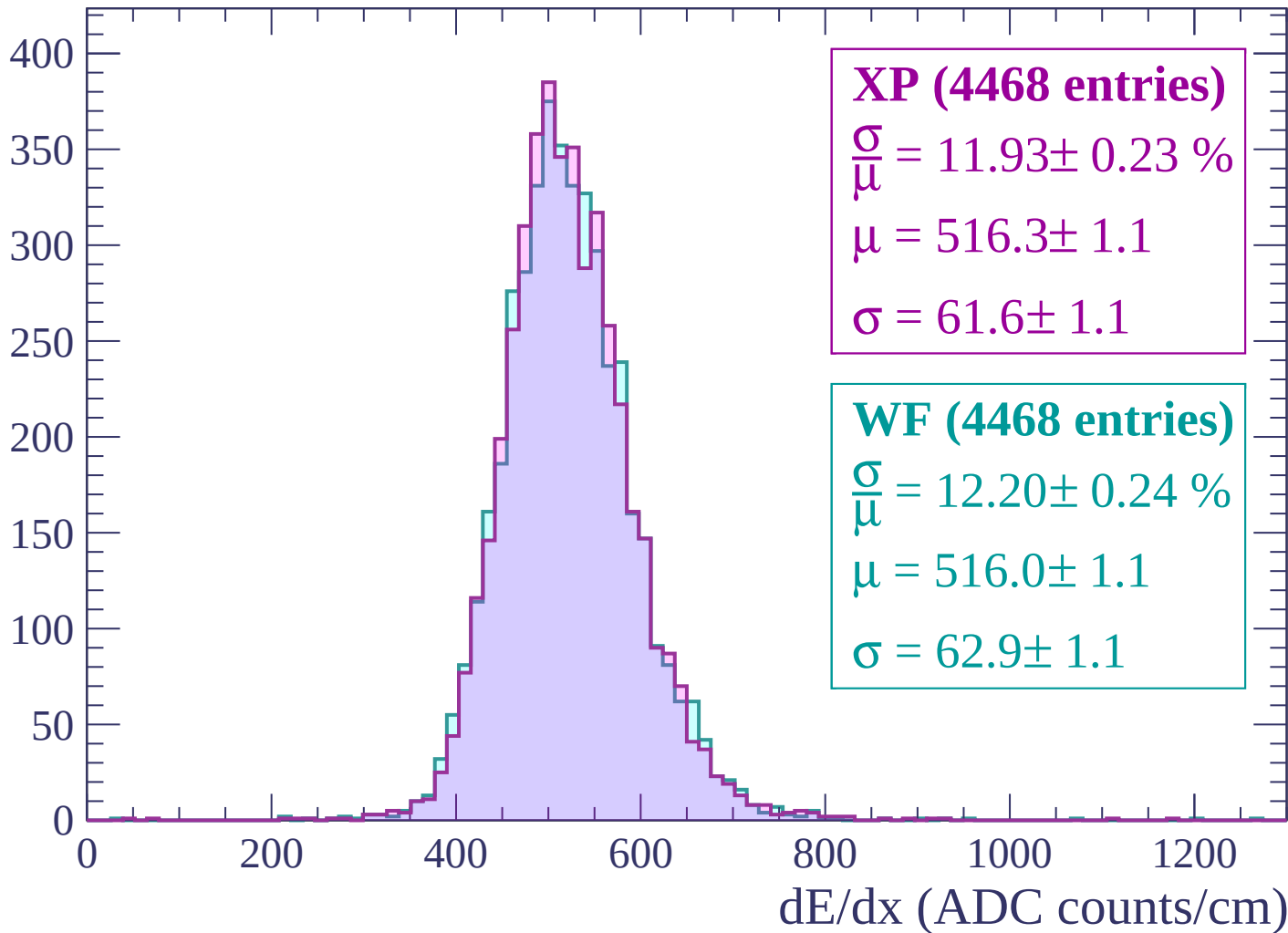
Energy loss | $-3000 < p < -2940$

Count



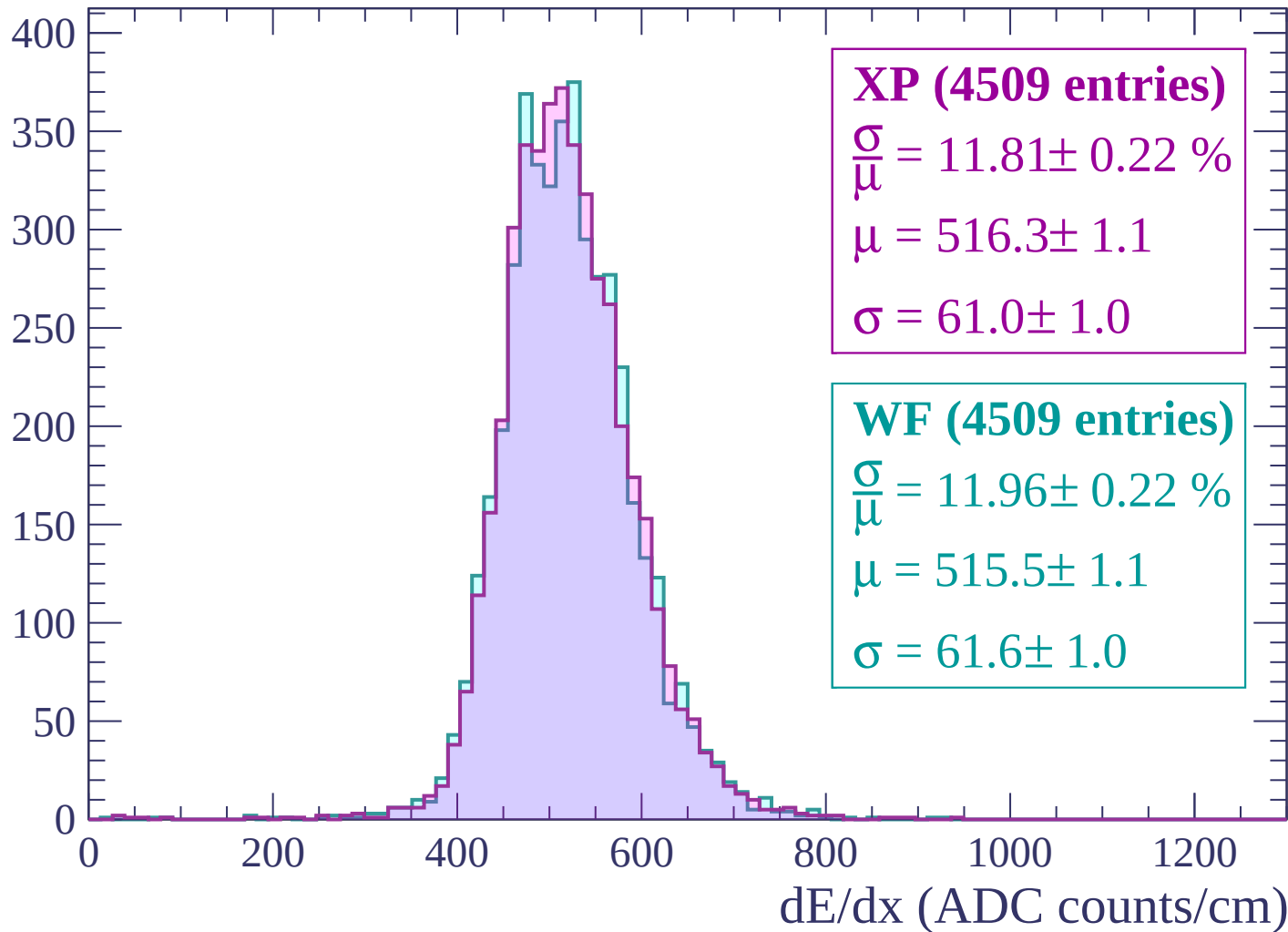
Energy loss | $-2940 < p < -2880$

Count



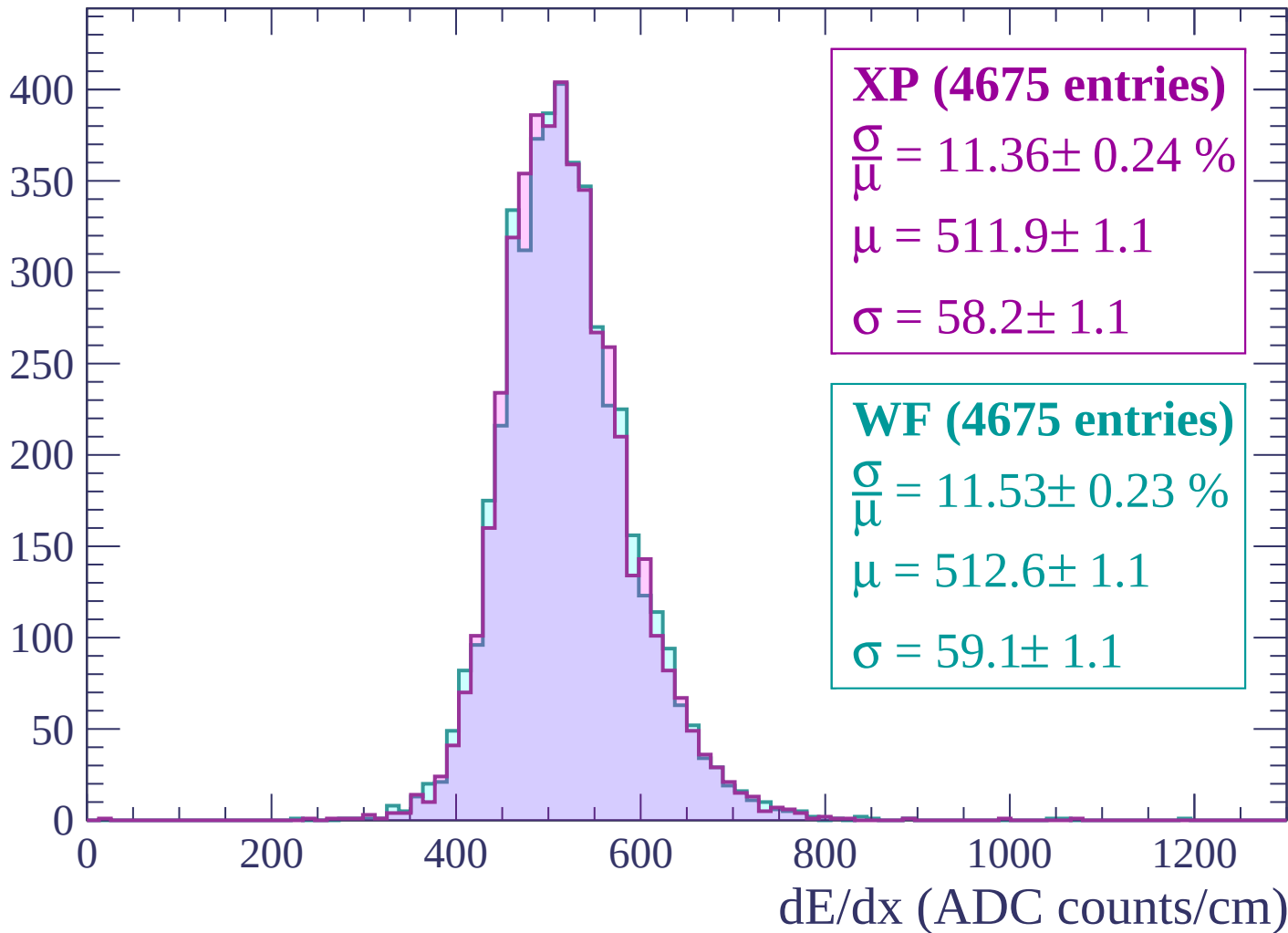
Energy loss | $-2880 < p < -2820$

Count



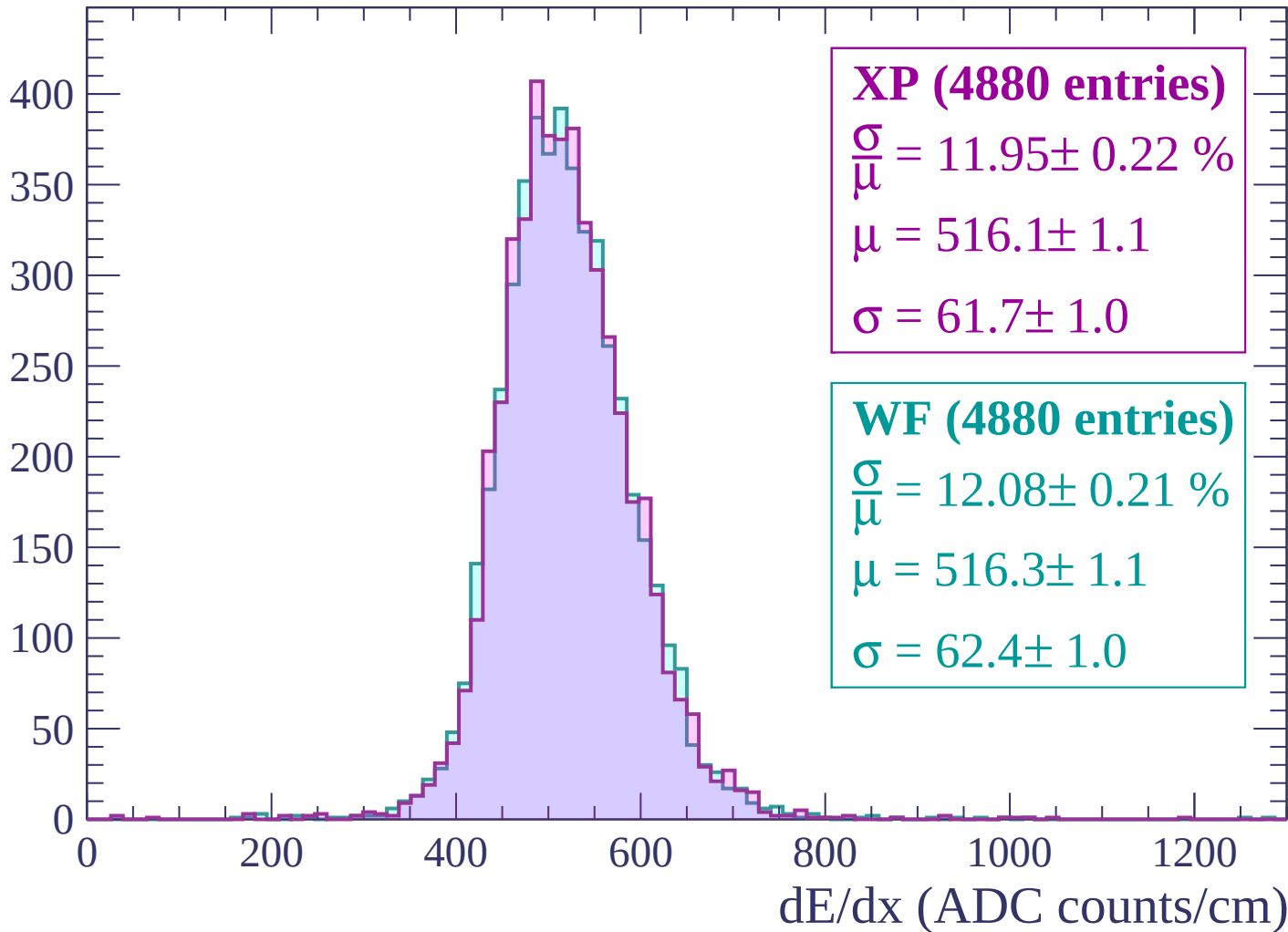
Energy loss | $-2820 < p < -2760$

Count



Energy loss | $-2760 < p < -2700$

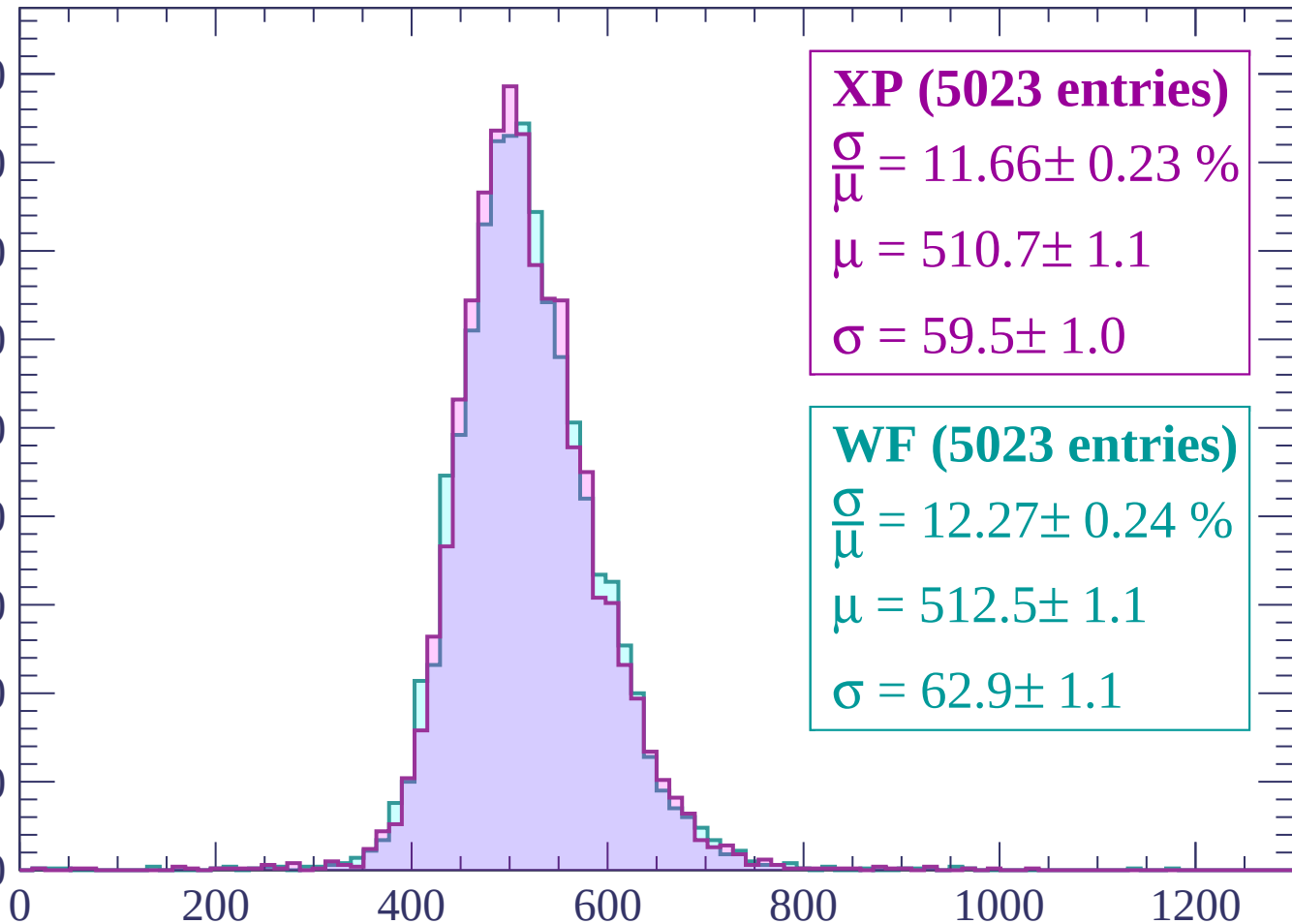
Count



Energy loss | -2700 < p < -2640

Count

450
400
350
300
250
200
150
100
50
0



XP (5023 entries)

$\frac{\sigma}{\mu} = 11.66 \pm 0.23 \%$

$\mu = 510.7 \pm 1.1$

$\sigma = 59.5 \pm 1.0$

WF (5023 entries)

$\frac{\sigma}{\mu} = 12.27 \pm 0.24 \%$

$\mu = 512.5 \pm 1.1$

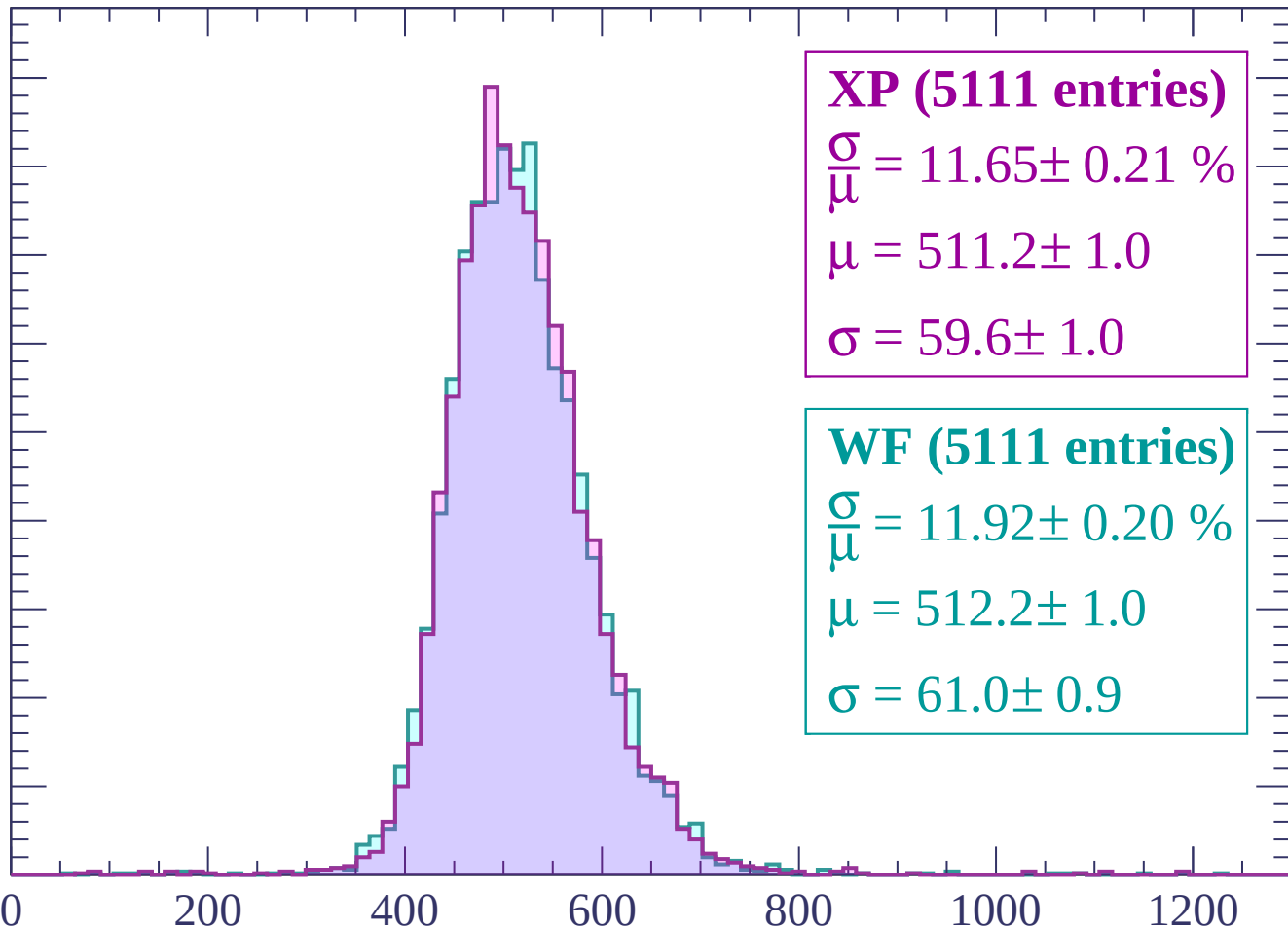
$\sigma = 62.9 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | $-2640 < p < -2580$

Count

450
400
350
300
250
200
150
100
50
0



XP (5111 entries)

$$\frac{\sigma}{\mu} = 11.65 \pm 0.21 \%$$

$$\mu = 511.2 \pm 1.0$$

$$\sigma = 59.6 \pm 1.0$$

WF (5111 entries)

$$\frac{\sigma}{\mu} = 11.92 \pm 0.20 \%$$

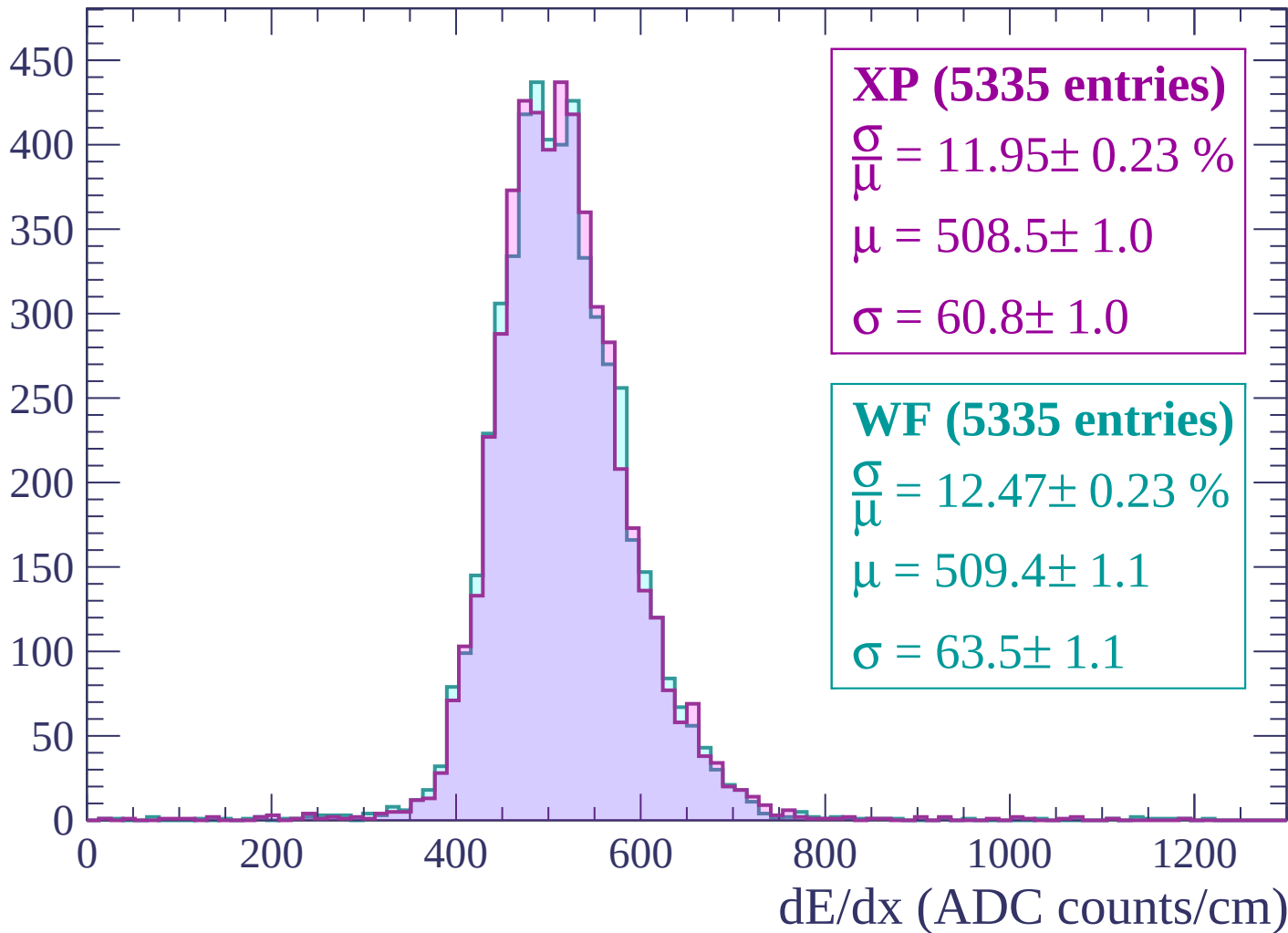
$$\mu = 512.2 \pm 1.0$$

$$\sigma = 61.0 \pm 0.9$$

dE/dx (ADC counts/cm)

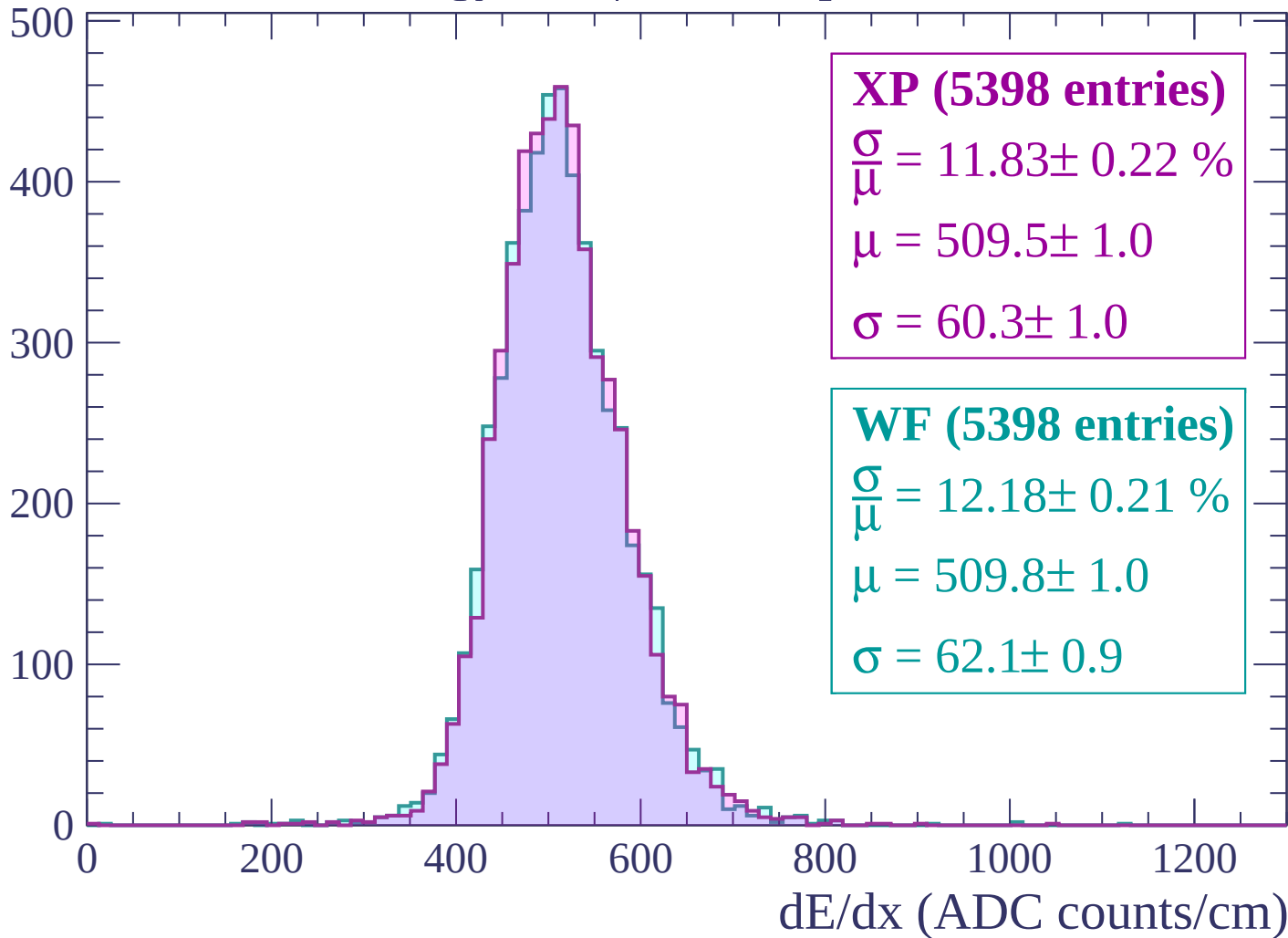
Energy loss | $-2580 < p < -2520$

Count



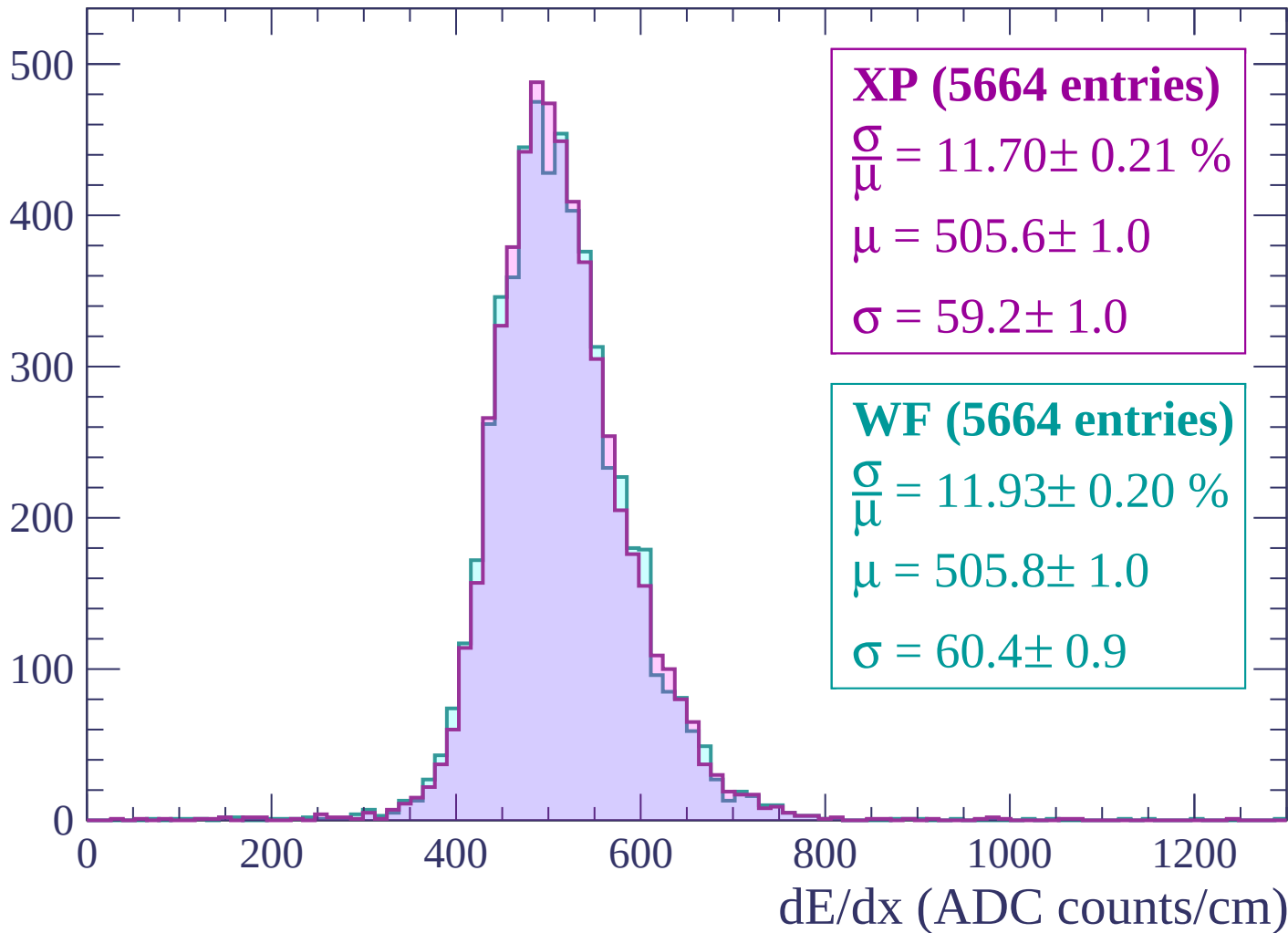
Energy loss | $-2520 < p < -2460$

Count



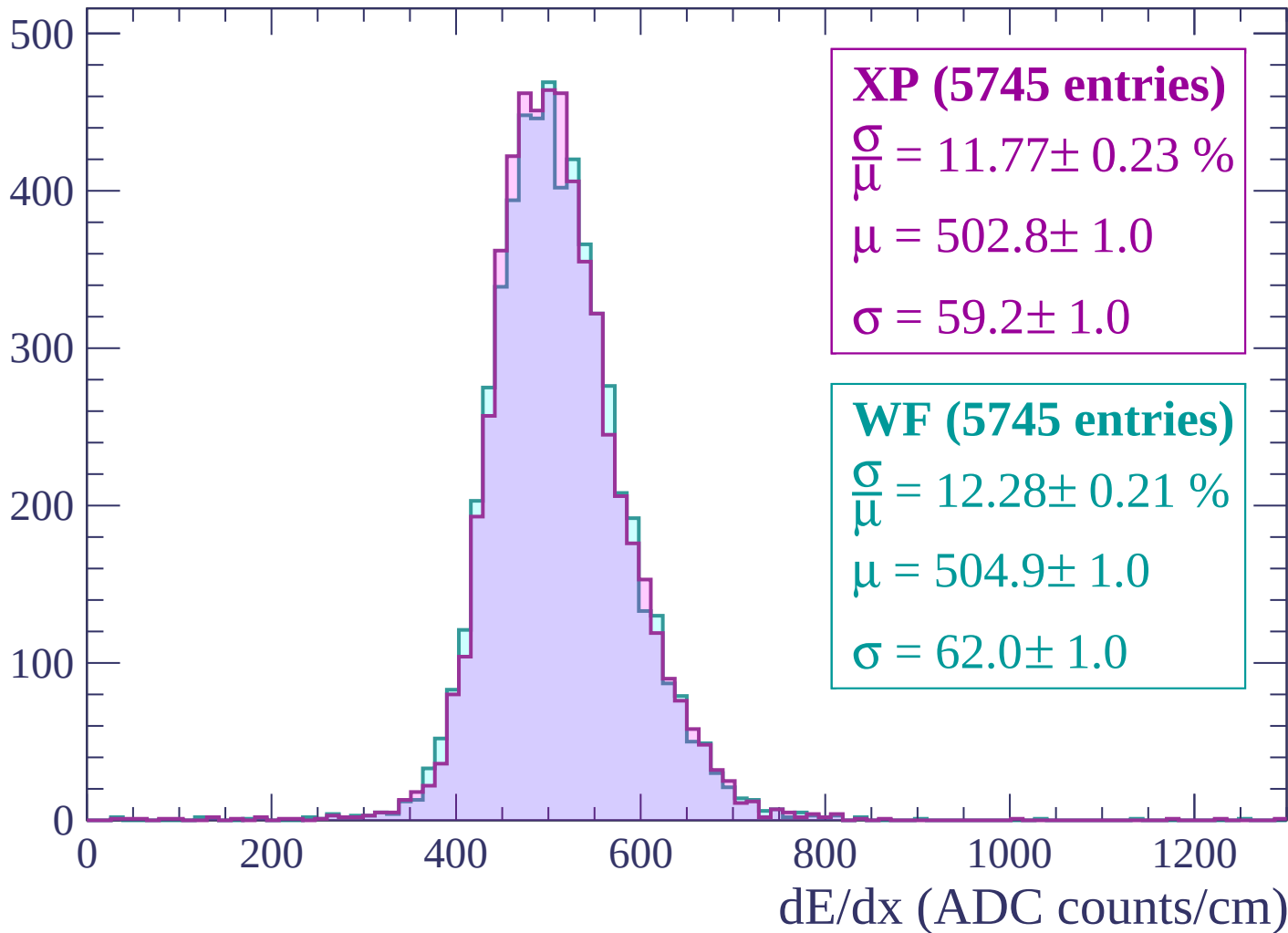
Energy loss | -2460 < p < -2400

Count



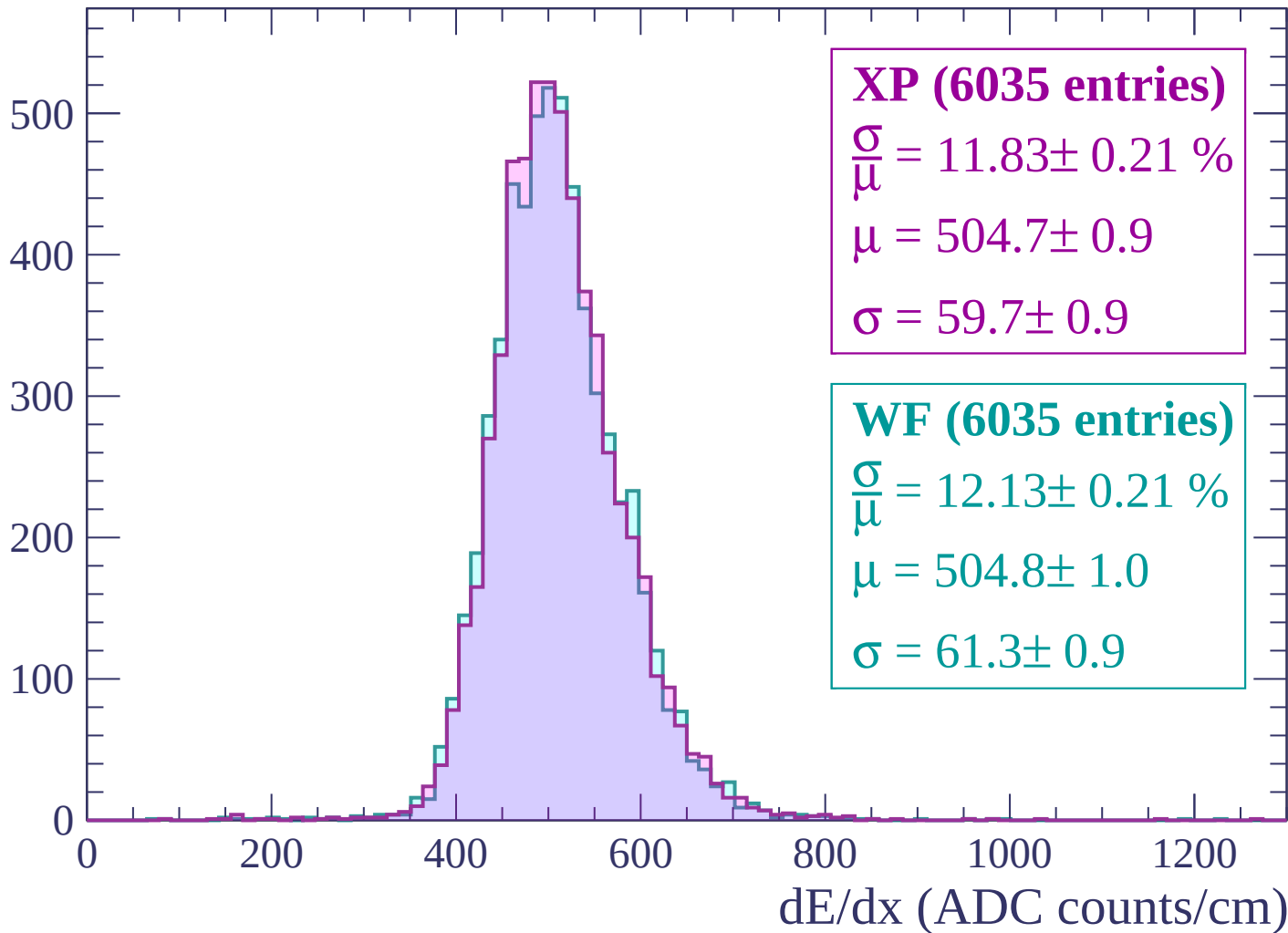
Energy loss | $-2400 < p < -2340$

Count



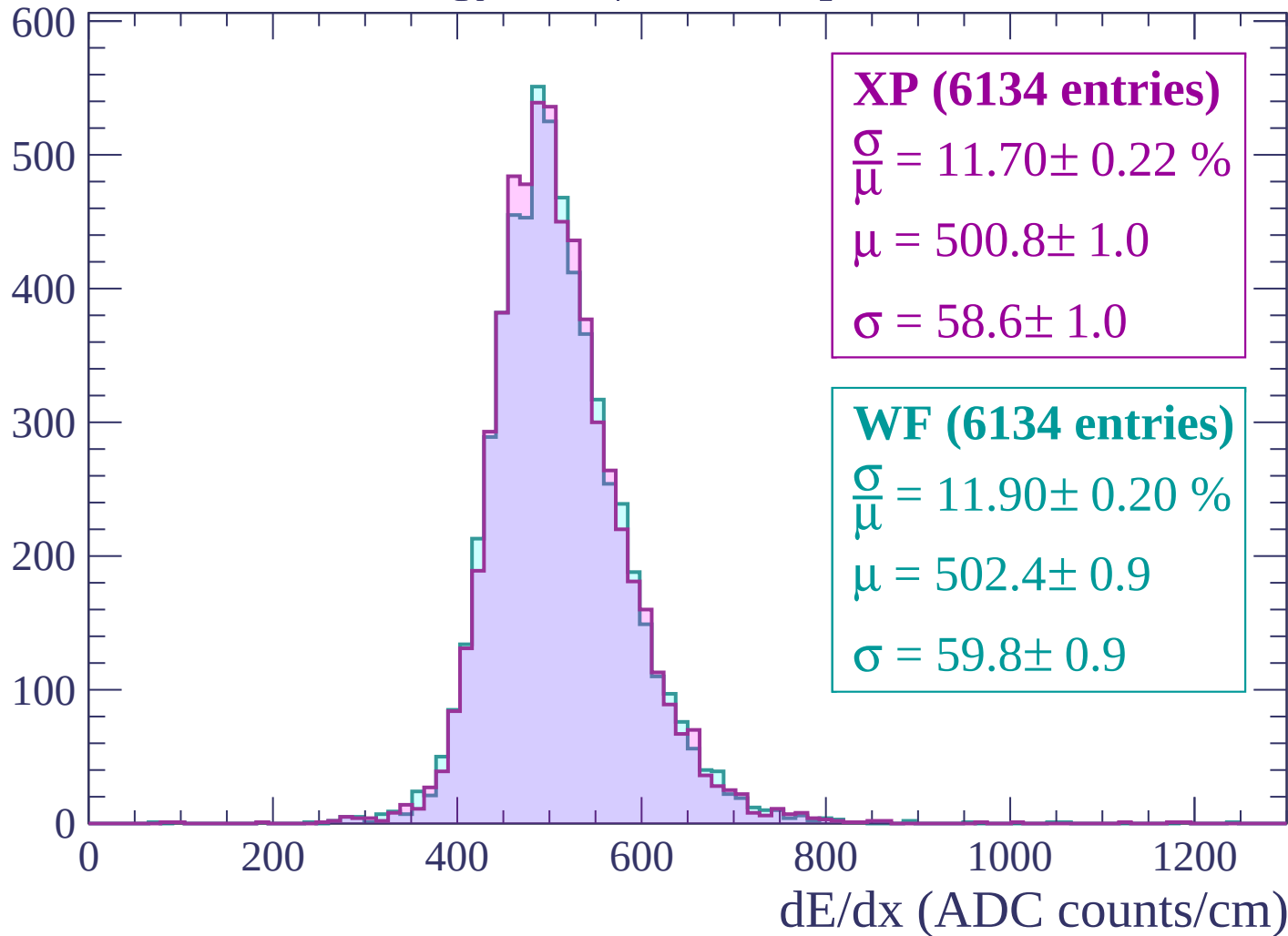
Energy loss | $-2340 < p < -2280$

Count



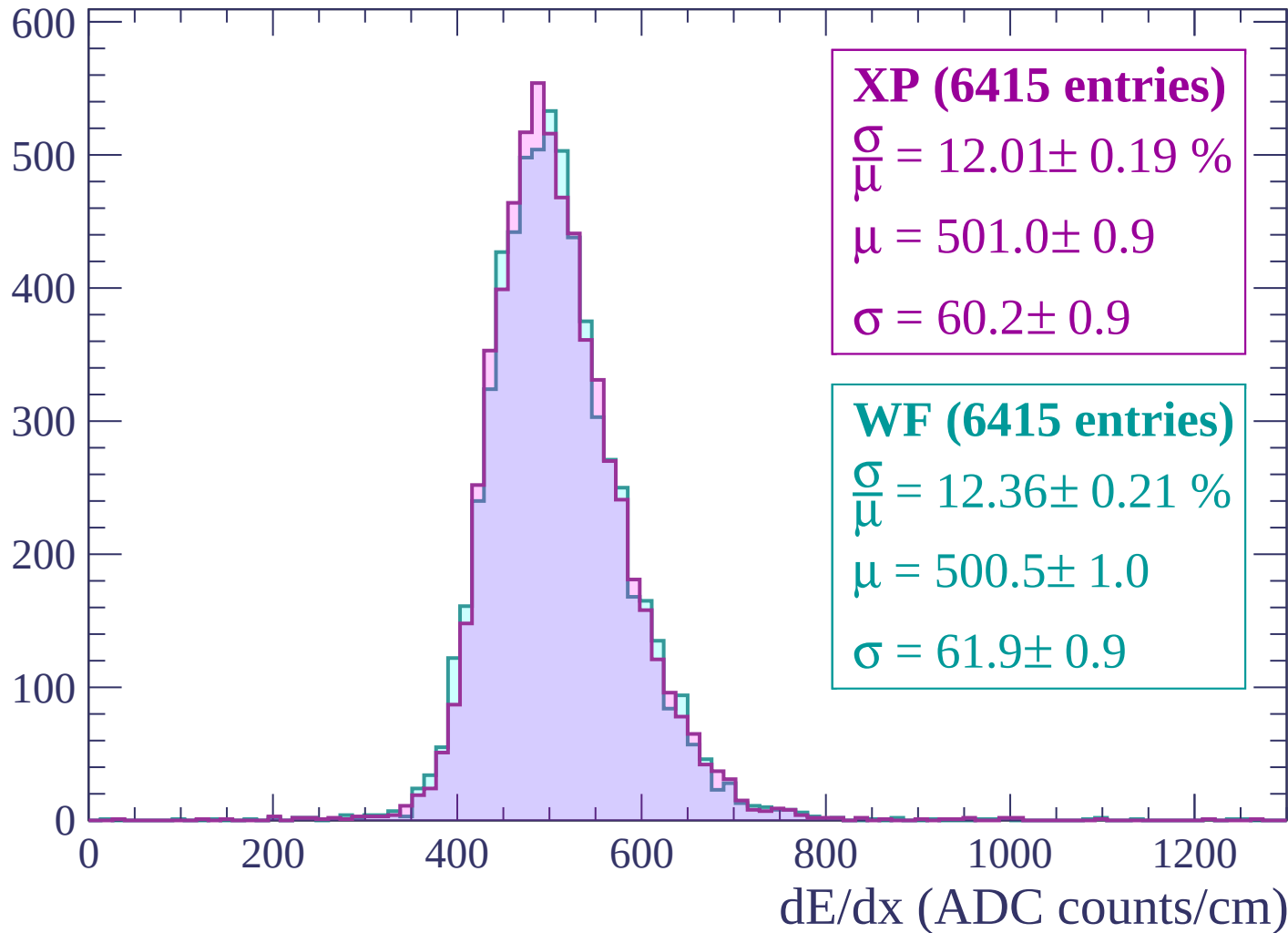
Energy loss | -2280 < p < -2220

Count



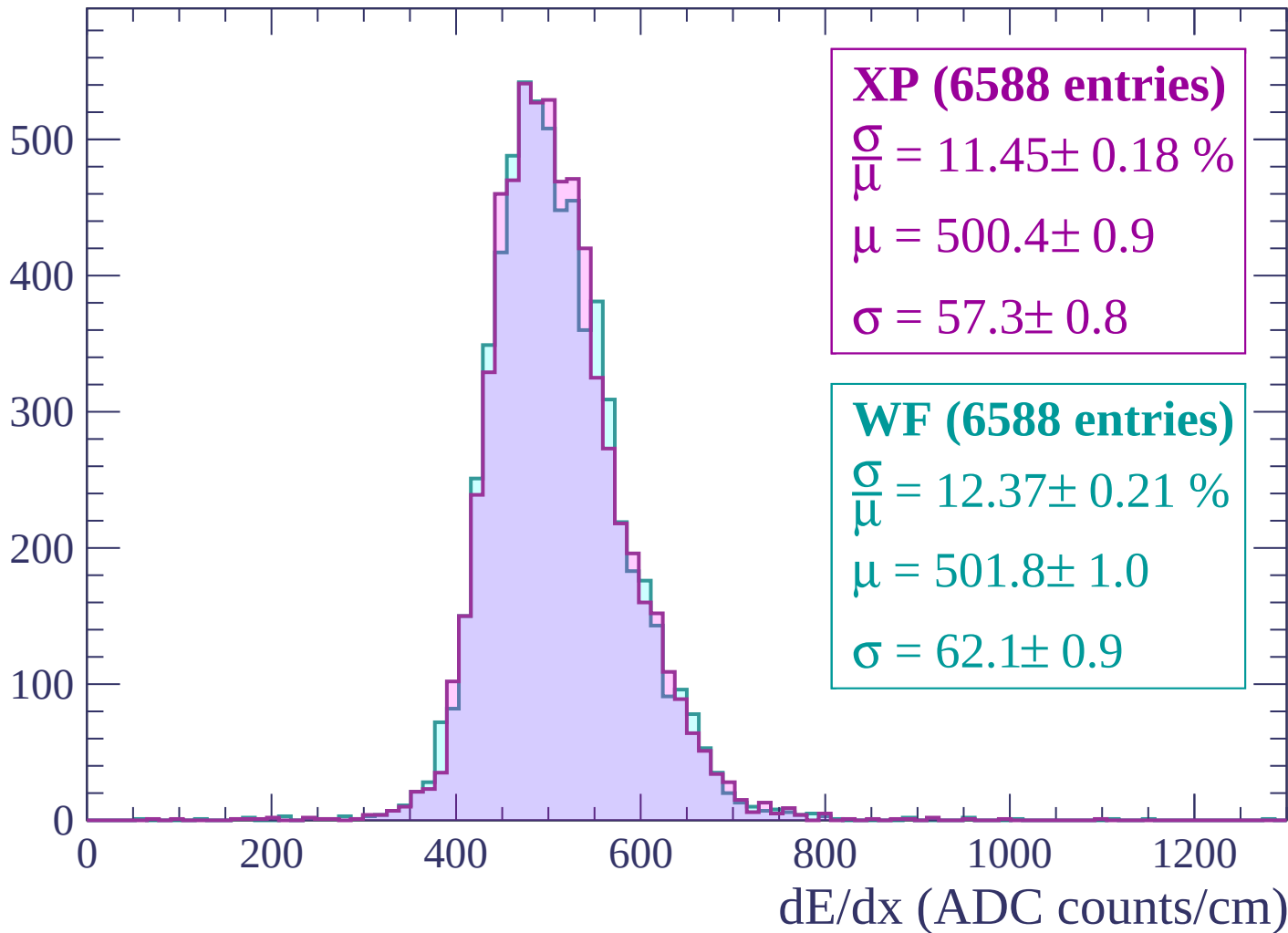
Energy loss | -2220 < p < -2160

Count



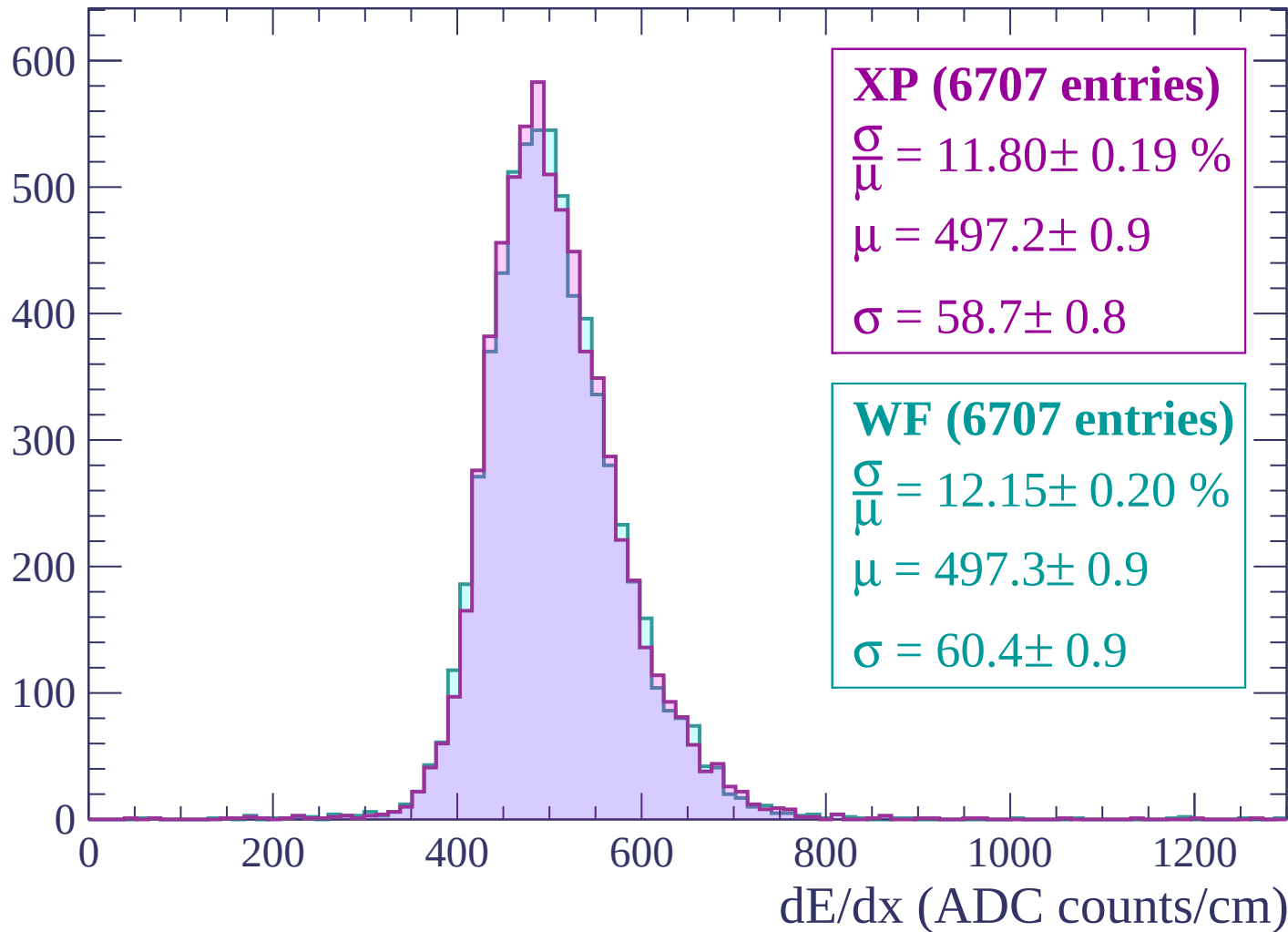
Energy loss | -2160 < p < -2100

Count



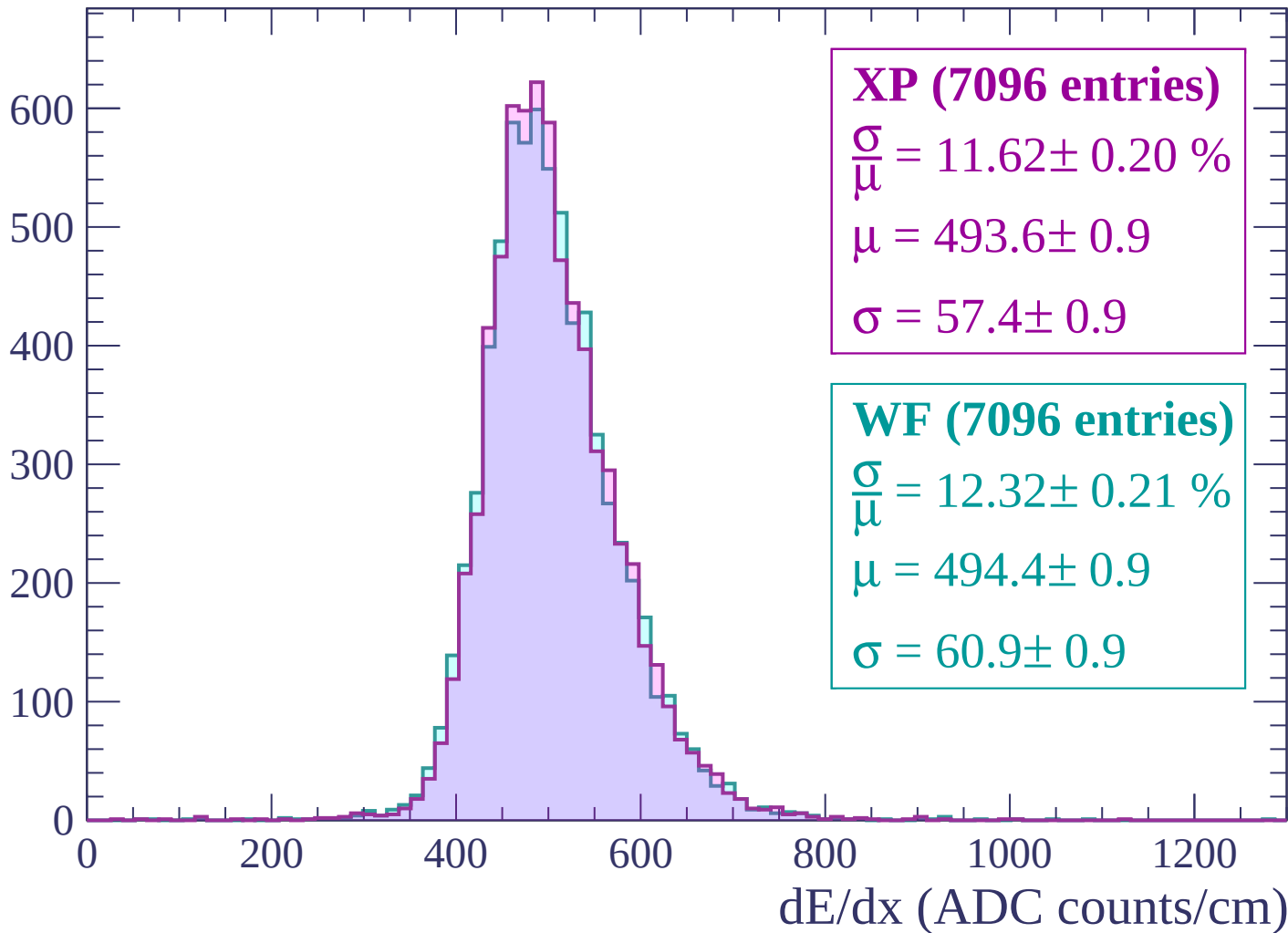
Energy loss | -2100 < p < -2040

Count



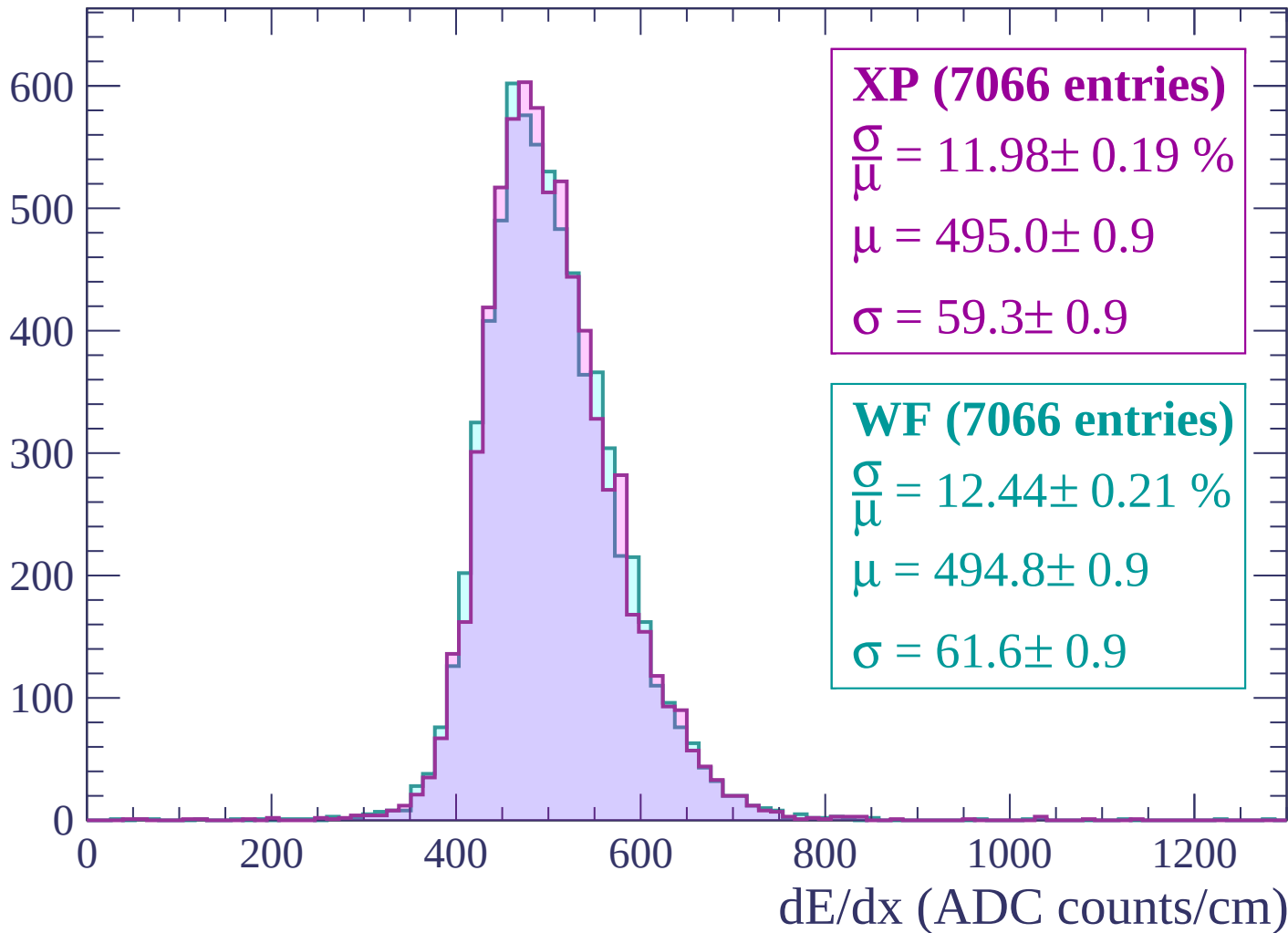
Energy loss | -2040 < p < -1980

Count



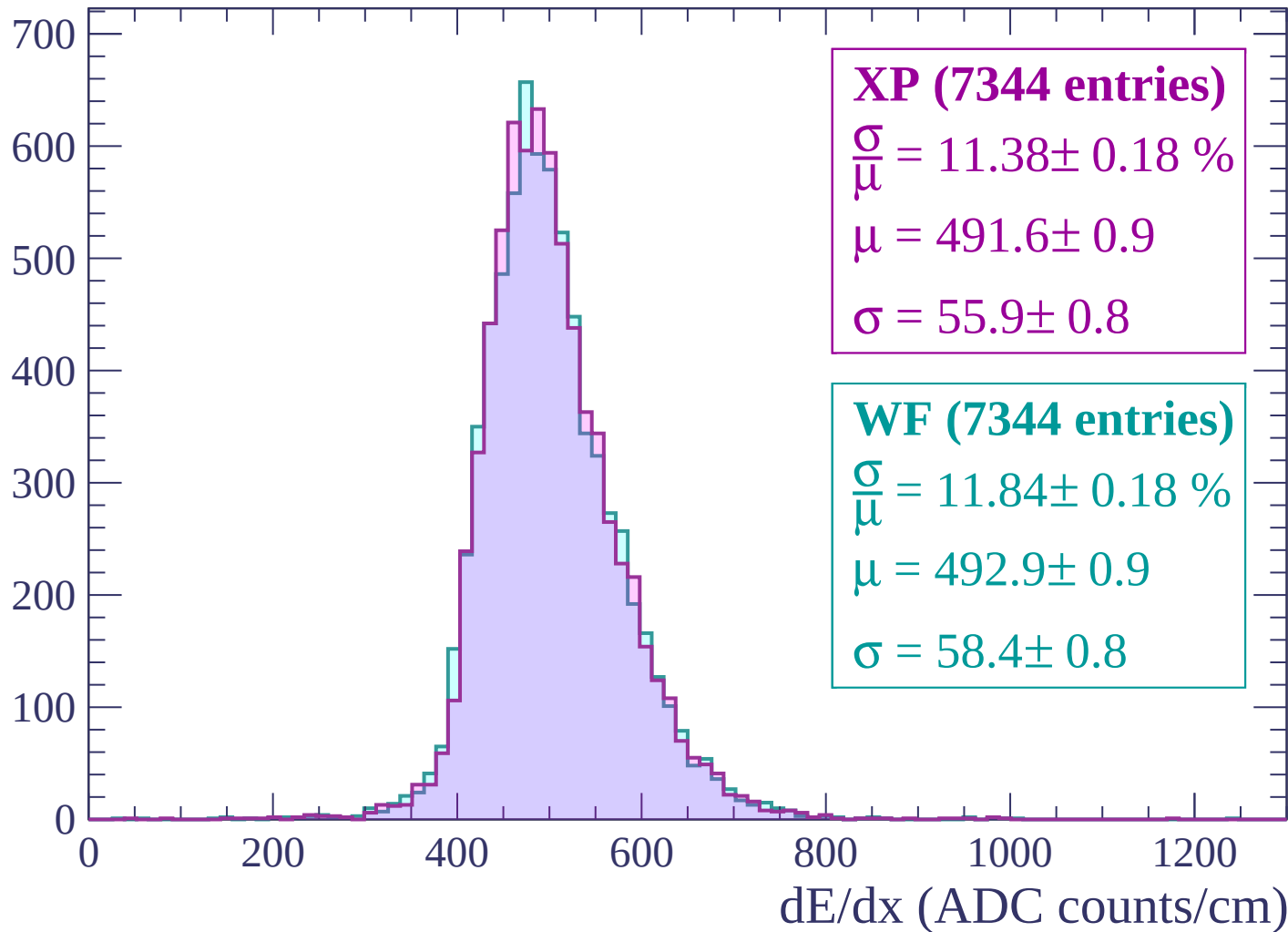
Energy loss | -1980 < p < -1920

Count



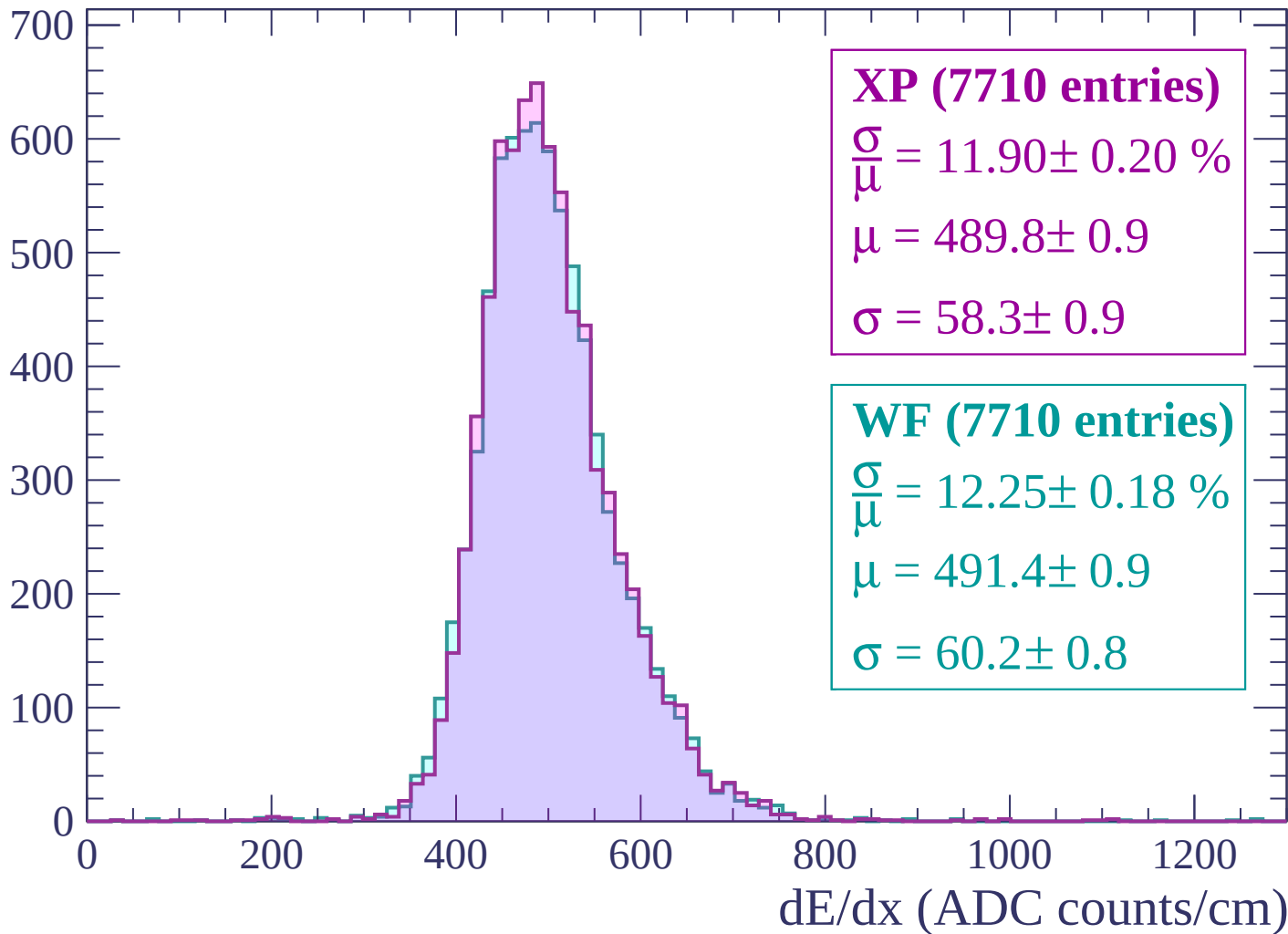
Energy loss | -1920 < p < -1860

Count



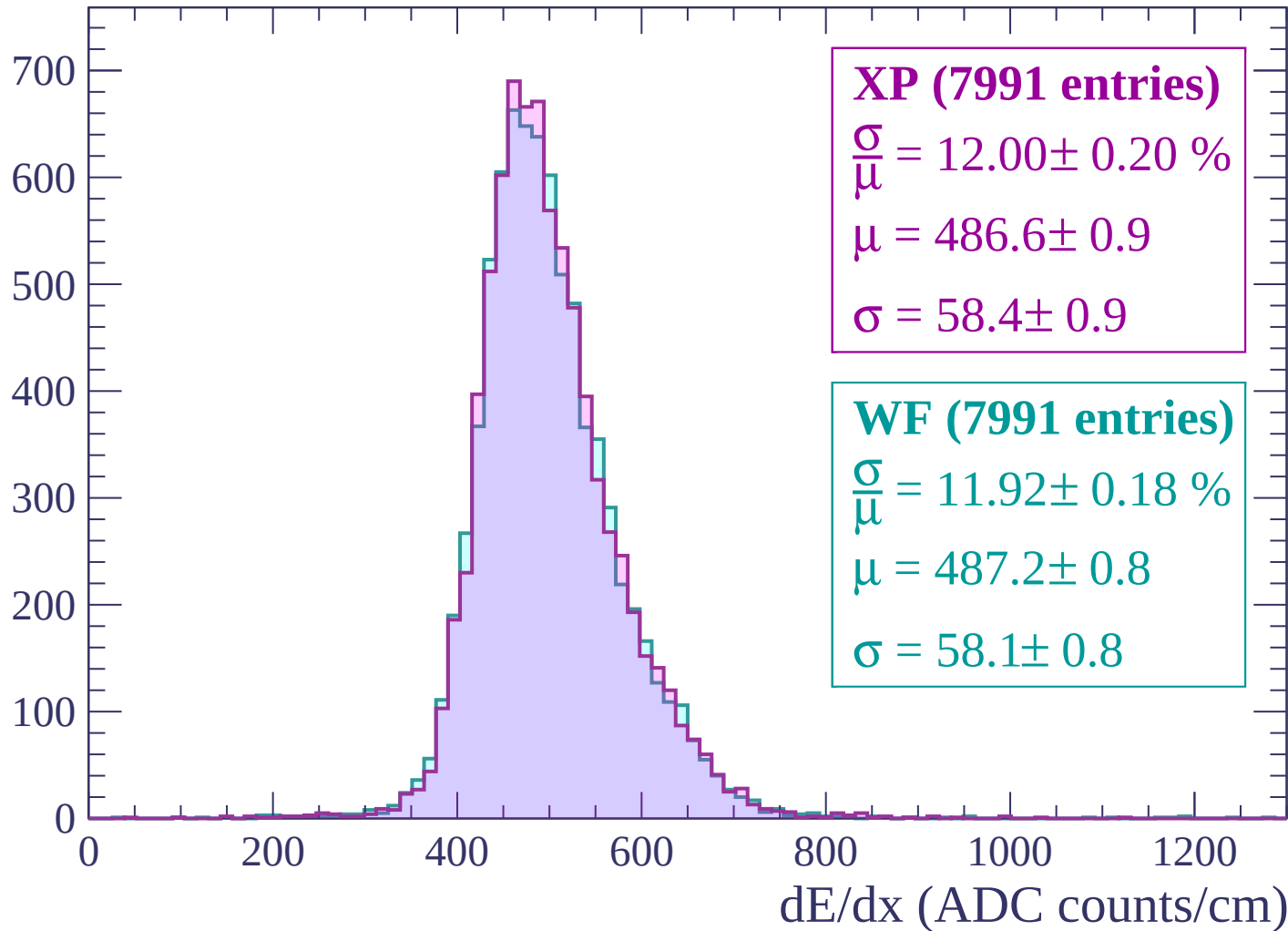
Energy loss | $-1860 < p < -1800$

Count



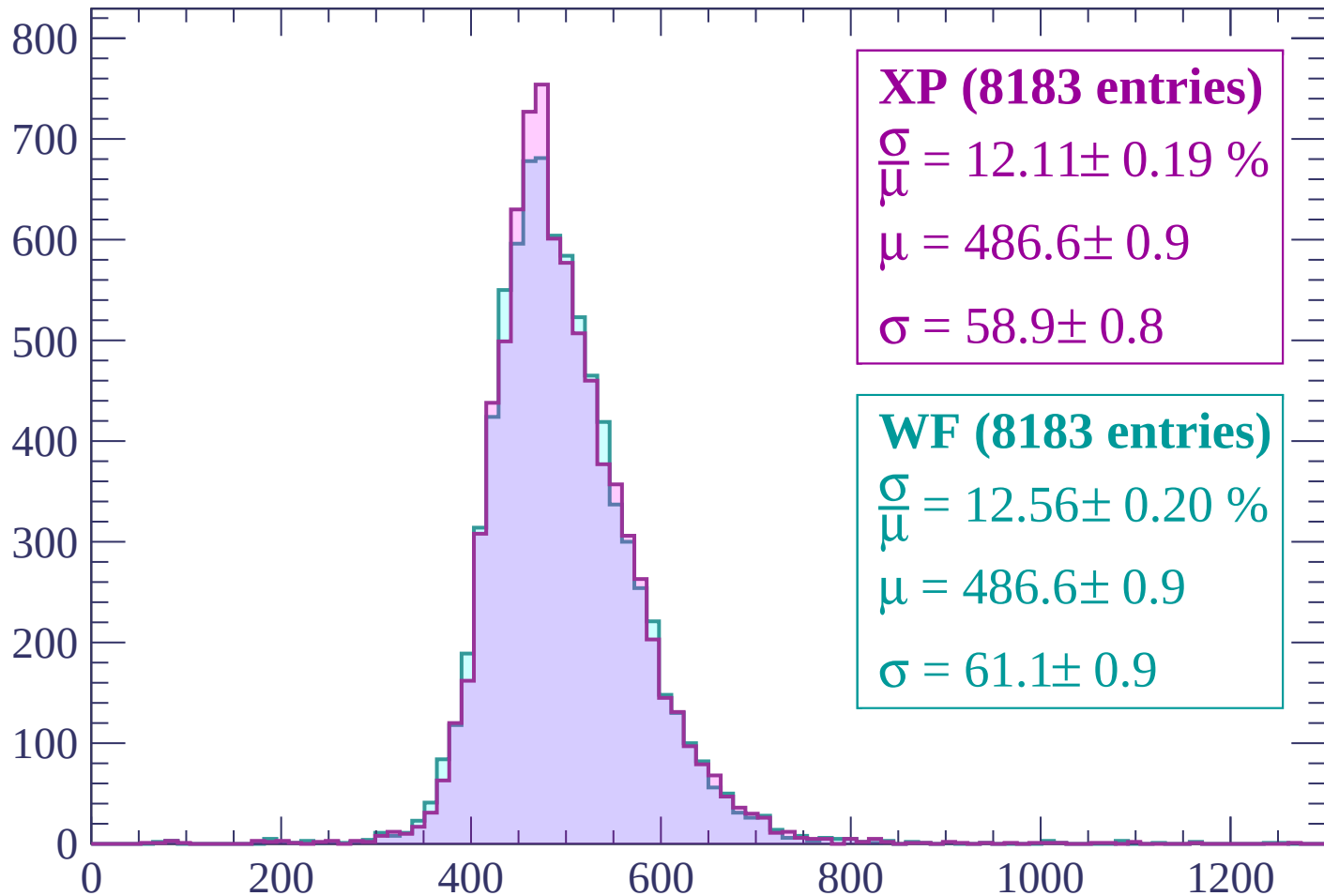
Energy loss | $-1800 < p < -1740$

Count



Energy loss | $-1740 < p < -1680$

Count



XP (8183 entries)

$$\frac{\sigma}{\mu} = 12.11 \pm 0.19 \%$$

$$\mu = 486.6 \pm 0.9$$

$$\sigma = 58.9 \pm 0.8$$

WF (8183 entries)

$$\frac{\sigma}{\mu} = 12.56 \pm 0.20 \%$$

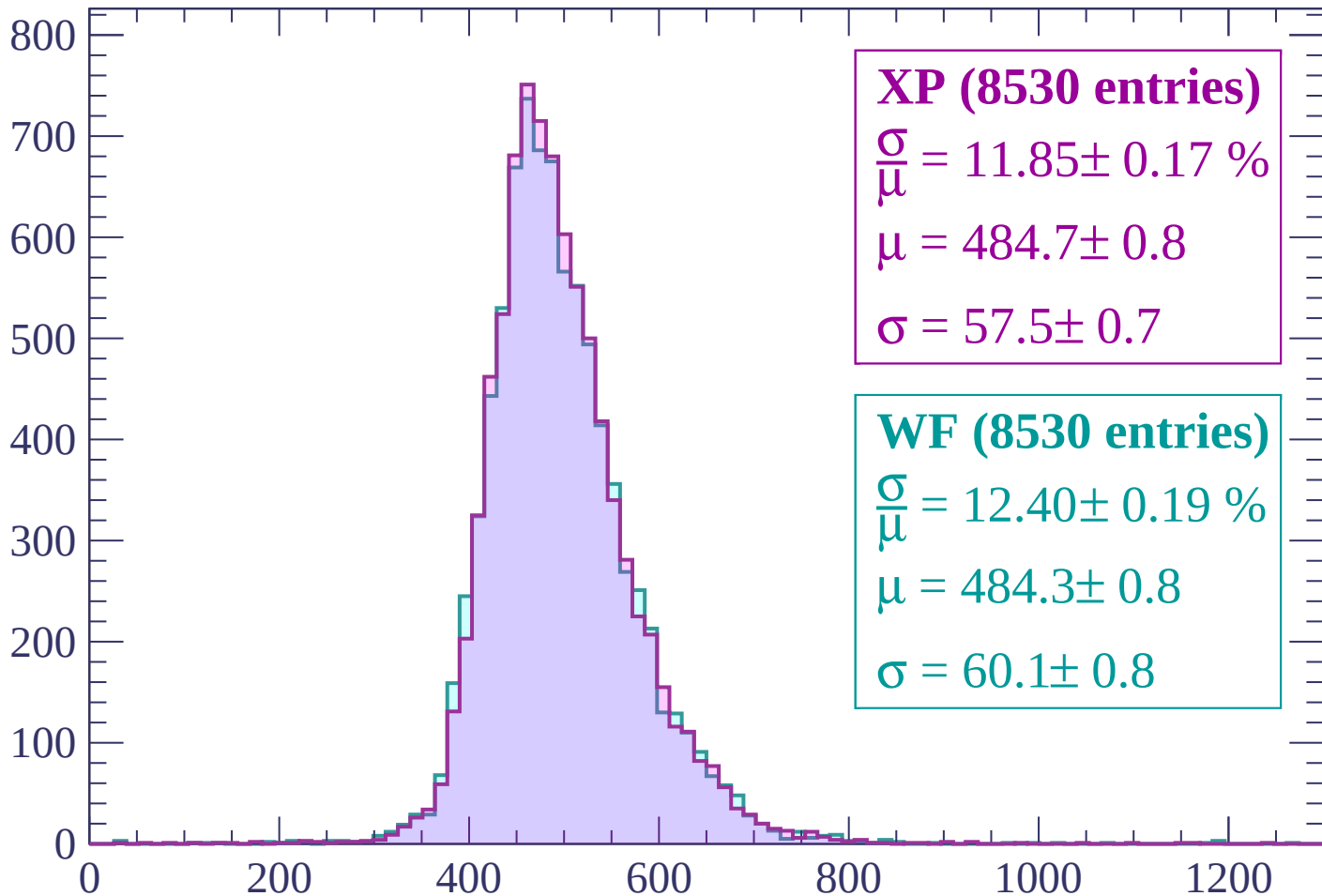
$$\mu = 486.6 \pm 0.9$$

$$\sigma = 61.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1620$

Count



XP (8530 entries)

$$\frac{\sigma}{\mu} = 11.85 \pm 0.17 \%$$

$$\mu = 484.7 \pm 0.8$$

$$\sigma = 57.5 \pm 0.7$$

WF (8530 entries)

$$\frac{\sigma}{\mu} = 12.40 \pm 0.19 \%$$

$$\mu = 484.3 \pm 0.8$$

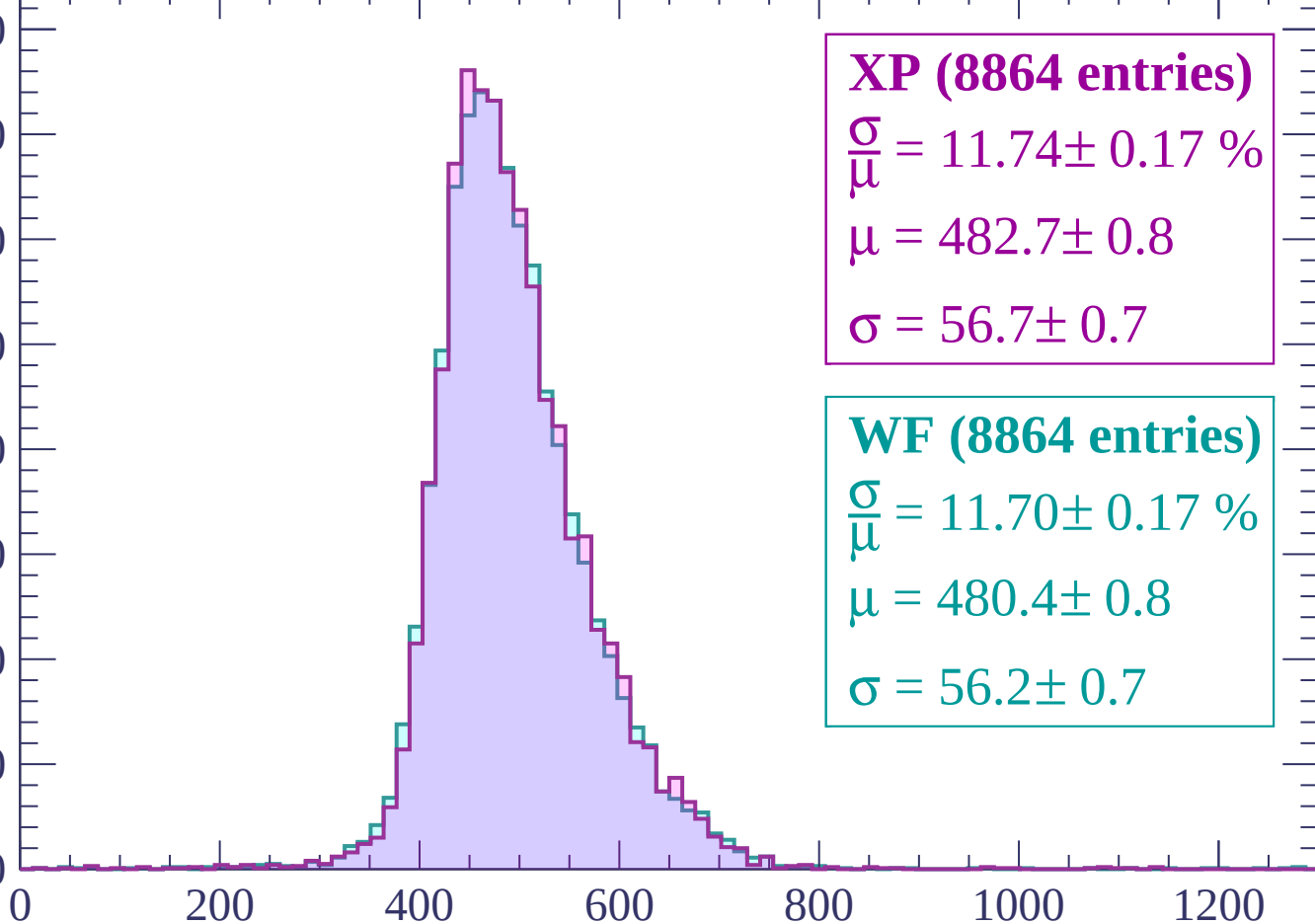
$$\sigma = 60.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1620 < p < -1560

Count

800
700
600
500
400
300
200
100
0



XP (8864 entries)

$\frac{\sigma}{\mu} = 11.74 \pm 0.17 \%$

$\mu = 482.7 \pm 0.8$

$\sigma = 56.7 \pm 0.7$

WF (8864 entries)

$\frac{\sigma}{\mu} = 11.70 \pm 0.17 \%$

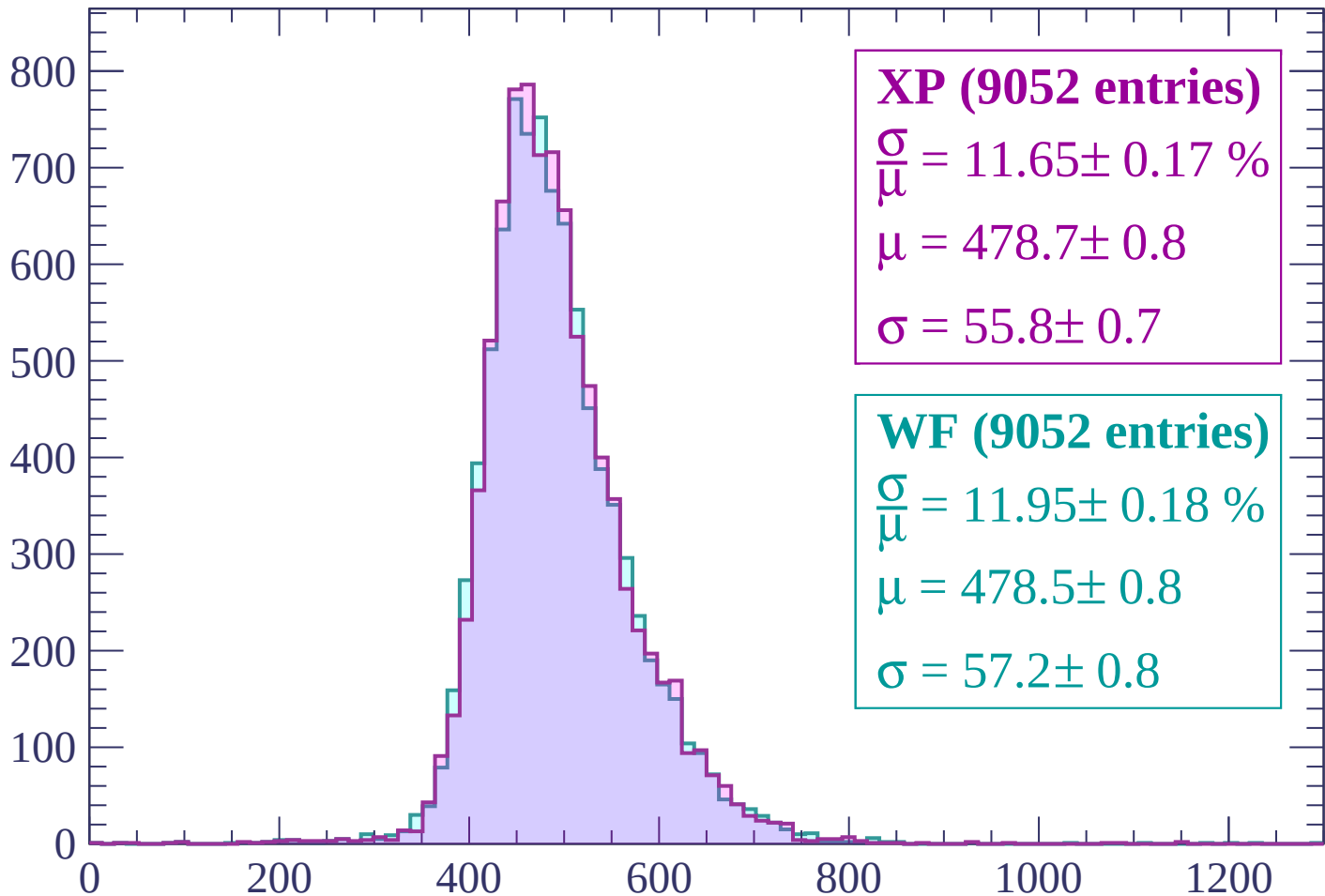
$\mu = 480.4 \pm 0.8$

$\sigma = 56.2 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

Count



XP (9052 entries)

$$\frac{\sigma}{\mu} = 11.65 \pm 0.17 \%$$

$$\mu = 478.7 \pm 0.8$$

$$\sigma = 55.8 \pm 0.7$$

WF (9052 entries)

$$\frac{\sigma}{\mu} = 11.95 \pm 0.18 \%$$

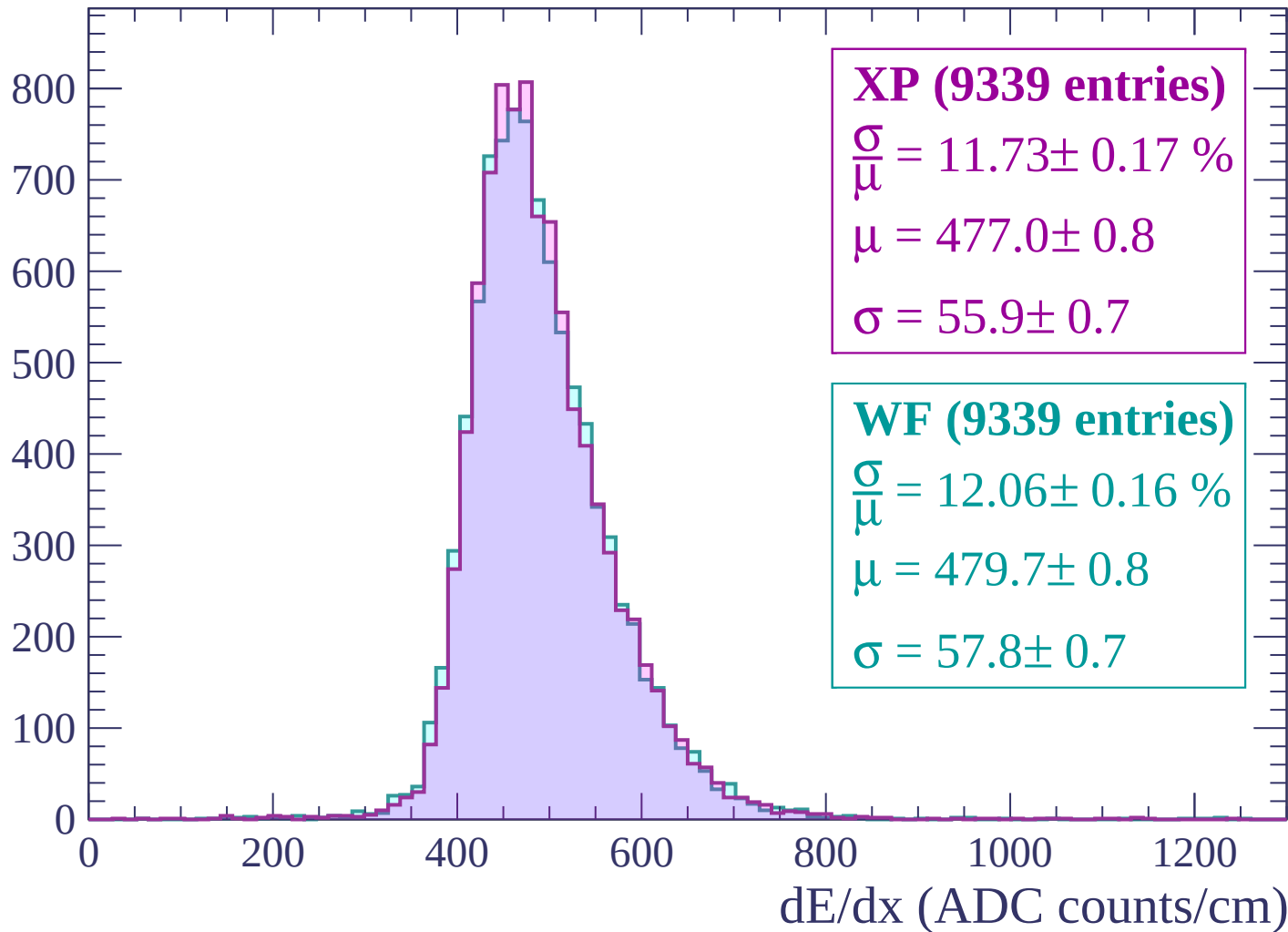
$$\mu = 478.5 \pm 0.8$$

$$\sigma = 57.2 \pm 0.8$$

dE/dx (ADC counts/cm)

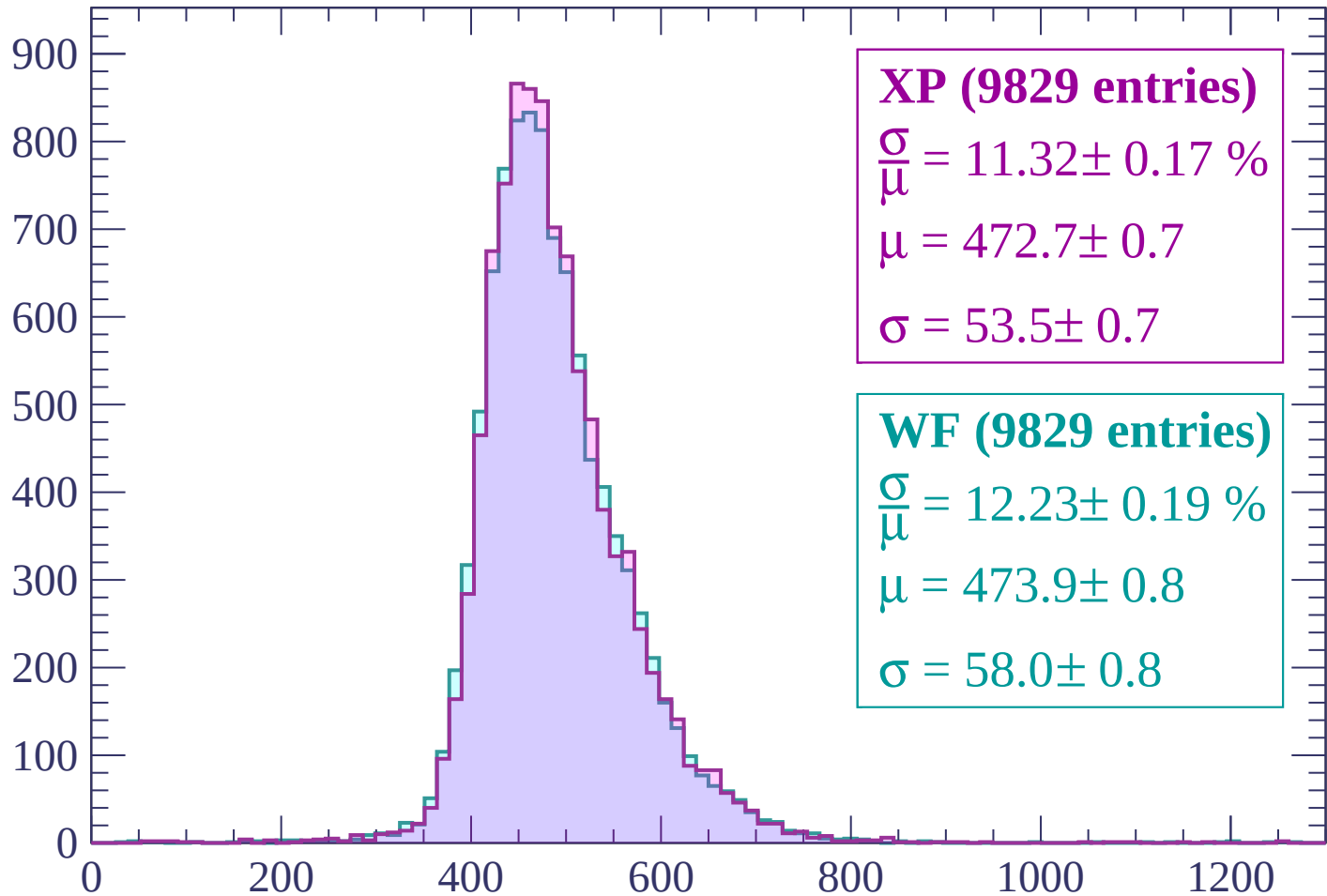
Energy loss | $-1500 < p < -1440$

Count



Energy loss | $-1440 < p < -1380$

Count



XP (9829 entries)

$$\frac{\sigma}{\mu} = 11.32 \pm 0.17 \%$$

$$\mu = 472.7 \pm 0.7$$

$$\sigma = 53.5 \pm 0.7$$

WF (9829 entries)

$$\frac{\sigma}{\mu} = 12.23 \pm 0.19 \%$$

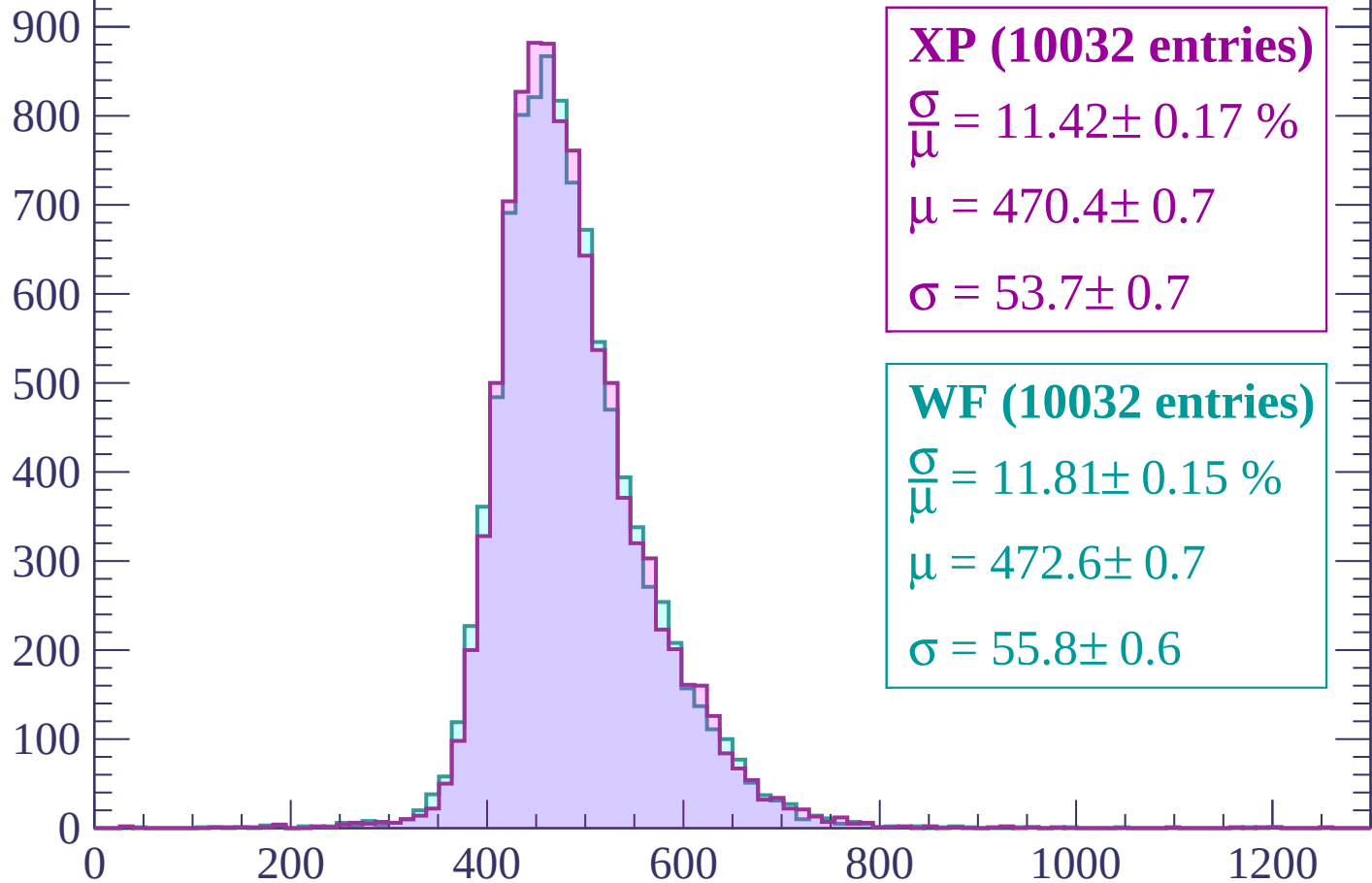
$$\mu = 473.9 \pm 0.8$$

$$\sigma = 58.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1380 < p < -1320$

Count



XP (10032 entries)

$$\frac{\sigma}{\mu} = 11.42 \pm 0.17 \%$$

$$\mu = 470.4 \pm 0.7$$

$$\sigma = 53.7 \pm 0.7$$

WF (10032 entries)

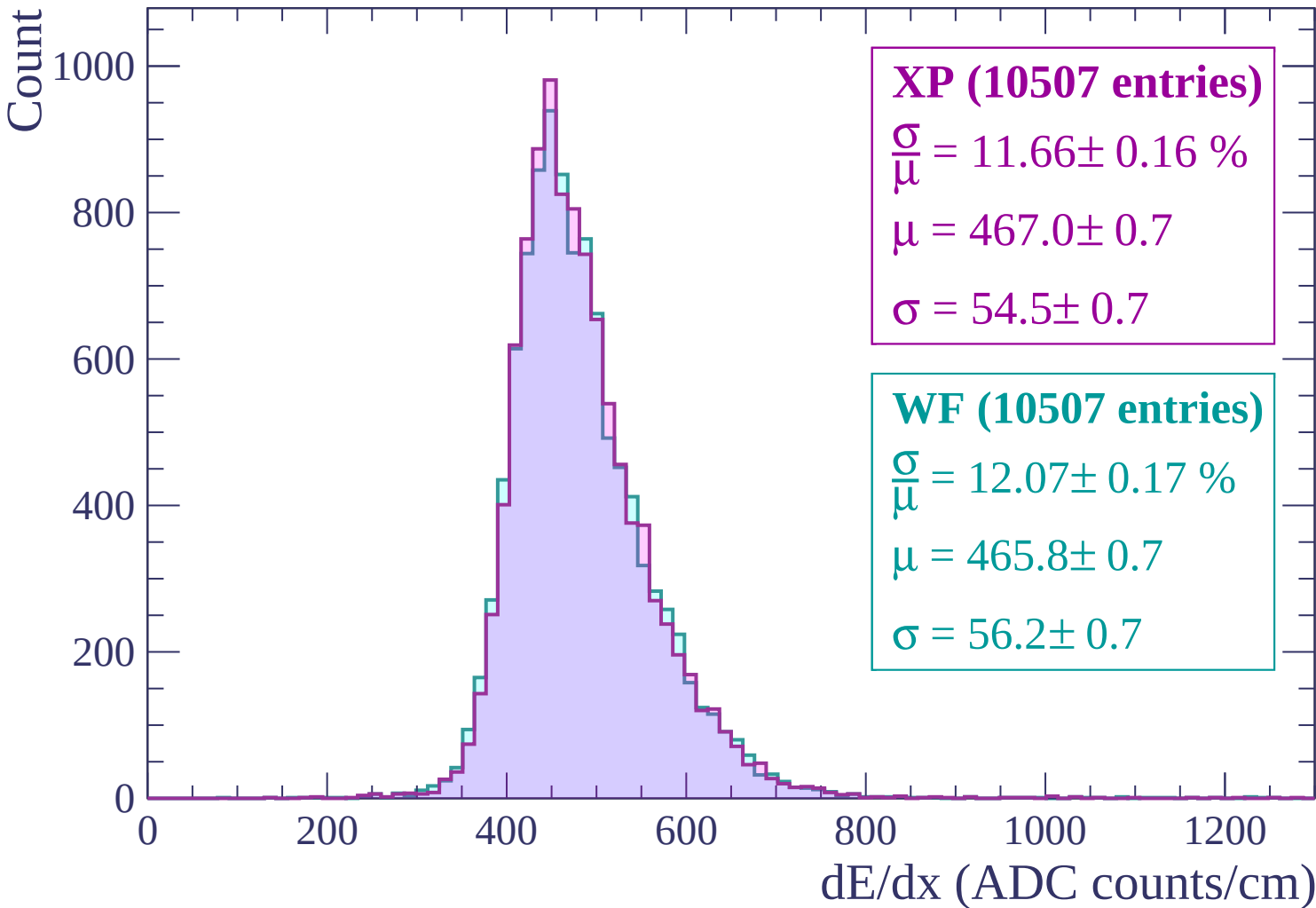
$$\frac{\sigma}{\mu} = 11.81 \pm 0.15 \%$$

$$\mu = 472.6 \pm 0.7$$

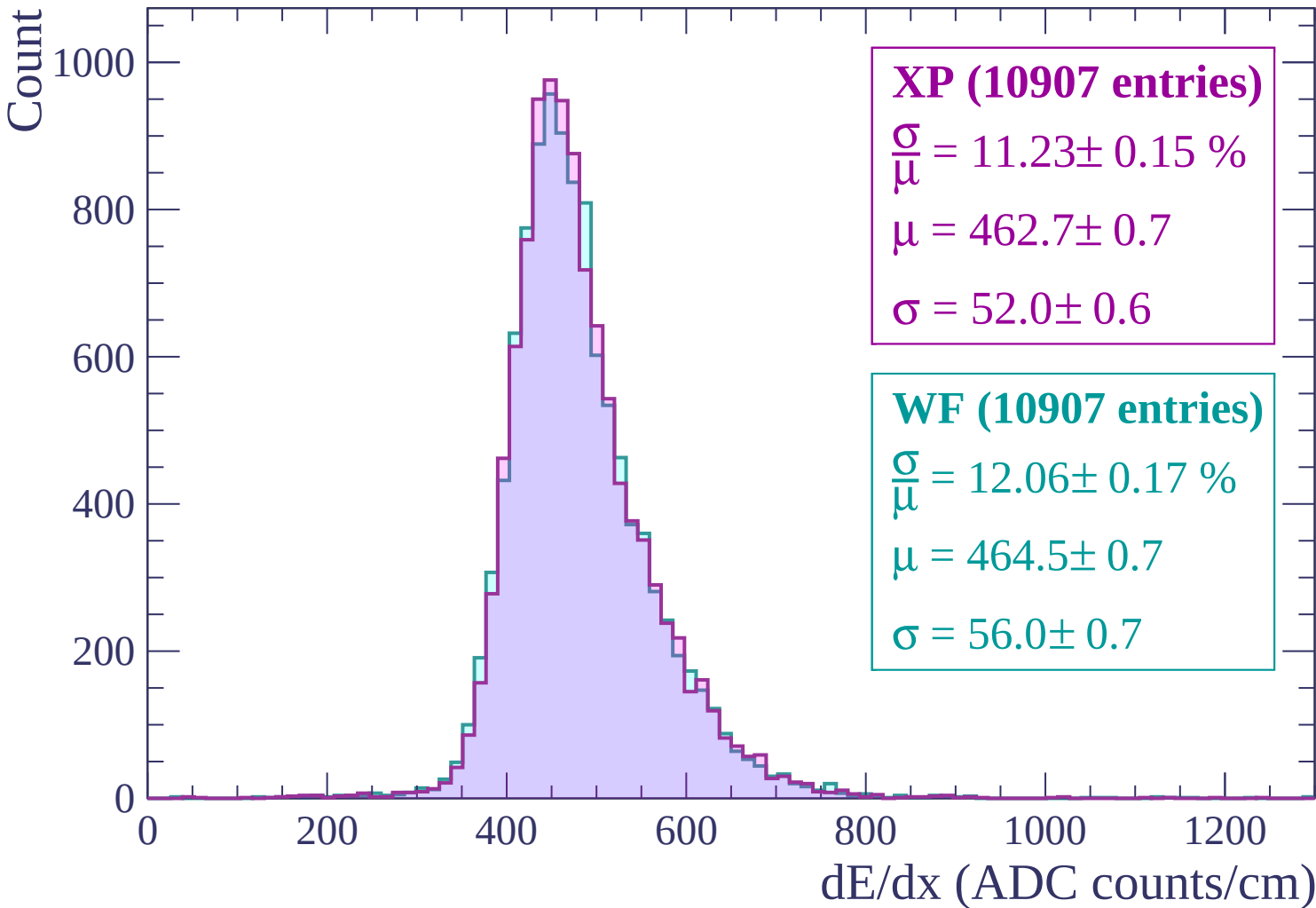
$$\sigma = 55.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1260



Energy loss | $-1260 < p < -1200$



Energy loss | -1200 < p < -1140

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (11400 entries)

$$\frac{\sigma}{\mu} = 11.38 \pm 0.15 \%$$

$$\mu = 460.4 \pm 0.7$$

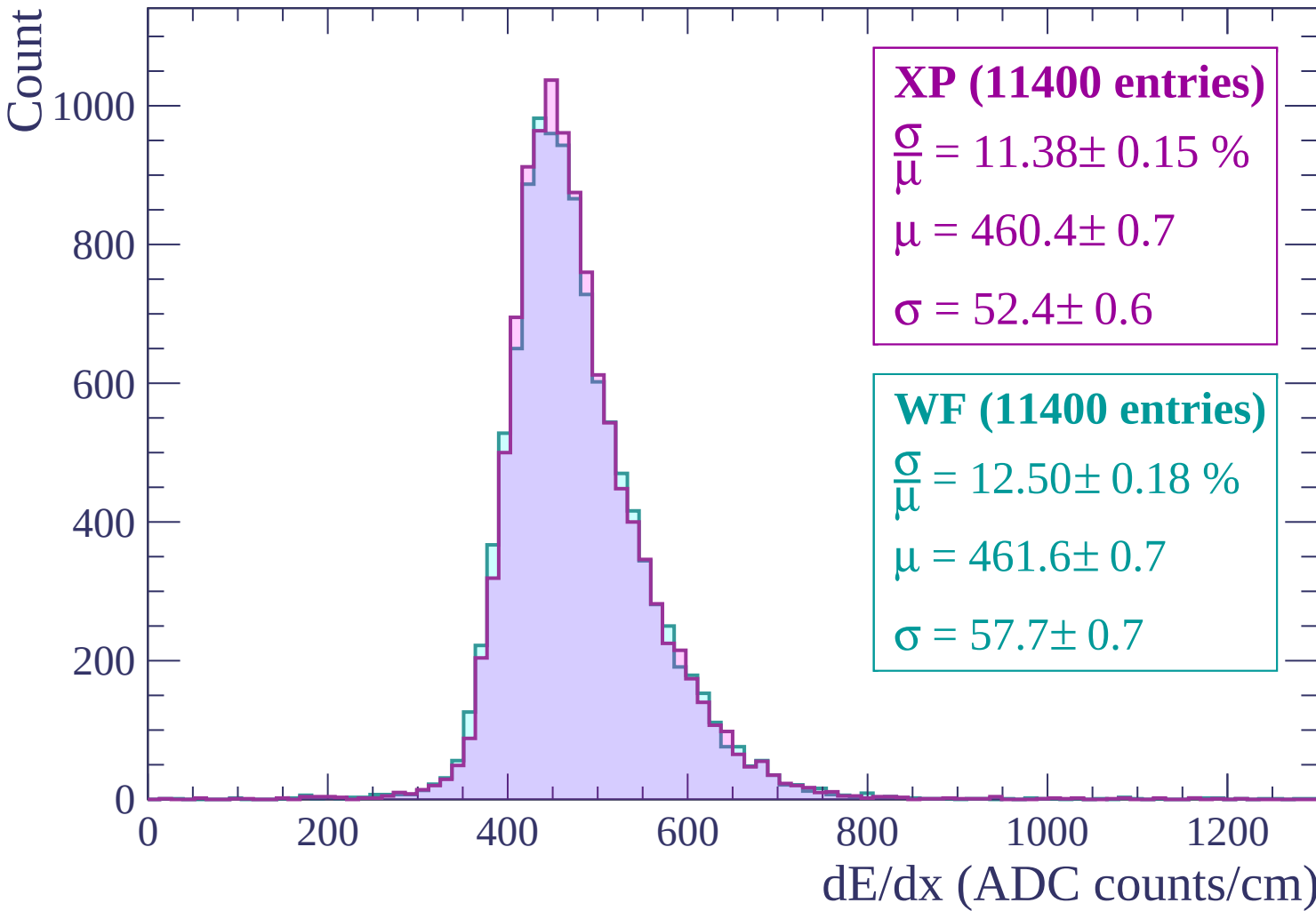
$$\sigma = 52.4 \pm 0.6$$

WF (11400 entries)

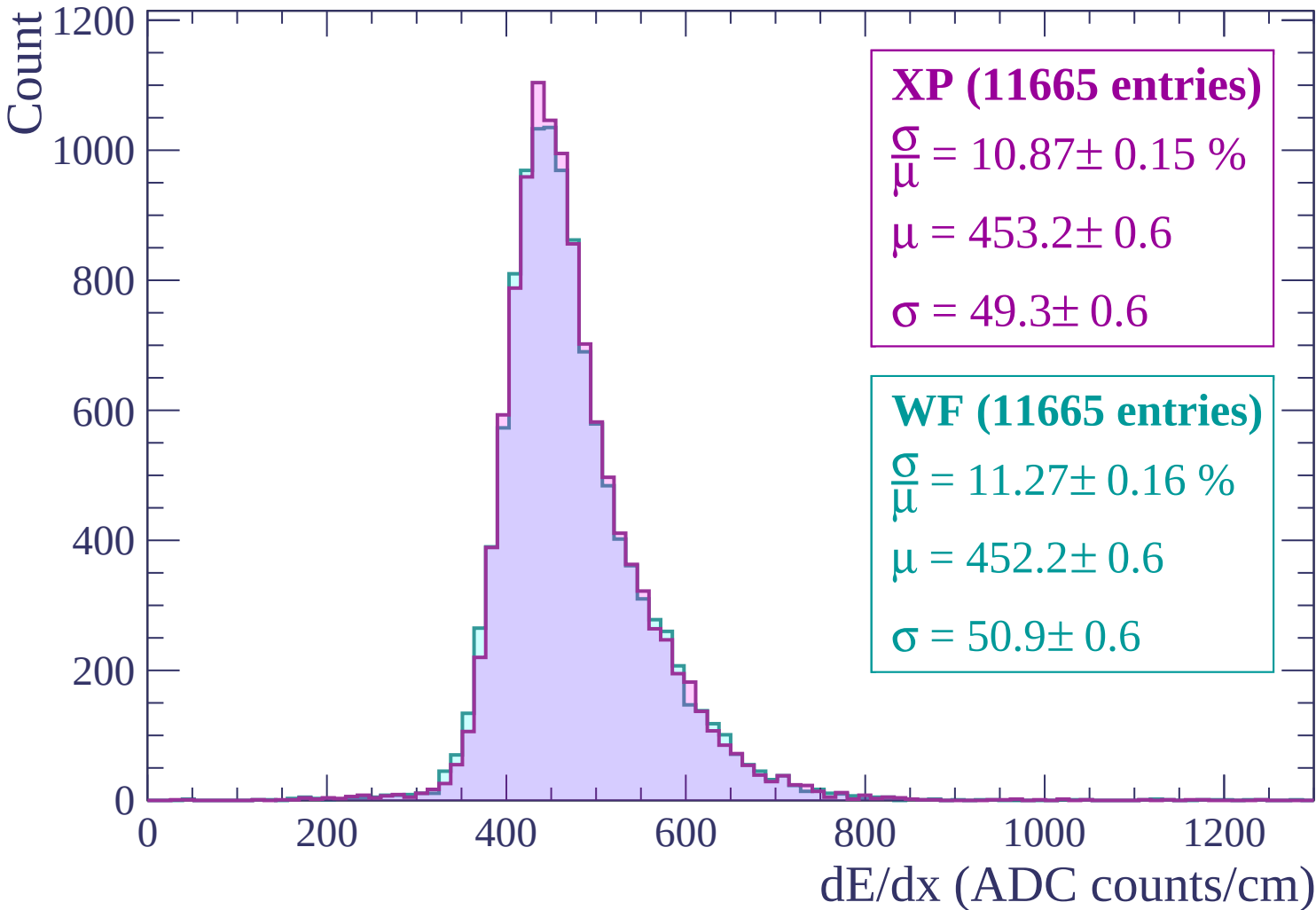
$$\frac{\sigma}{\mu} = 12.50 \pm 0.18 \%$$

$$\mu = 461.6 \pm 0.7$$

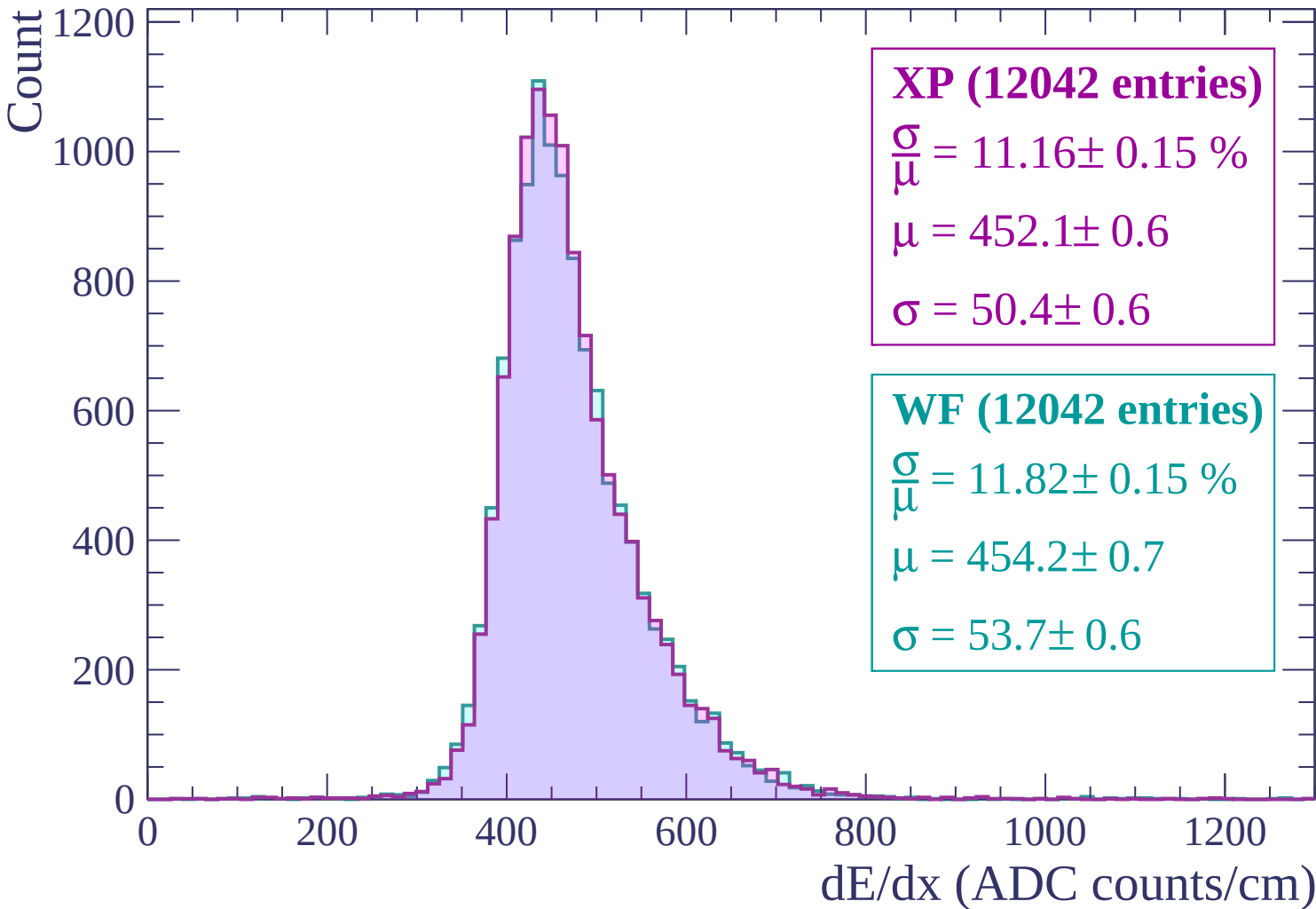
$$\sigma = 57.7 \pm 0.7$$



Energy loss | $-1140 < p < -1080$



Energy loss | -1080 < p < -1020



Energy loss | $-1020 < p < -960$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (12694 entries)

$\frac{\sigma}{\mu} = 10.81 \pm 0.15 \%$

$\mu = 445.3 \pm 0.6$

$\sigma = 48.1 \pm 0.6$

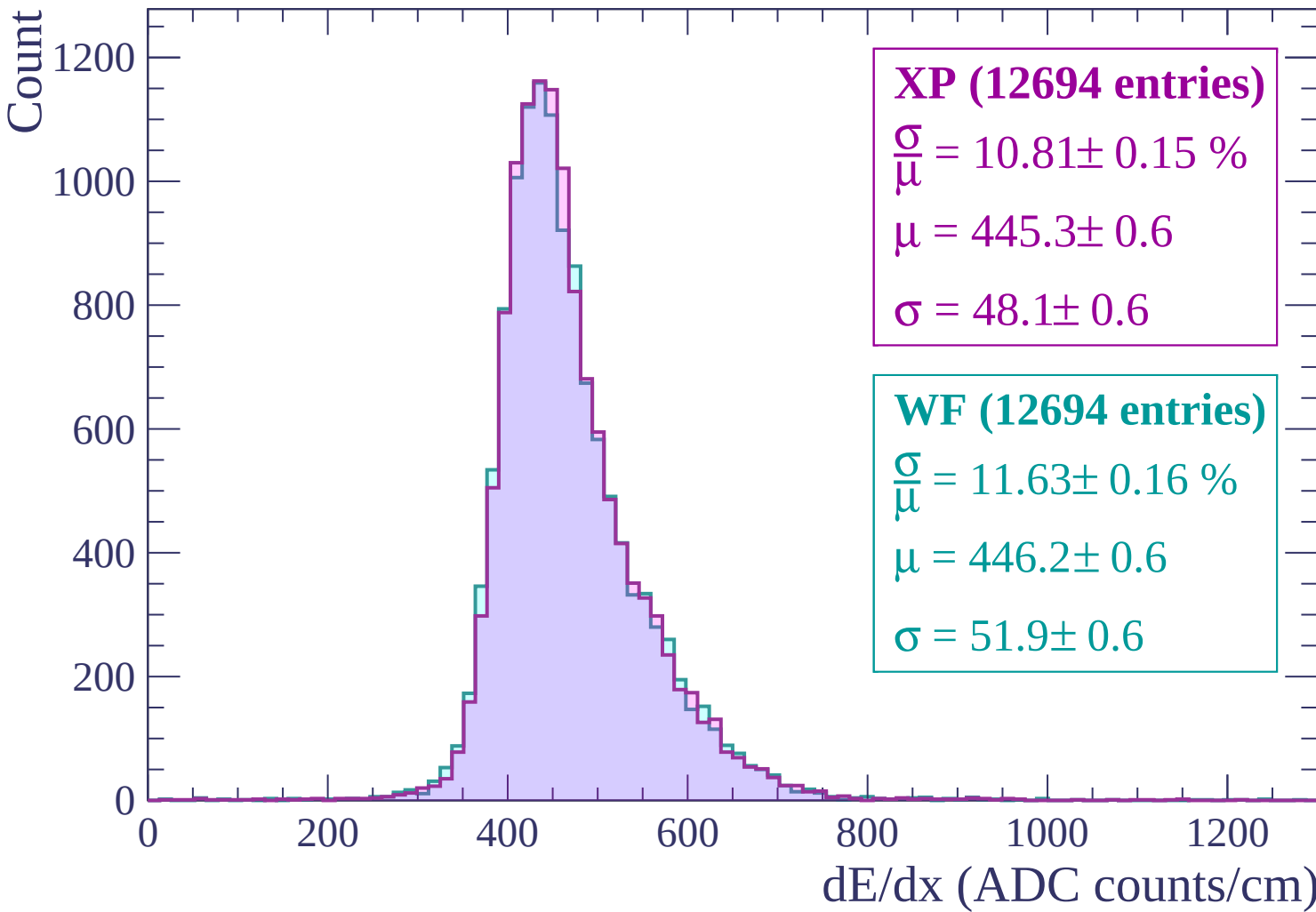
WF (12694 entries)

$\frac{\sigma}{\mu} = 11.63 \pm 0.16 \%$

$\mu = 446.2 \pm 0.6$

$\sigma = 51.9 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $-960 < p < -900$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (12930 entries)

$\frac{\sigma}{\mu} = 10.77 \pm 0.14 \%$

$\mu = 442.9 \pm 0.6$

$\sigma = 47.7 \pm 0.6$

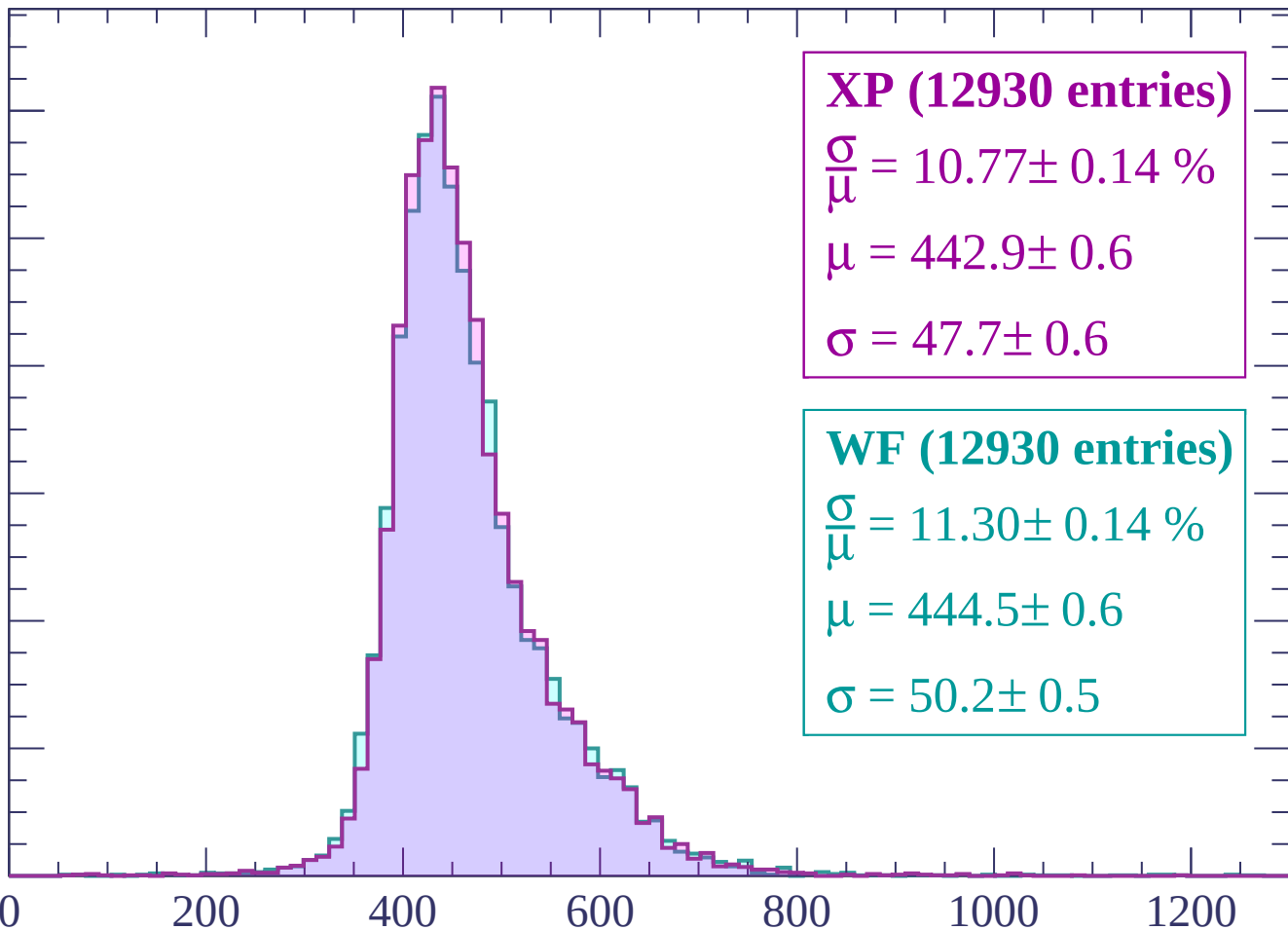
WF (12930 entries)

$\frac{\sigma}{\mu} = 11.30 \pm 0.14 \%$

$\mu = 444.5 \pm 0.6$

$\sigma = 50.2 \pm 0.5$

dE/dx (ADC counts/cm)



Energy loss | $-900 < p < -840$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (13322 entries)

$$\frac{\sigma}{\mu} = 10.55 \pm 0.12 \%$$

$$\mu = 439.5 \pm 0.5$$

$$\sigma = 46.4 \pm 0.5$$

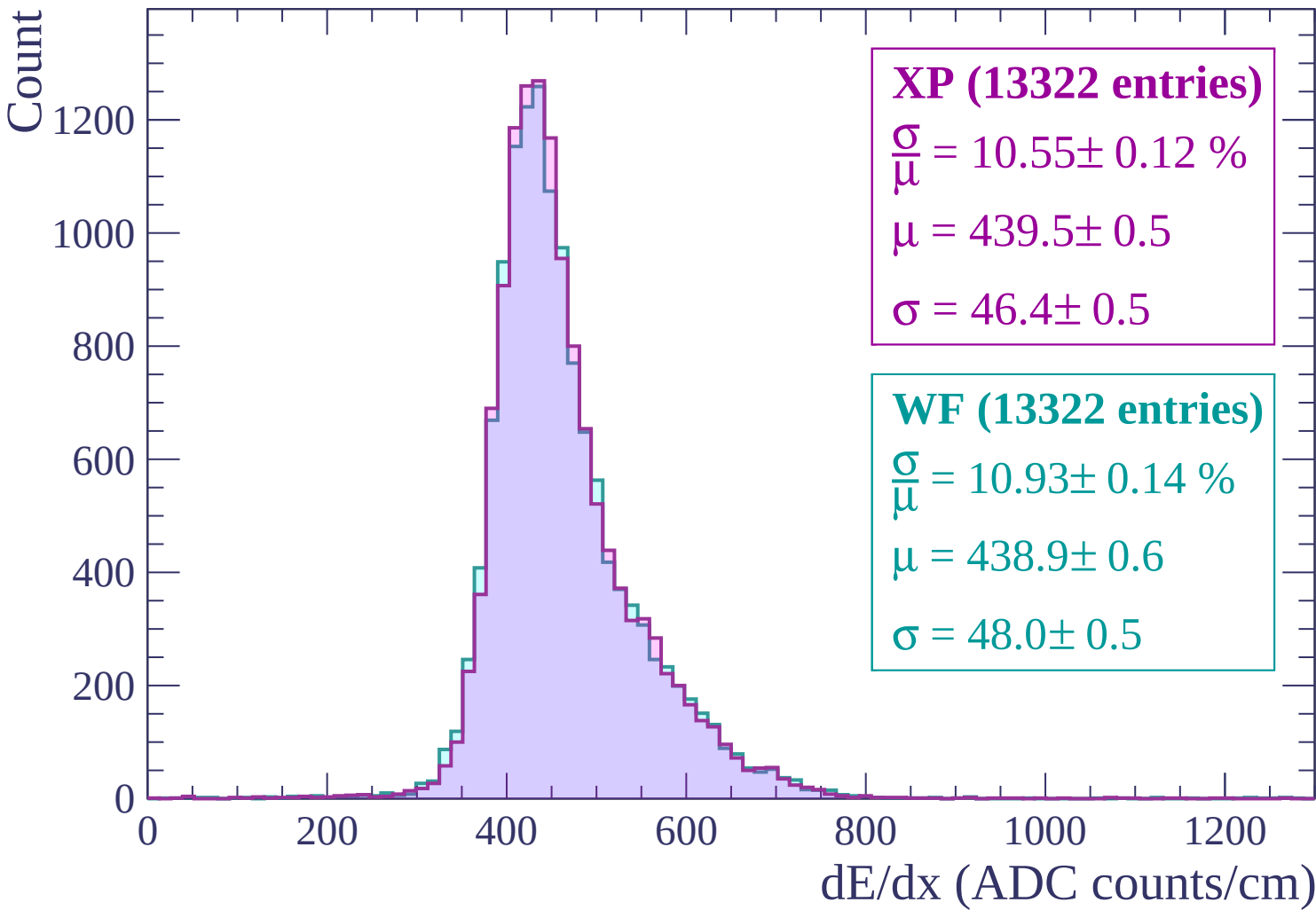
WF (13322 entries)

$$\frac{\sigma}{\mu} = 10.93 \pm 0.14 \%$$

$$\mu = 438.9 \pm 0.6$$

$$\sigma = 48.0 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss | $-840 < p < -780$

Count

1400
1200
1000
800
600
400
200
0

XP (13950 entries)

$\frac{\sigma}{\mu} = 11.07 \pm 0.14 \%$

$\mu = 436.1 \pm 0.6$

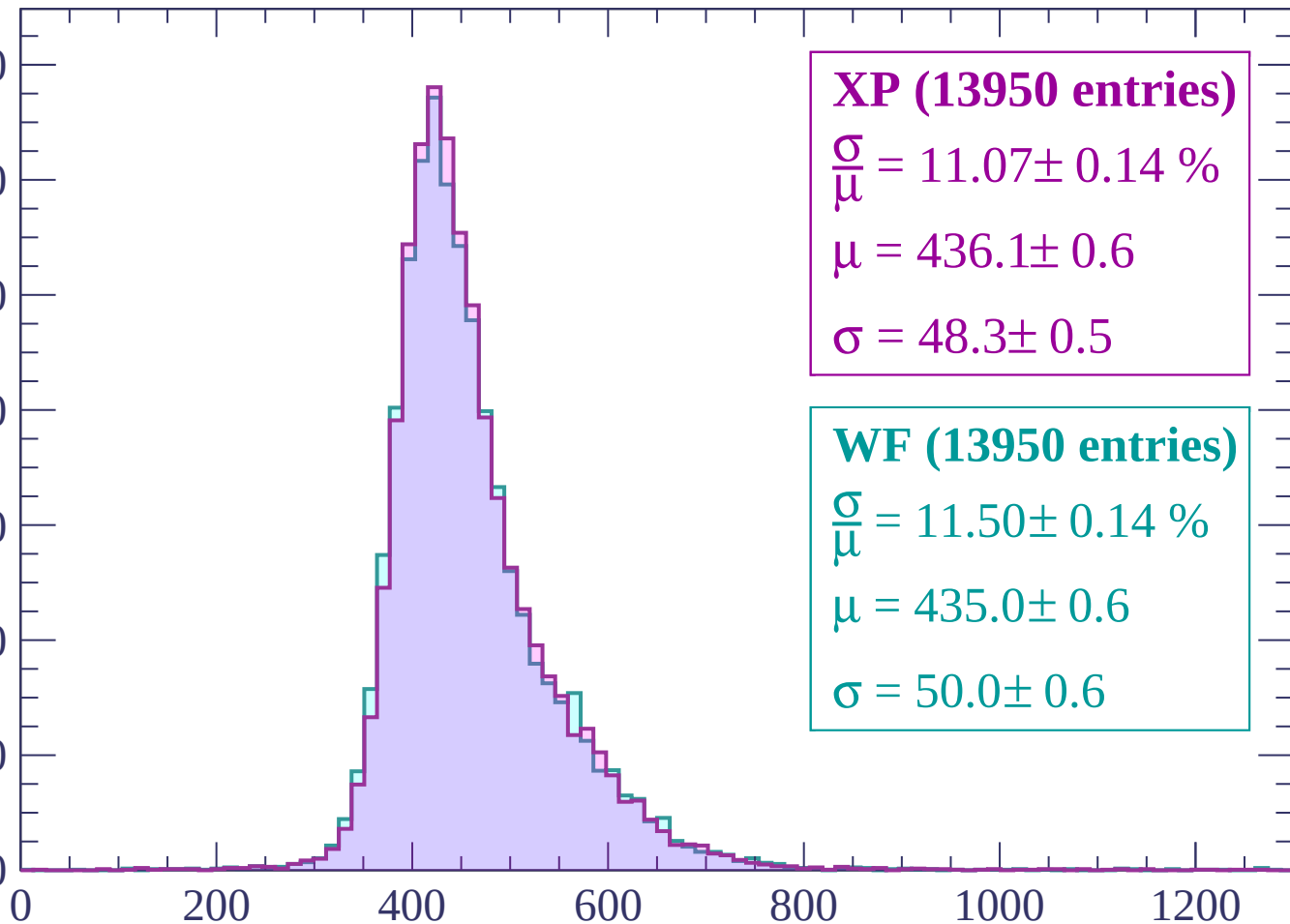
$\sigma = 48.3 \pm 0.5$

WF (13950 entries)

$\frac{\sigma}{\mu} = 11.50 \pm 0.14 \%$

$\mu = 435.0 \pm 0.6$

$\sigma = 50.0 \pm 0.6$



dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count

1400
1200
1000
800
600
400
200
0

XP (14607 entries)

$\frac{\sigma}{\mu} = 10.52 \pm 0.13 \%$

$\mu = 431.2 \pm 0.5$

$\sigma = 45.4 \pm 0.5$

WF (14607 entries)

$\frac{\sigma}{\mu} = 11.14 \pm 0.14 \%$

$\mu = 430.1 \pm 0.5$

$\sigma = 47.9 \pm 0.6$

0

200

400

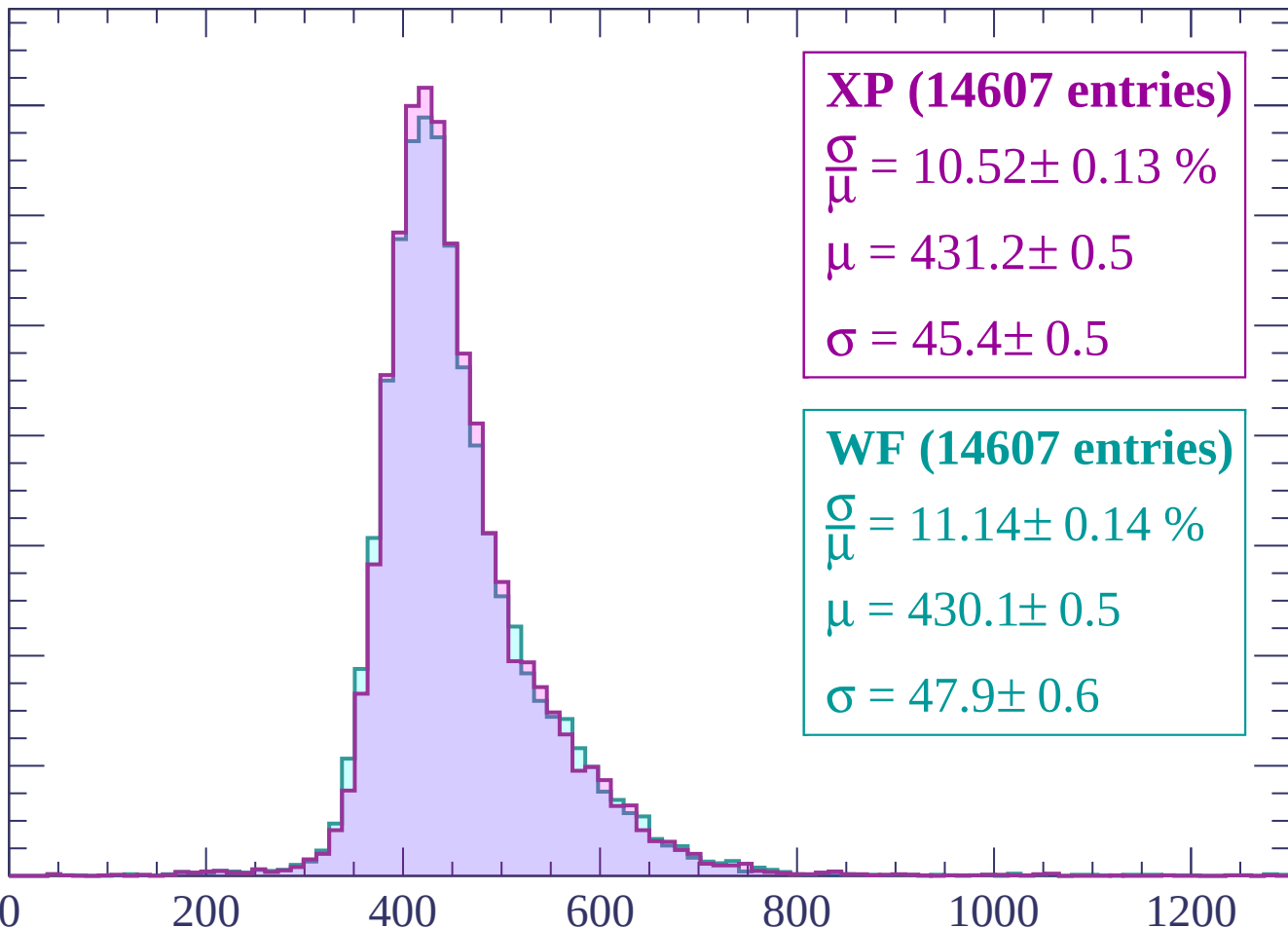
600

800

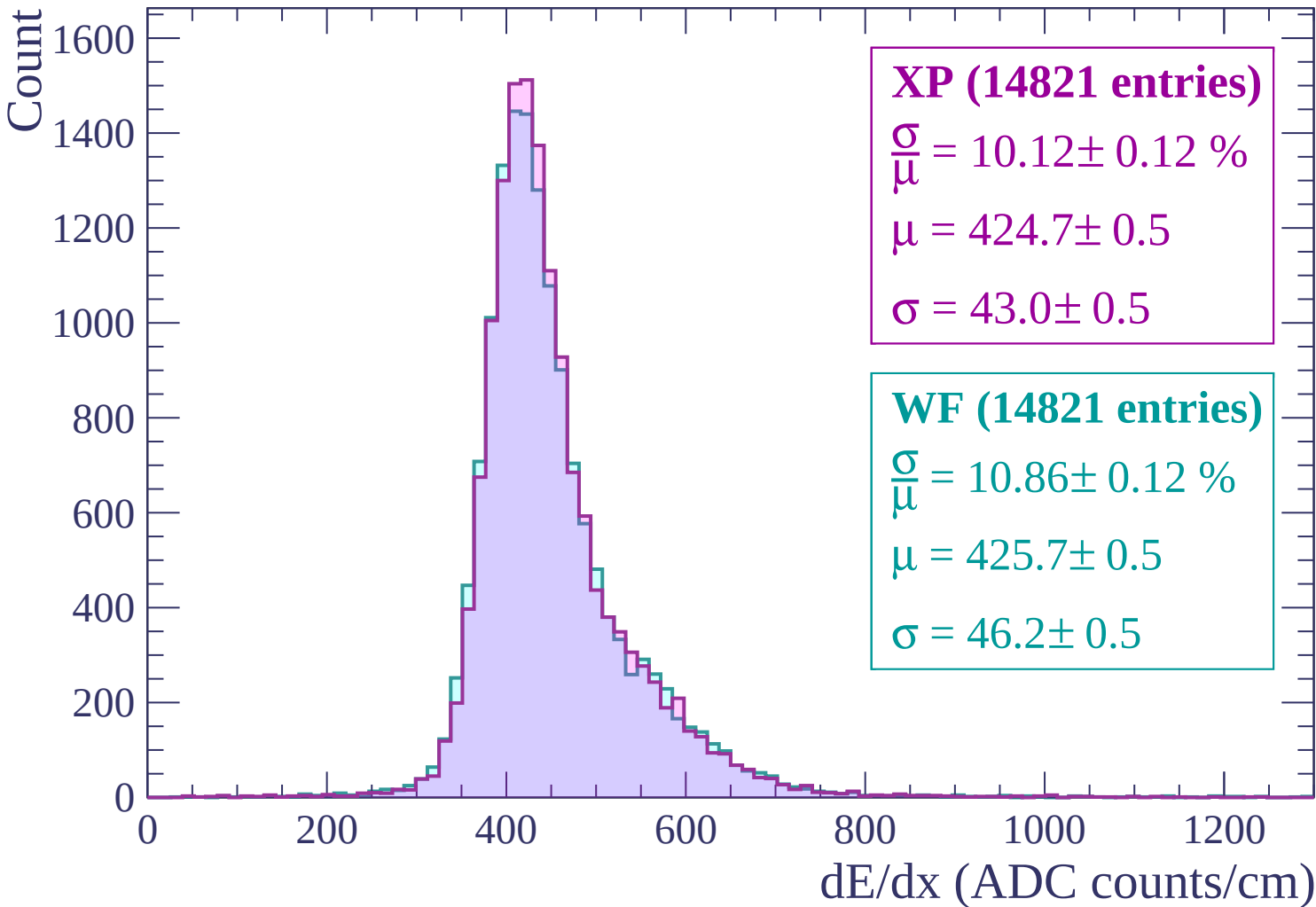
1000

1200

dE/dx (ADC counts/cm)



Energy loss | $-720 < p < -660$



Energy loss | $-660 < p < -600$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (15561 entries)

$\frac{\sigma}{\mu} = 10.43 \pm 0.13 \%$

$\mu = 420.7 \pm 0.5$

$\sigma = 43.9 \pm 0.5$

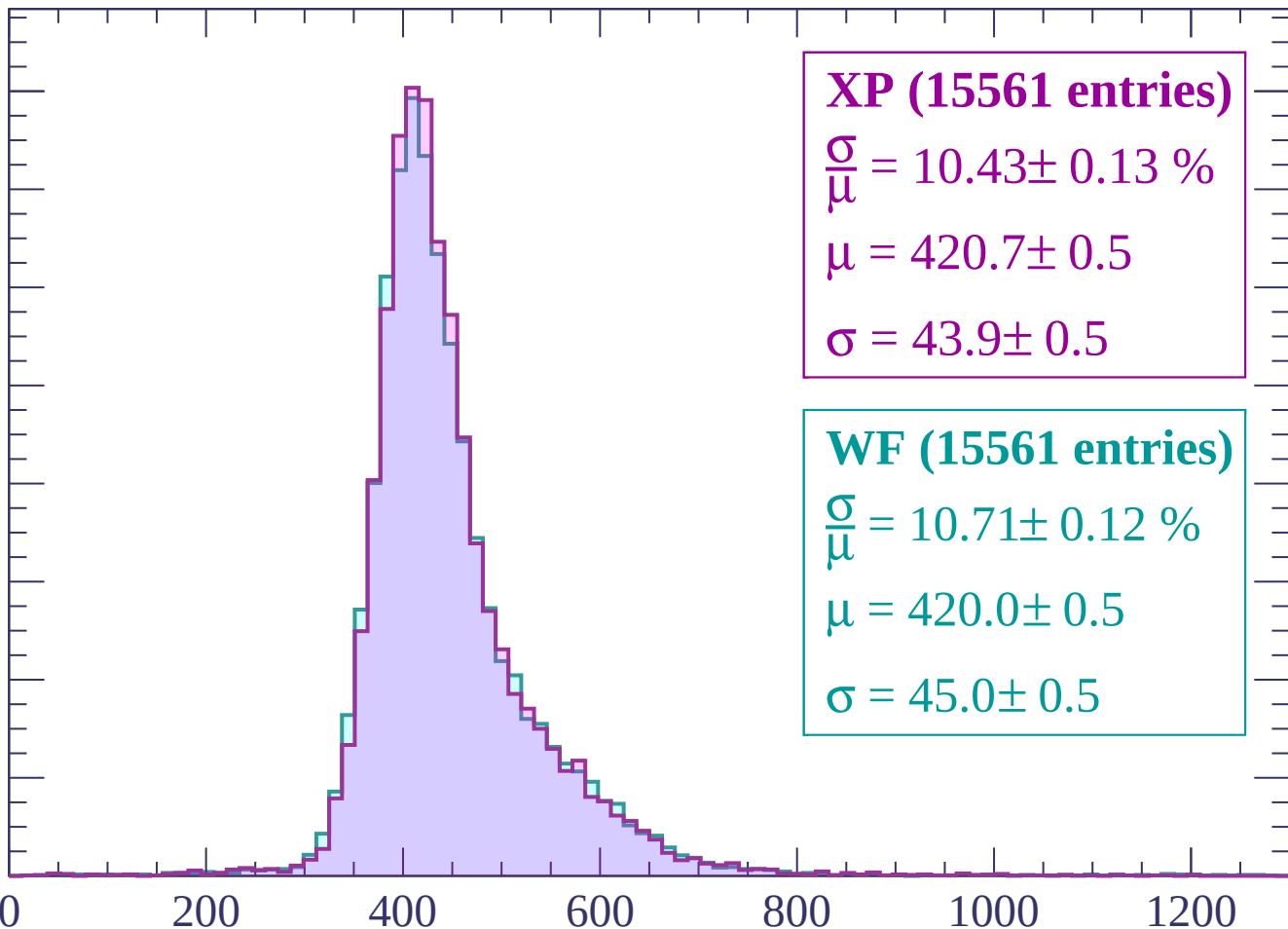
WF (15561 entries)

$\frac{\sigma}{\mu} = 10.71 \pm 0.12 \%$

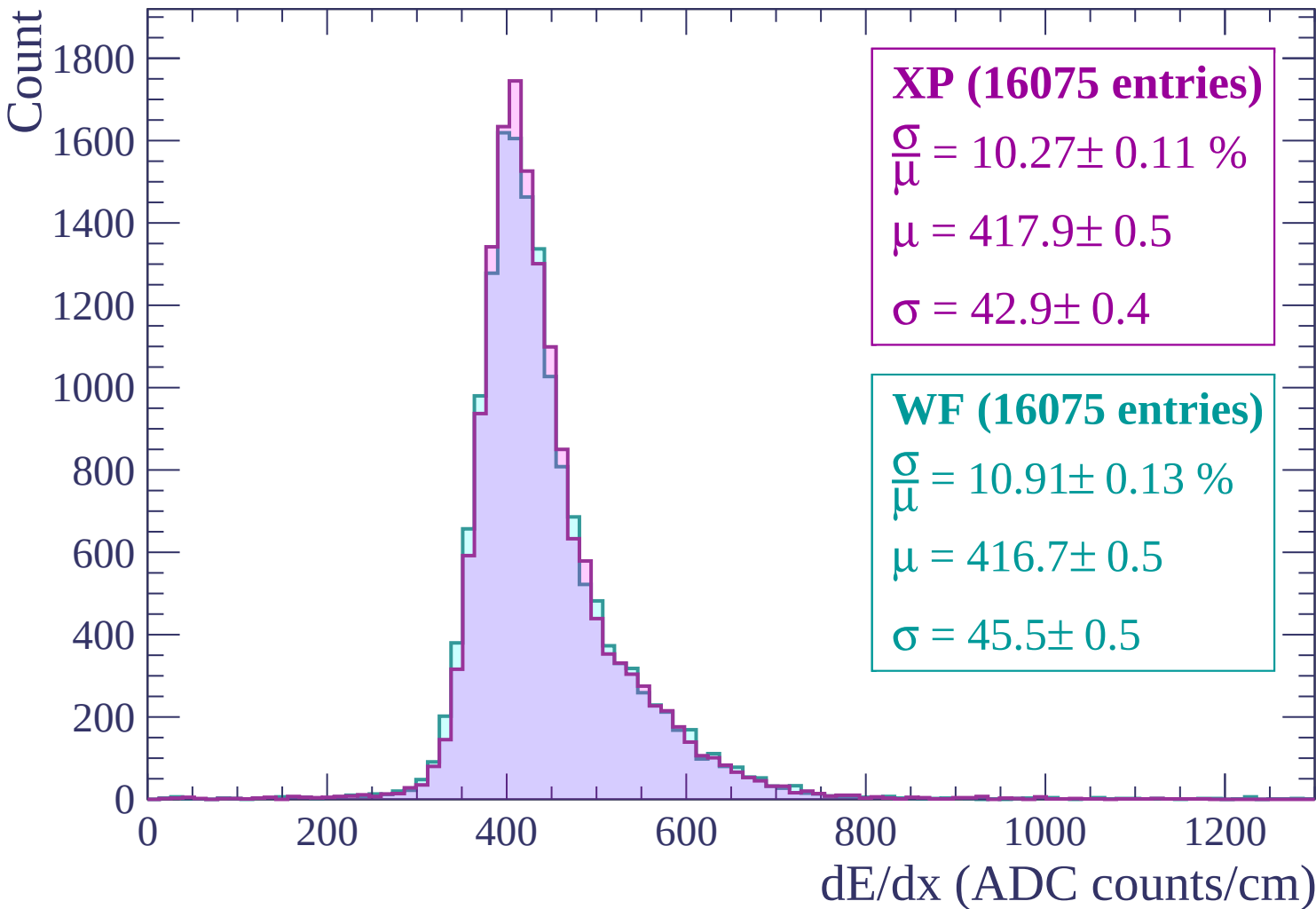
$\mu = 420.0 \pm 0.5$

$\sigma = 45.0 \pm 0.5$

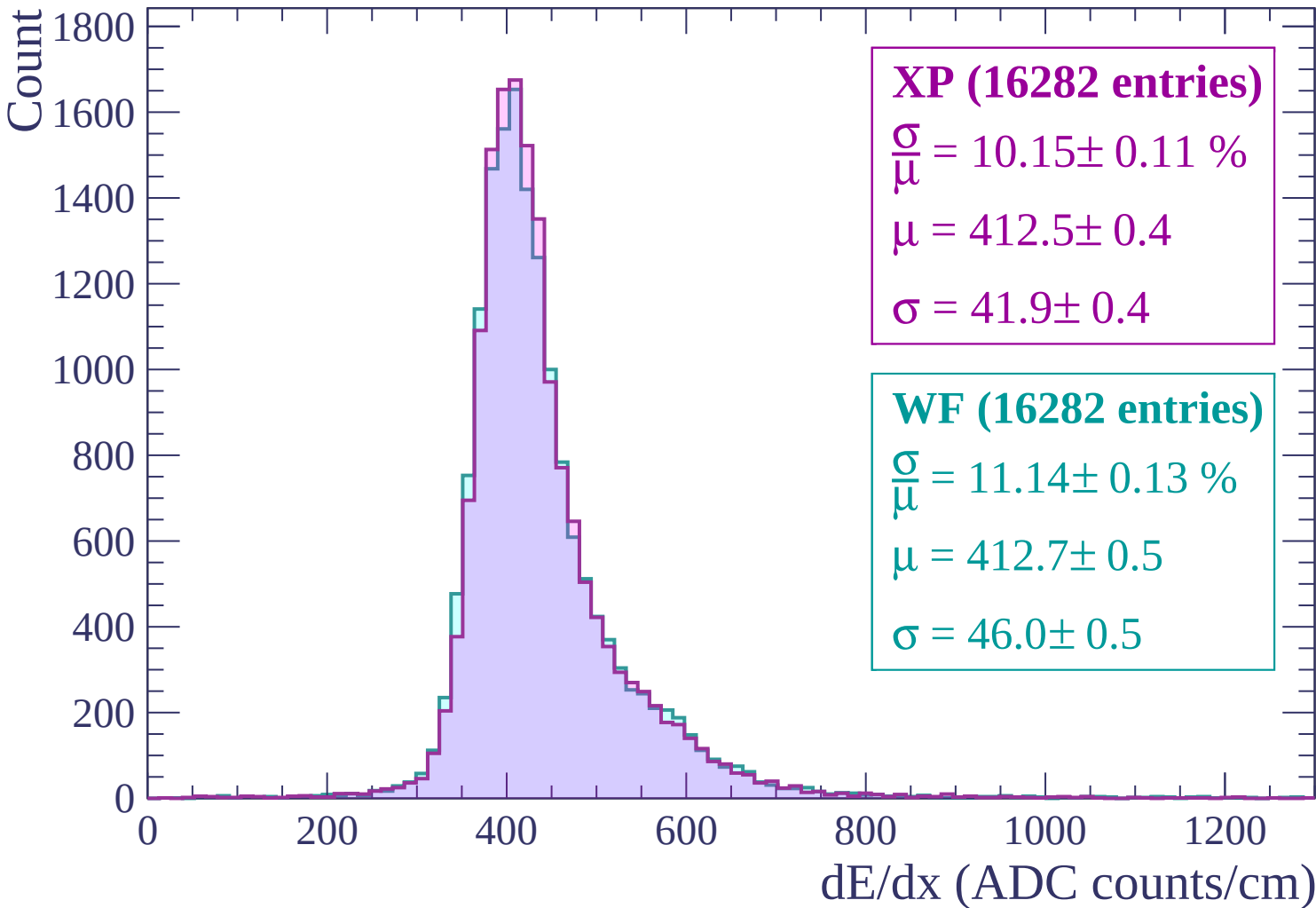
0 200 400 600 800 1000 1200
dE/dx (ADC counts/cm)



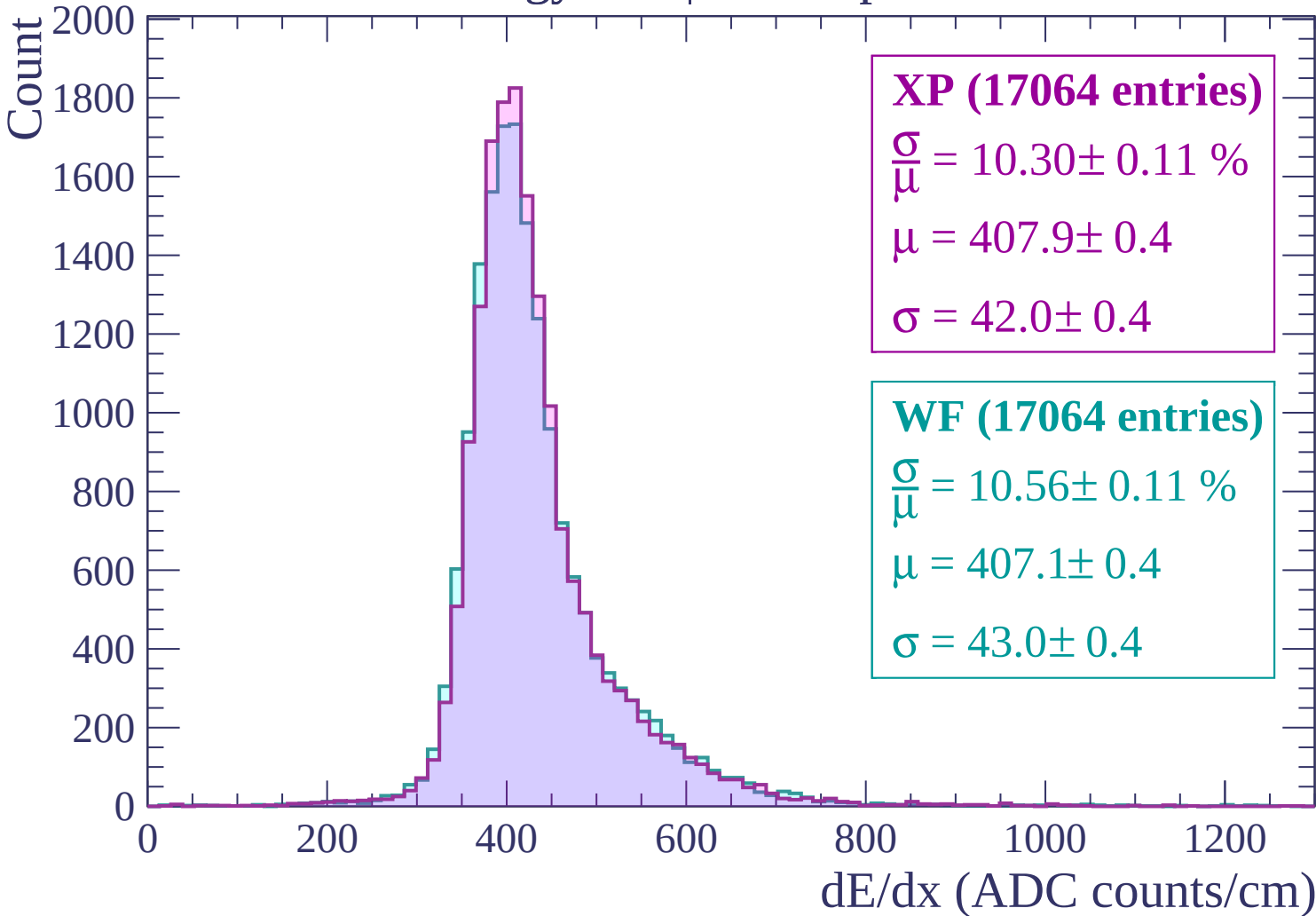
Energy loss | $-600 < p < -540$



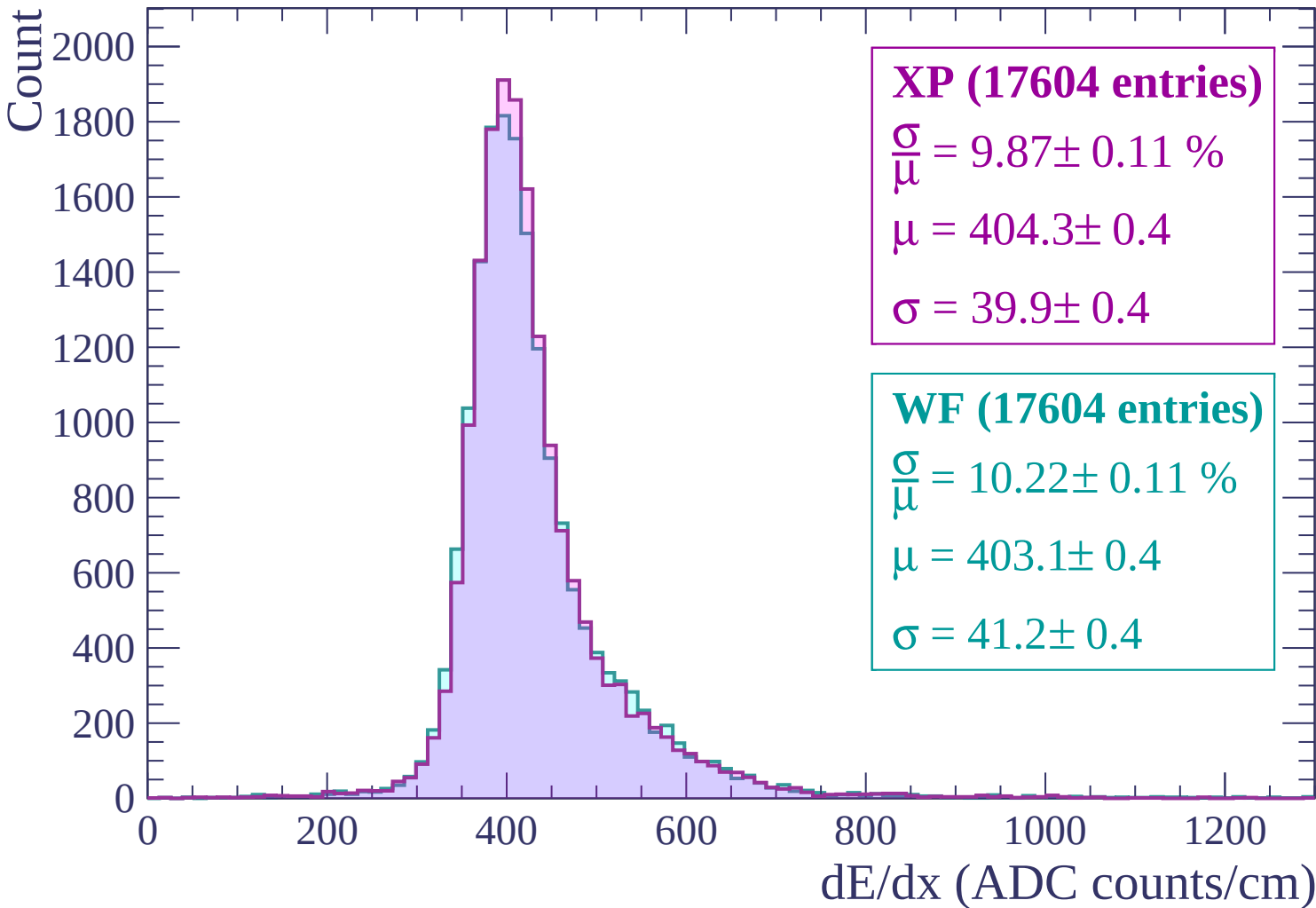
Energy loss | $-540 < p < -480$



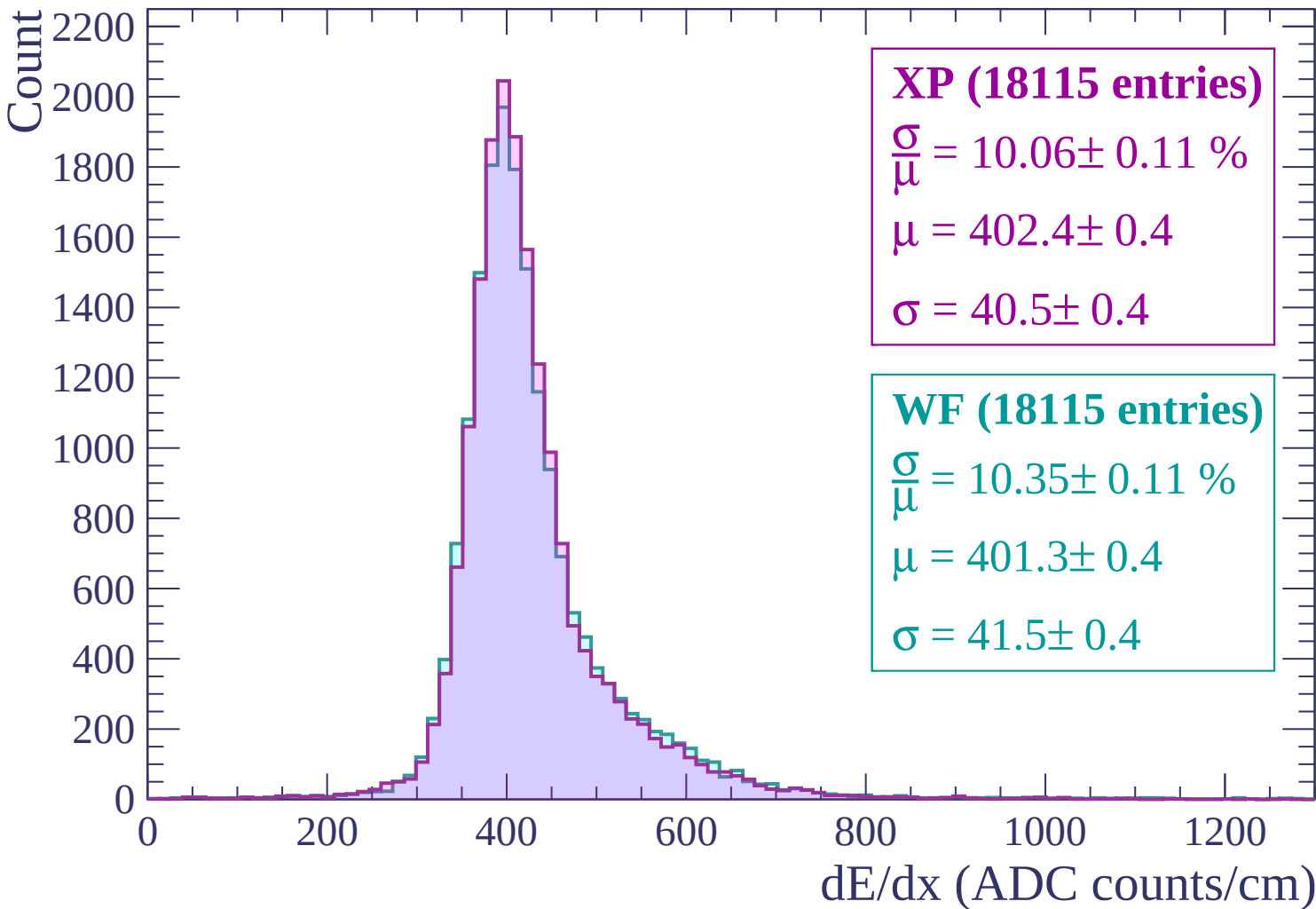
Energy loss | $-480 < p < -420$



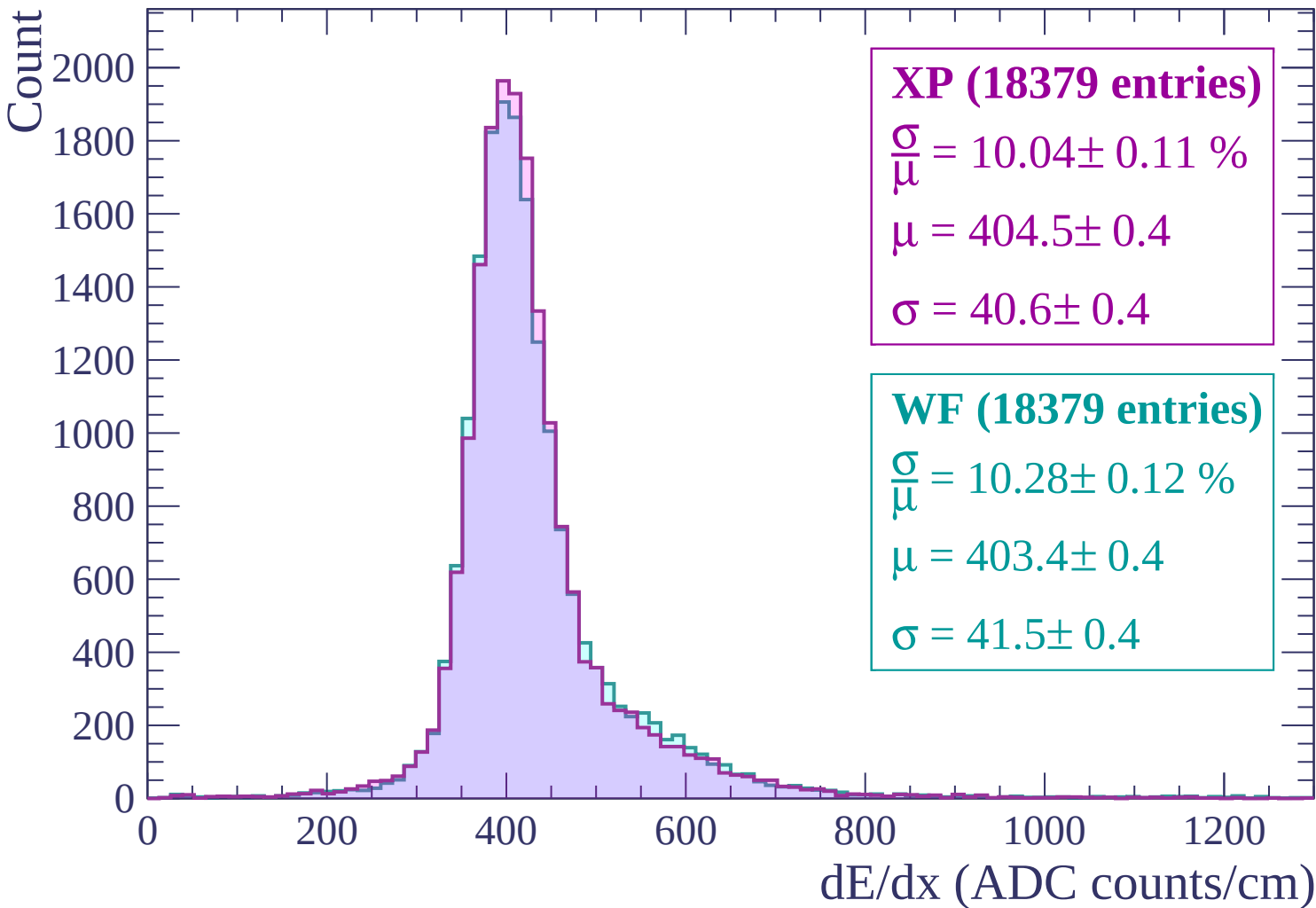
Energy loss | $-420 < p < -360$



Energy loss | -360 < p < -300



Energy loss | $-300 < p < -240$



Energy loss | $-240 < p < -180$

Count

2000
1800
1600
1400
1200
1000
800
600
400
200
0

XP (18412 entries)

$$\frac{\sigma}{\mu} = 10.33 \pm 0.11 \%$$

$$\mu = 414.4 \pm 0.4$$

$$\sigma = 42.8 \pm 0.4$$

WF (18412 entries)

$$\frac{\sigma}{\mu} = 10.57 \pm 0.12 \%$$

$$\mu = 413.7 \pm 0.4$$

$$\sigma = 43.7 \pm 0.4$$

0

200

400

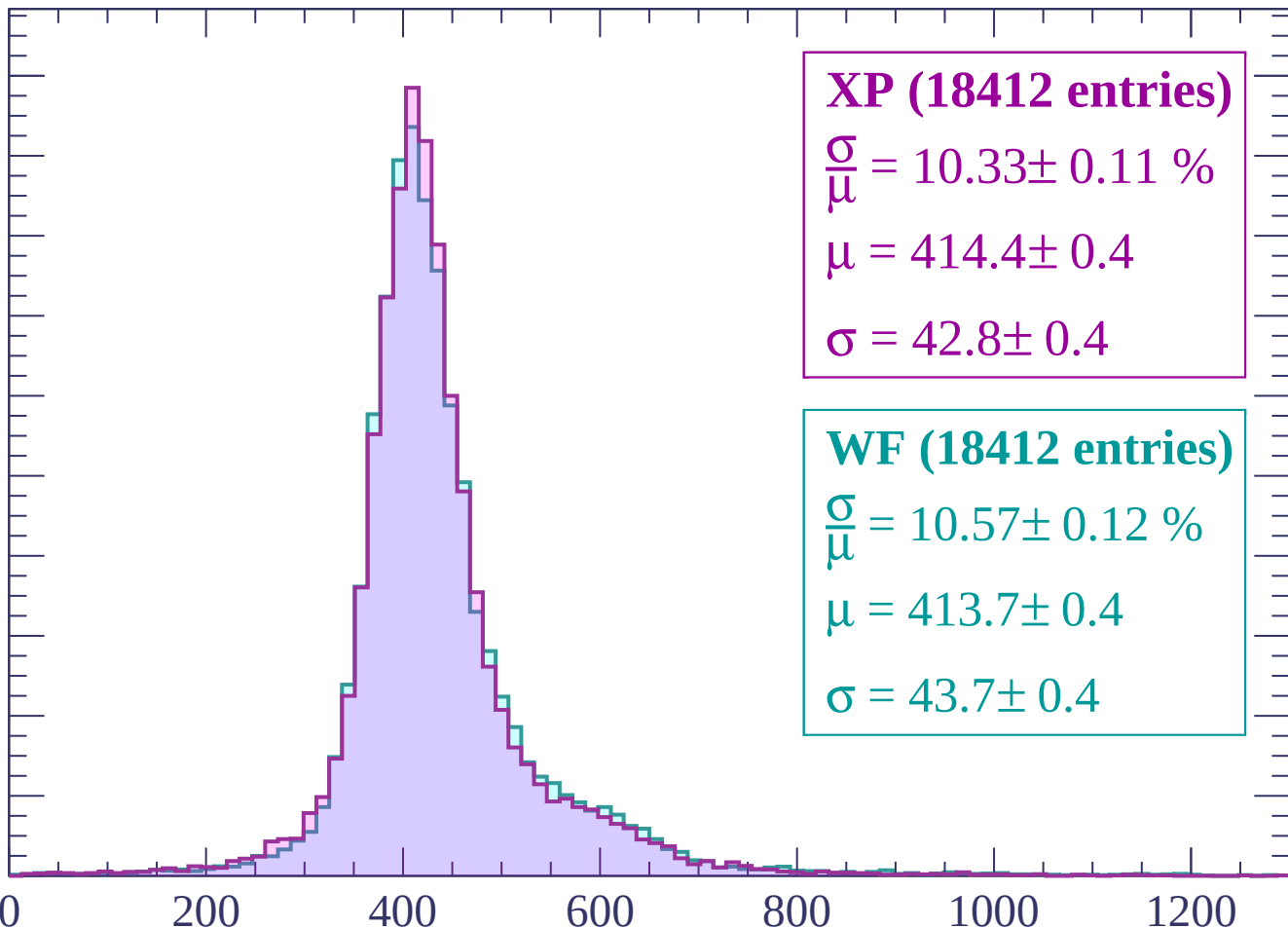
600

800

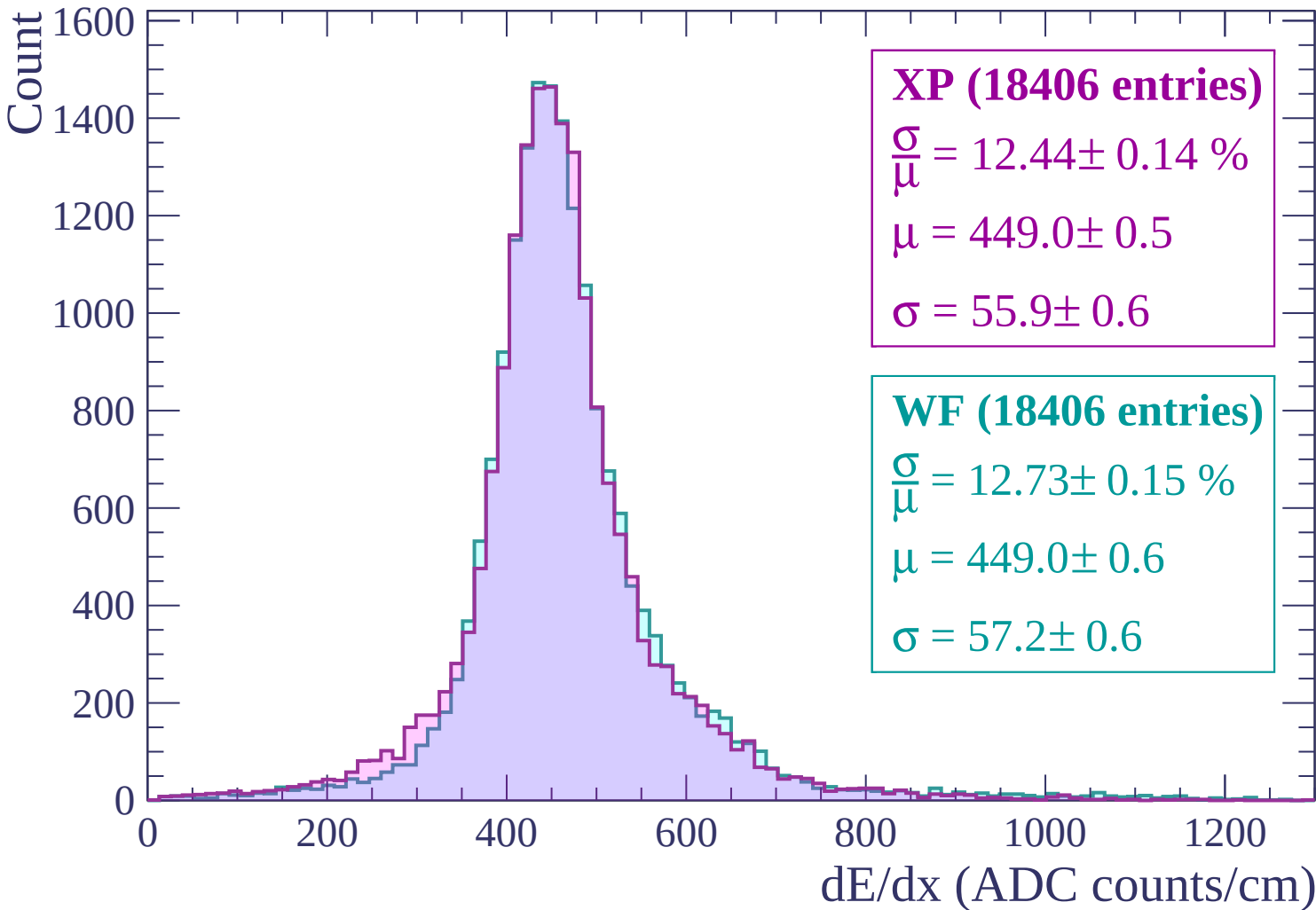
1000

1200

dE/dx (ADC counts/cm)



Energy loss | $-180 < p < -120$



Energy loss | $-120 < p < -60$

Count

1000

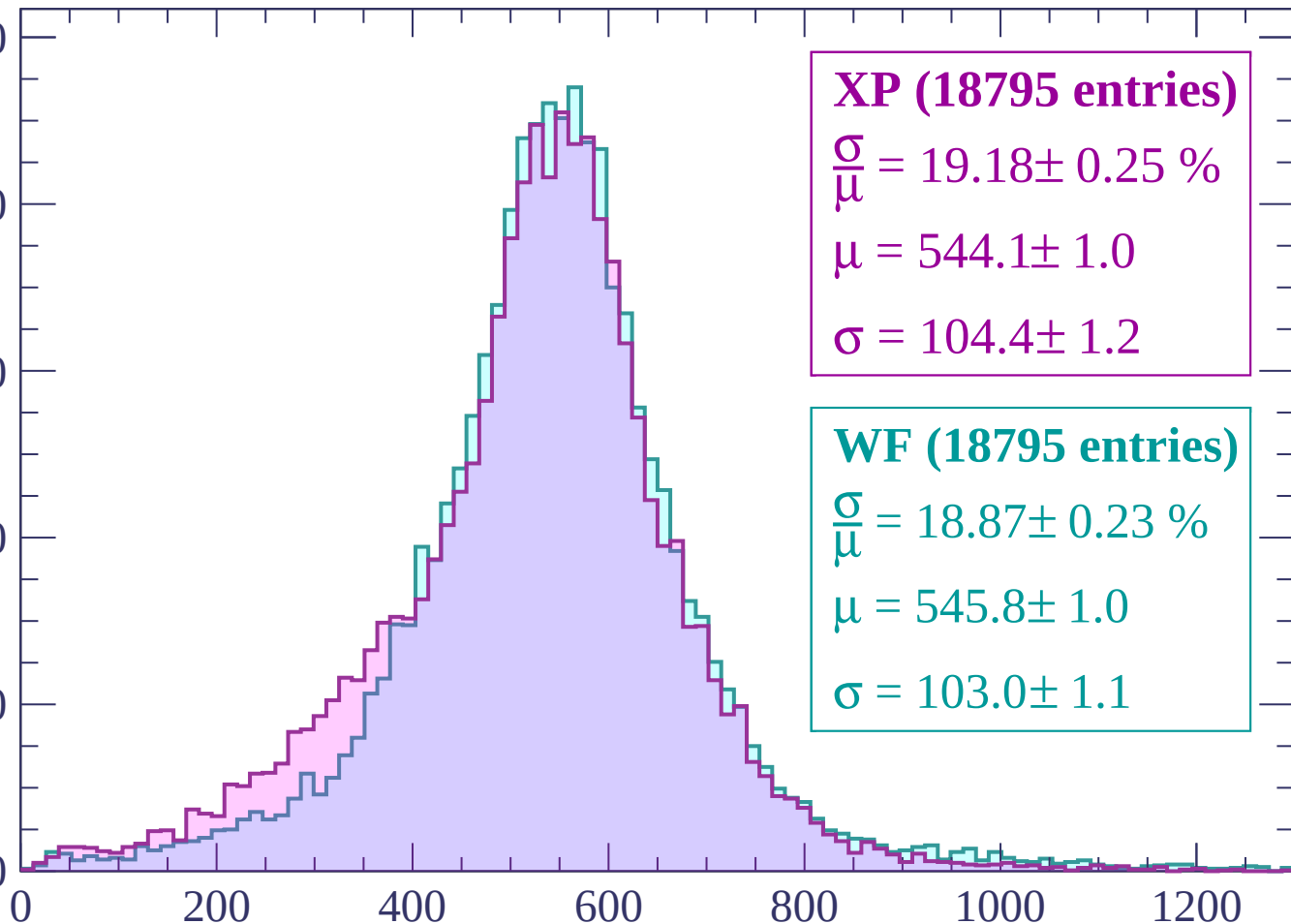
800

600

400

200

0



XP (18795 entries)

$\frac{\sigma}{\mu} = 19.18 \pm 0.25 \%$

$\mu = 544.1 \pm 1.0$

$\sigma = 104.4 \pm 1.2$

WF (18795 entries)

$\frac{\sigma}{\mu} = 18.87 \pm 0.23 \%$

$\mu = 545.8 \pm 1.0$

$\sigma = 103.0 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | $-60 < p < 0$

Count

2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0

XP (34336 entries)

$$\frac{\sigma}{\mu} = 11.46 \pm 0.11 \%$$

$$\mu = 550.0 \pm 0.5$$

$$\sigma = 63.0 \pm 0.5$$

WF (34336 entries)

$$\frac{\sigma}{\mu} = 12.30 \pm 0.11 \%$$

$$\mu = 545.8 \pm 0.5$$

$$\sigma = 67.1 \pm 0.5$$

0

200

400

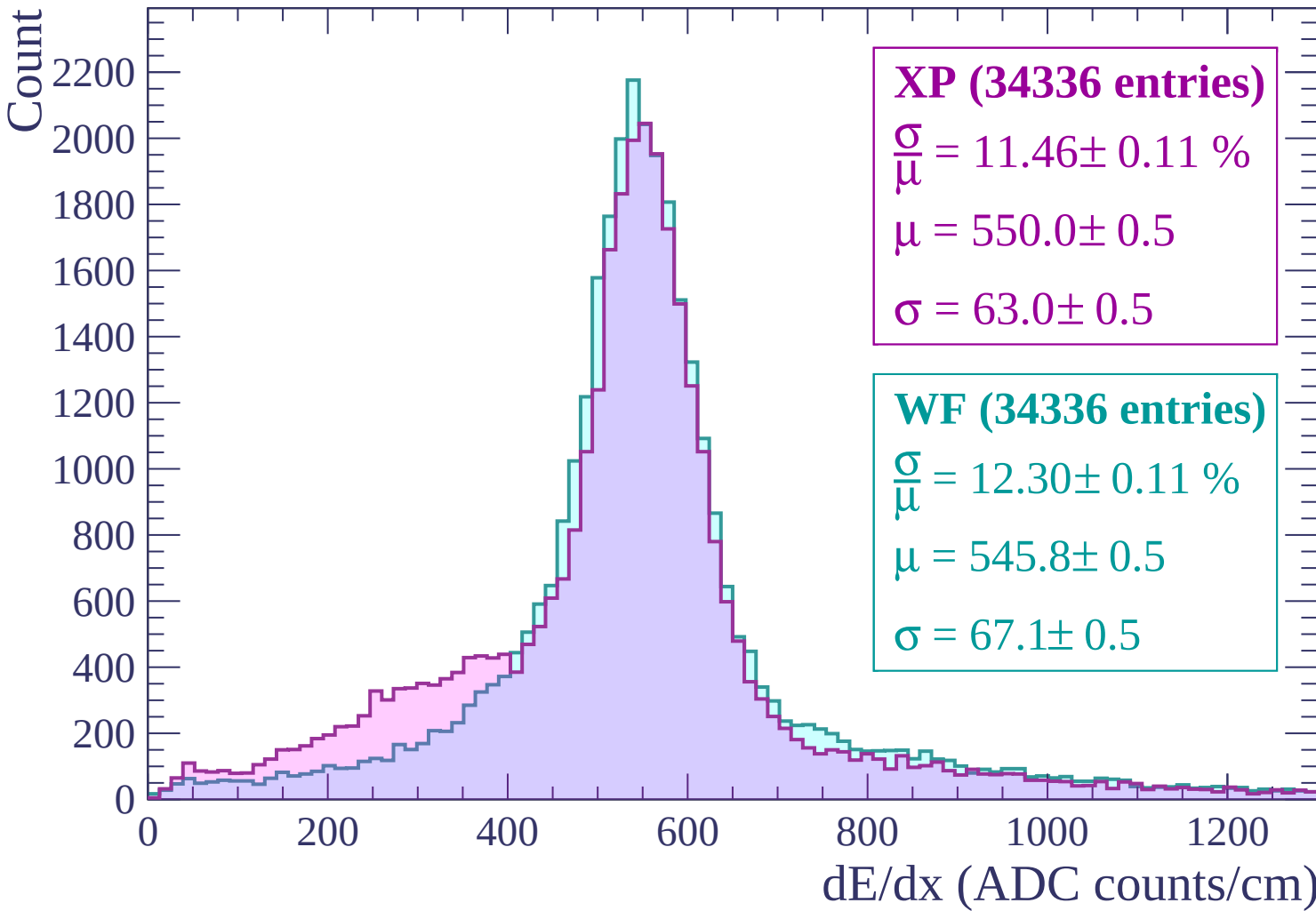
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $0 < p < 60$

Count

5000

4000

3000

2000

1000

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (78449 entries)

$\frac{\sigma}{\mu} = 15.48 \pm 0.09 \%$

$\mu = 496.5 \pm 0.4$

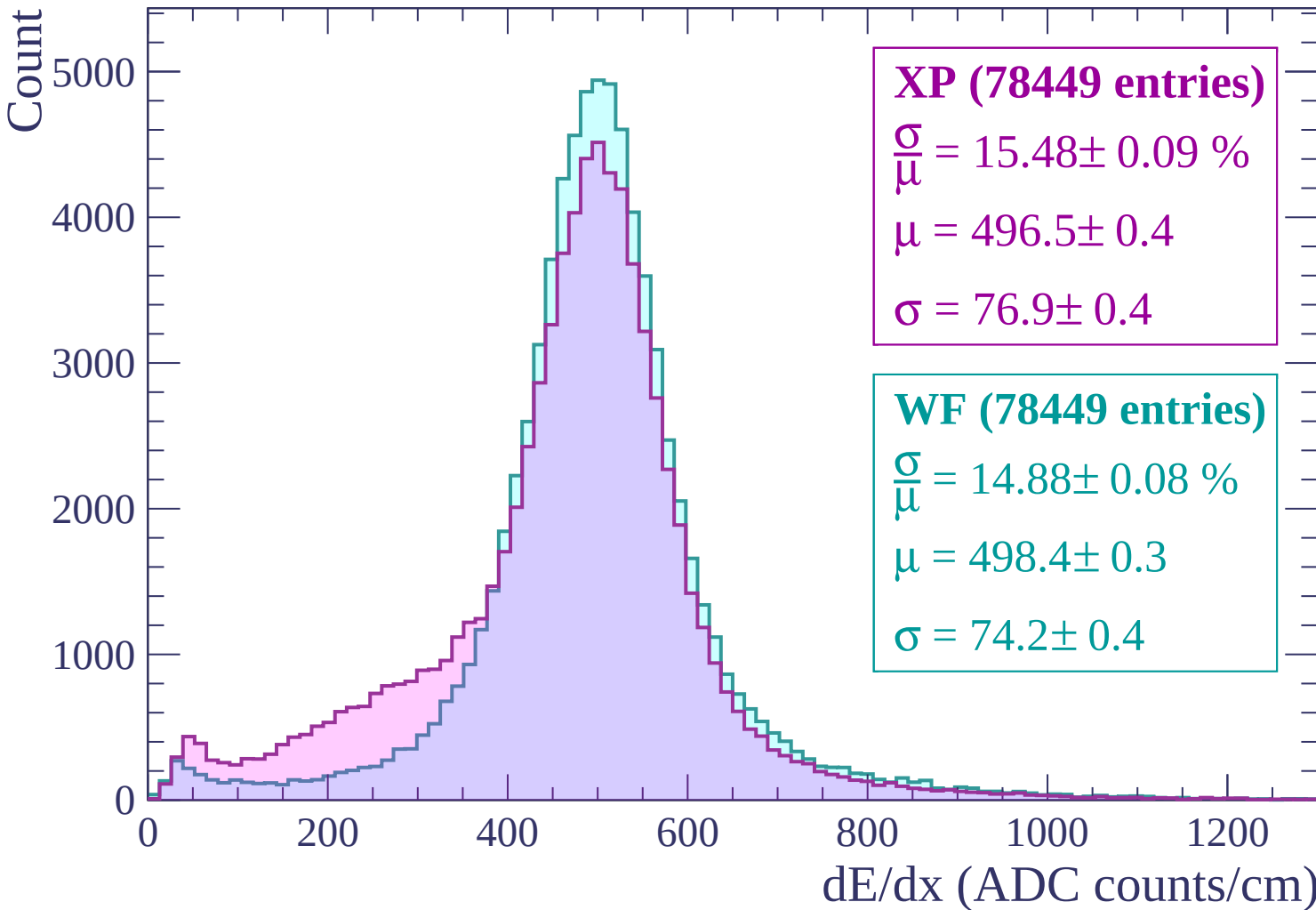
$\sigma = 76.9 \pm 0.4$

WF (78449 entries)

$\frac{\sigma}{\mu} = 14.88 \pm 0.08 \%$

$\mu = 498.4 \pm 0.3$

$\sigma = 74.2 \pm 0.4$



Energy loss | $60 < p < 120$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (20872 entries)

$\frac{\sigma}{\mu} = 12.64 \pm 0.15 \%$

$\mu = 546.1 \pm 0.7$

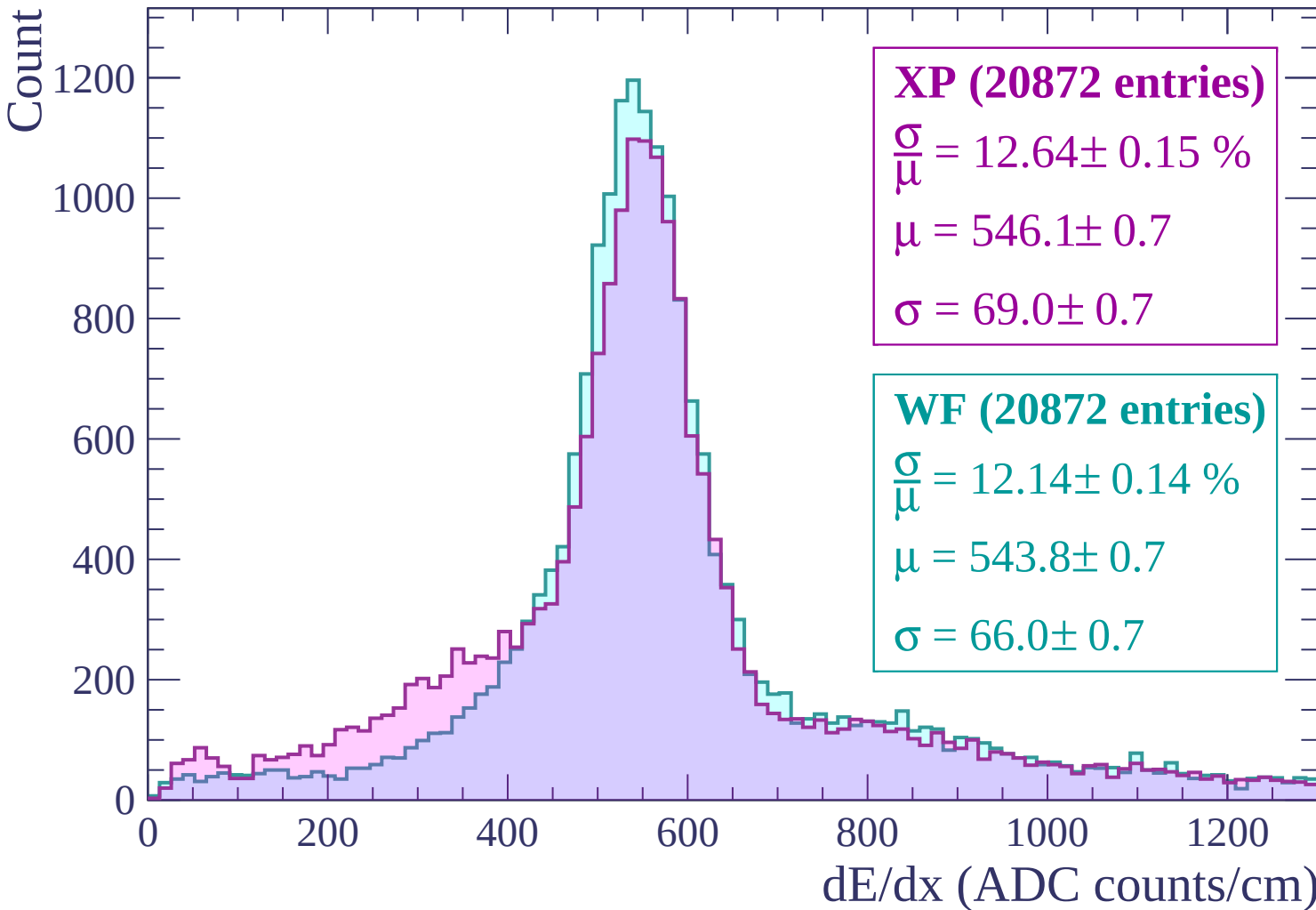
$\sigma = 69.0 \pm 0.7$

WF (20872 entries)

$\frac{\sigma}{\mu} = 12.14 \pm 0.14 \%$

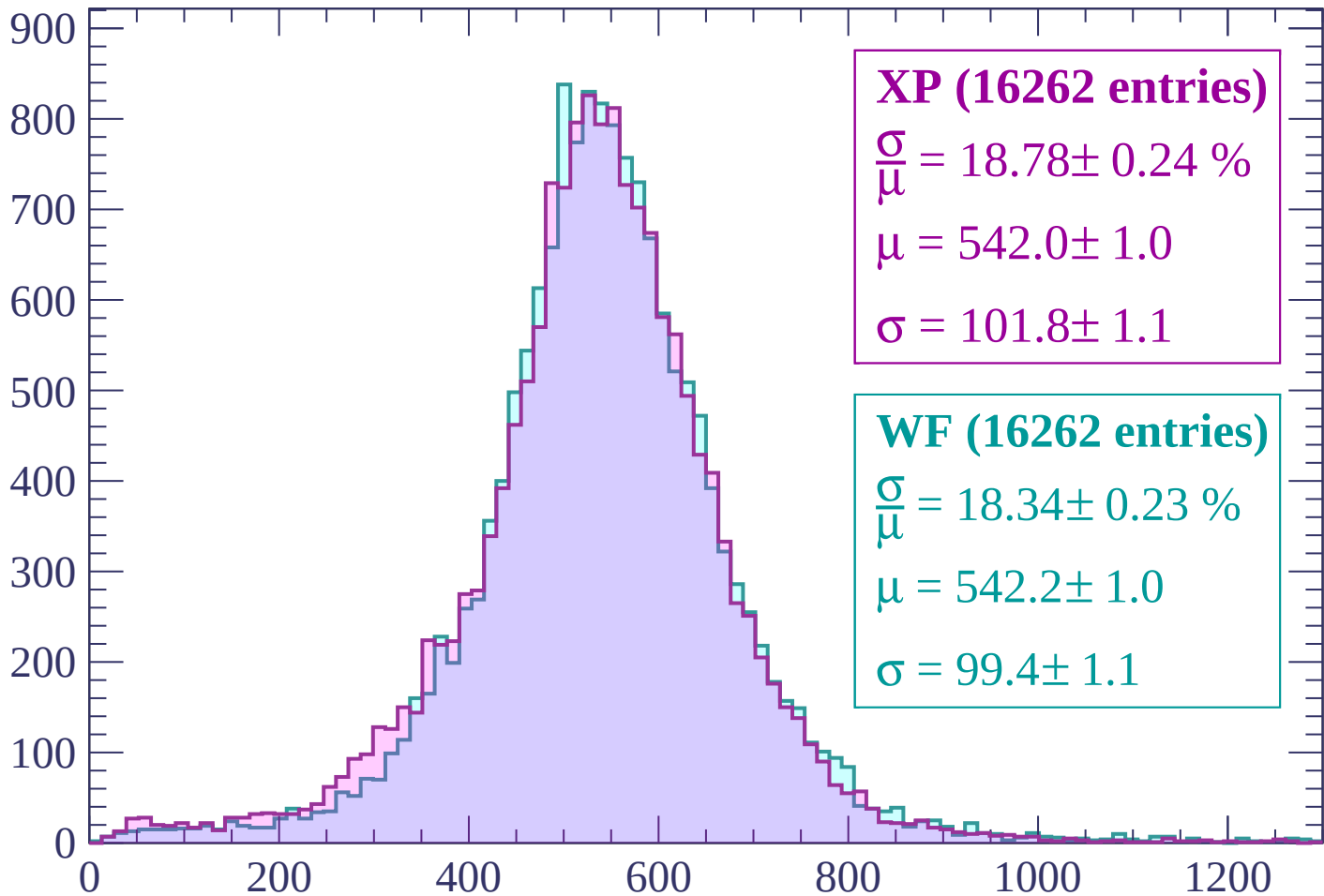
$\mu = 543.8 \pm 0.7$

$\sigma = 66.0 \pm 0.7$



Energy loss | $120 < p < 180$

Count



XP (16262 entries)

$$\frac{\sigma}{\mu} = 18.78 \pm 0.24 \%$$

$$\mu = 542.0 \pm 1.0$$

$$\sigma = 101.8 \pm 1.1$$

WF (16262 entries)

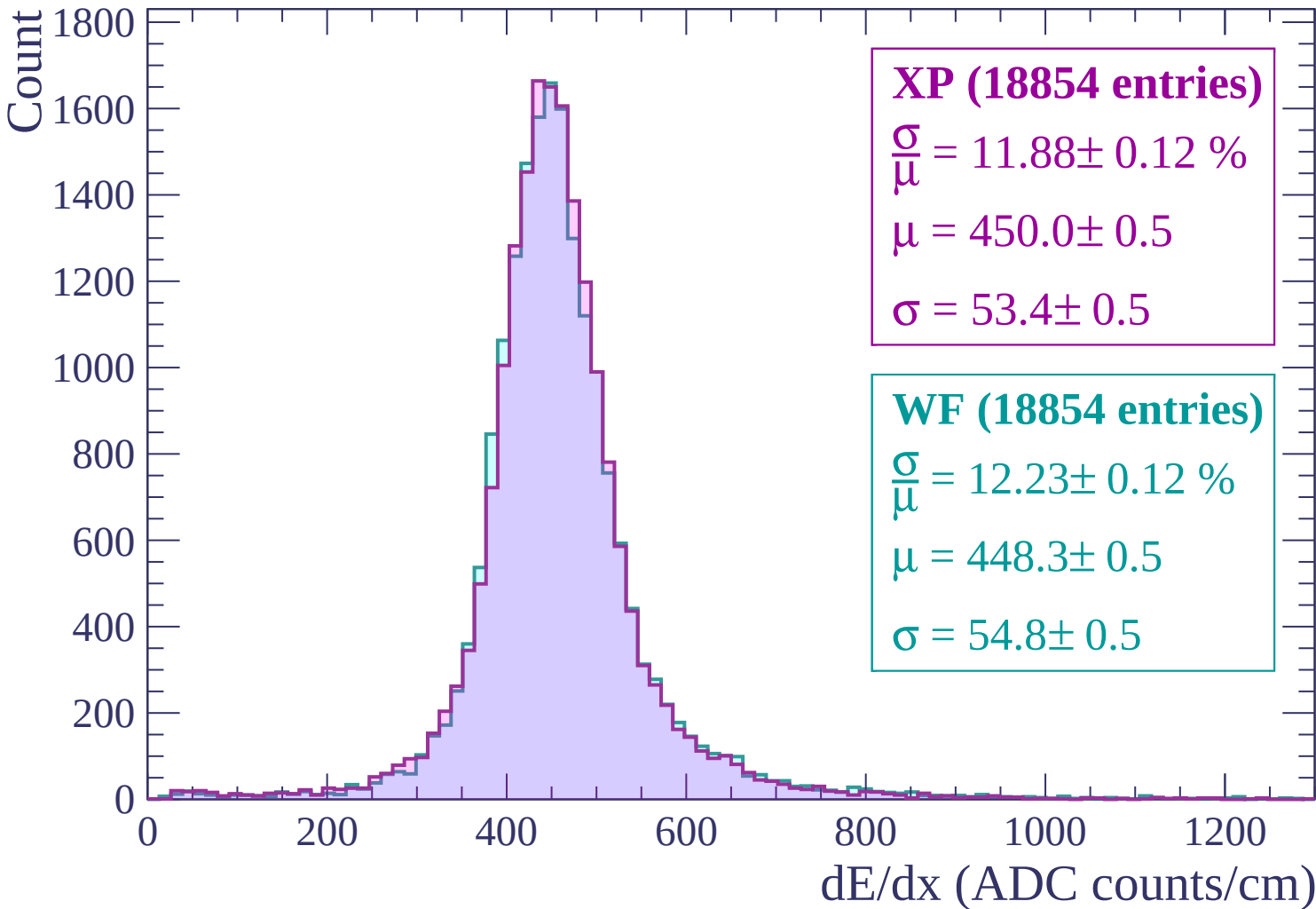
$$\frac{\sigma}{\mu} = 18.34 \pm 0.23 \%$$

$$\mu = 542.2 \pm 1.0$$

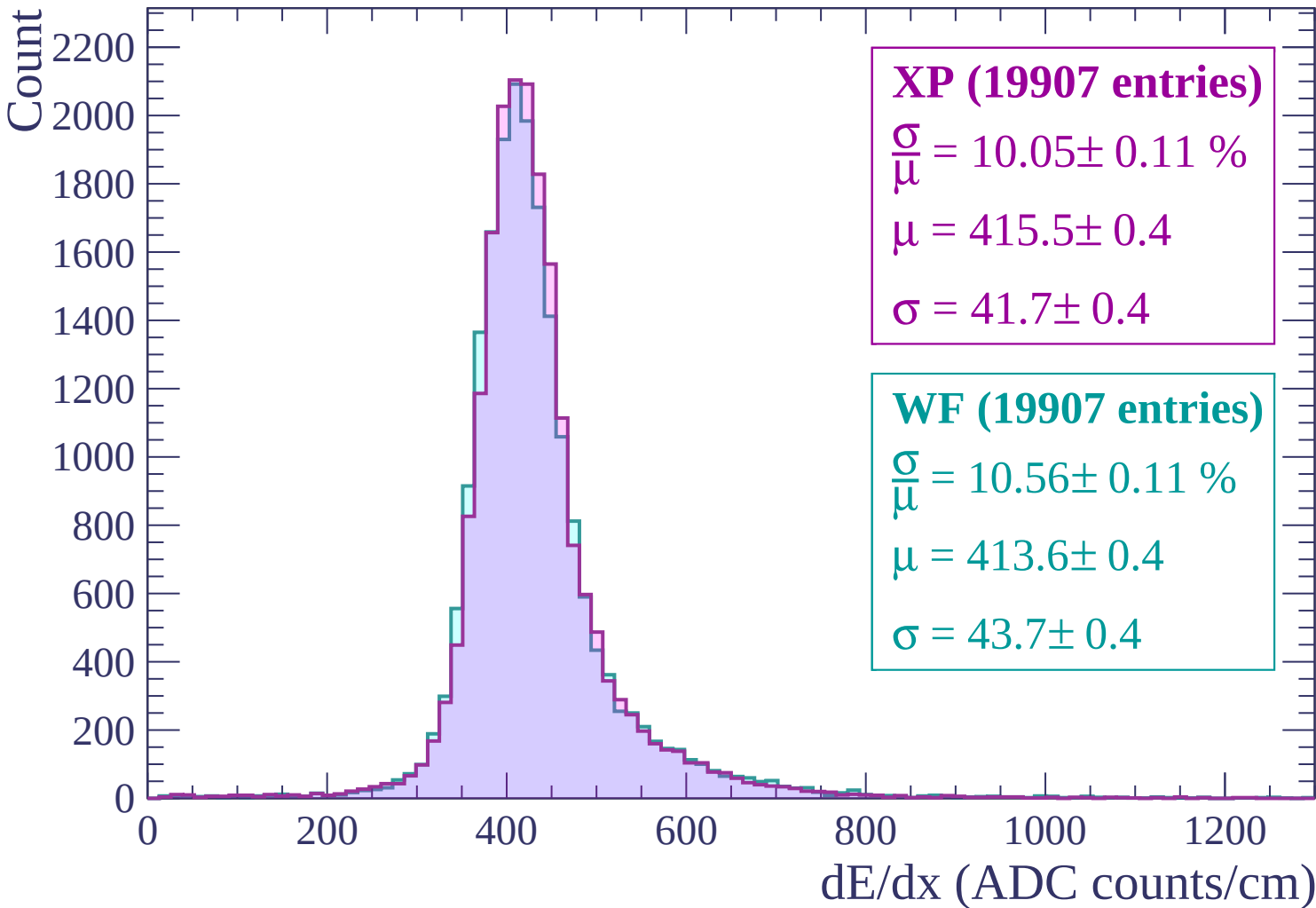
$$\sigma = 99.4 \pm 1.1$$

dE/dx (ADC counts/cm)

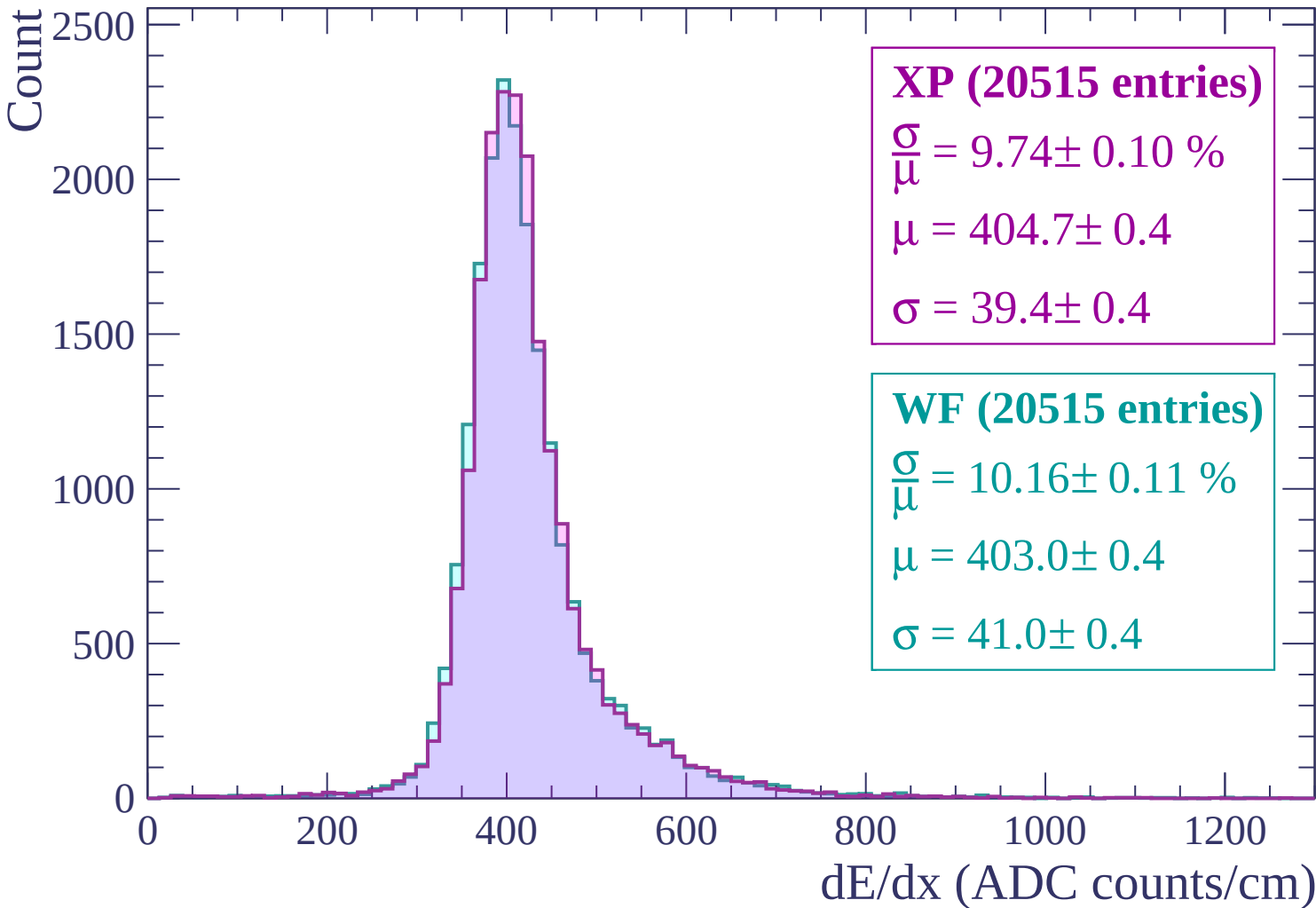
Energy loss | $180 < p < 240$



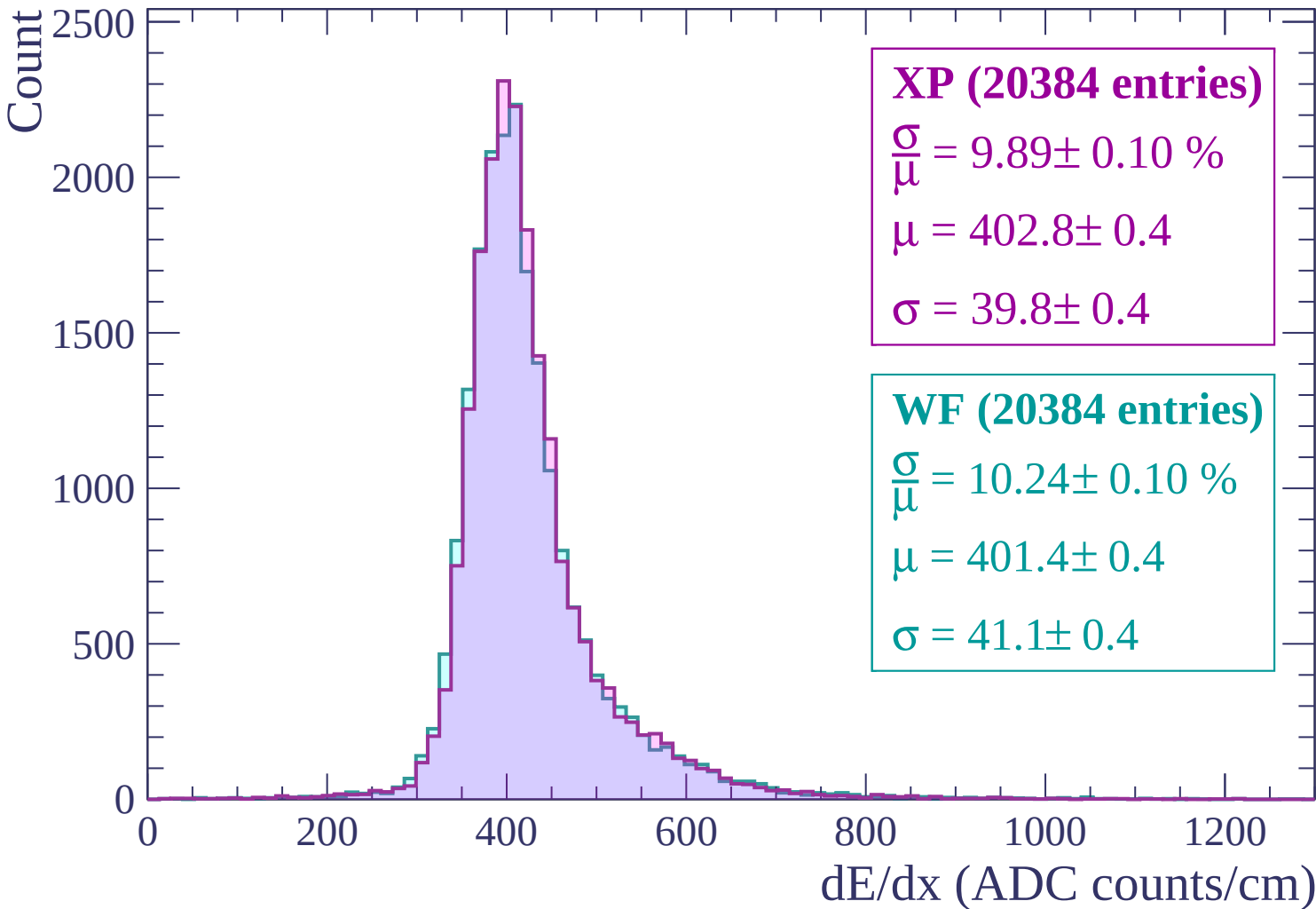
Energy loss | $240 < p < 300$



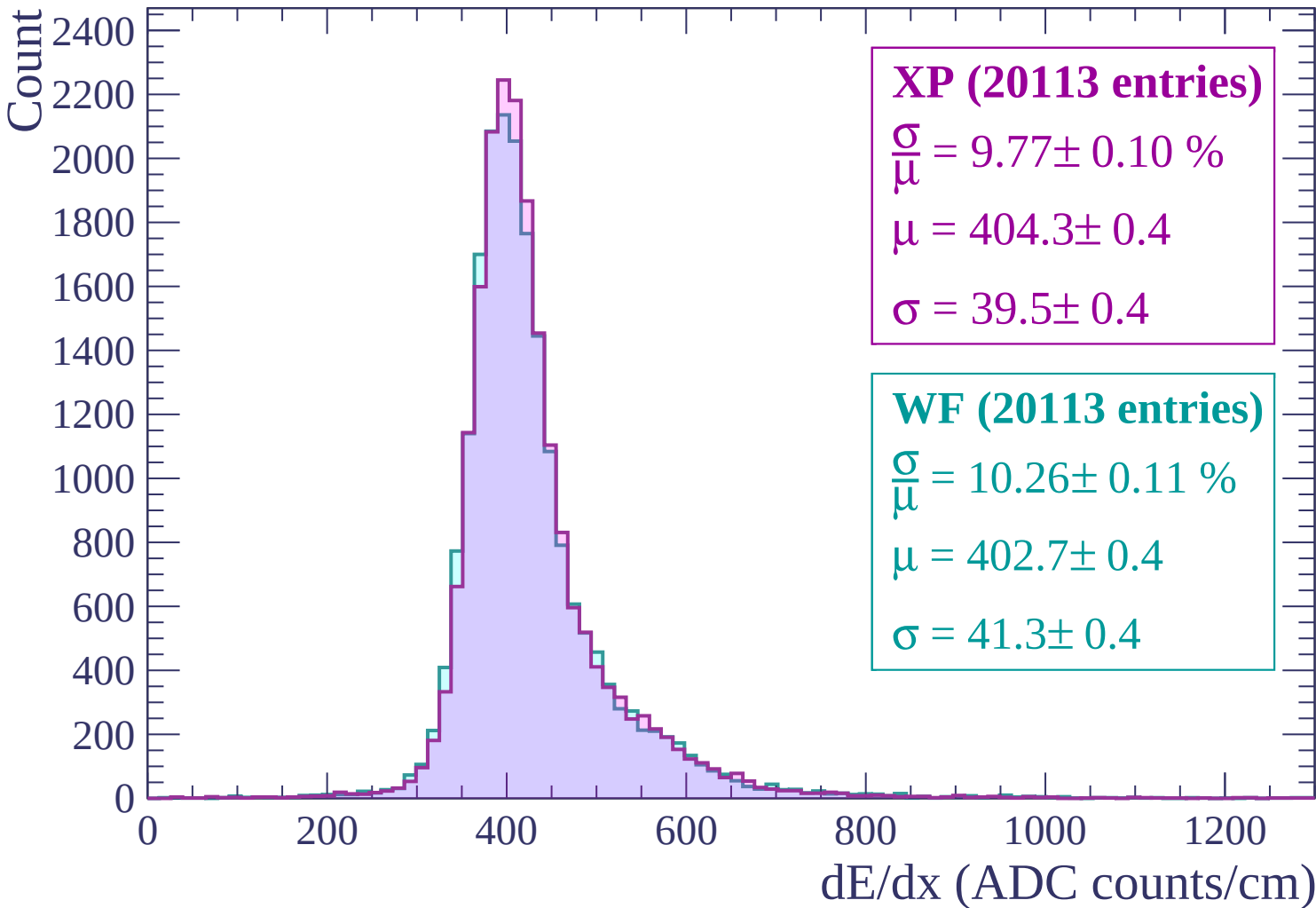
Energy loss | $300 < p < 360$



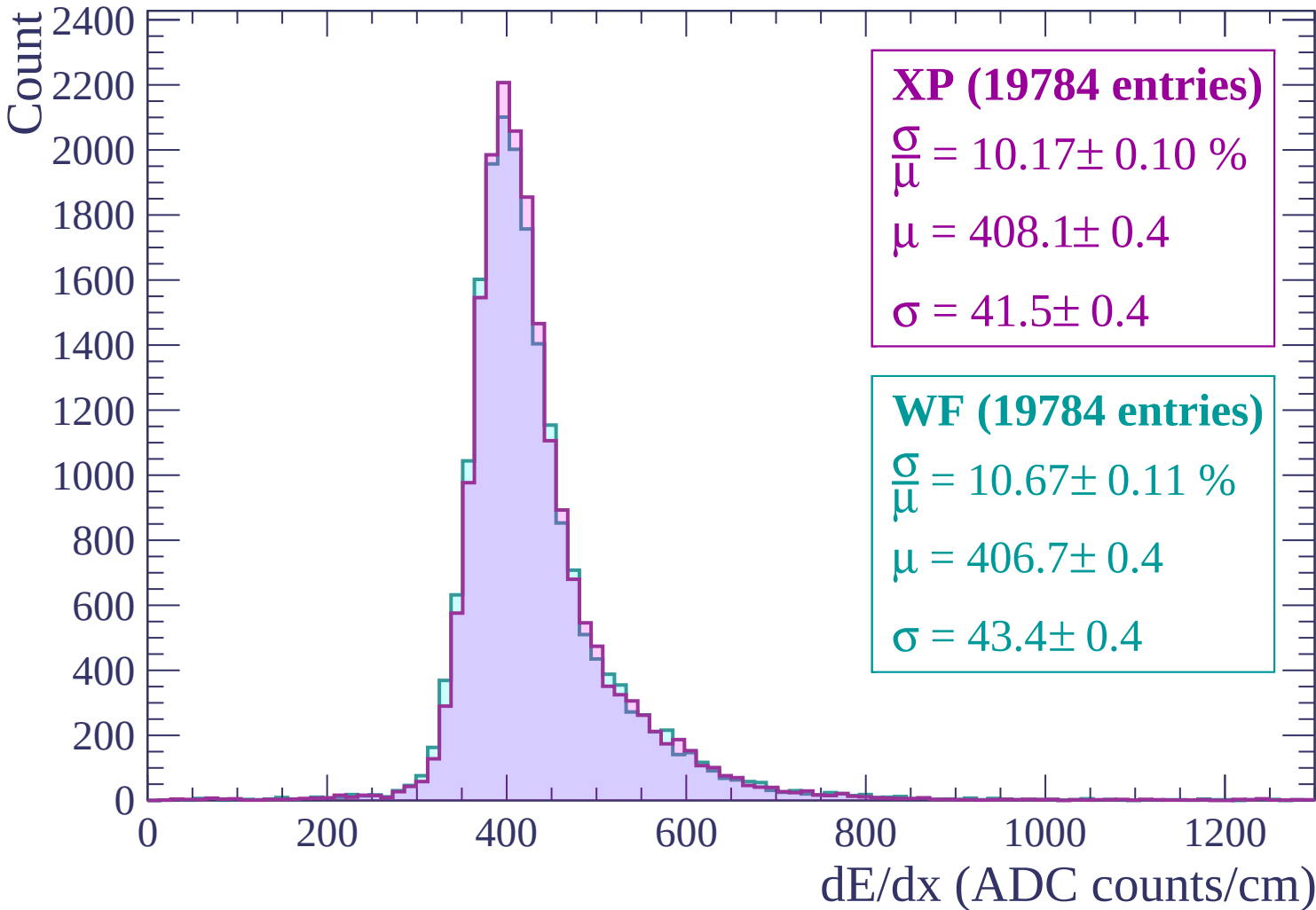
Energy loss | $360 < p < 420$



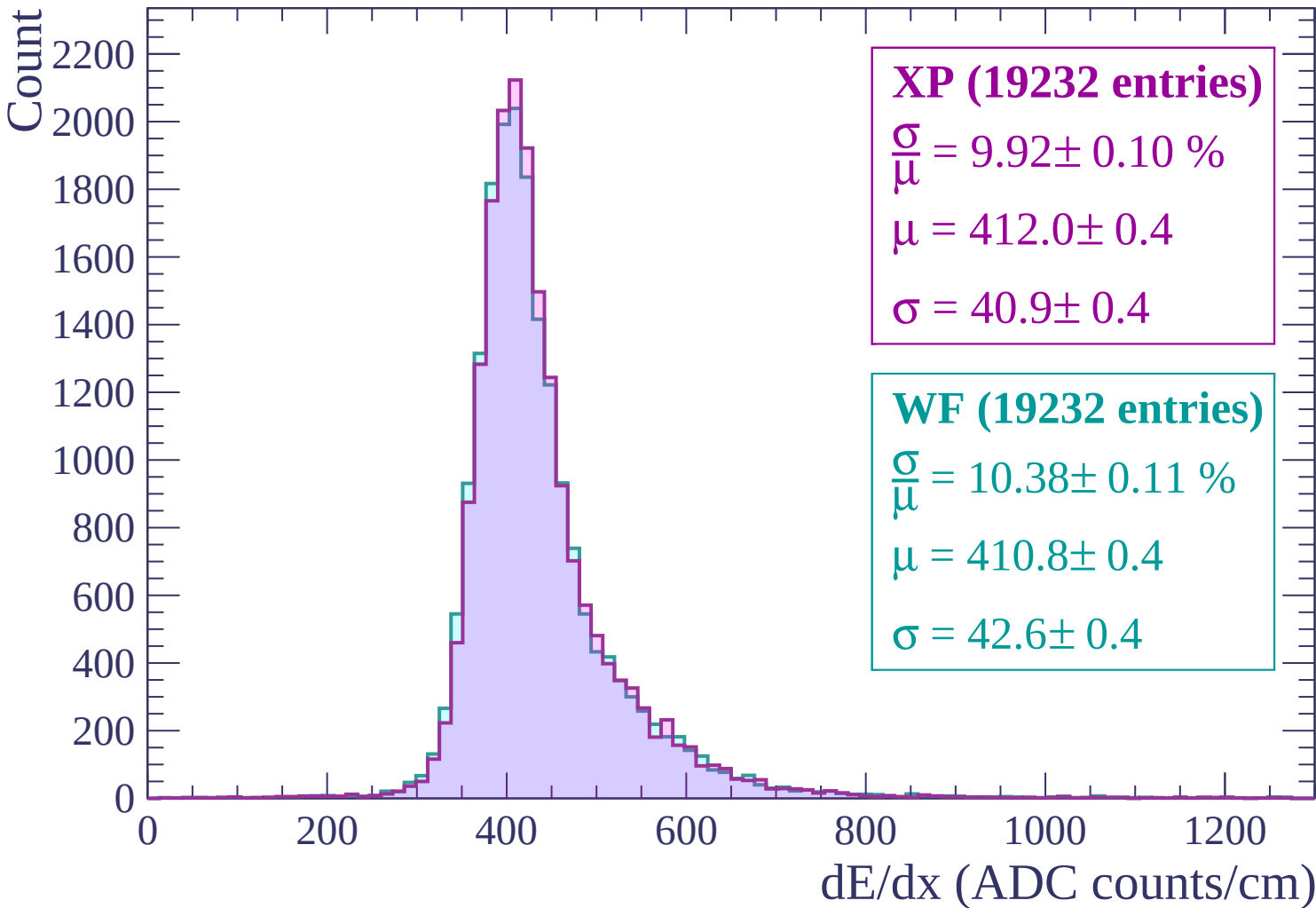
Energy loss | $420 < p < 480$



Energy loss | $480 < p < 540$



Energy loss | $540 < p < 600$



Energy loss | $600 < p < 660$

Count

2000
1800
1600
1400
1200
1000
800
600
400
200
0

XP (18595 entries)

$$\frac{\sigma}{\mu} = 10.39 \pm 0.10 \%$$

$$\mu = 417.8 \pm 0.4$$

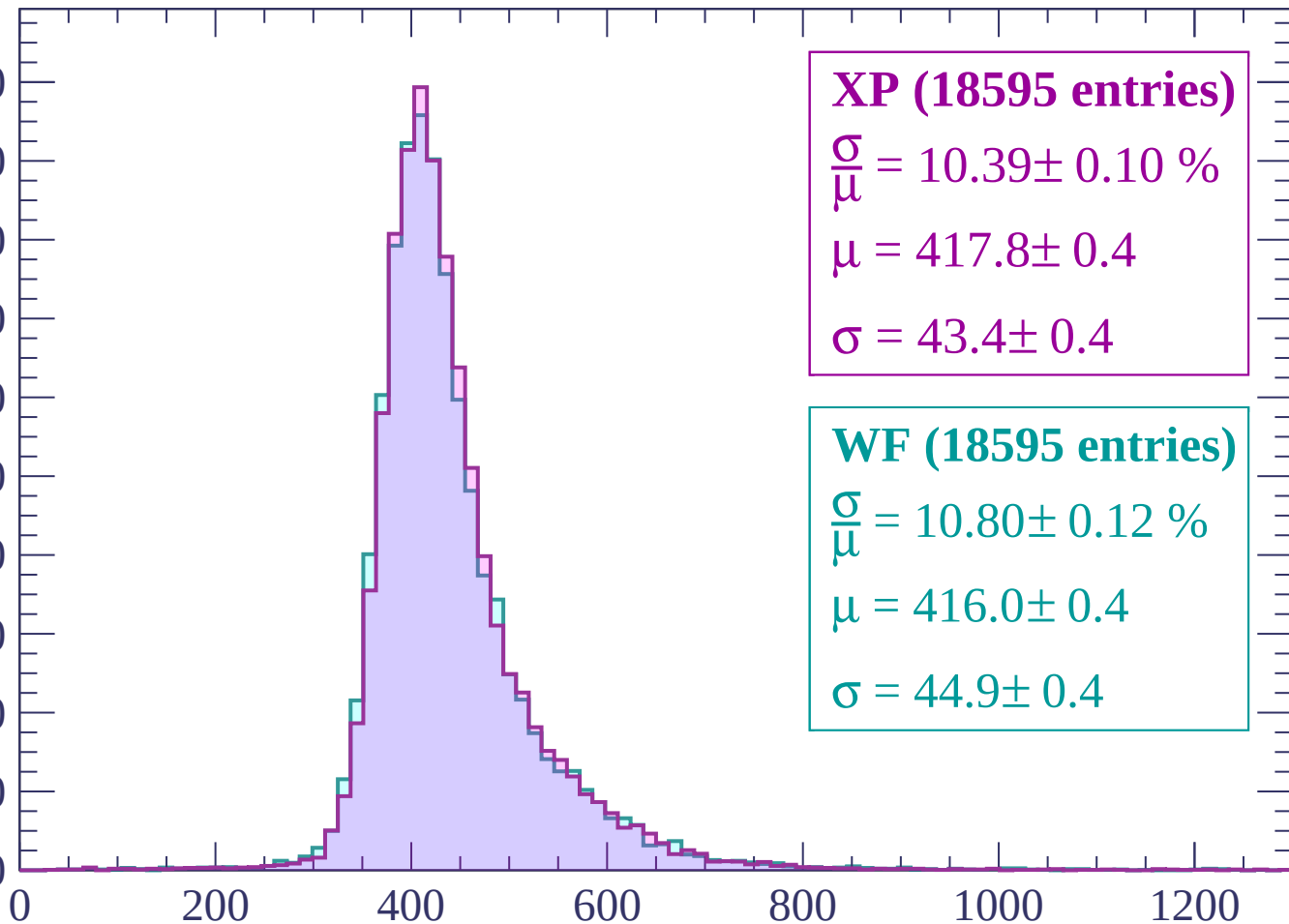
$$\sigma = 43.4 \pm 0.4$$

WF (18595 entries)

$$\frac{\sigma}{\mu} = 10.80 \pm 0.12 \%$$

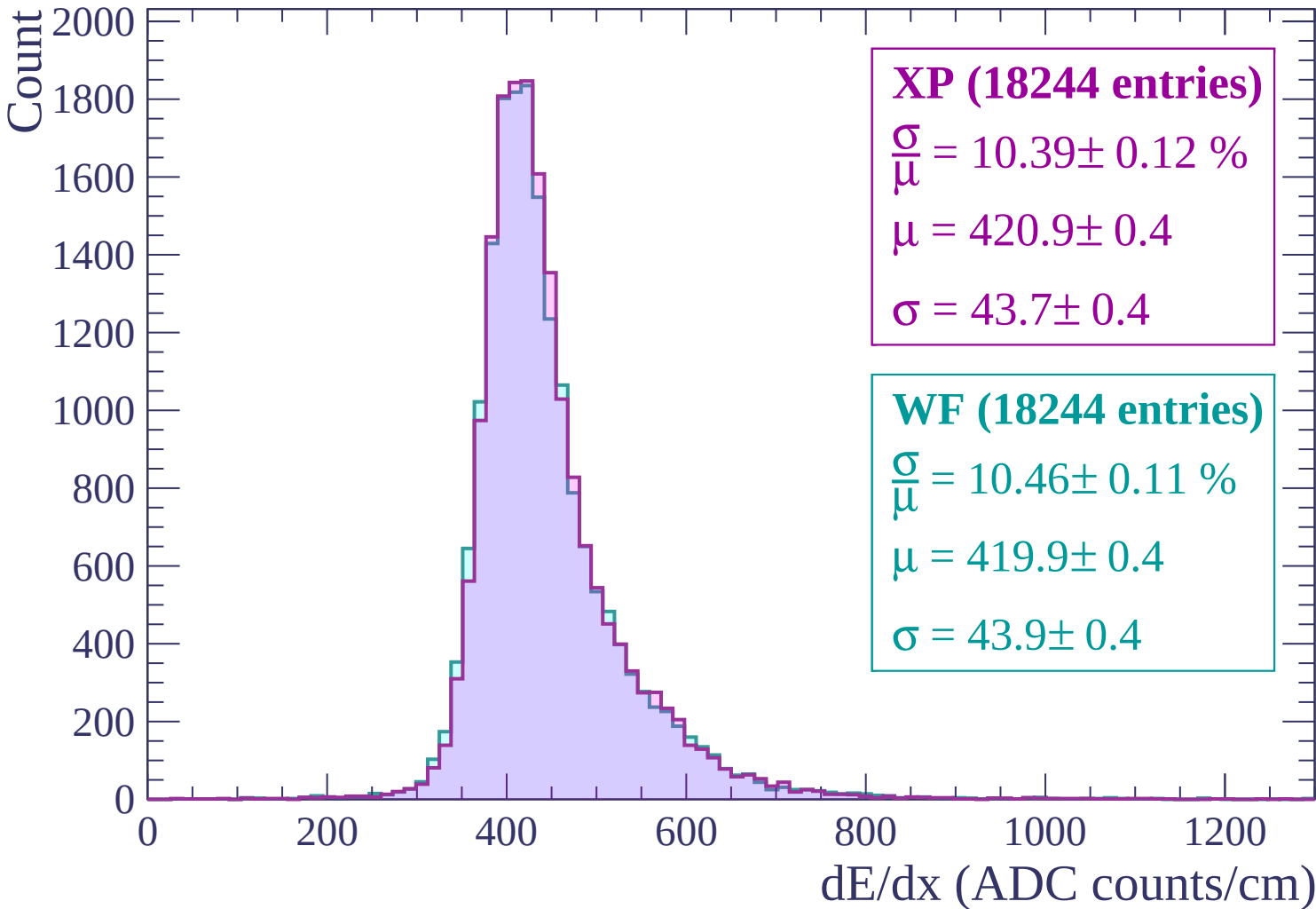
$$\mu = 416.0 \pm 0.4$$

$$\sigma = 44.9 \pm 0.4$$



dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$



Energy loss | $720 < p < 780$

Count

1800
1600
1400
1200
1000
800
600
400
200
0

XP (17759 entries)

$$\frac{\sigma}{\mu} = 10.26 \pm 0.11 \%$$

$$\mu = 425.8 \pm 0.4$$

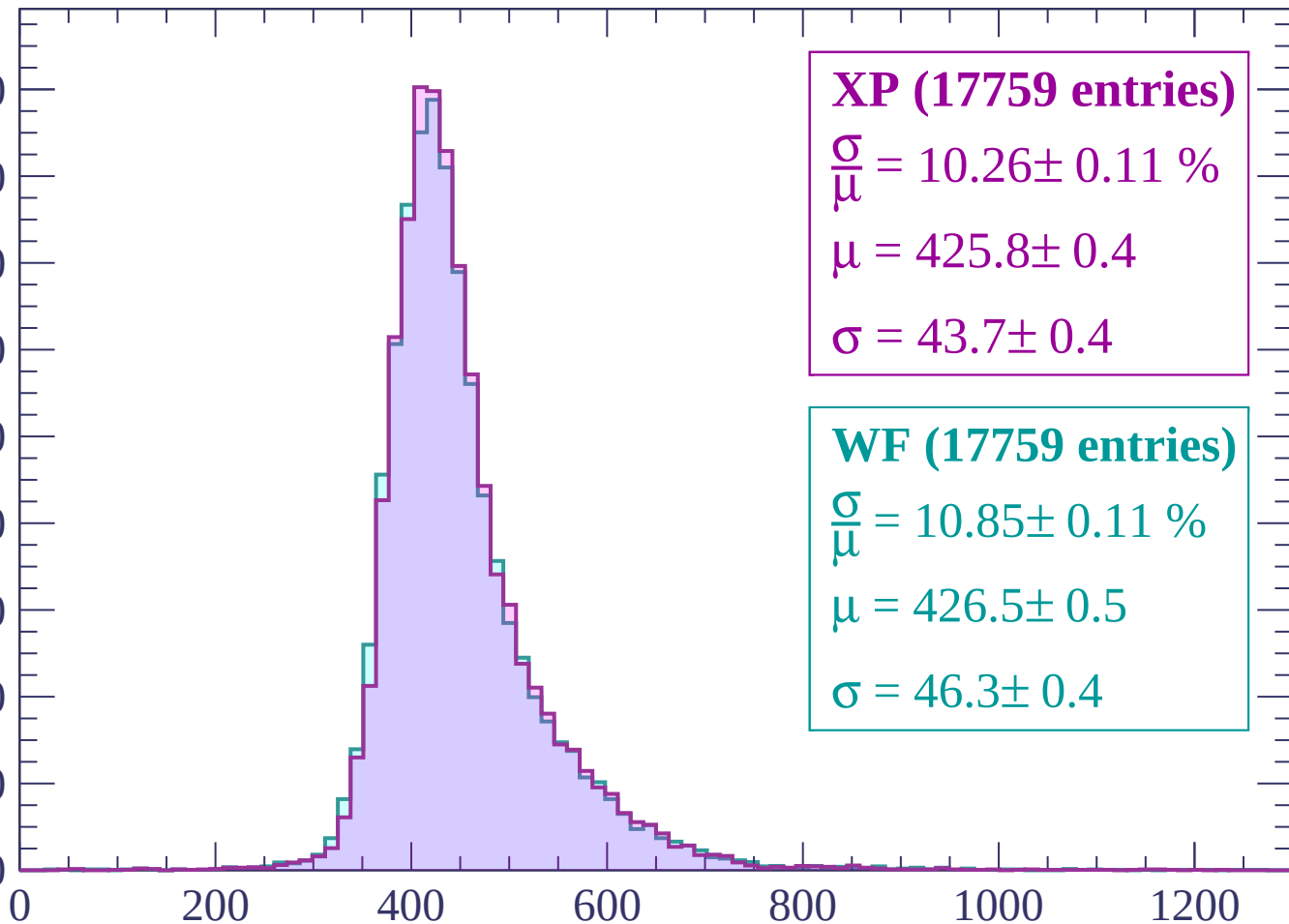
$$\sigma = 43.7 \pm 0.4$$

WF (17759 entries)

$$\frac{\sigma}{\mu} = 10.85 \pm 0.11 \%$$

$$\mu = 426.5 \pm 0.5$$

$$\sigma = 46.3 \pm 0.4$$



dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count

1800
1600
1400
1200
1000
800
600
400
200
0

XP (17106 entries)

$\frac{\sigma}{\mu} = 10.49 \pm 0.11 \%$

$\mu = 429.9 \pm 0.5$

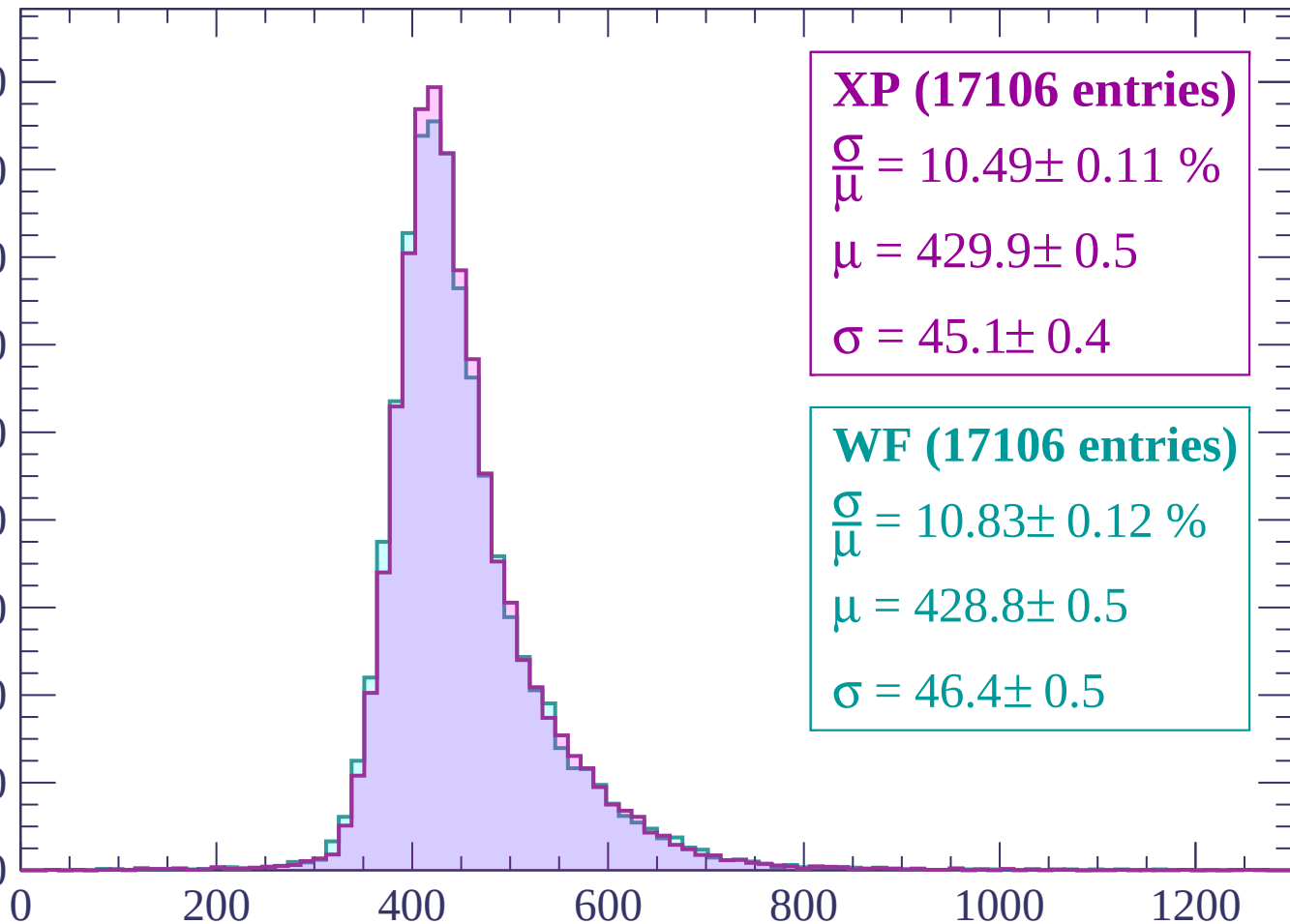
$\sigma = 45.1 \pm 0.4$

WF (17106 entries)

$\frac{\sigma}{\mu} = 10.83 \pm 0.12 \%$

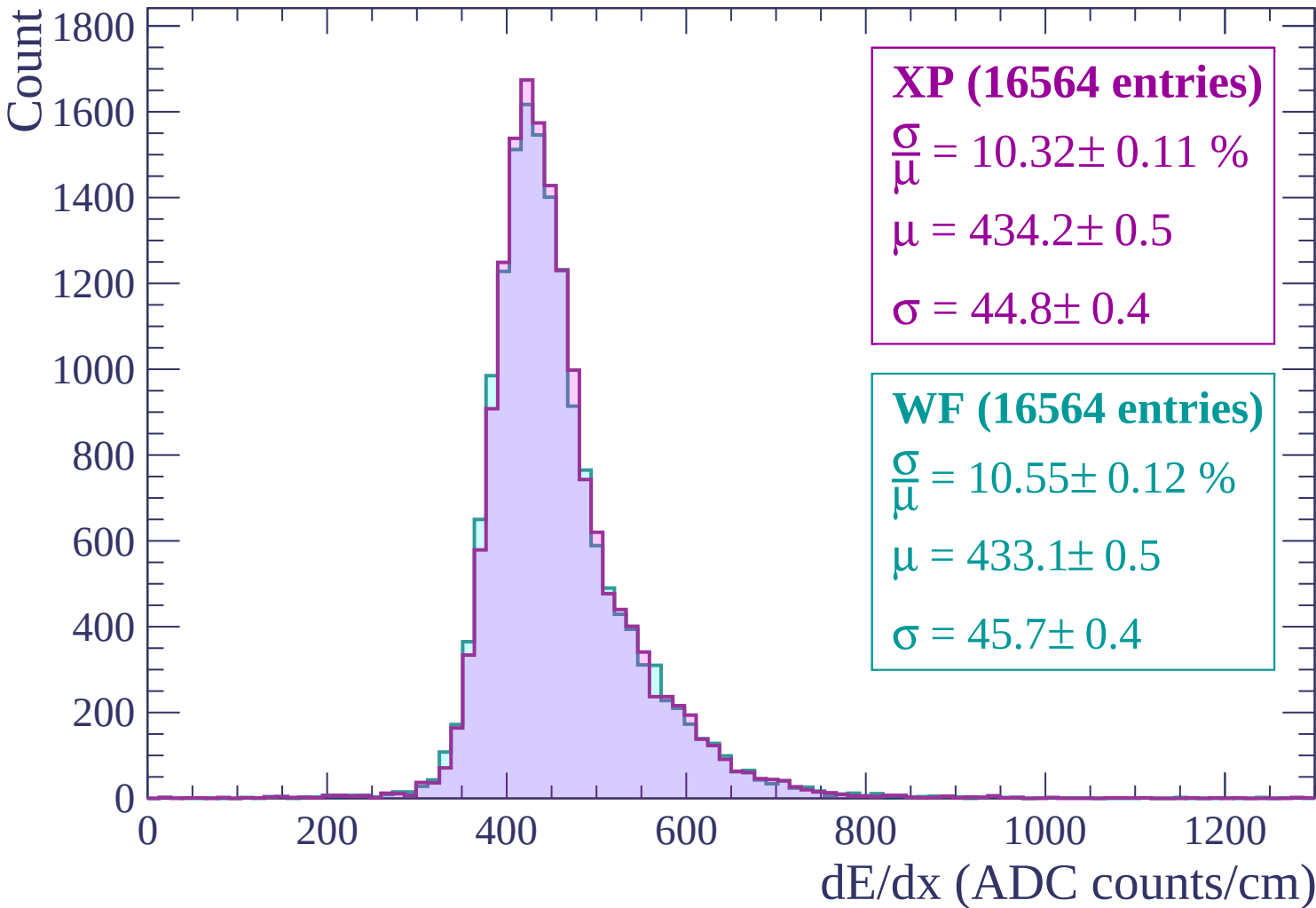
$\mu = 428.8 \pm 0.5$

$\sigma = 46.4 \pm 0.5$



dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$



Energy loss | $900 < p < 960$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (15902 entries)

$\frac{\sigma}{\mu} = 10.50 \pm 0.11 \%$

$\mu = 439.9 \pm 0.5$

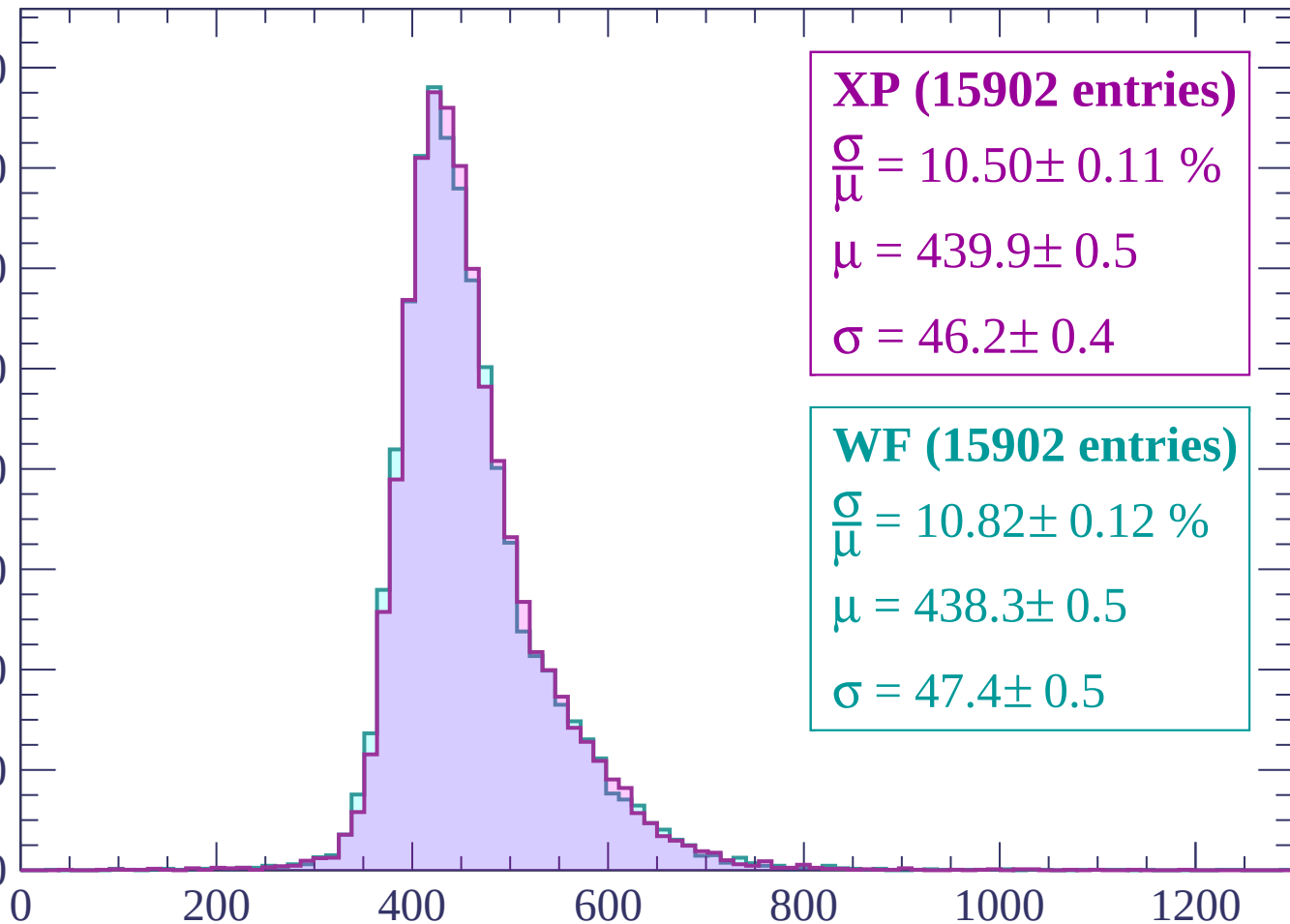
$\sigma = 46.2 \pm 0.4$

WF (15902 entries)

$\frac{\sigma}{\mu} = 10.82 \pm 0.12 \%$

$\mu = 438.3 \pm 0.5$

$\sigma = 47.4 \pm 0.5$

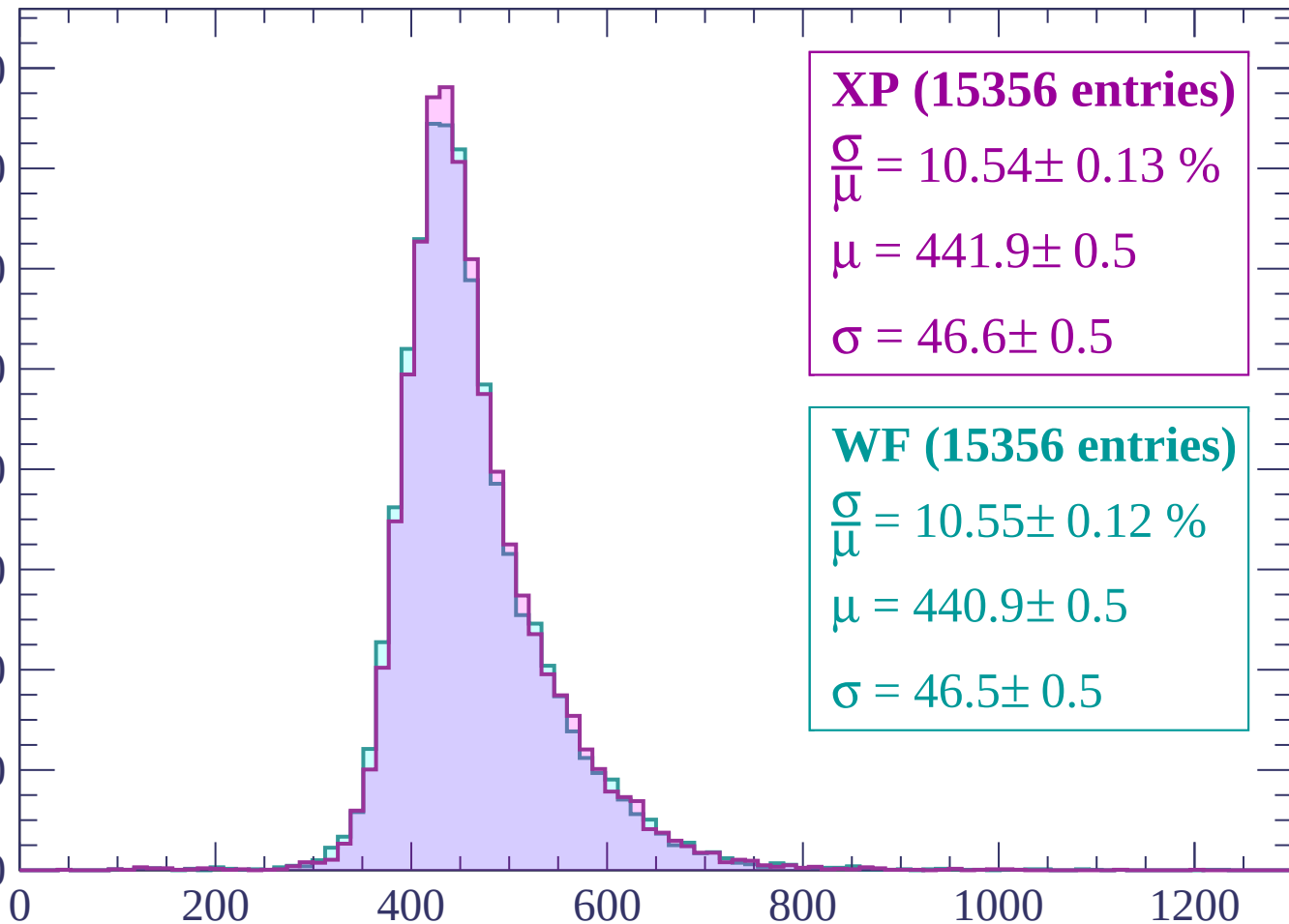


dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (15356 entries)

$\frac{\sigma}{\mu} = 10.54 \pm 0.13 \%$

$\mu = 441.9 \pm 0.5$

$\sigma = 46.6 \pm 0.5$

WF (15356 entries)

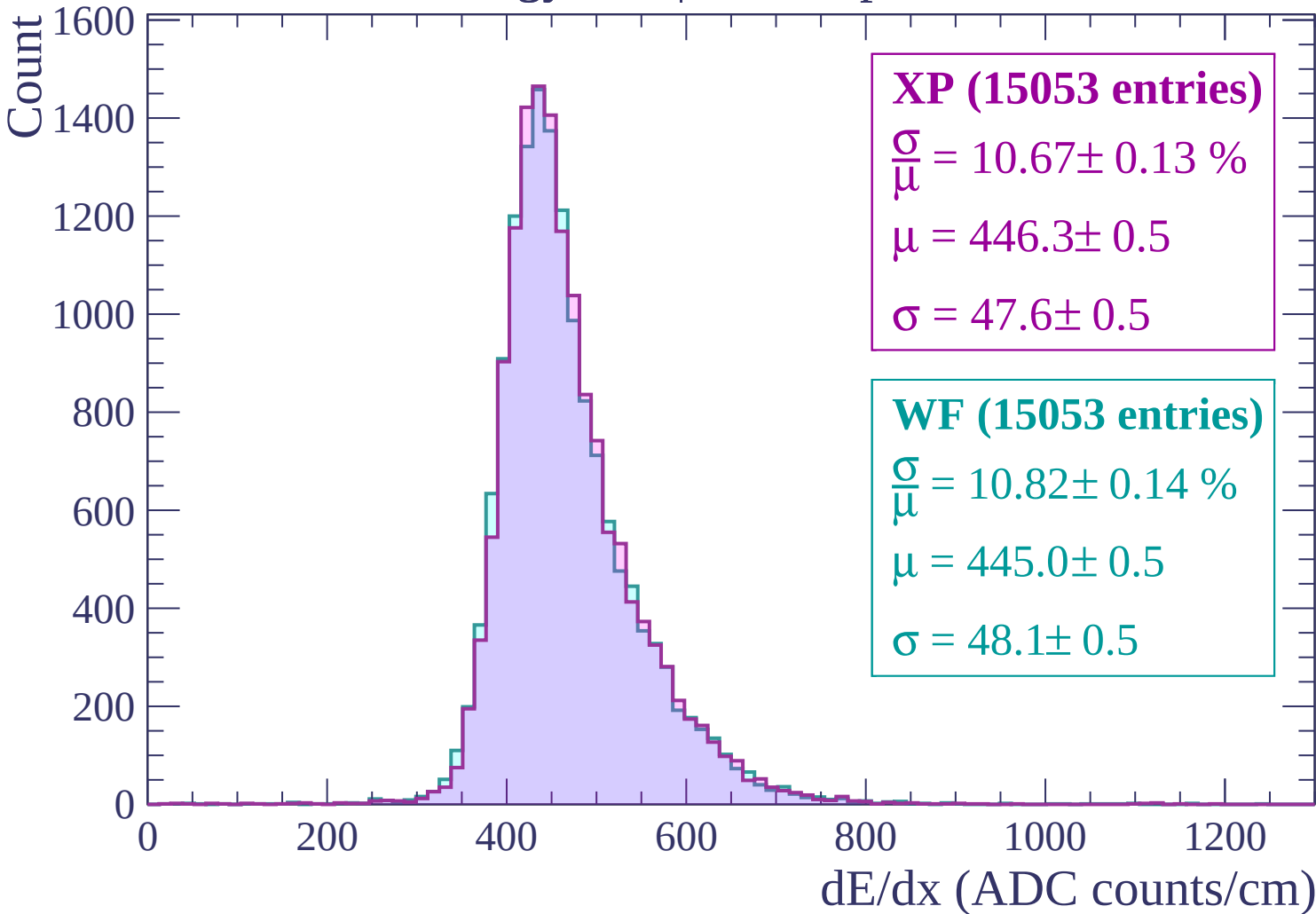
$\frac{\sigma}{\mu} = 10.55 \pm 0.12 \%$

$\mu = 440.9 \pm 0.5$

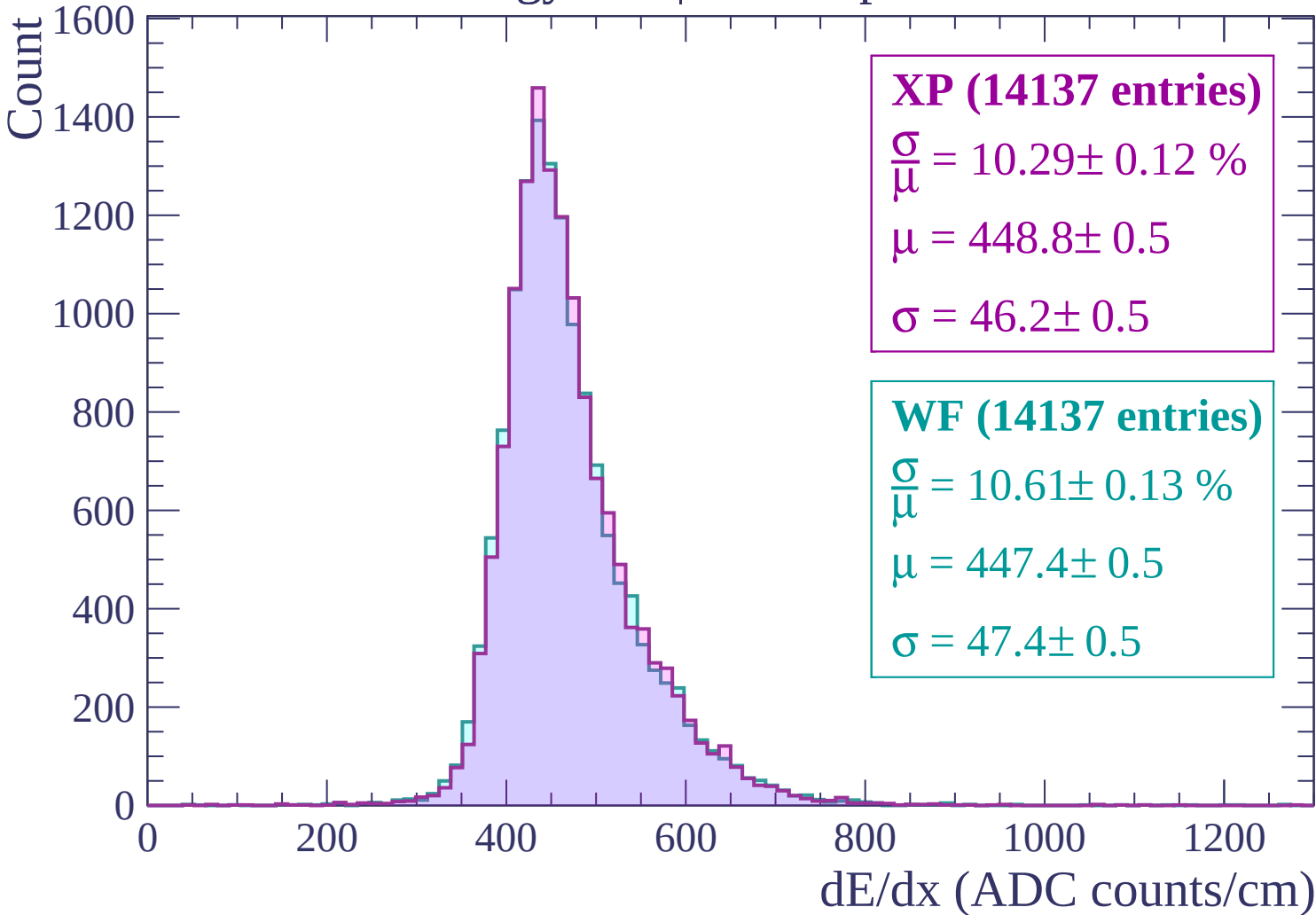
$\sigma = 46.5 \pm 0.5$

dE/dx (ADC counts/cm)

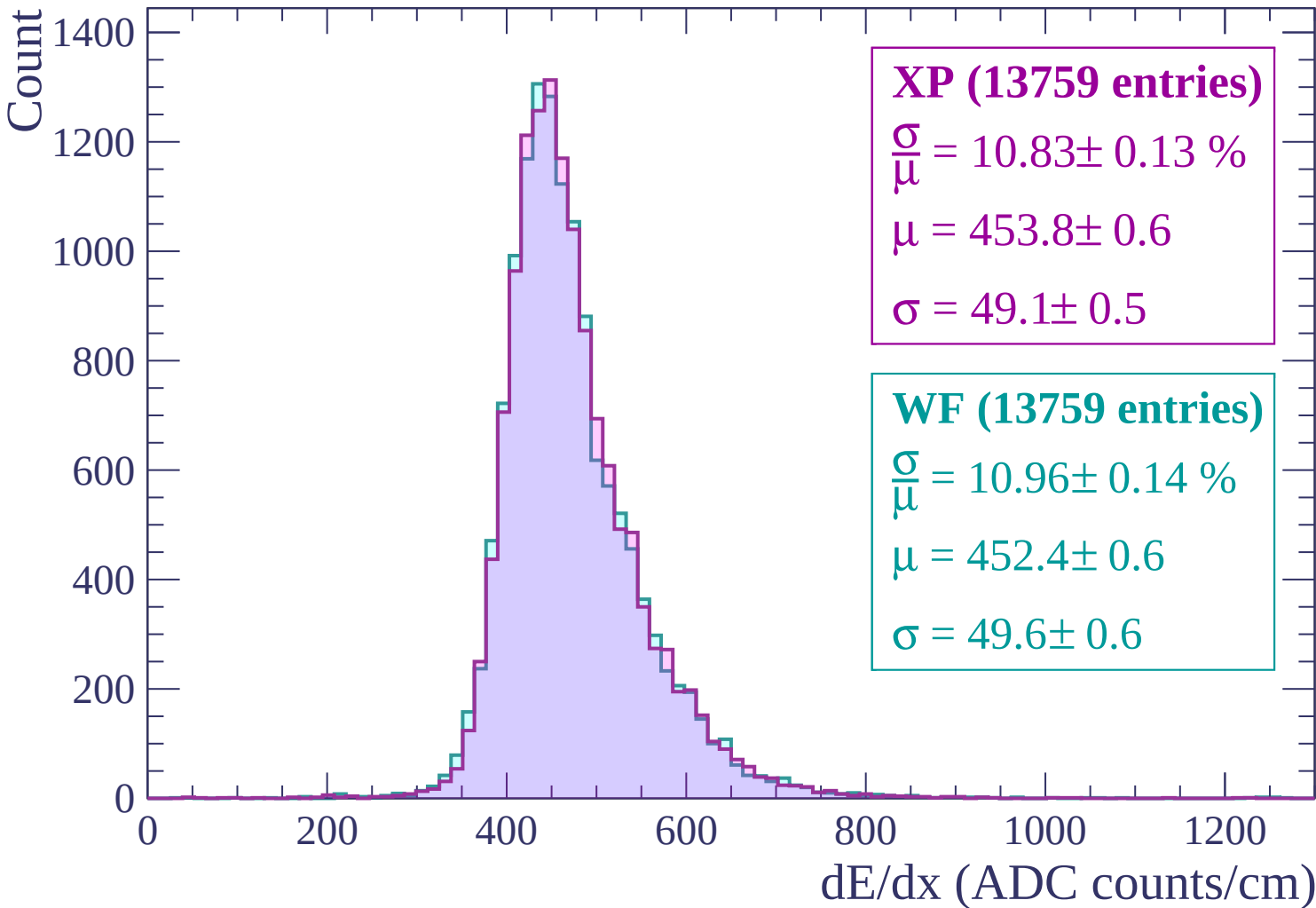
Energy loss | $1020 < p < 1080$



Energy loss | $1080 < p < 1140$

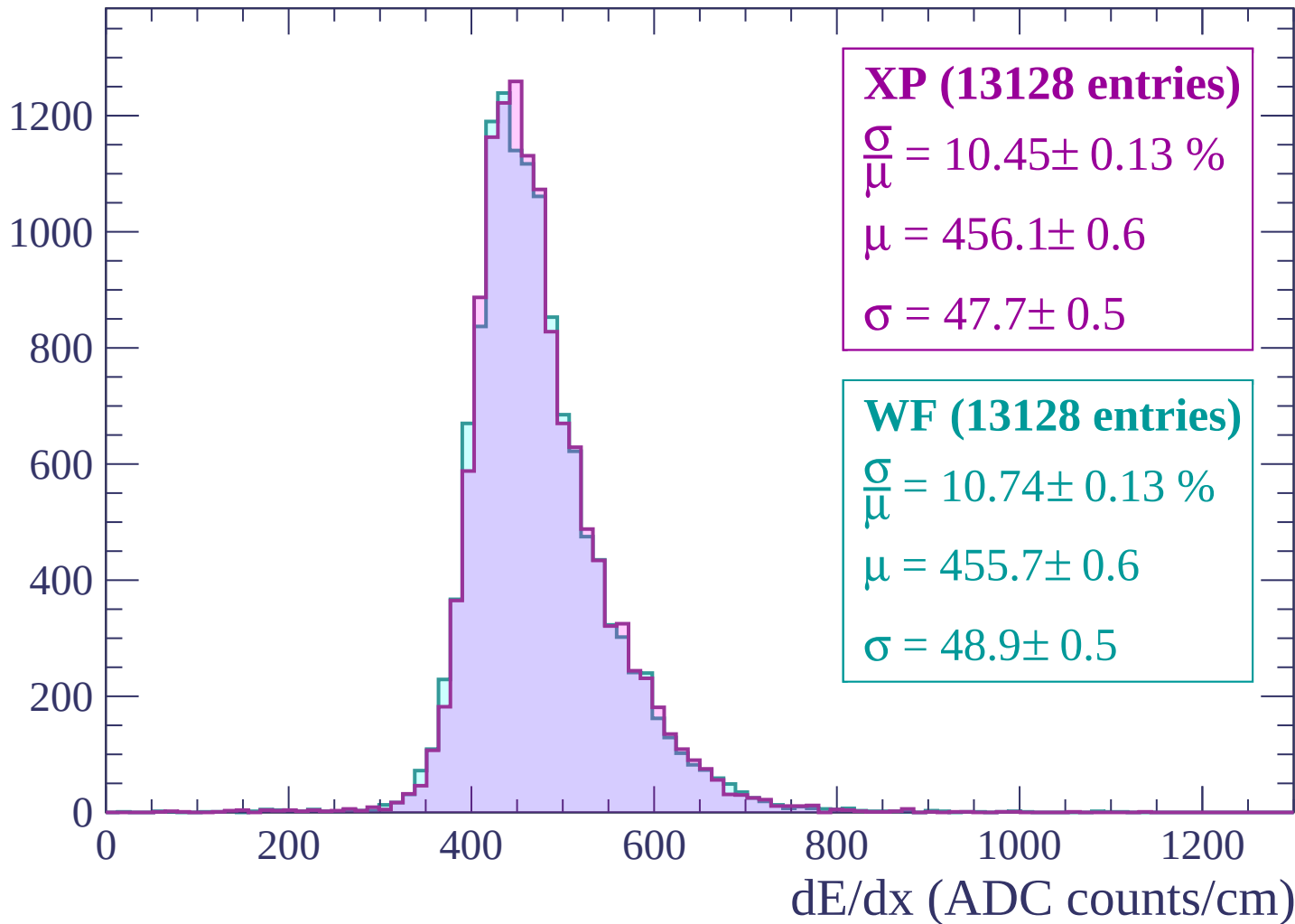


Energy loss | $1140 < p < 1200$



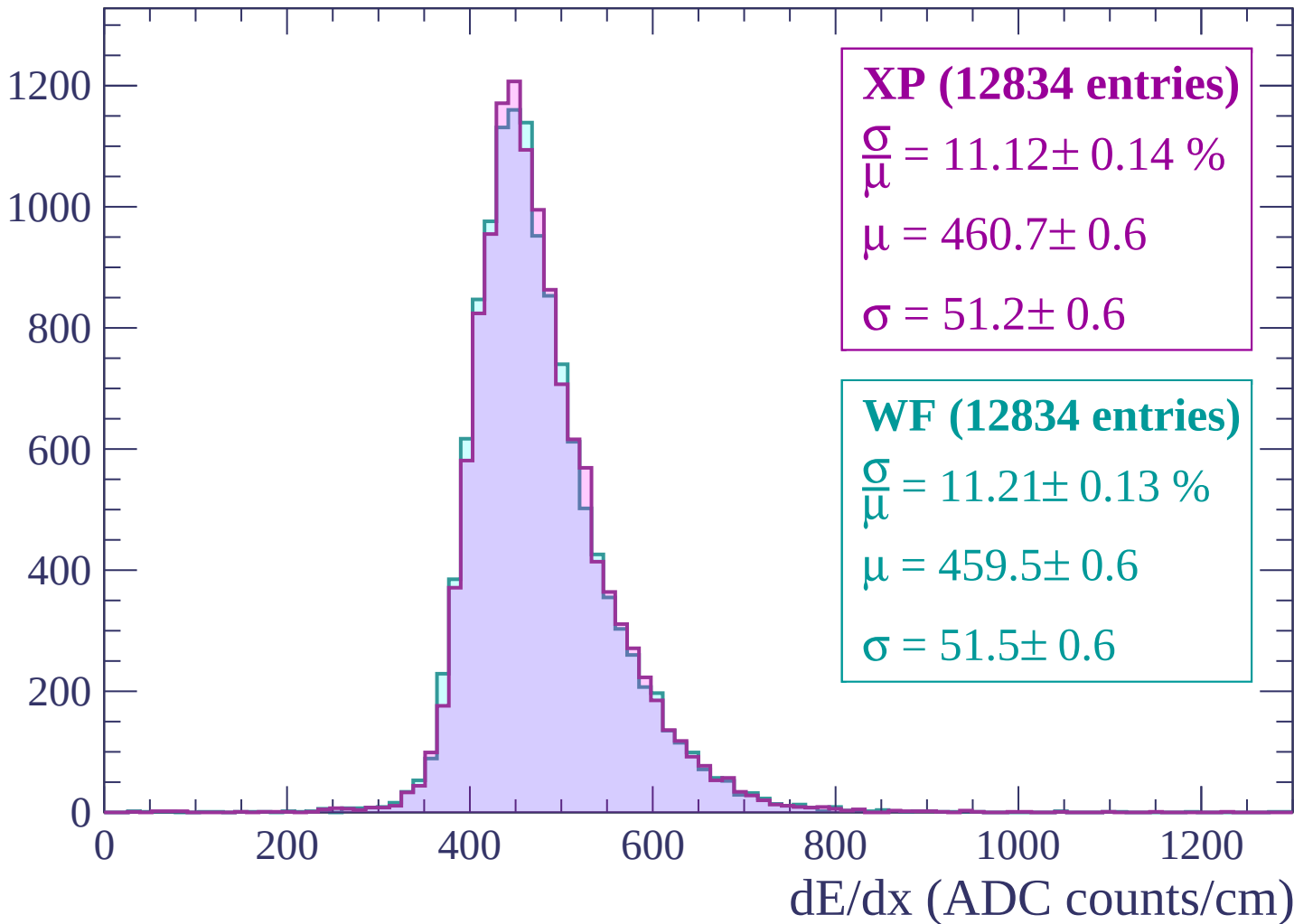
Energy loss | $1200 < p < 1260$

Count



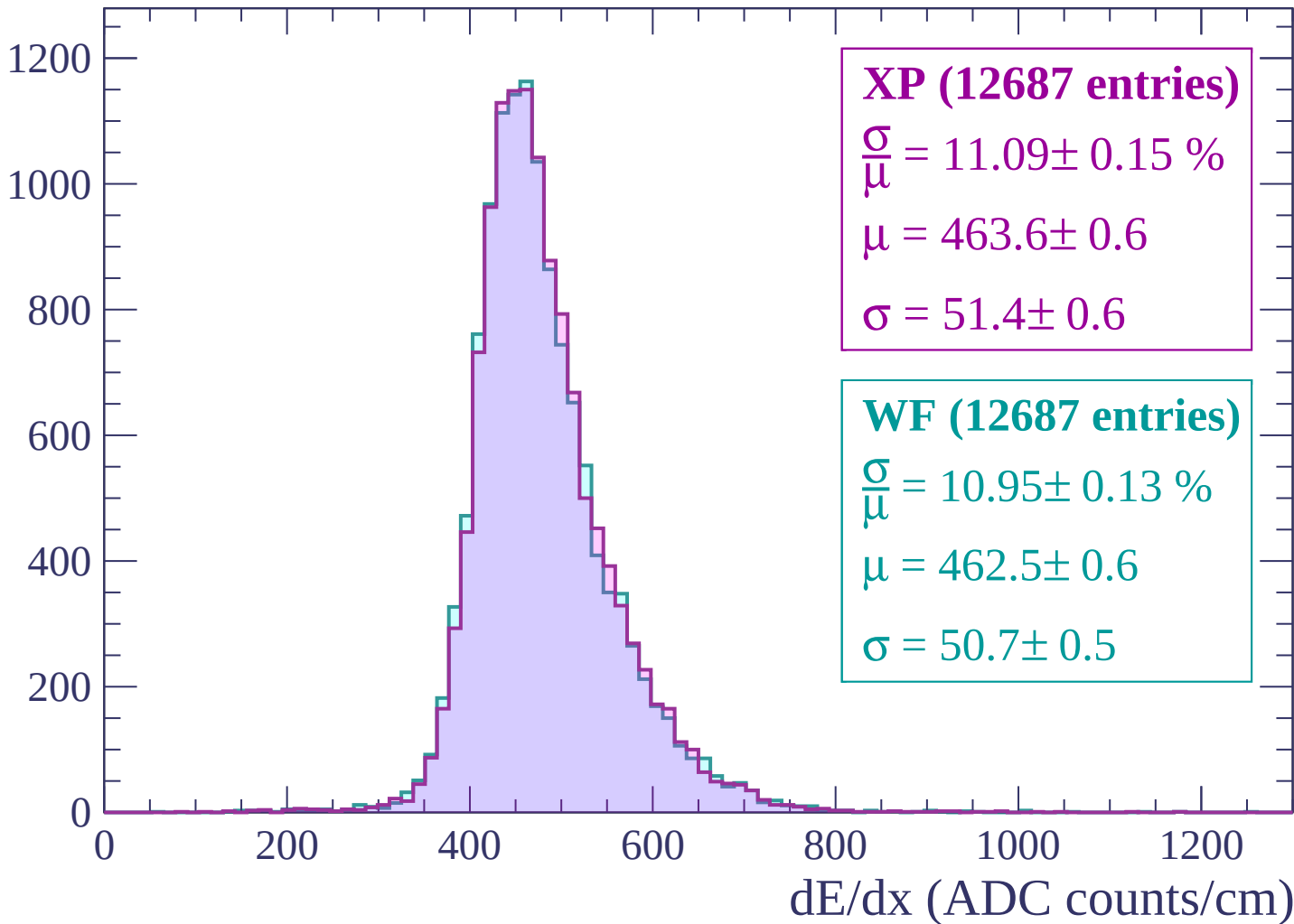
Energy loss | $1260 < p < 1320$

Count

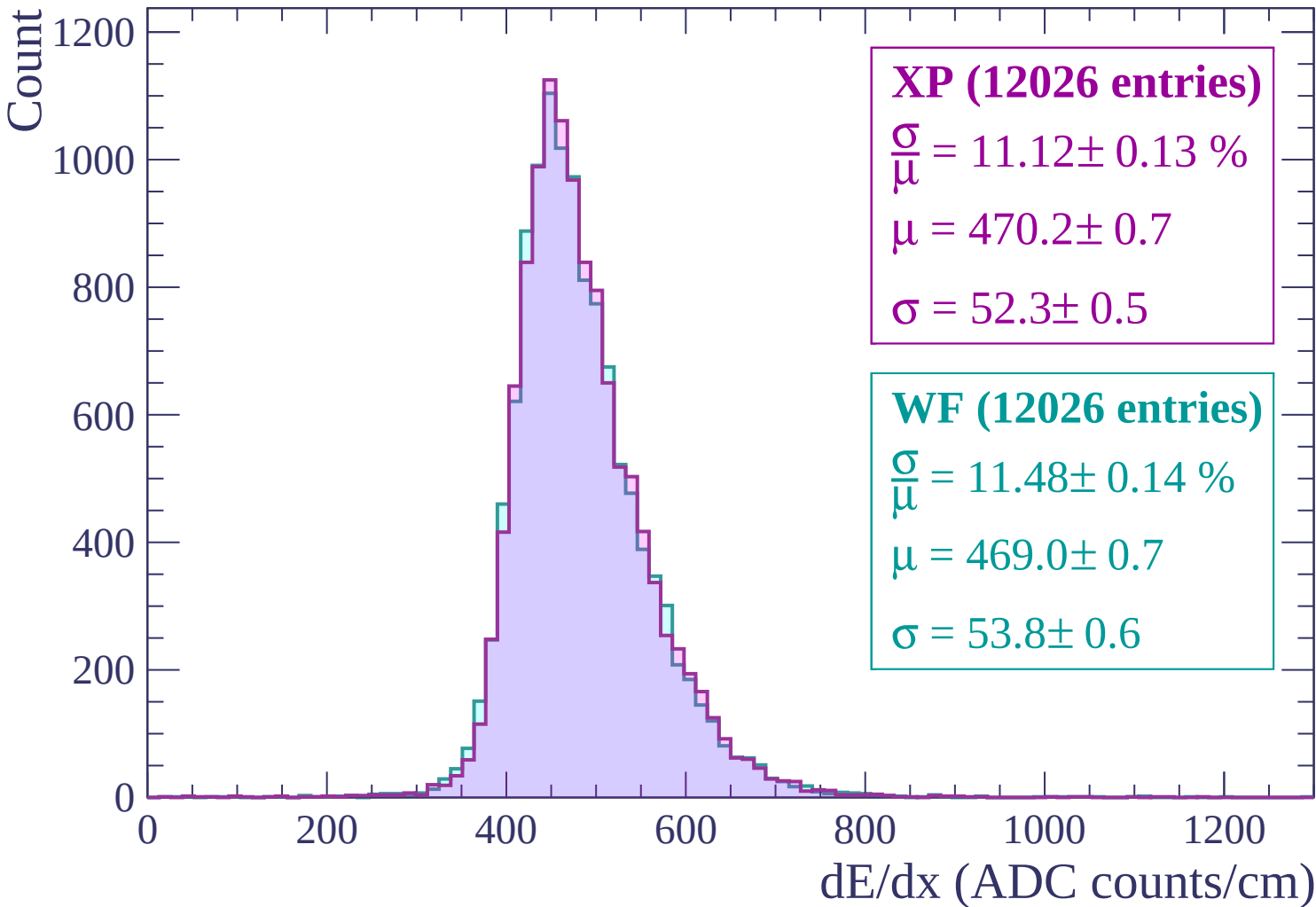


Energy loss | $1320 < p < 1380$

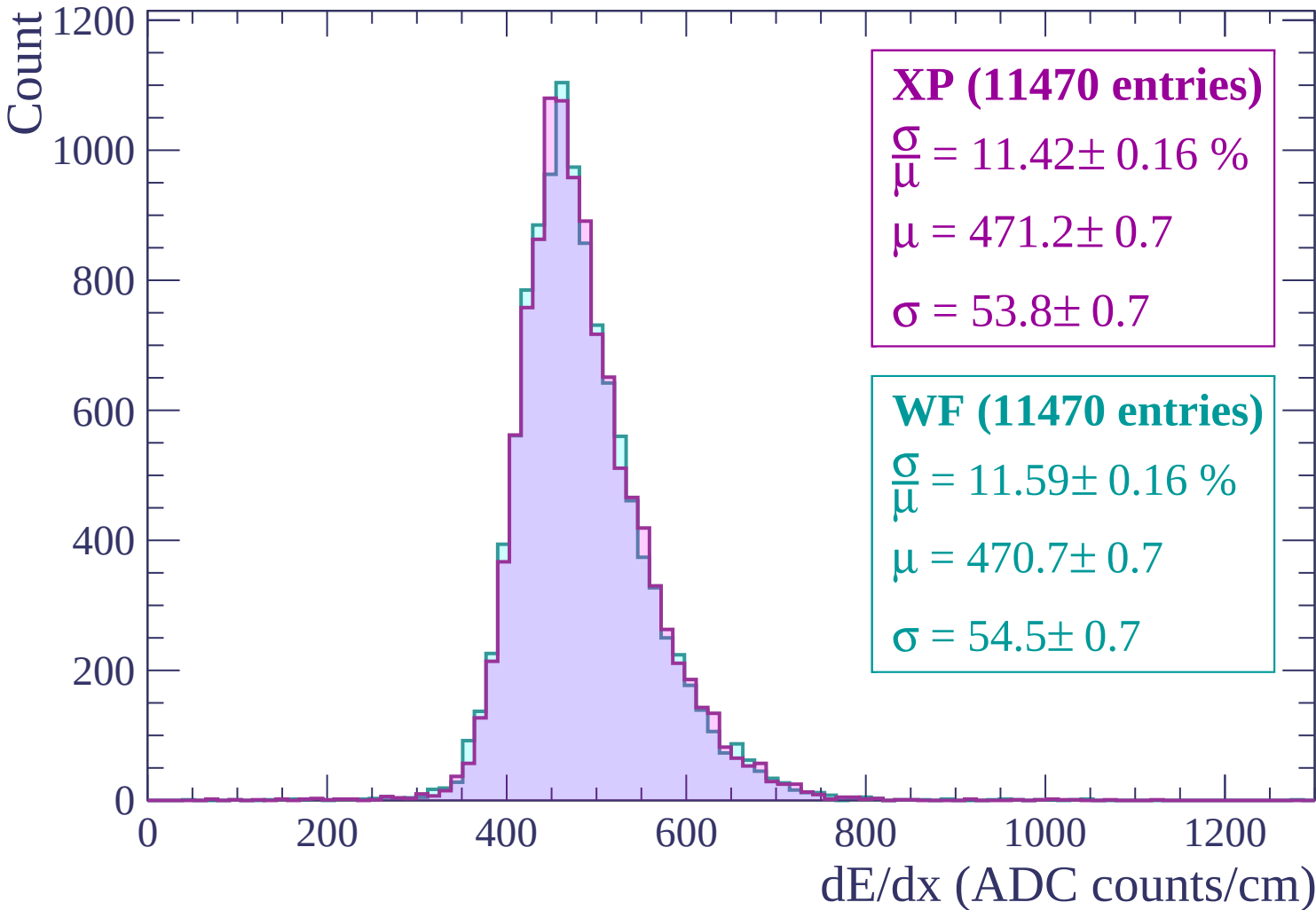
Count



Energy loss | $1380 < p < 1440$



Energy loss | $1440 < p < 1500$



Energy loss | $1500 < p < 1560$

Count

1000

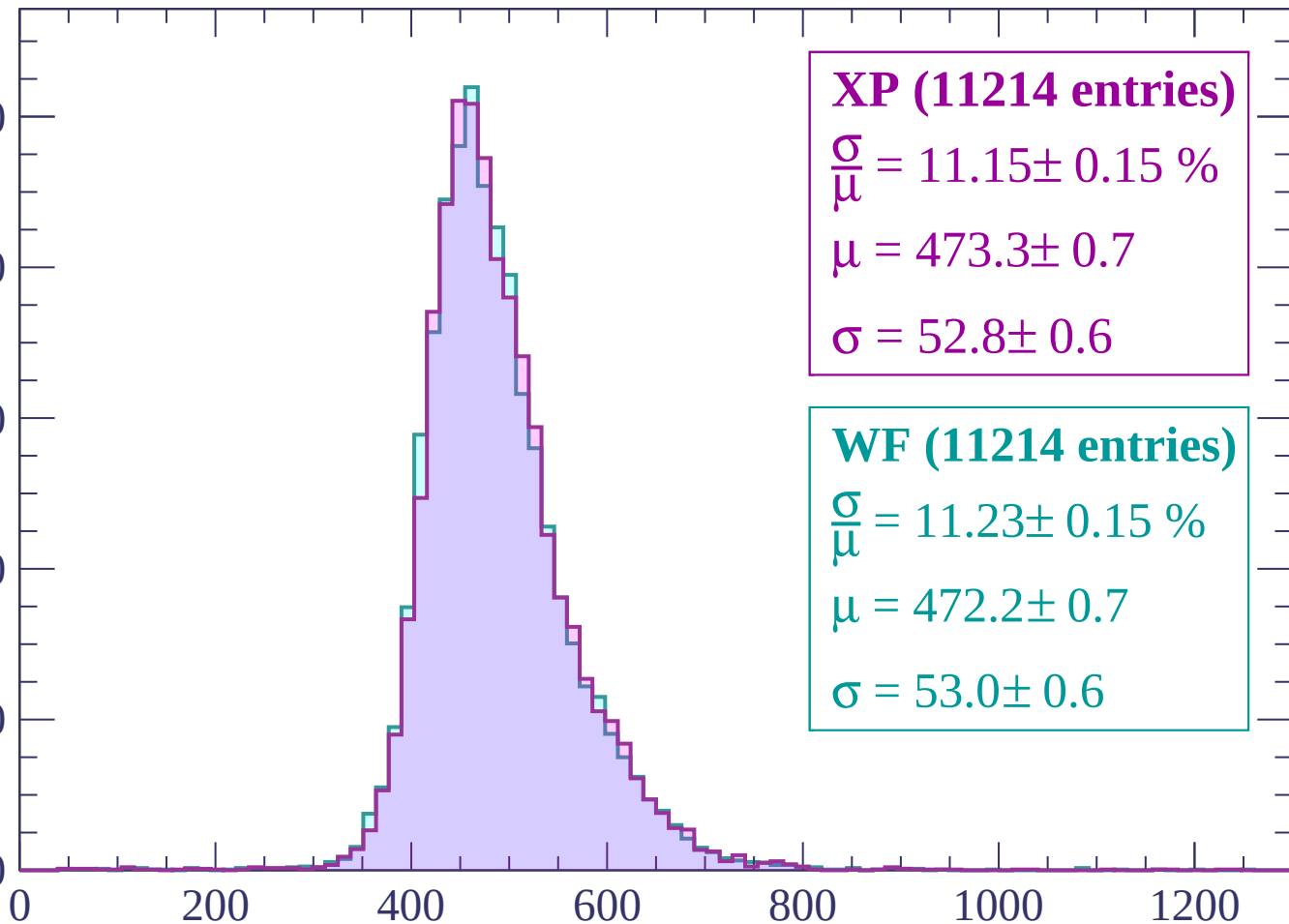
800

600

400

200

0



XP (11214 entries)

$\frac{\sigma}{\mu} = 11.15 \pm 0.15 \%$

$\mu = 473.3 \pm 0.7$

$\sigma = 52.8 \pm 0.6$

WF (11214 entries)

$\frac{\sigma}{\mu} = 11.23 \pm 0.15 \%$

$\mu = 472.2 \pm 0.7$

$\sigma = 53.0 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count

1000

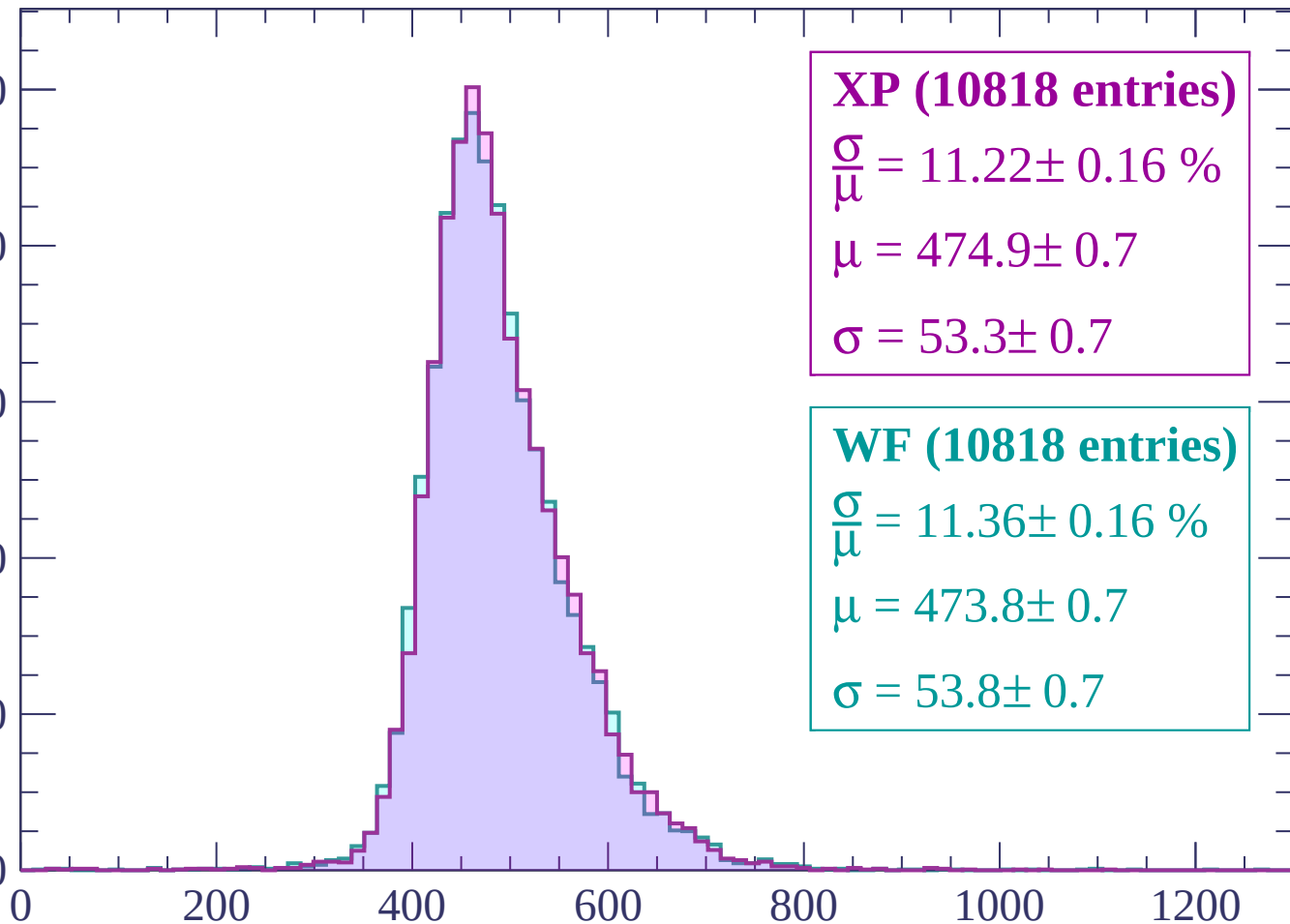
800

600

400

200

0



XP (10818 entries)

$\frac{\sigma}{\mu} = 11.22 \pm 0.16 \%$

$\mu = 474.9 \pm 0.7$

$\sigma = 53.3 \pm 0.7$

WF (10818 entries)

$\frac{\sigma}{\mu} = 11.36 \pm 0.16 \%$

$\mu = 473.8 \pm 0.7$

$\sigma = 53.8 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count

1000

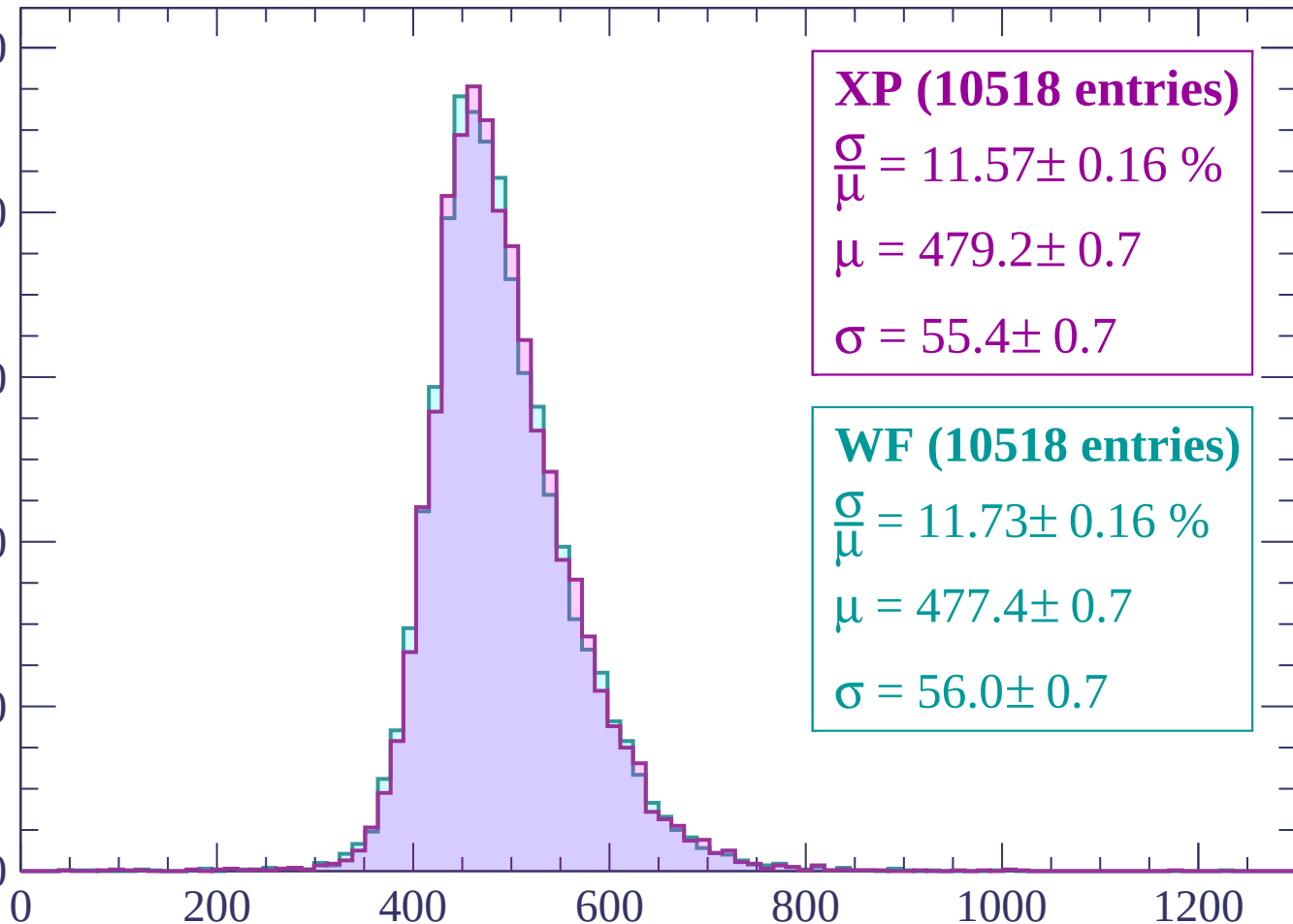
800

600

400

200

0



XP (10518 entries)

$\frac{\sigma}{\mu} = 11.57 \pm 0.16 \%$

$\mu = 479.2 \pm 0.7$

$\sigma = 55.4 \pm 0.7$

WF (10518 entries)

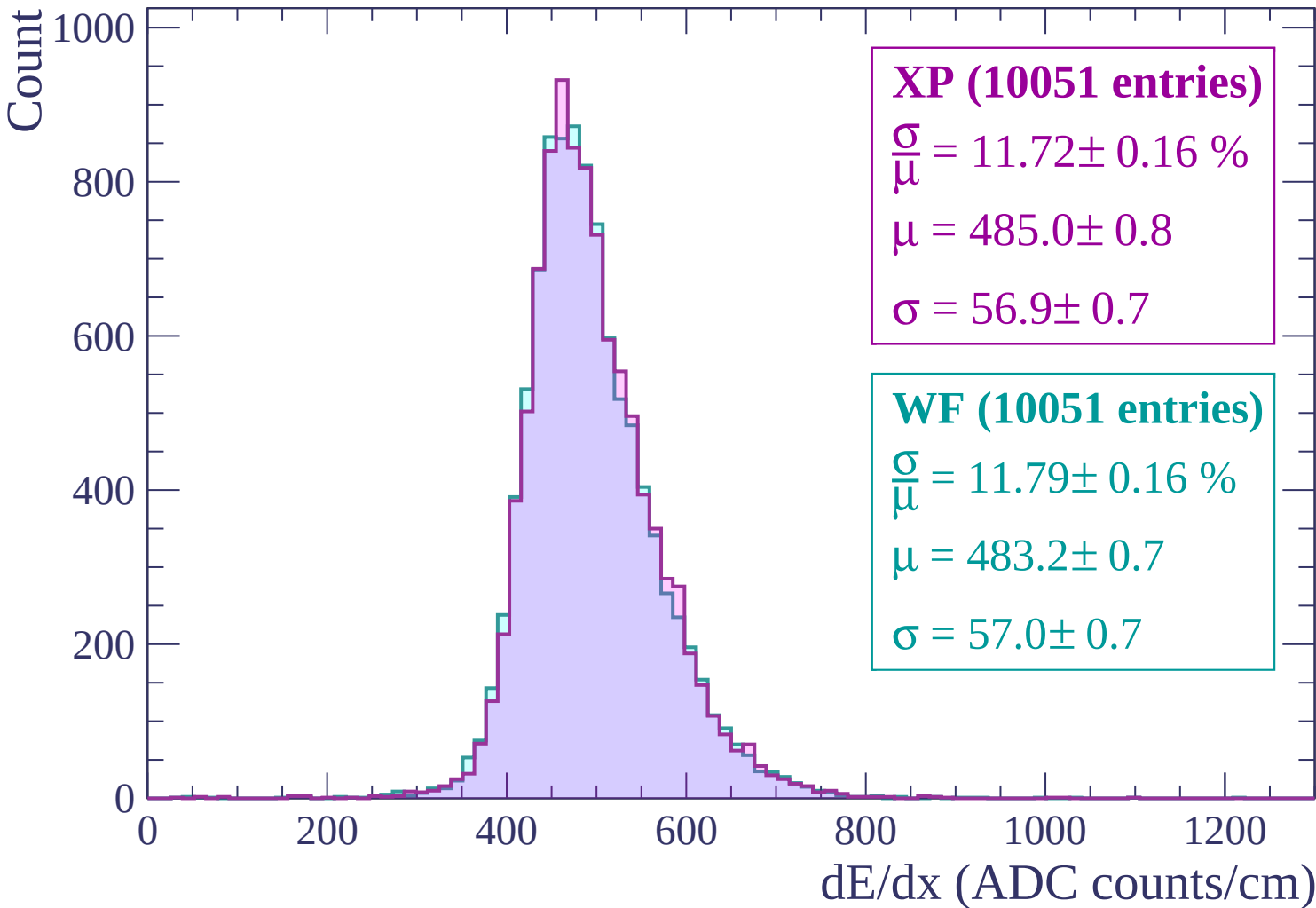
$\frac{\sigma}{\mu} = 11.73 \pm 0.16 \%$

$\mu = 477.4 \pm 0.7$

$\sigma = 56.0 \pm 0.7$

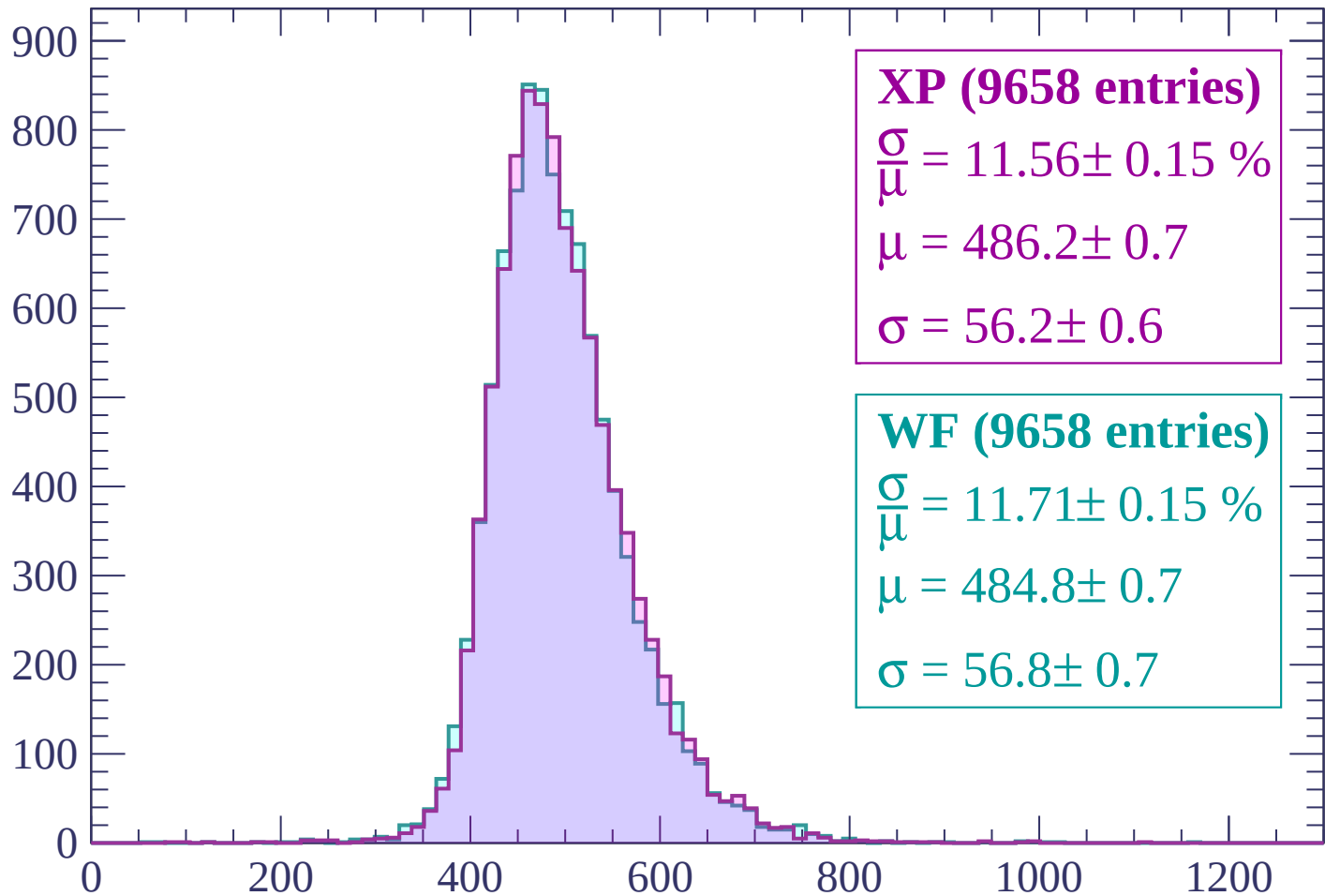
dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$



Energy loss | $1740 < p < 1800$

Count



XP (9658 entries)

$$\frac{\sigma}{\mu} = 11.56 \pm 0.15 \%$$

$$\mu = 486.2 \pm 0.7$$

$$\sigma = 56.2 \pm 0.6$$

WF (9658 entries)

$$\frac{\sigma}{\mu} = 11.71 \pm 0.15 \%$$

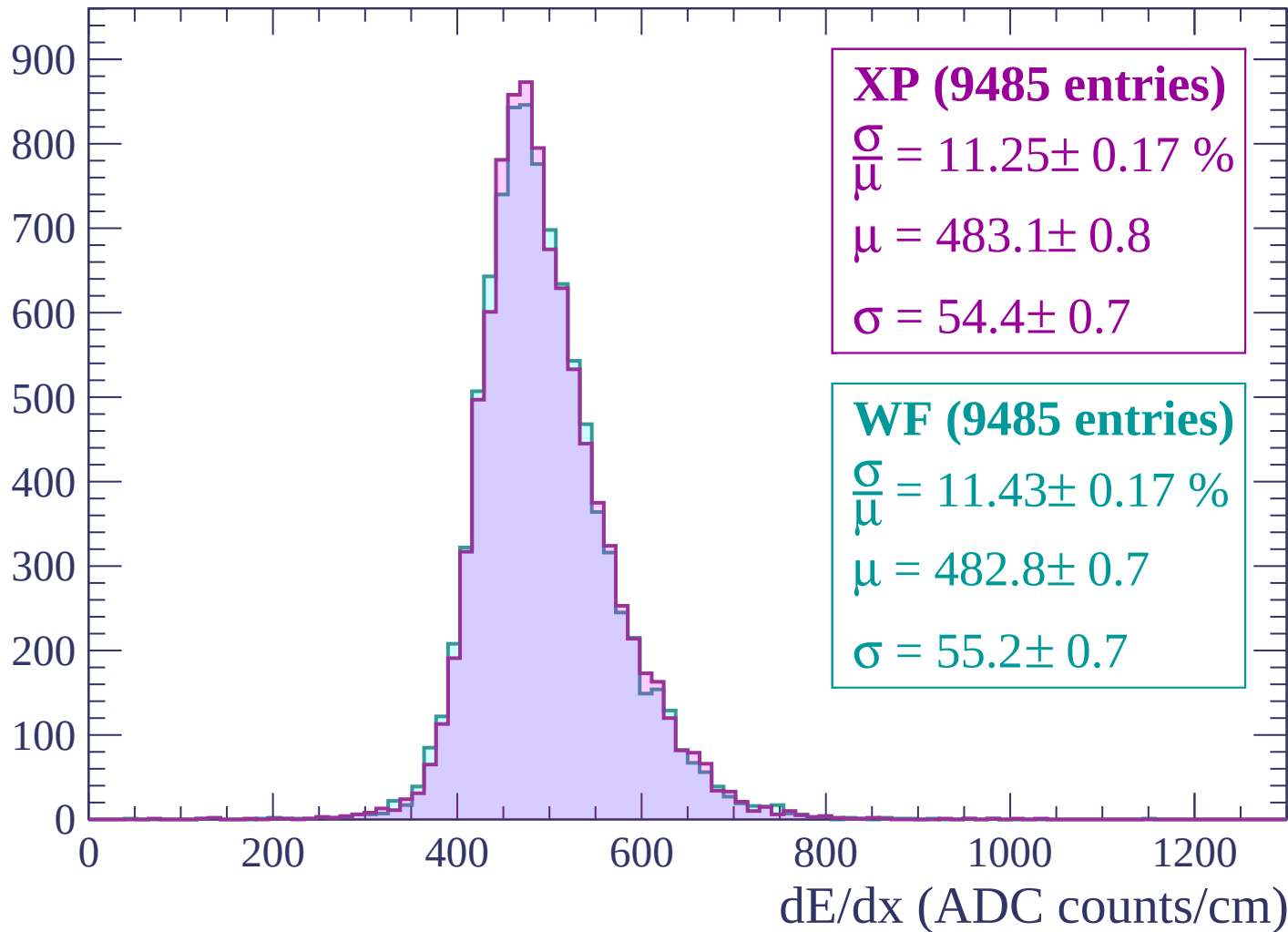
$$\mu = 484.8 \pm 0.7$$

$$\sigma = 56.8 \pm 0.7$$

dE/dx (ADC counts/cm)

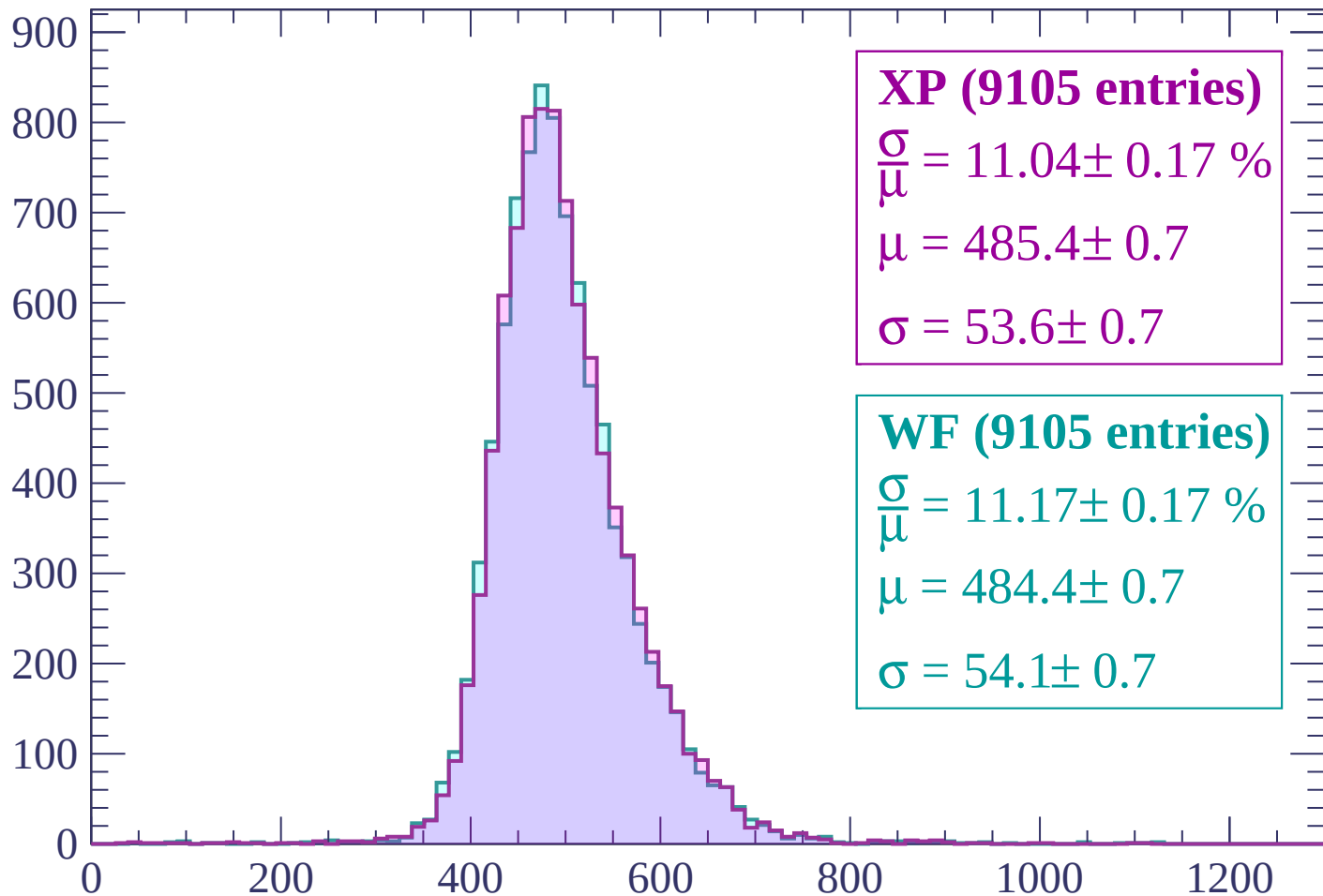
Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count



XP (9105 entries)

$$\frac{\sigma}{\mu} = 11.04 \pm 0.17 \%$$

$$\mu = 485.4 \pm 0.7$$

$$\sigma = 53.6 \pm 0.7$$

WF (9105 entries)

$$\frac{\sigma}{\mu} = 11.17 \pm 0.17 \%$$

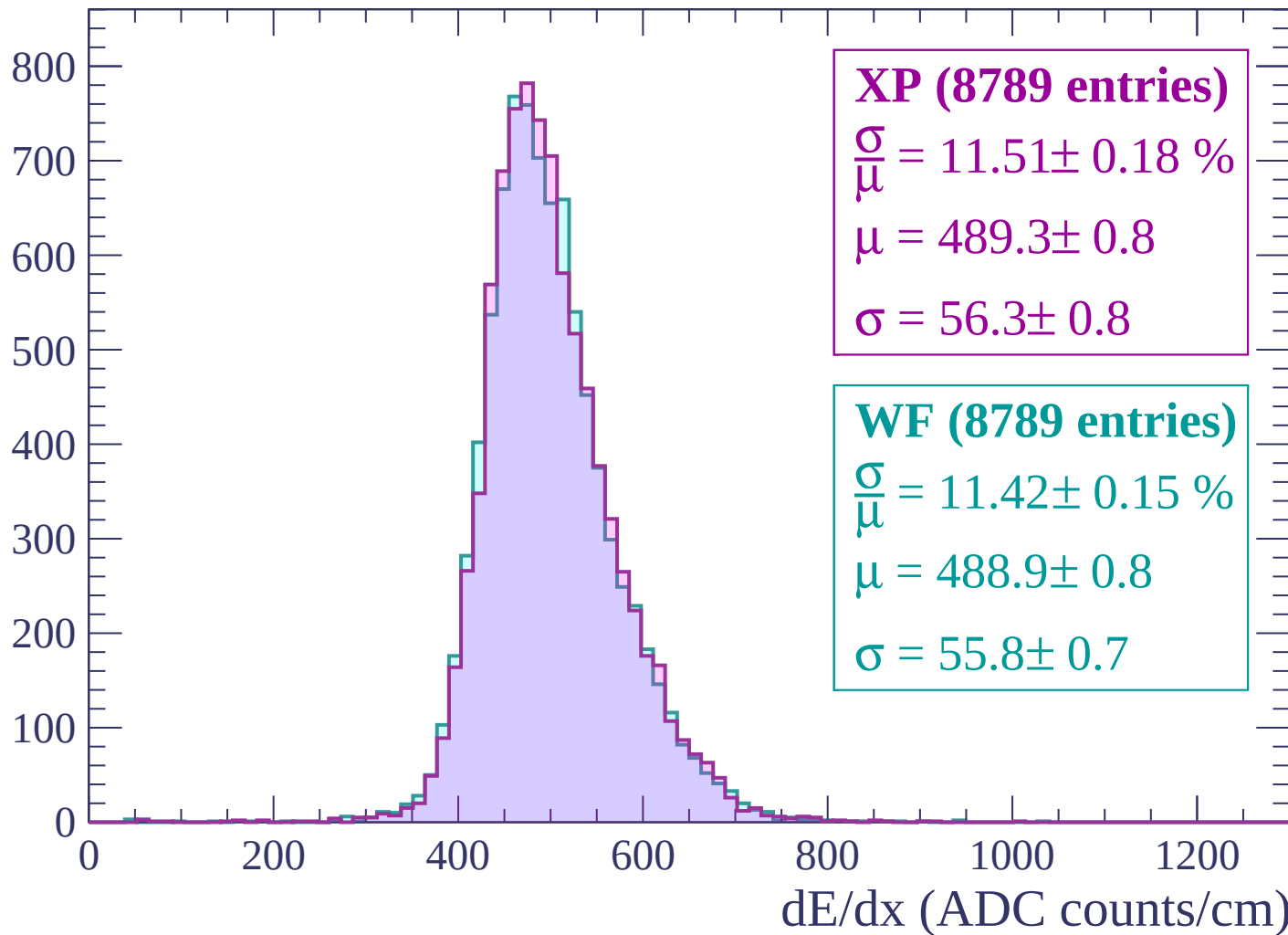
$$\mu = 484.4 \pm 0.7$$

$$\sigma = 54.1 \pm 0.7$$

dE/dx (ADC counts/cm)

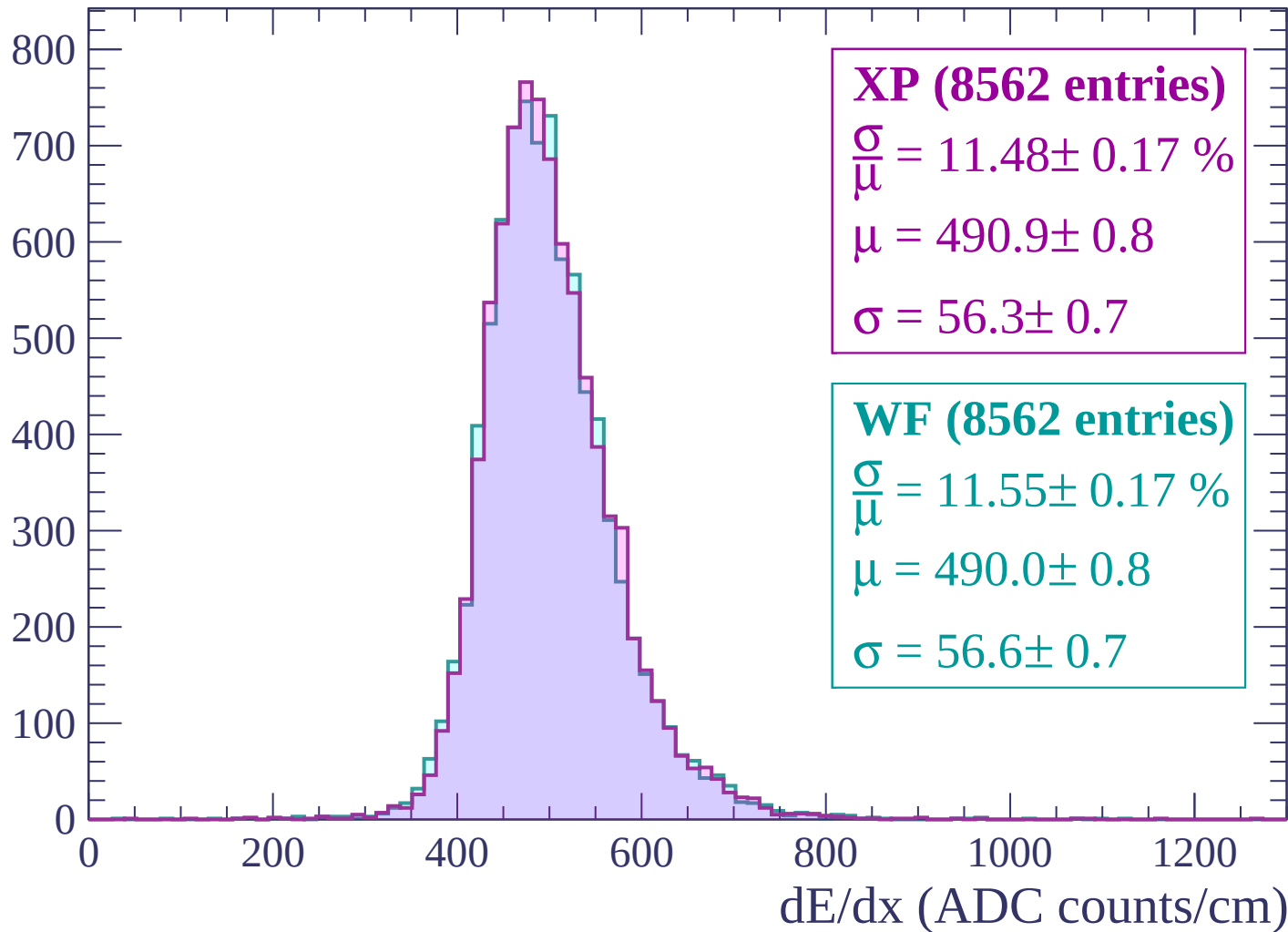
Energy loss | $1920 < p < 1980$

Count



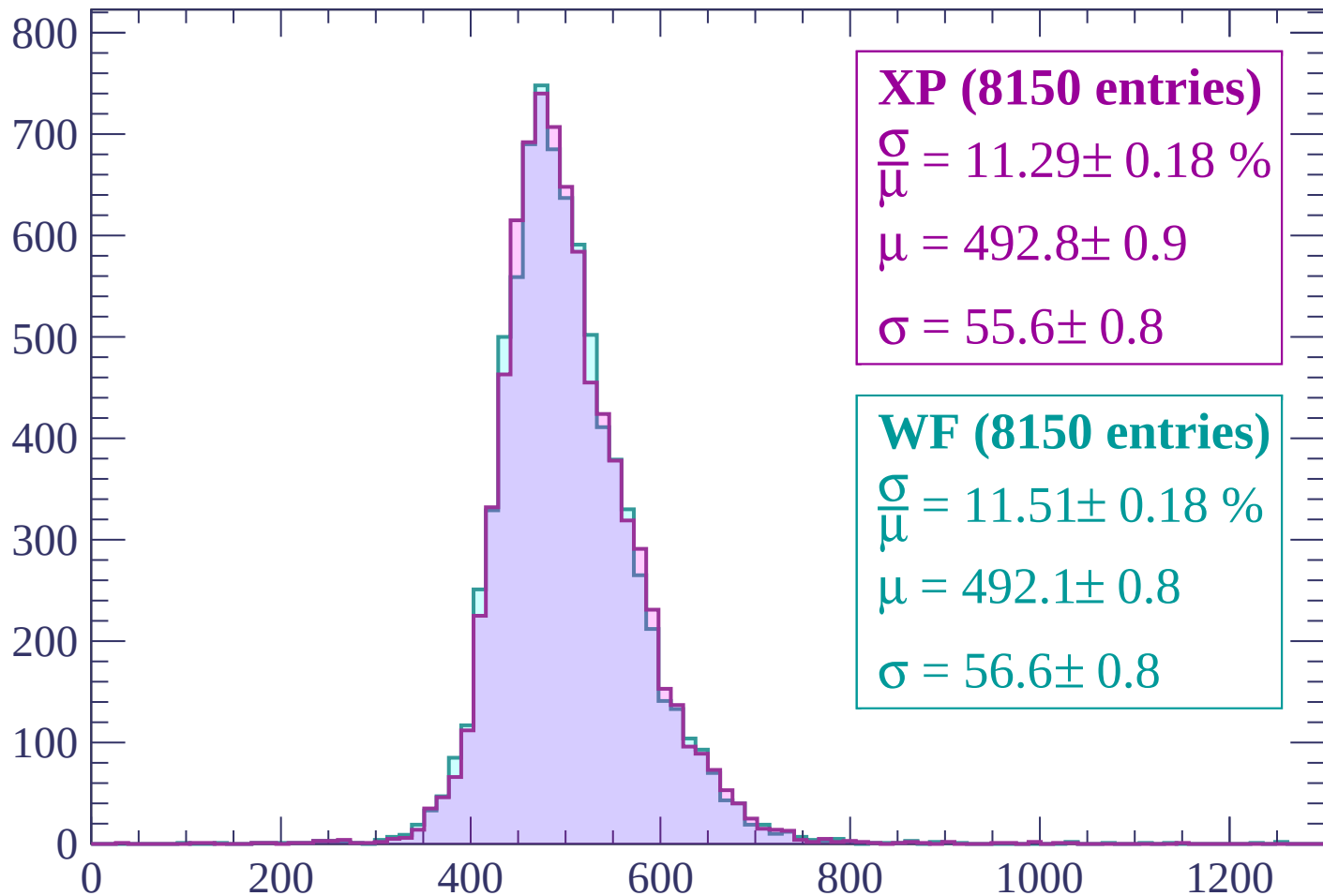
Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

Count



XP (8150 entries)

$$\frac{\sigma}{\mu} = 11.29 \pm 0.18 \%$$

$$\mu = 492.8 \pm 0.9$$

$$\sigma = 55.6 \pm 0.8$$

WF (8150 entries)

$$\frac{\sigma}{\mu} = 11.51 \pm 0.18 \%$$

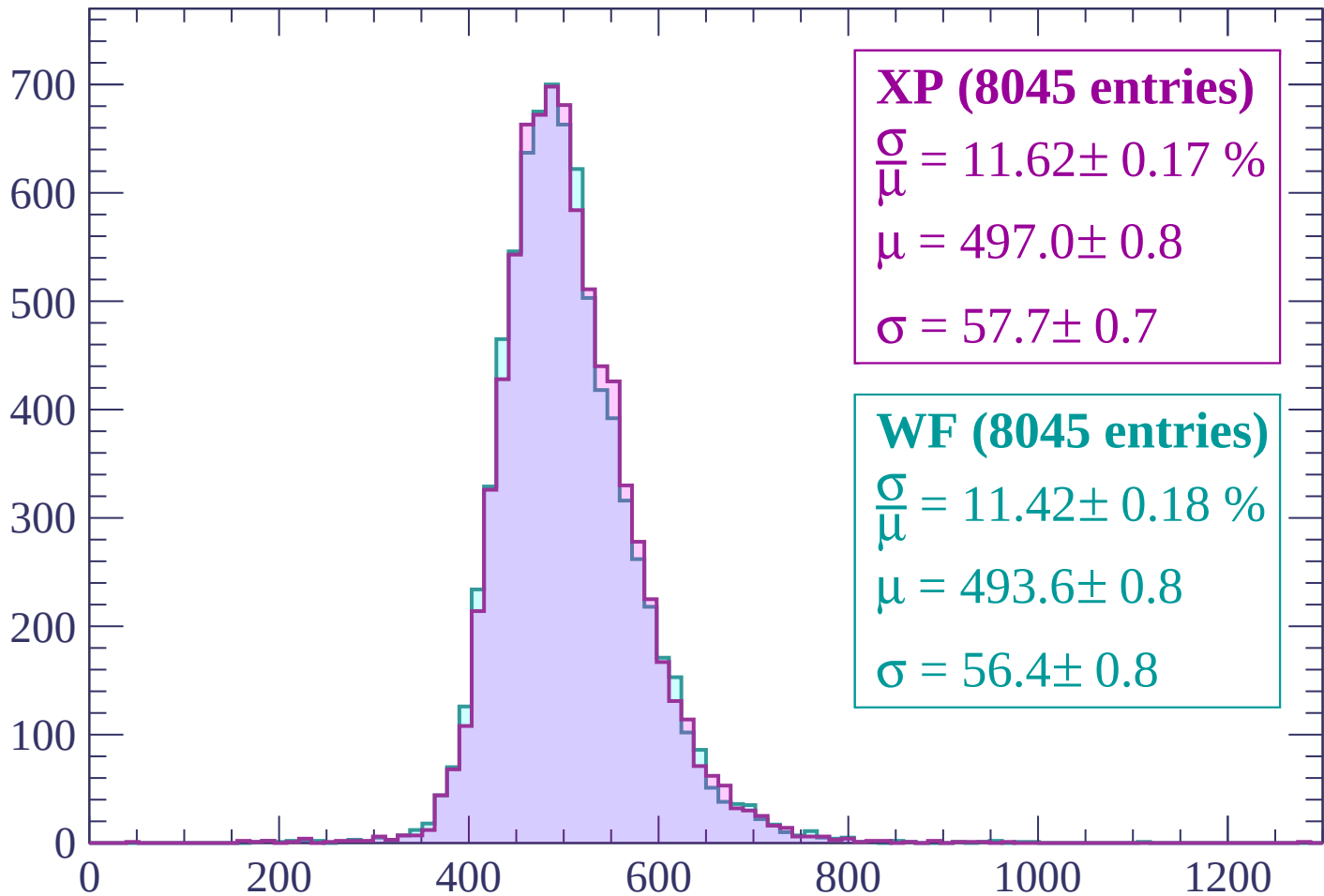
$$\mu = 492.1 \pm 0.8$$

$$\sigma = 56.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP (8045 entries)

$$\frac{\sigma}{\mu} = 11.62 \pm 0.17 \%$$

$$\mu = 497.0 \pm 0.8$$

$$\sigma = 57.7 \pm 0.7$$

WF (8045 entries)

$$\frac{\sigma}{\mu} = 11.42 \pm 0.18 \%$$

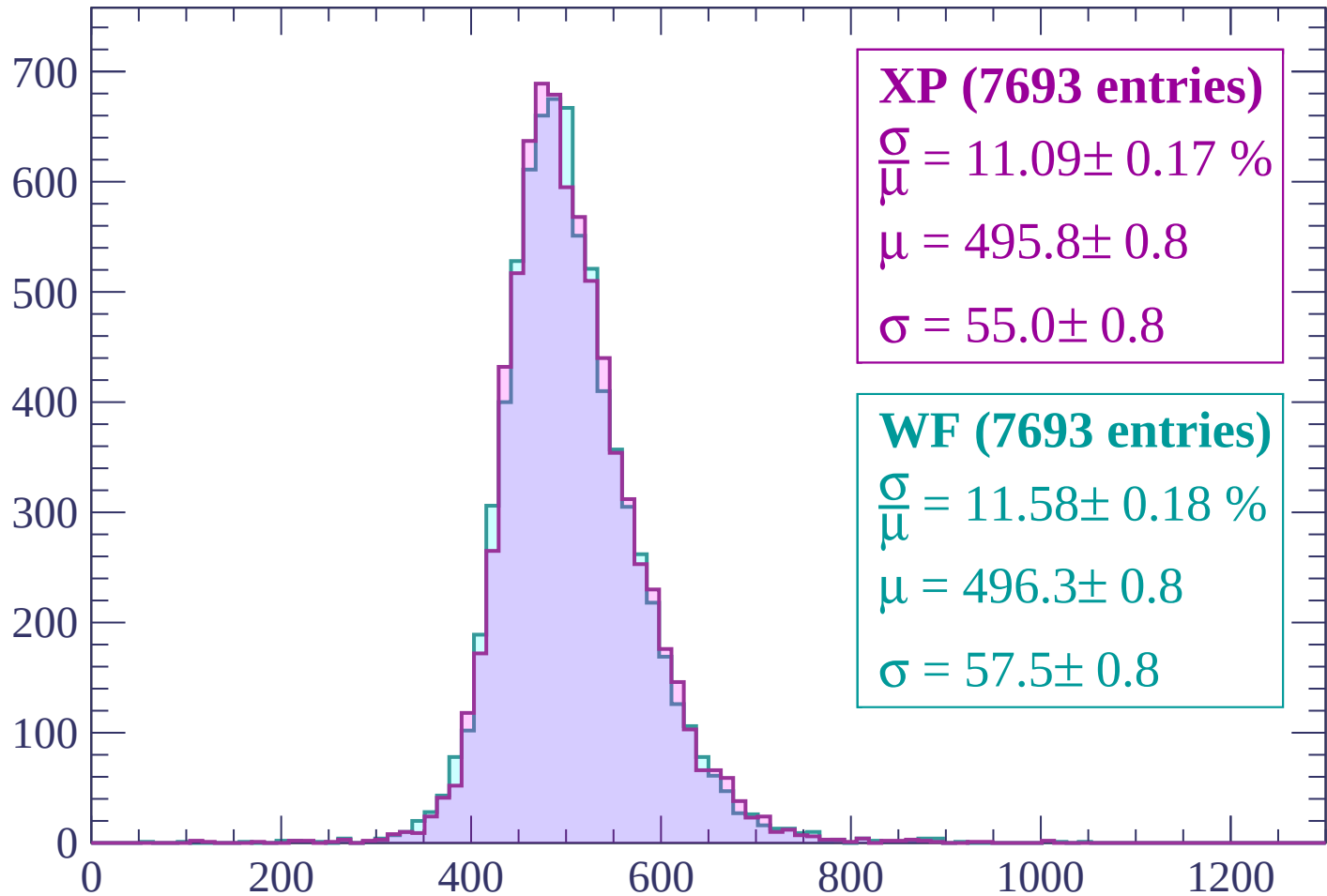
$$\mu = 493.6 \pm 0.8$$

$$\sigma = 56.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP (7693 entries)

$$\frac{\sigma}{\mu} = 11.09 \pm 0.17 \%$$

$$\mu = 495.8 \pm 0.8$$

$$\sigma = 55.0 \pm 0.8$$

WF (7693 entries)

$$\frac{\sigma}{\mu} = 11.58 \pm 0.18 \%$$

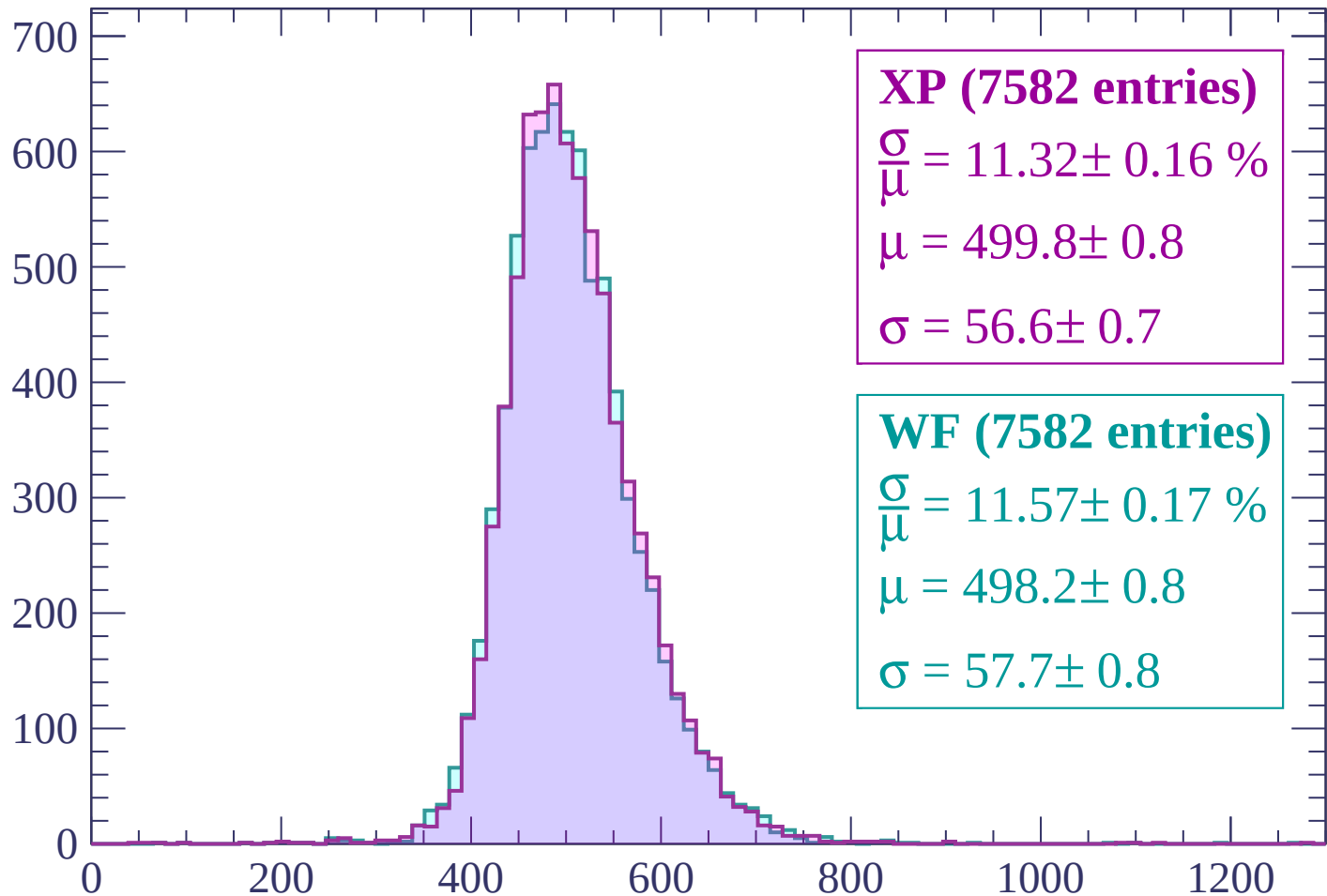
$$\mu = 496.3 \pm 0.8$$

$$\sigma = 57.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (7582 entries)

$$\frac{\sigma}{\mu} = 11.32 \pm 0.16 \%$$

$$\mu = 499.8 \pm 0.8$$

$$\sigma = 56.6 \pm 0.7$$

WF (7582 entries)

$$\frac{\sigma}{\mu} = 11.57 \pm 0.17 \%$$

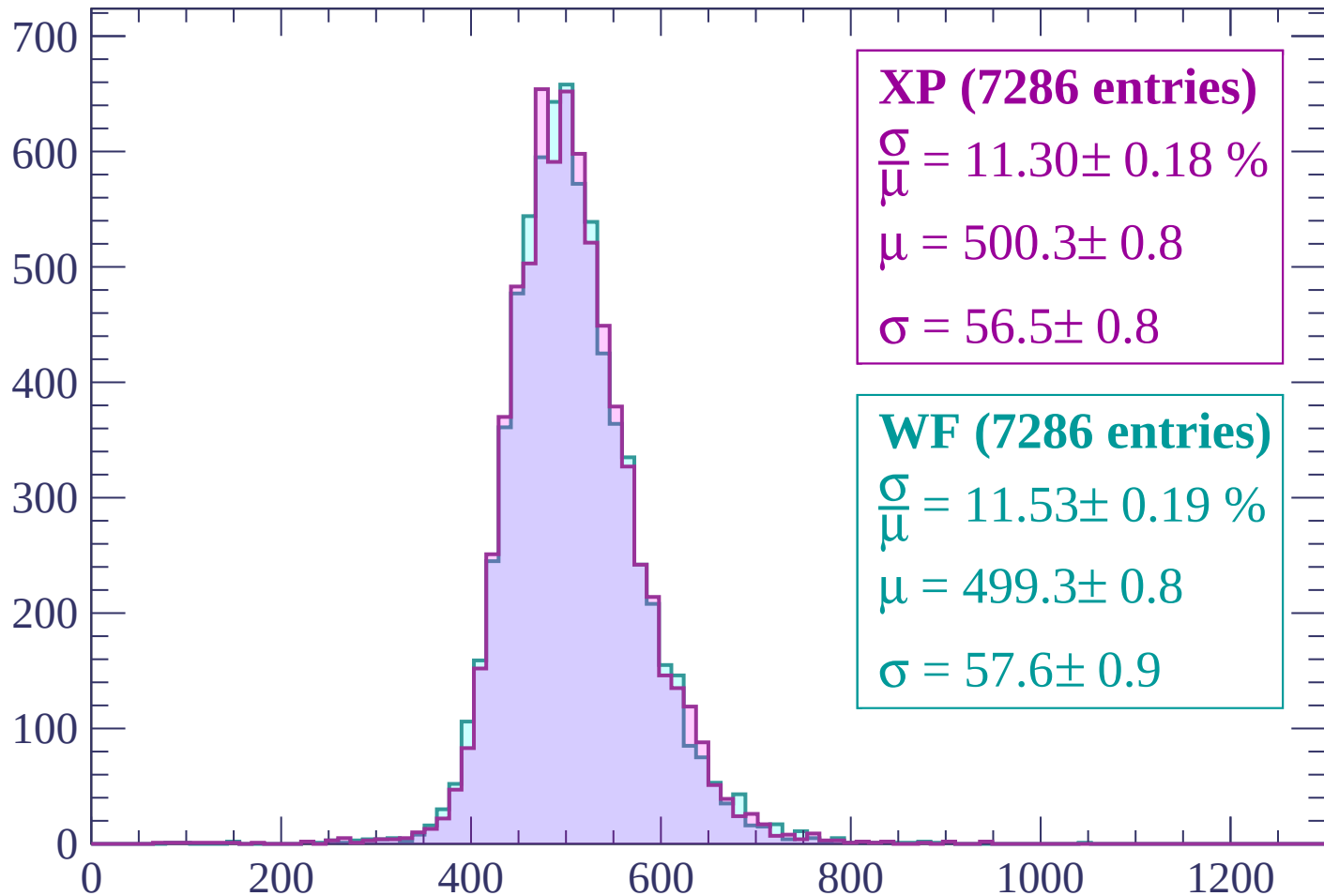
$$\mu = 498.2 \pm 0.8$$

$$\sigma = 57.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count



XP (7286 entries)

$$\frac{\sigma}{\mu} = 11.30 \pm 0.18 \%$$

$$\mu = 500.3 \pm 0.8$$

$$\sigma = 56.5 \pm 0.8$$

WF (7286 entries)

$$\frac{\sigma}{\mu} = 11.53 \pm 0.19 \%$$

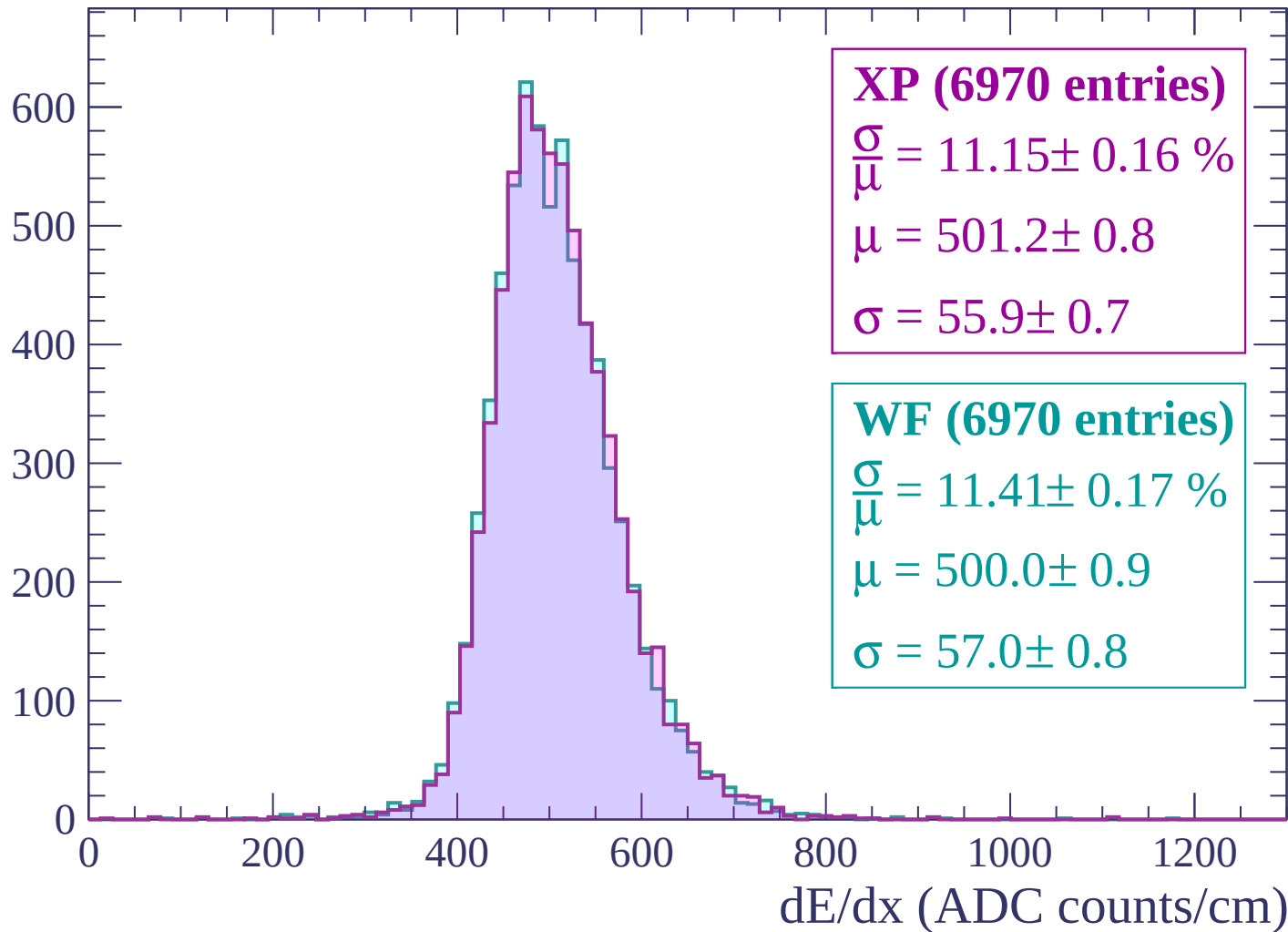
$$\mu = 499.3 \pm 0.8$$

$$\sigma = 57.6 \pm 0.9$$

dE/dx (ADC counts/cm)

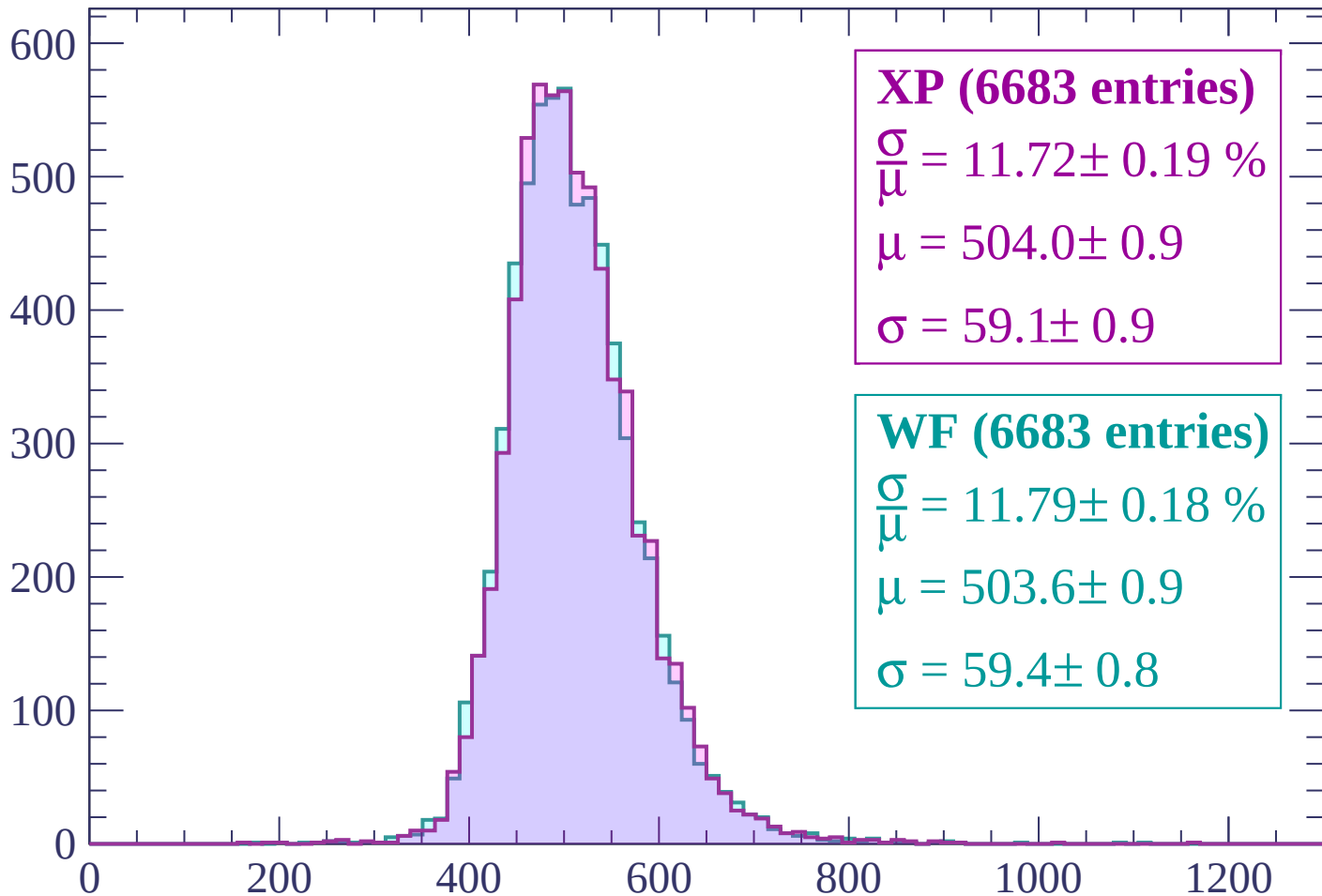
Energy loss | $2340 < p < 2400$

Count



Energy loss | $2400 < p < 2460$

Count



XP (6683 entries)

$\frac{\sigma}{\mu} = 11.72 \pm 0.19 \%$

$\mu = 504.0 \pm 0.9$

$\sigma = 59.1 \pm 0.9$

WF (6683 entries)

$\frac{\sigma}{\mu} = 11.79 \pm 0.18 \%$

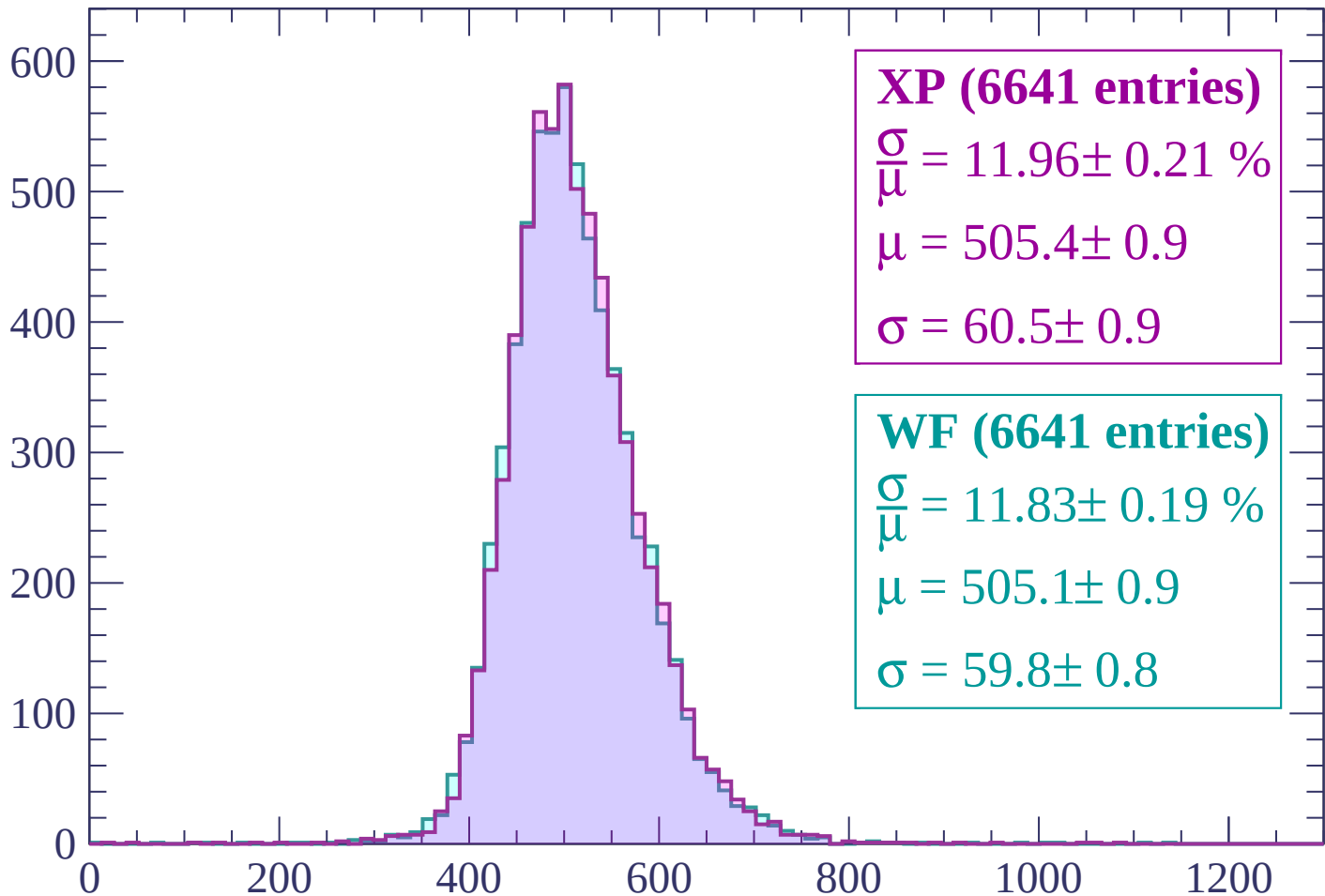
$\mu = 503.6 \pm 0.9$

$\sigma = 59.4 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $2460 < p < 2520$

Count



XP (6641 entries)

$$\frac{\sigma}{\mu} = 11.96 \pm 0.21 \%$$

$$\mu = 505.4 \pm 0.9$$

$$\sigma = 60.5 \pm 0.9$$

WF (6641 entries)

$$\frac{\sigma}{\mu} = 11.83 \pm 0.19 \%$$

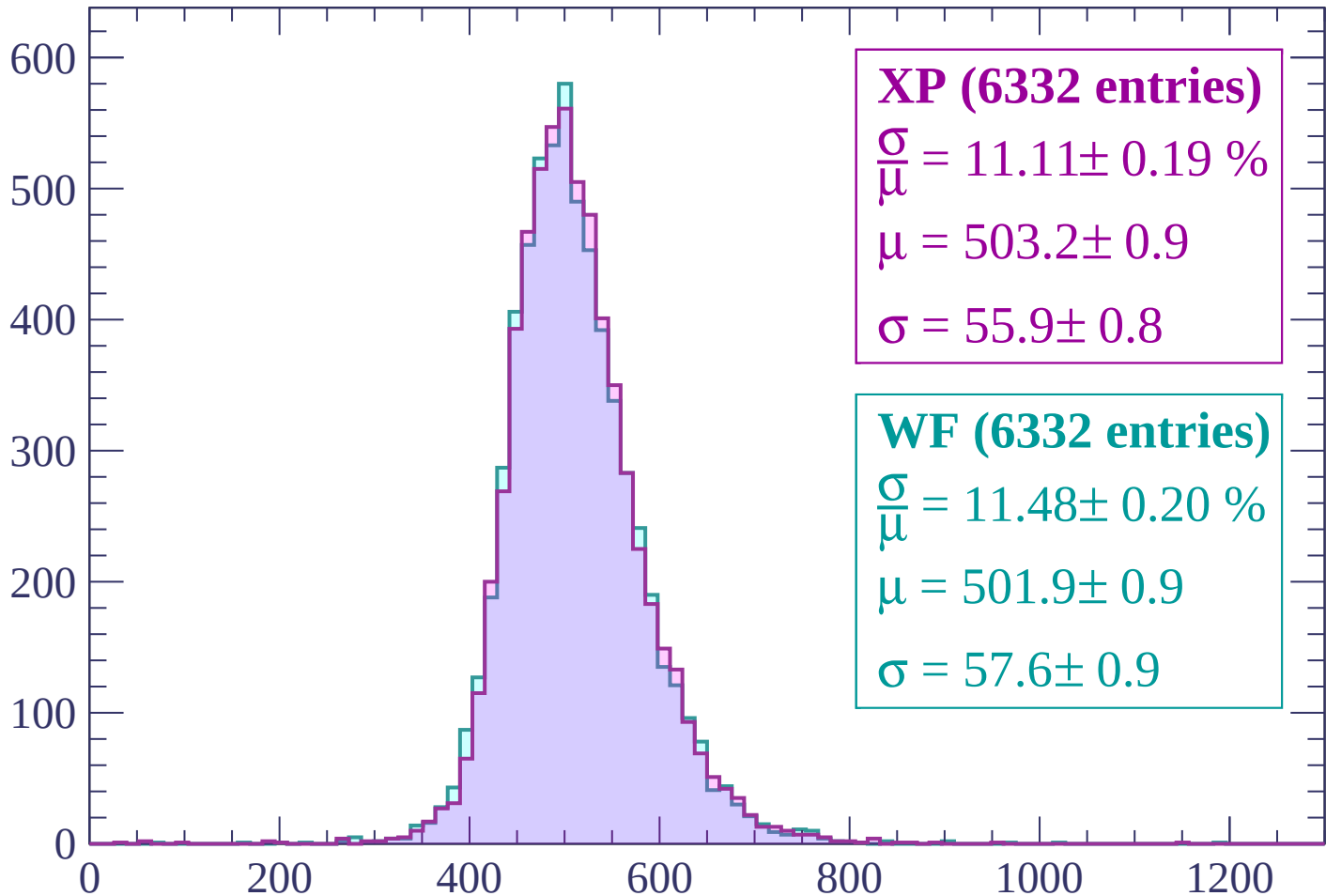
$$\mu = 505.1 \pm 0.9$$

$$\sigma = 59.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2520 < p < 2580$

Count



XP (6332 entries)

$\frac{\sigma}{\mu} = 11.11 \pm 0.19 \%$

$\mu = 503.2 \pm 0.9$

$\sigma = 55.9 \pm 0.8$

WF (6332 entries)

$\frac{\sigma}{\mu} = 11.48 \pm 0.20 \%$

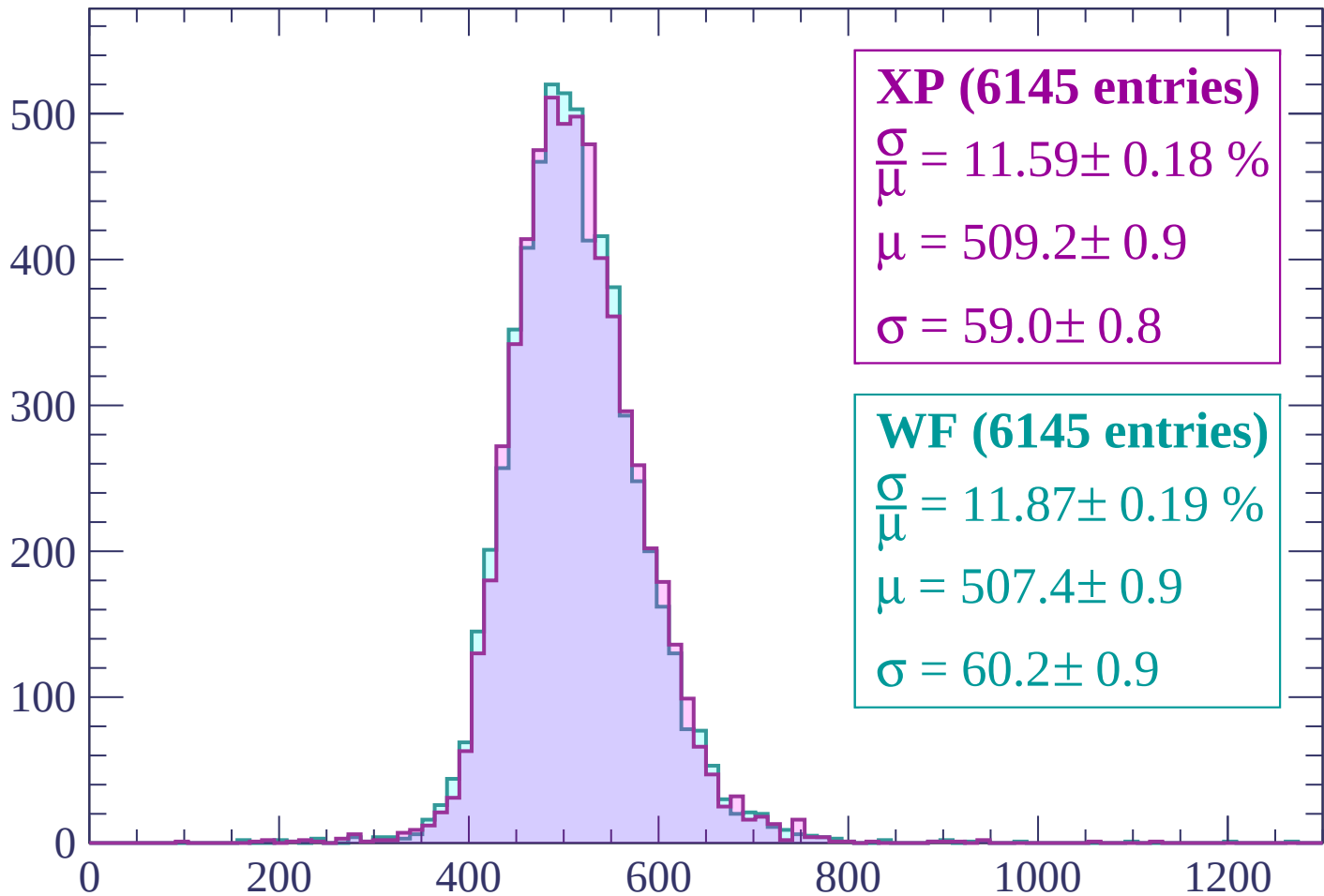
$\mu = 501.9 \pm 0.9$

$\sigma = 57.6 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP (6145 entries)

$$\frac{\sigma}{\mu} = 11.59 \pm 0.18 \%$$

$$\mu = 509.2 \pm 0.9$$

$$\sigma = 59.0 \pm 0.8$$

WF (6145 entries)

$$\frac{\sigma}{\mu} = 11.87 \pm 0.19 \%$$

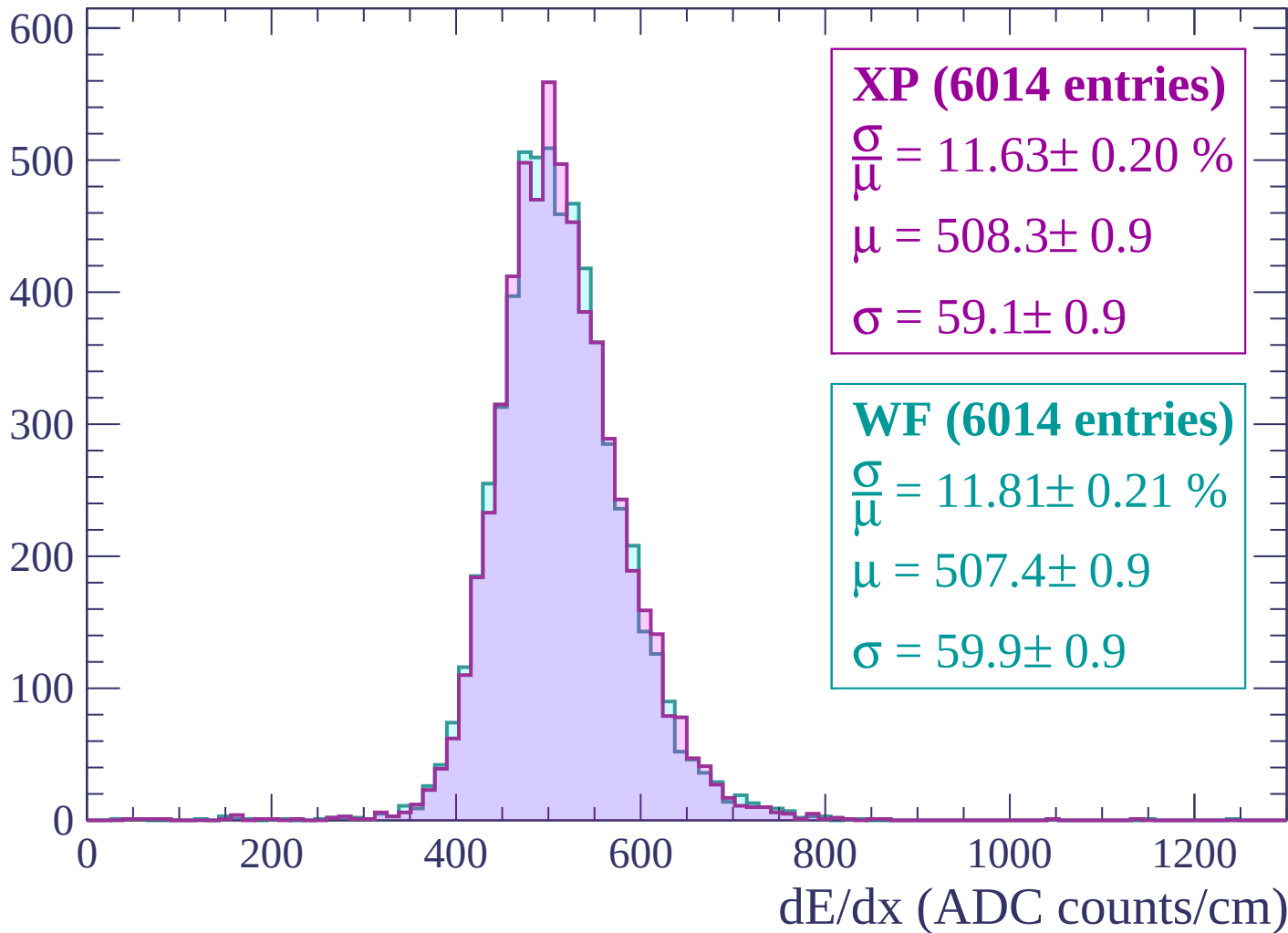
$$\mu = 507.4 \pm 0.9$$

$$\sigma = 60.2 \pm 0.9$$

dE/dx (ADC counts/cm)

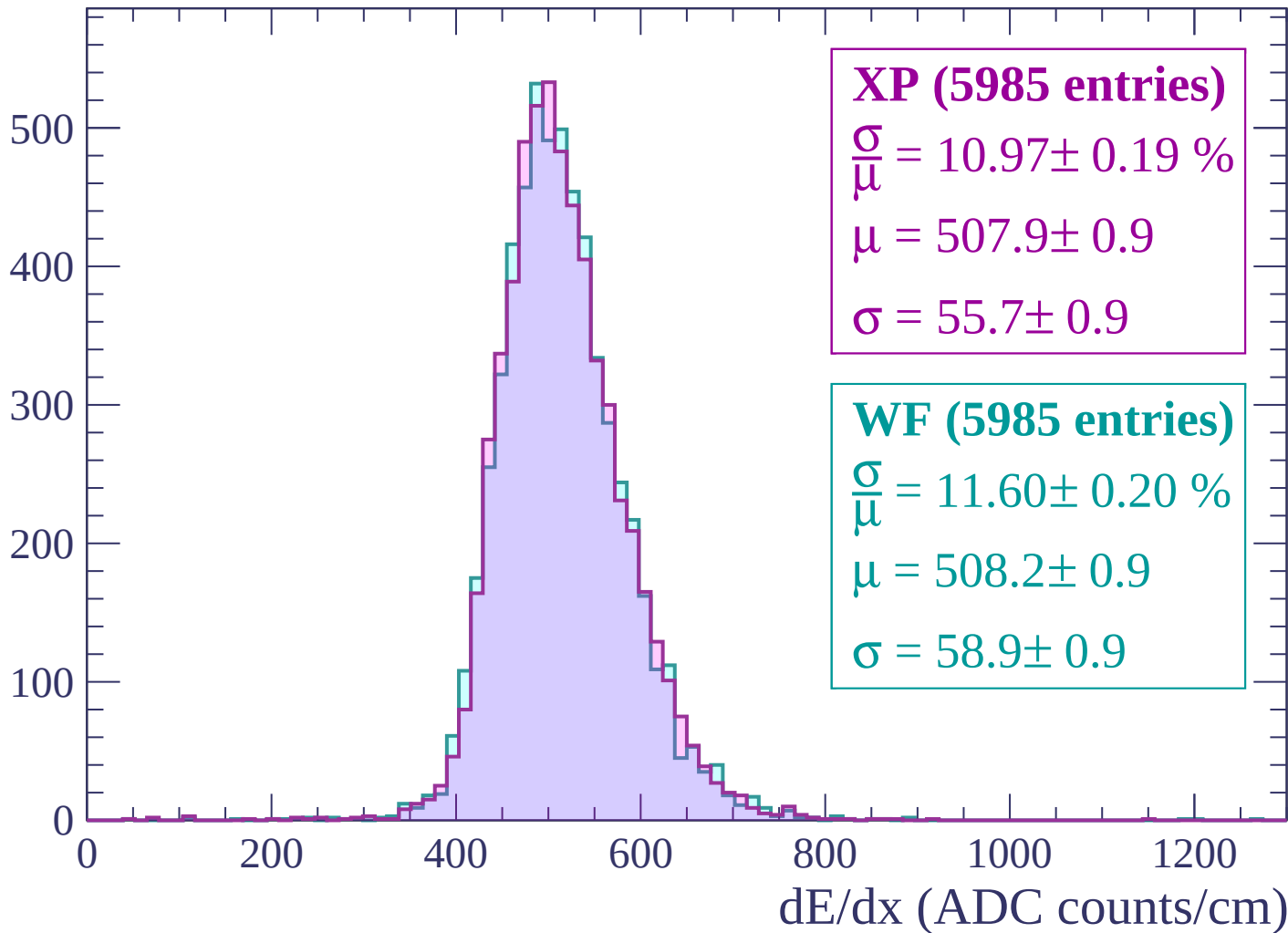
Energy loss | $2640 < p < 2700$

Count



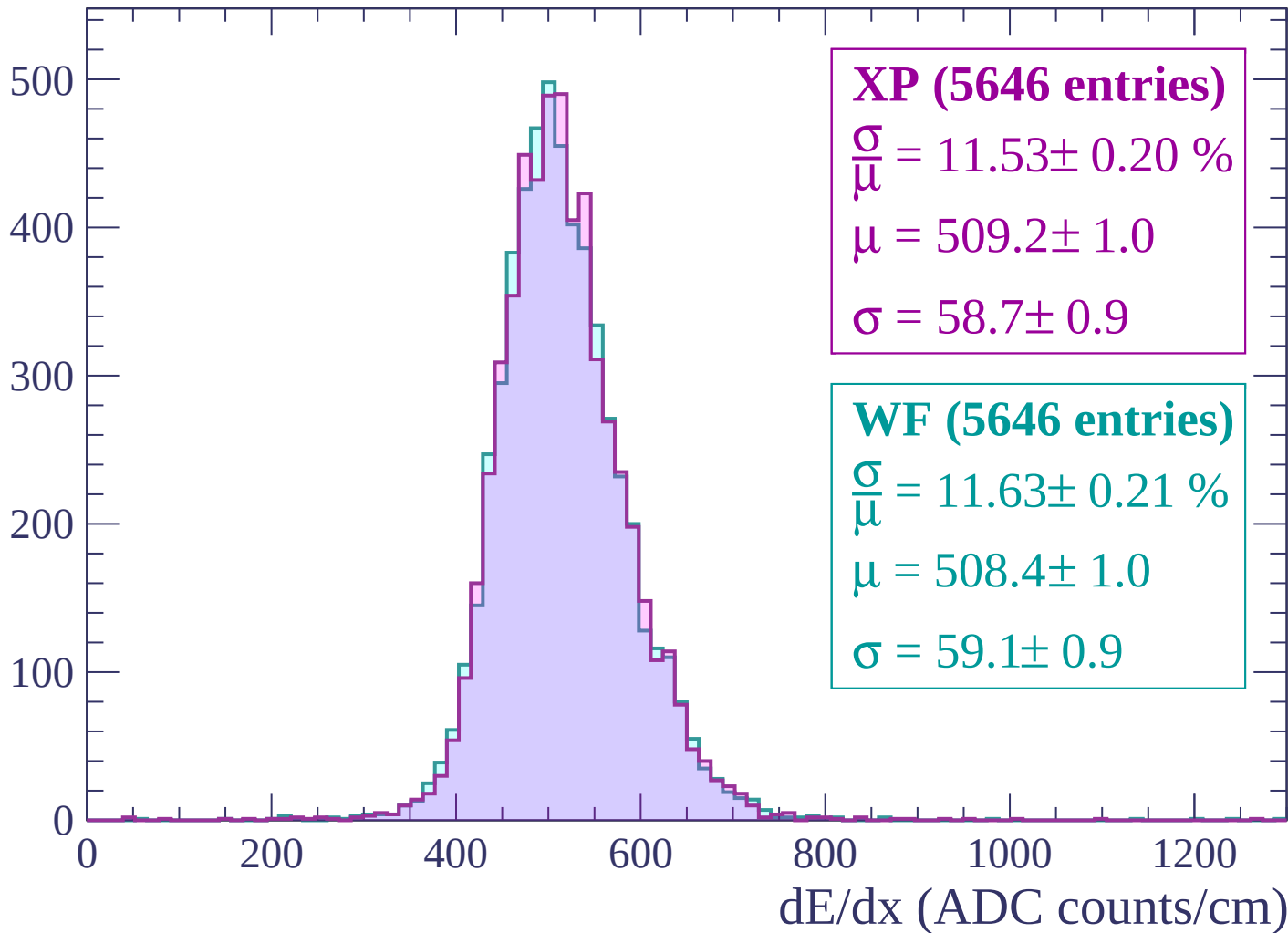
Energy loss | $2700 < p < 2760$

Count



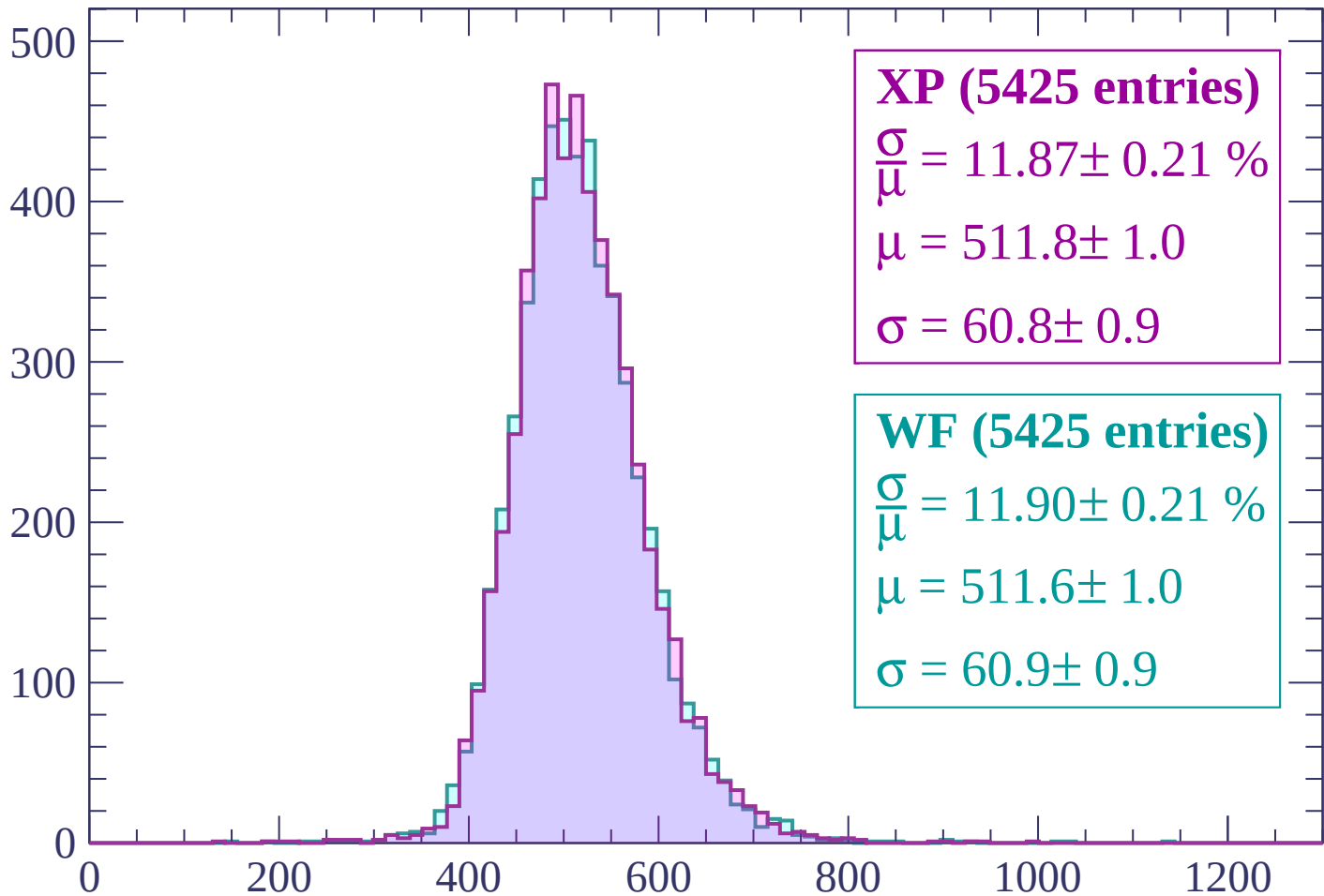
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP (5425 entries)

$$\frac{\sigma}{\mu} = 11.87 \pm 0.21 \%$$

$$\mu = 511.8 \pm 1.0$$

$$\sigma = 60.8 \pm 0.9$$

WF (5425 entries)

$$\frac{\sigma}{\mu} = 11.90 \pm 0.21 \%$$

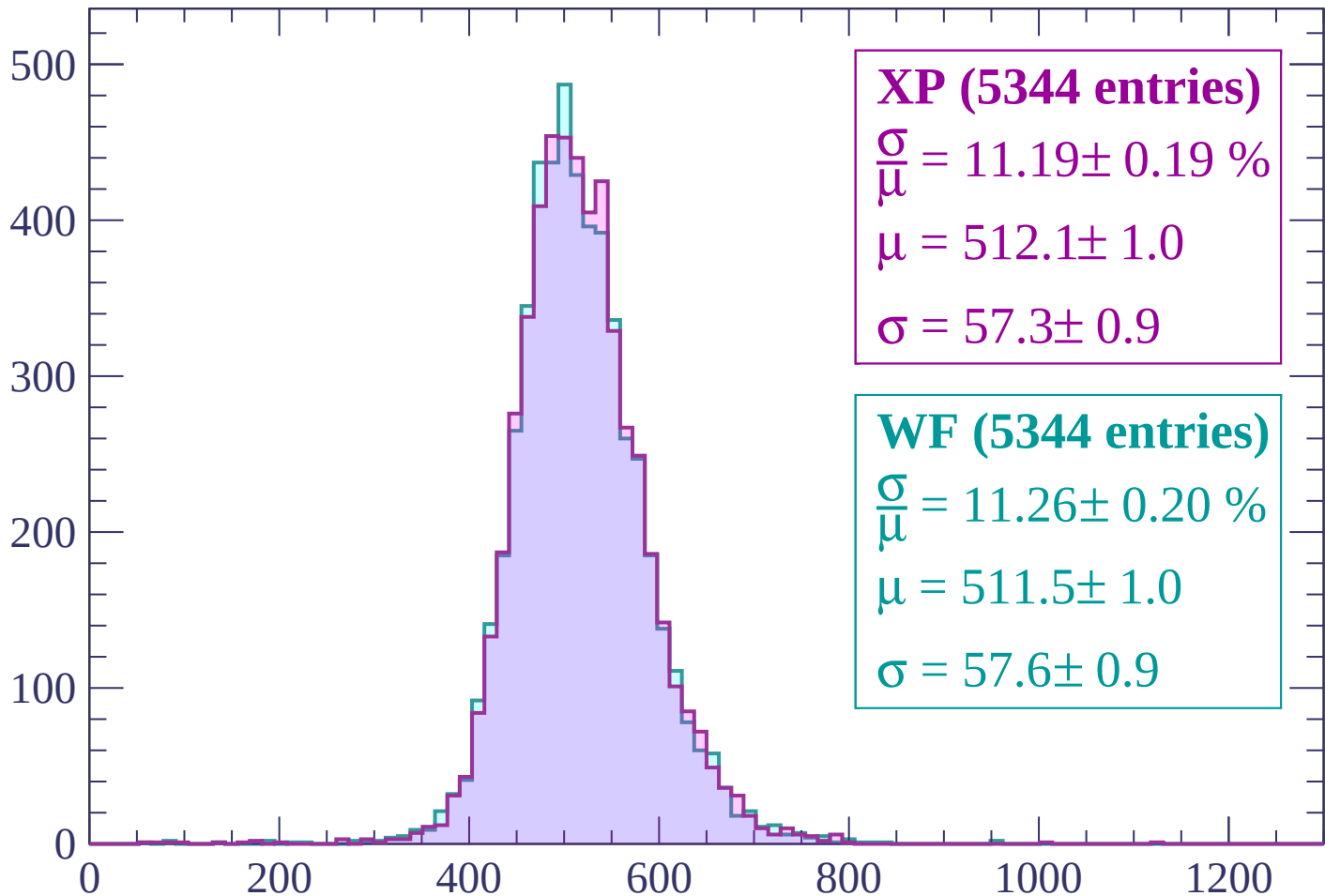
$$\mu = 511.6 \pm 1.0$$

$$\sigma = 60.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2880 < p < 2940$

Count



XP (5344 entries)

$\frac{\sigma}{\mu} = 11.19 \pm 0.19 \%$

$\mu = 512.1 \pm 1.0$

$\sigma = 57.3 \pm 0.9$

WF (5344 entries)

$\frac{\sigma}{\mu} = 11.26 \pm 0.20 \%$

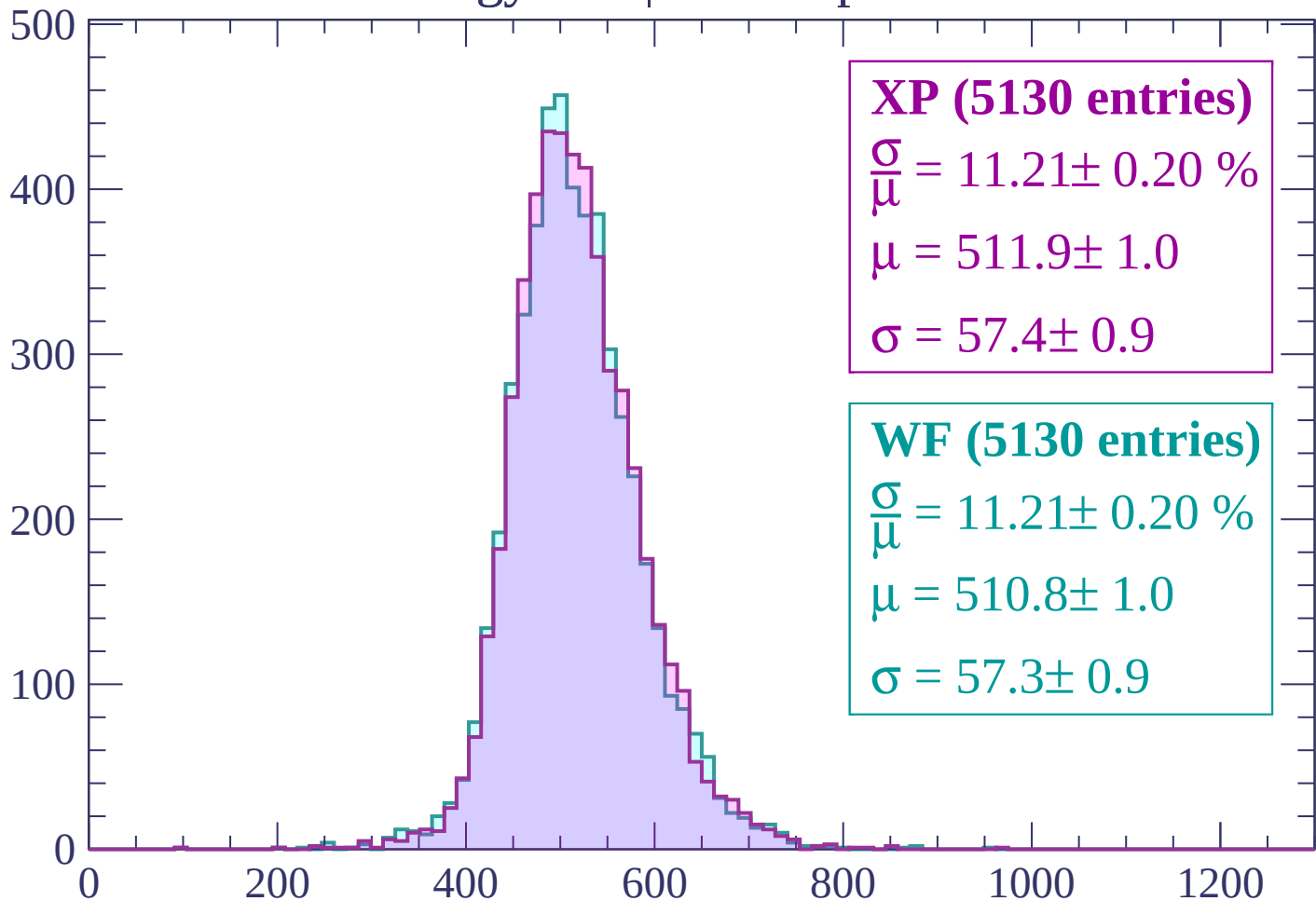
$\mu = 511.5 \pm 1.0$

$\sigma = 57.6 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $2940 < p < 3000$

Count



XP (5130 entries)

$\frac{\sigma}{\mu} = 11.21 \pm 0.20 \%$

$\mu = 511.9 \pm 1.0$

$\sigma = 57.4 \pm 0.9$

WF (5130 entries)

$\frac{\sigma}{\mu} = 11.21 \pm 0.20 \%$

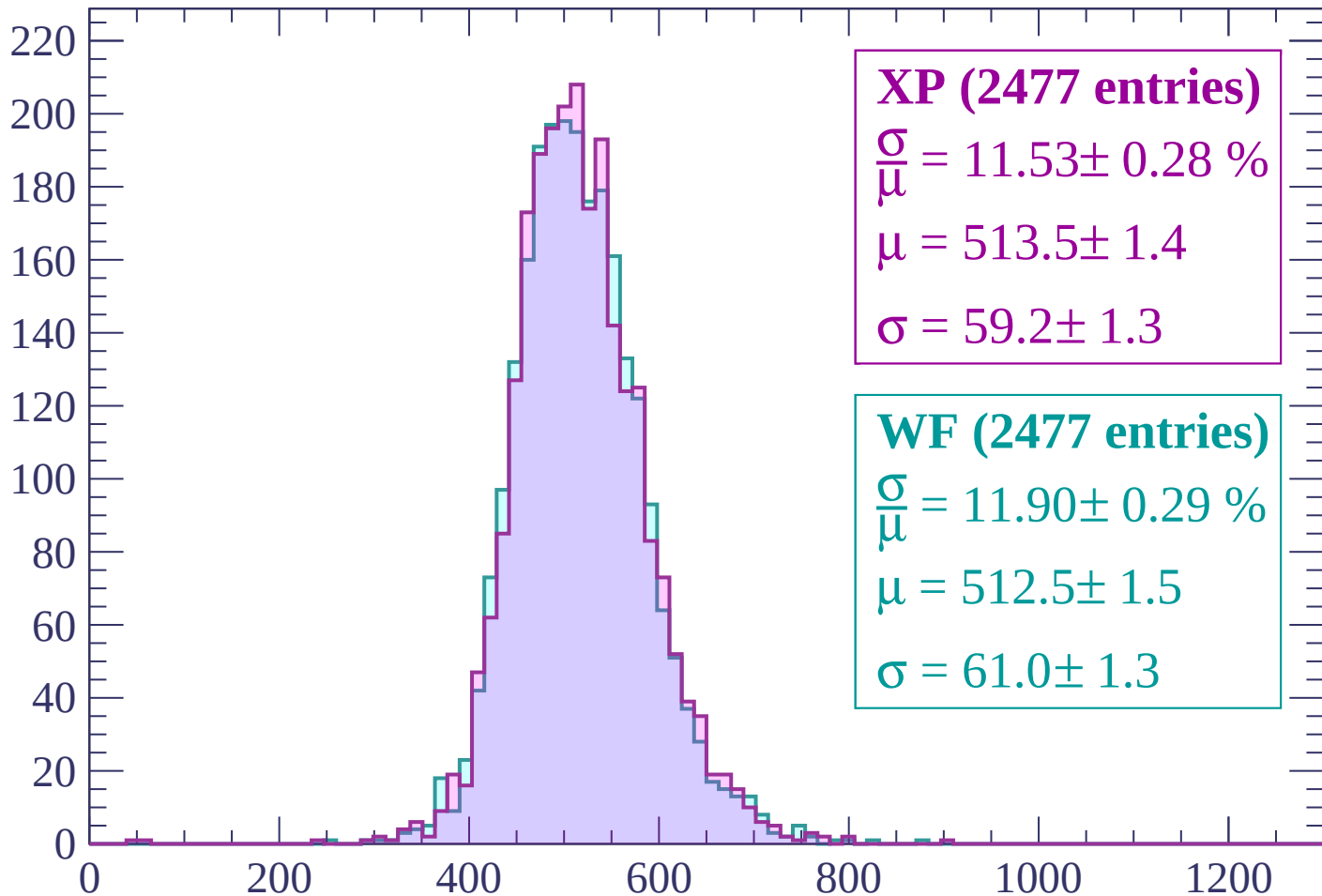
$\mu = 510.8 \pm 1.0$

$\sigma = 57.3 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count



XP (2477 entries)

$$\frac{\sigma}{\mu} = 11.53 \pm 0.28 \%$$

$$\mu = 513.5 \pm 1.4$$

$$\sigma = 59.2 \pm 1.3$$

WF (2477 entries)

$$\frac{\sigma}{\mu} = 11.90 \pm 0.29 \%$$

$$\mu = 512.5 \pm 1.5$$

$$\sigma = 61.0 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-90 < \theta < -88$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$

$\mu = 6.2 \pm 2.4$

$\sigma = 0.3 \pm 1.3$

WF (1 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

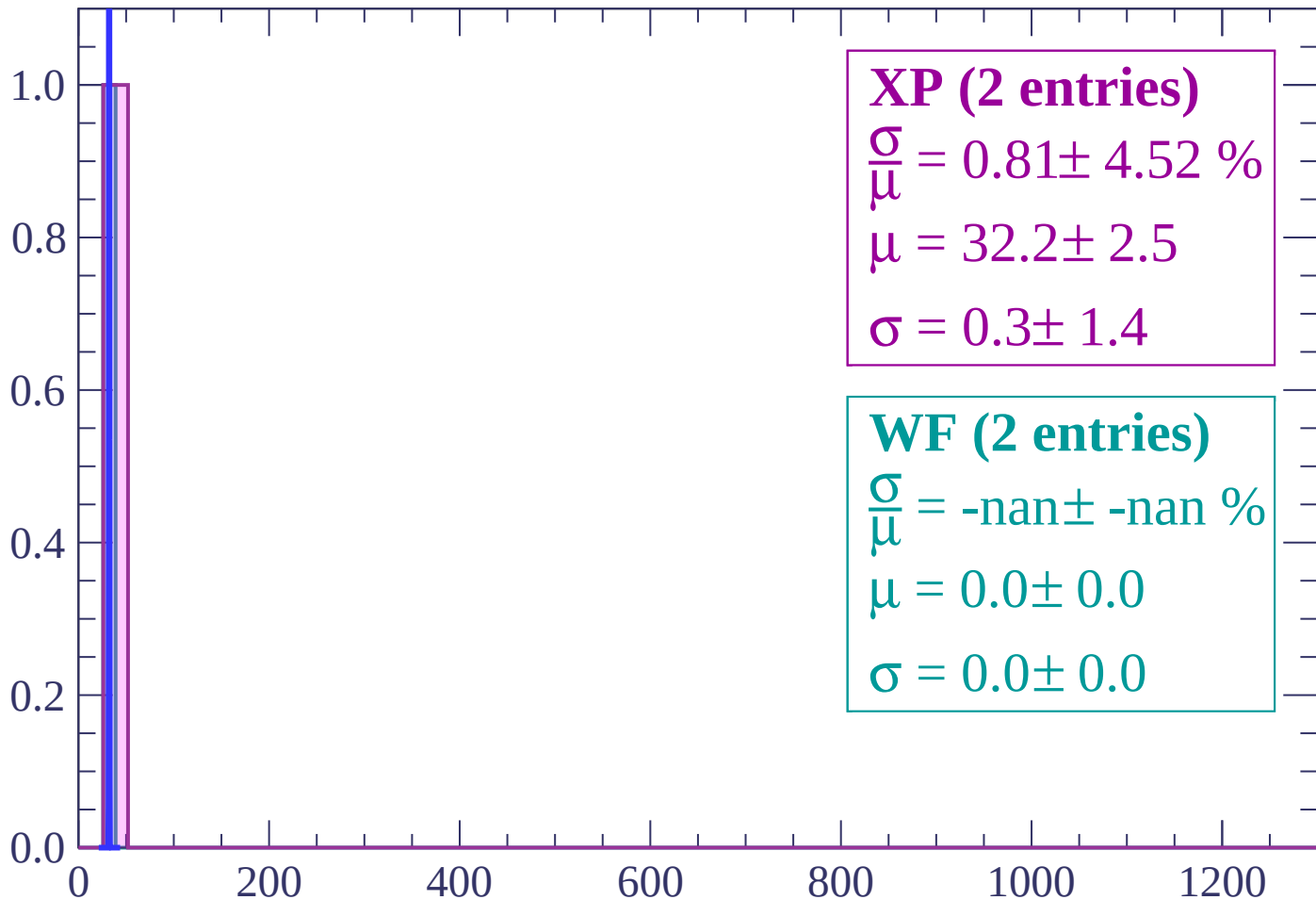
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count



XP (2 entries)

$\frac{\sigma}{\mu} = 0.81 \pm 4.52 \%$

$\mu = 32.2 \pm 2.5$

$\sigma = 0.3 \pm 1.4$

WF (2 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

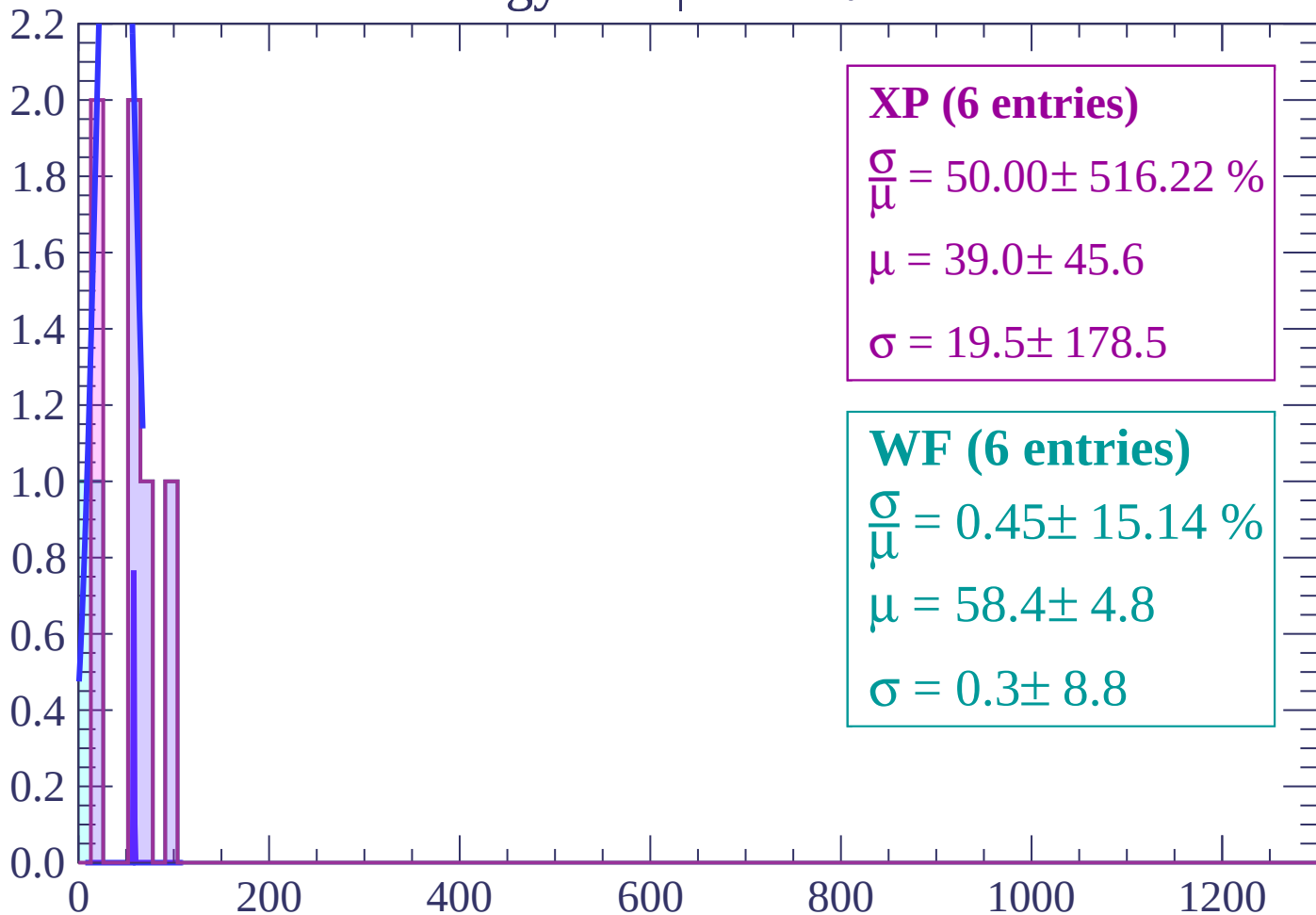
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

Count



XP (6 entries)

$\frac{\sigma}{\mu} = 50.00 \pm 516.22 \%$

$\mu = 39.0 \pm 45.6$

$\sigma = 19.5 \pm 178.5$

WF (6 entries)

$\frac{\sigma}{\mu} = 0.45 \pm 15.14 \%$

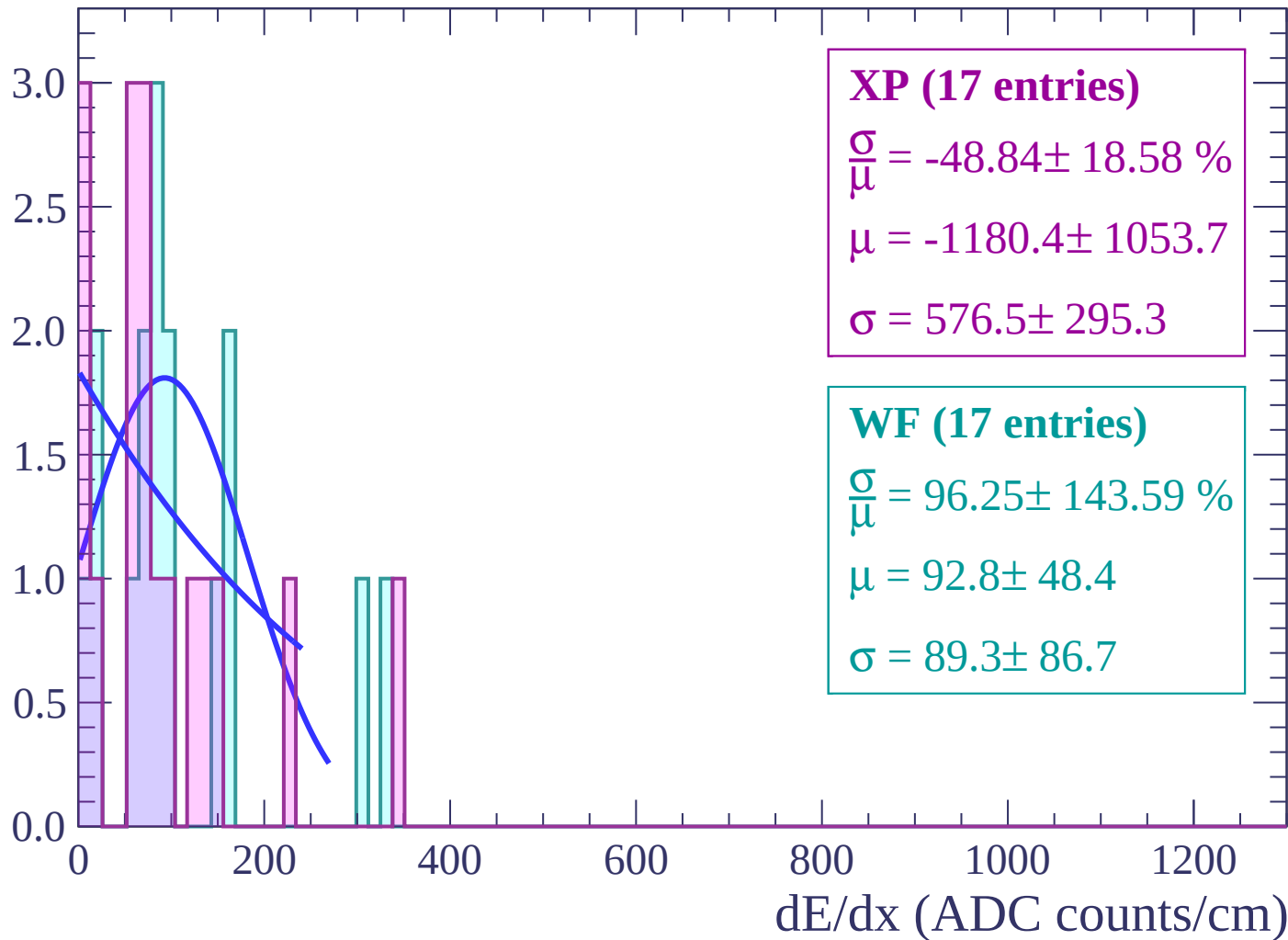
$\mu = 58.4 \pm 4.8$

$\sigma = 0.3 \pm 8.8$

dE/dx (ADC counts/cm)

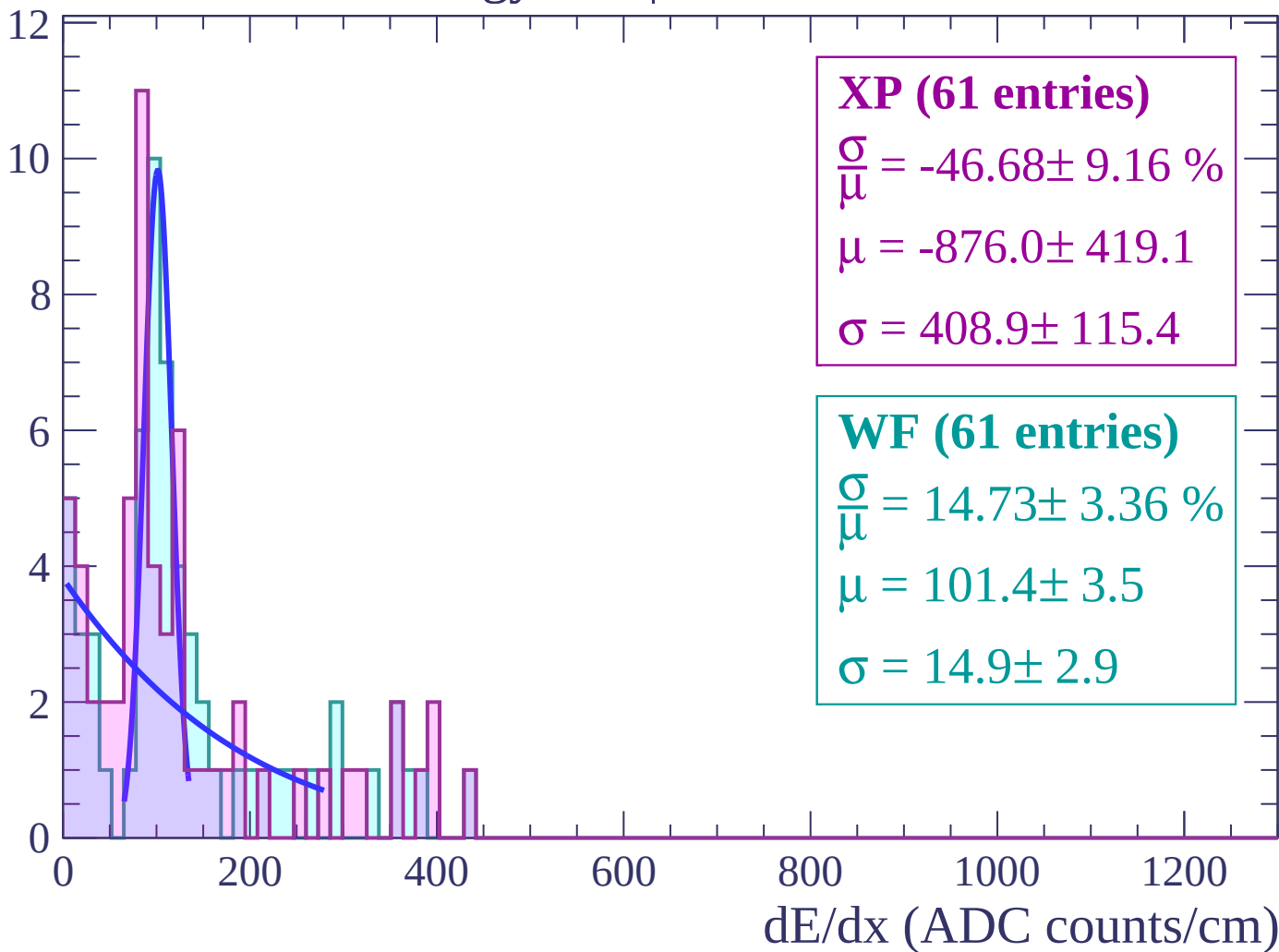
Energy loss | $-80 < \theta < -78$

Count



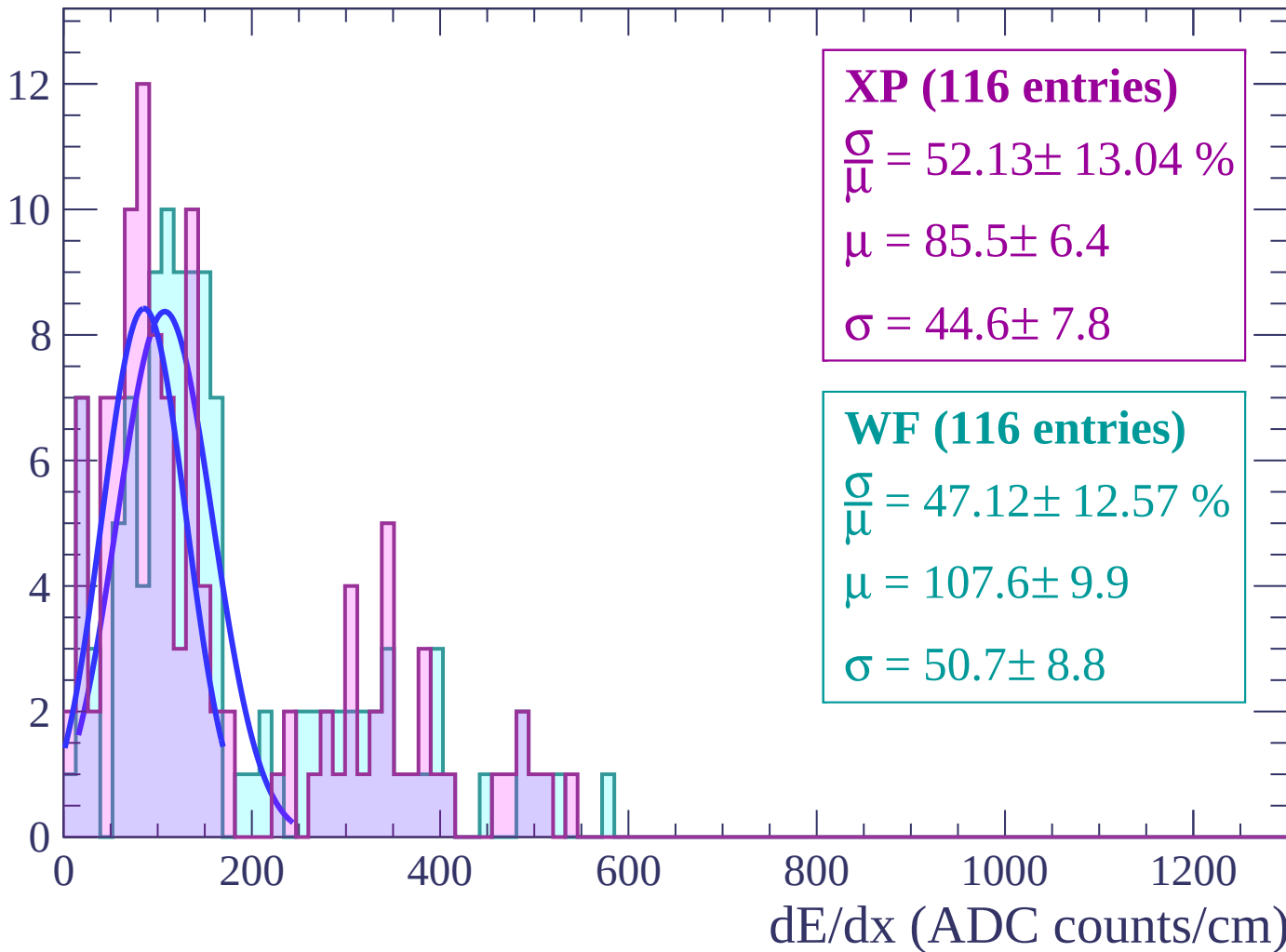
Energy loss | $-78 < \theta < -76$

Count



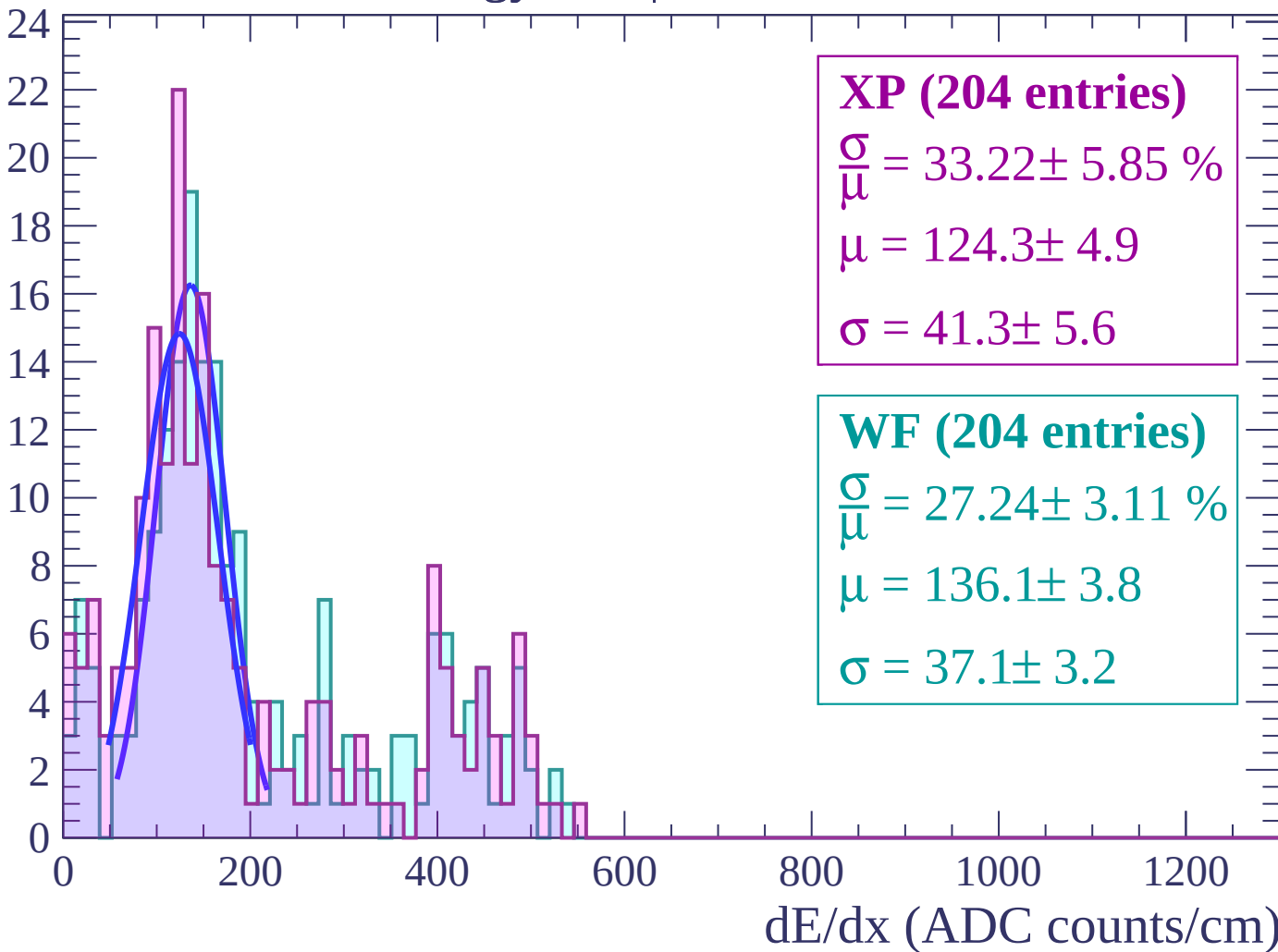
Energy loss | $-76 < \theta < -74$

Count



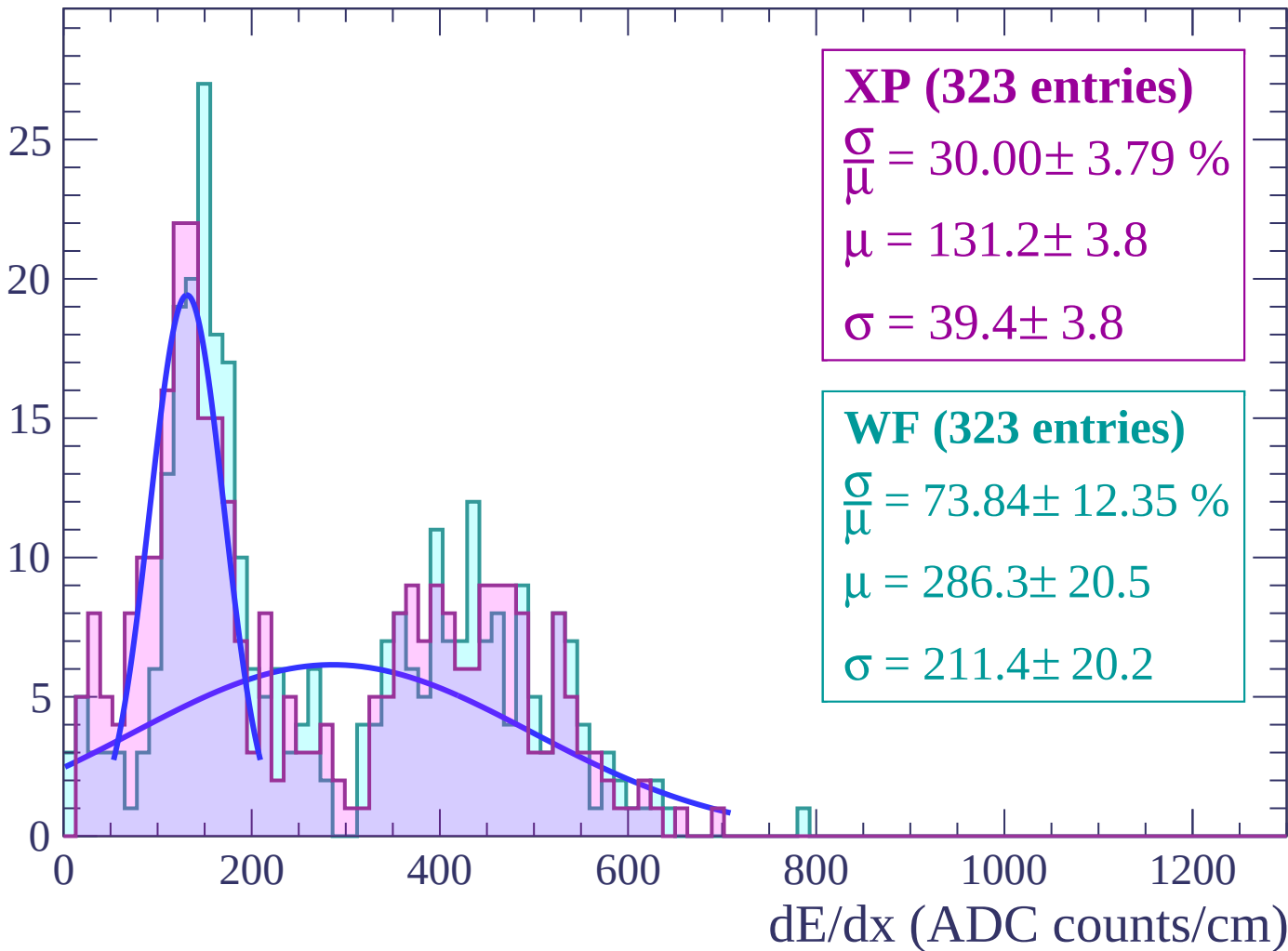
Energy loss | $-74 < \theta < -72$

Count



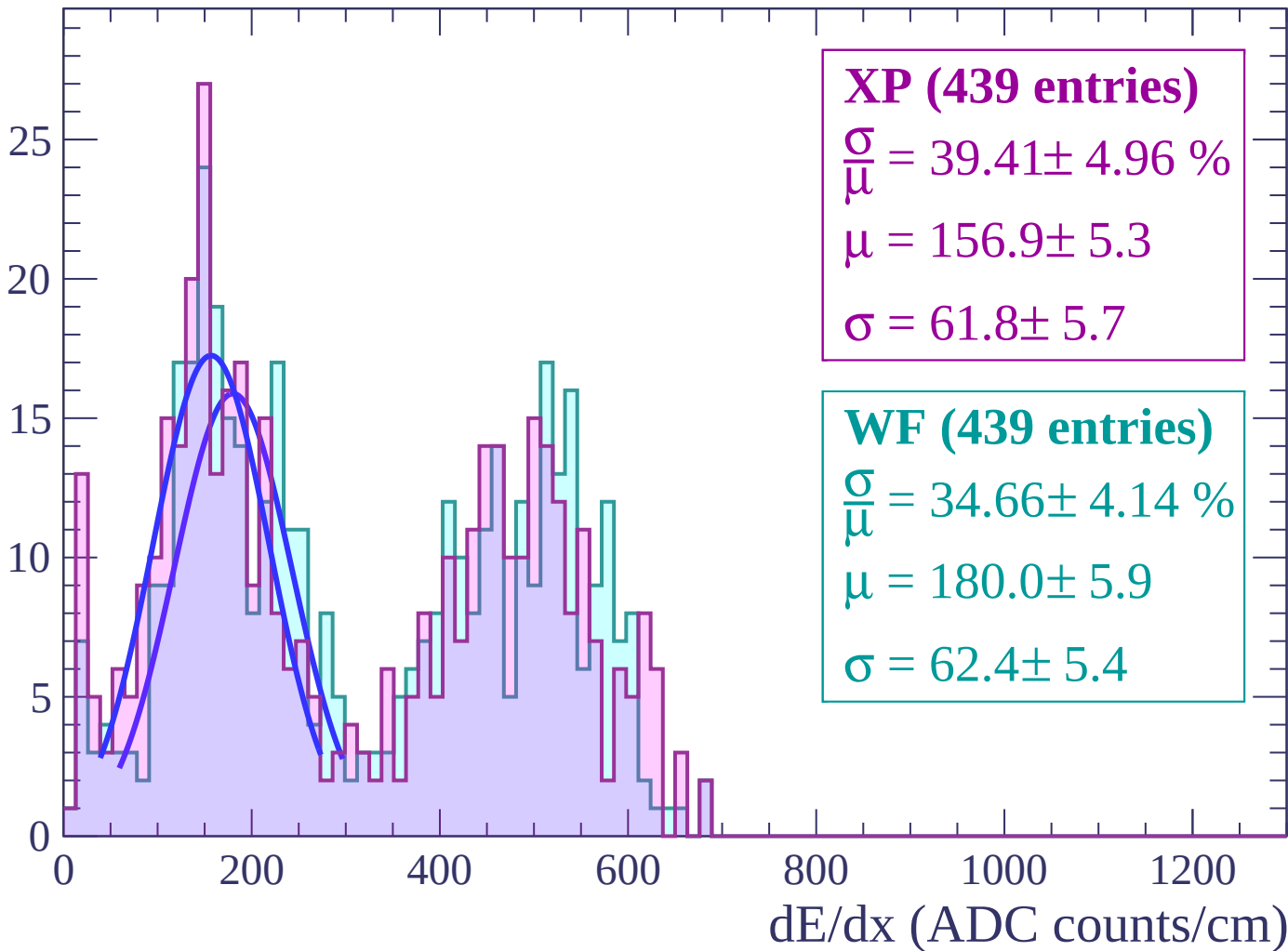
Energy loss | $-72 < \theta < -70$

Count



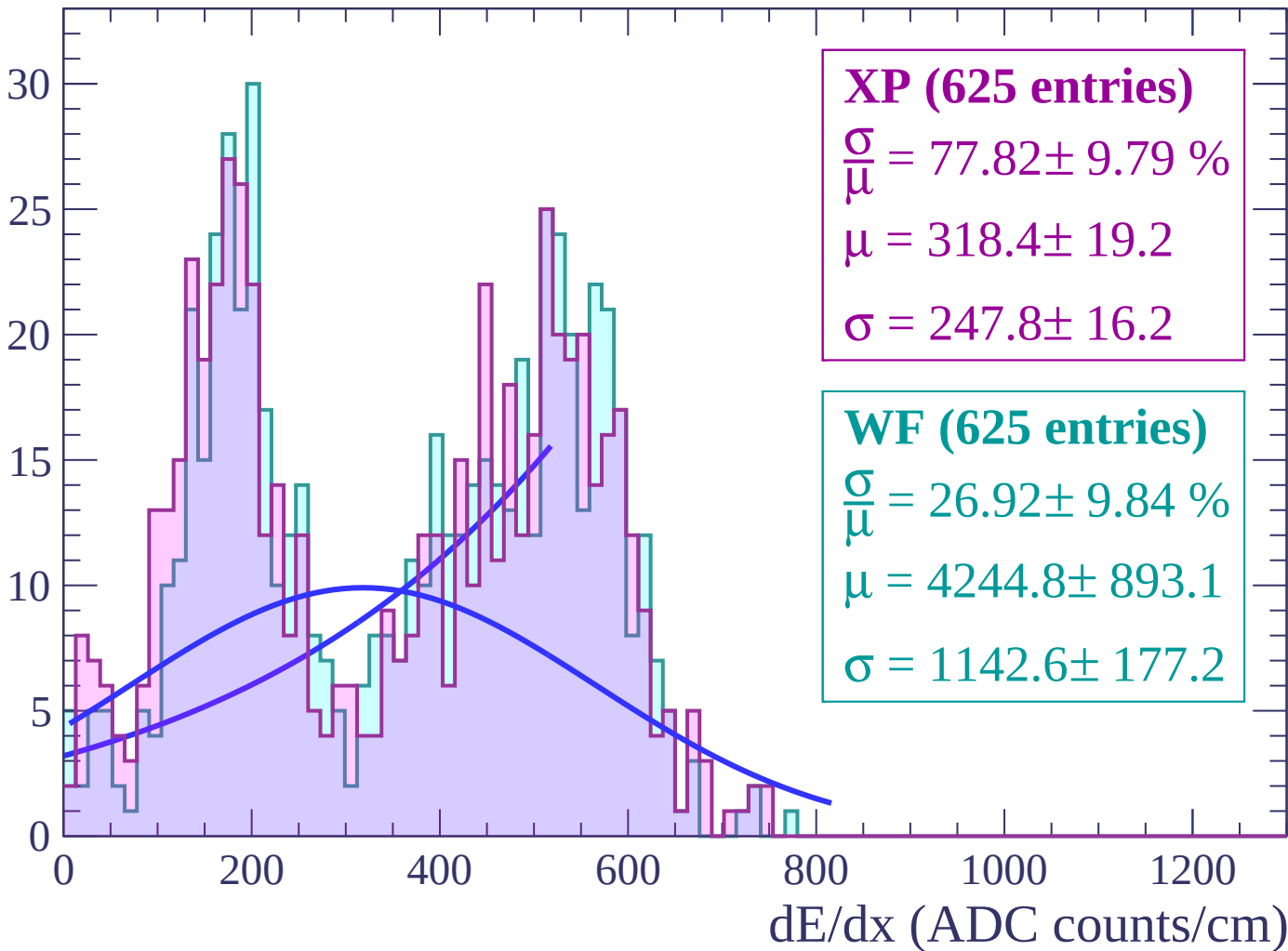
Energy loss | $-70 < \theta < -68$

Count



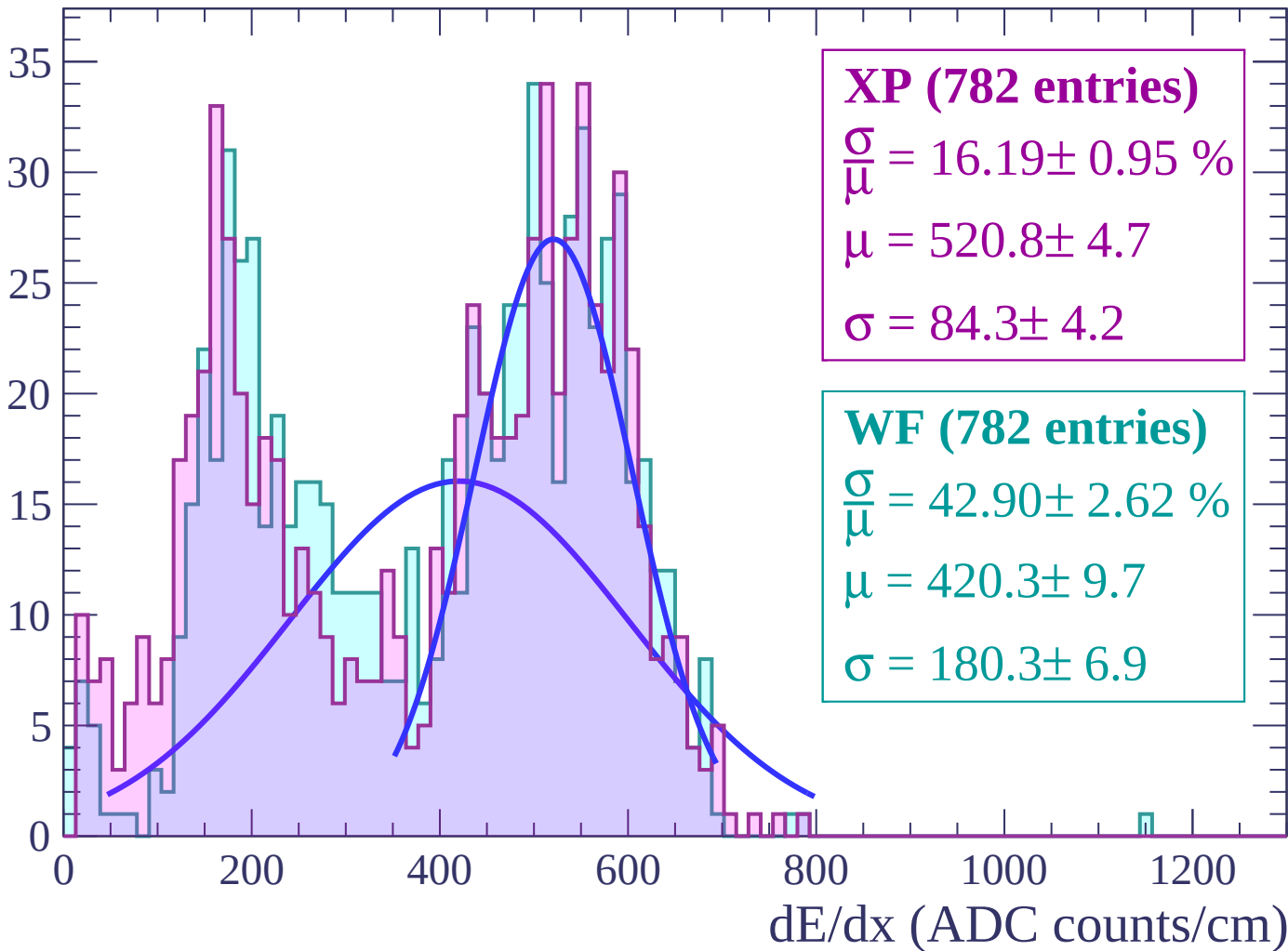
Energy loss | $-68 < \theta < -66$

Count



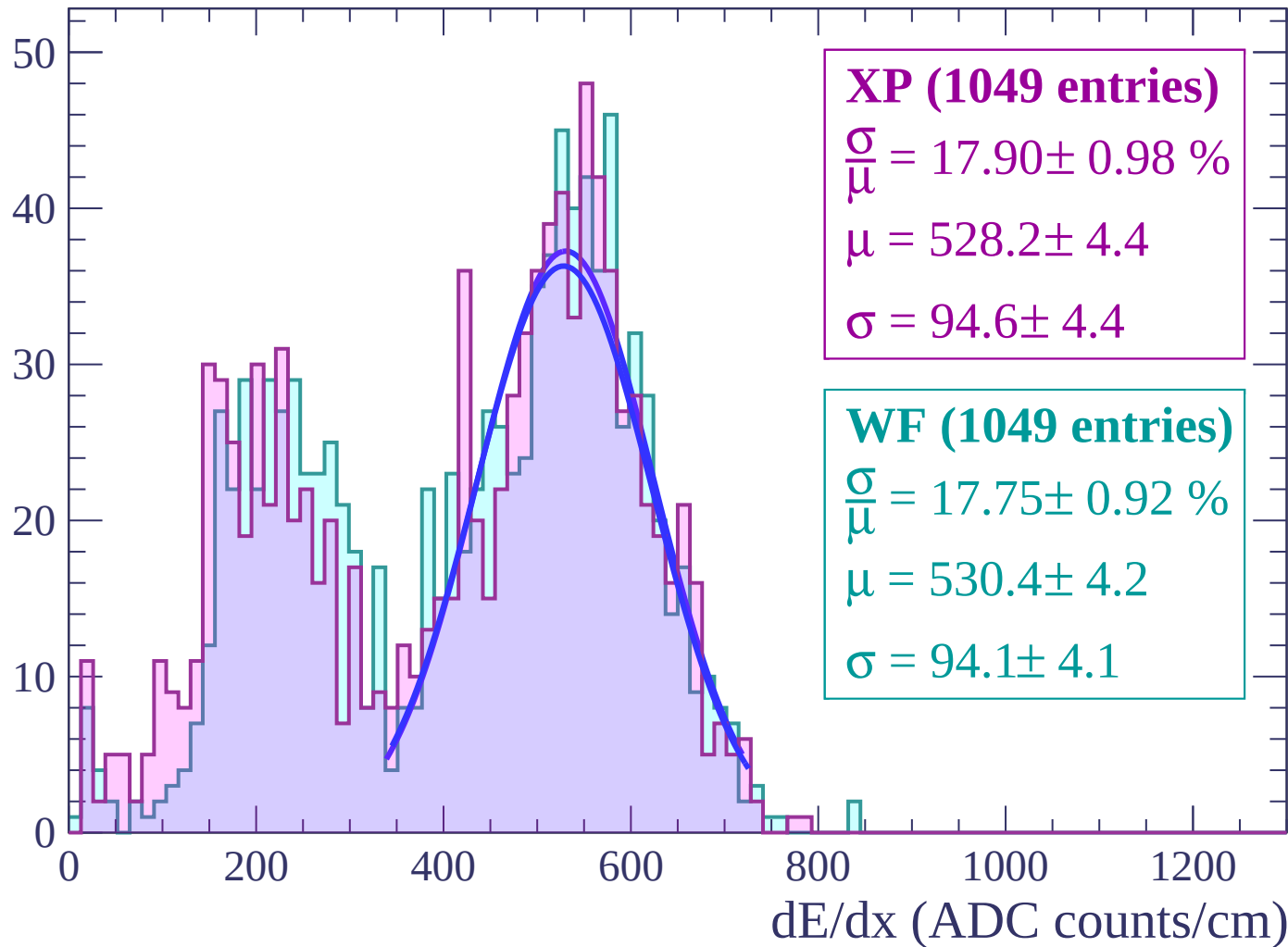
Energy loss | $-66 < \theta < -64$

Count



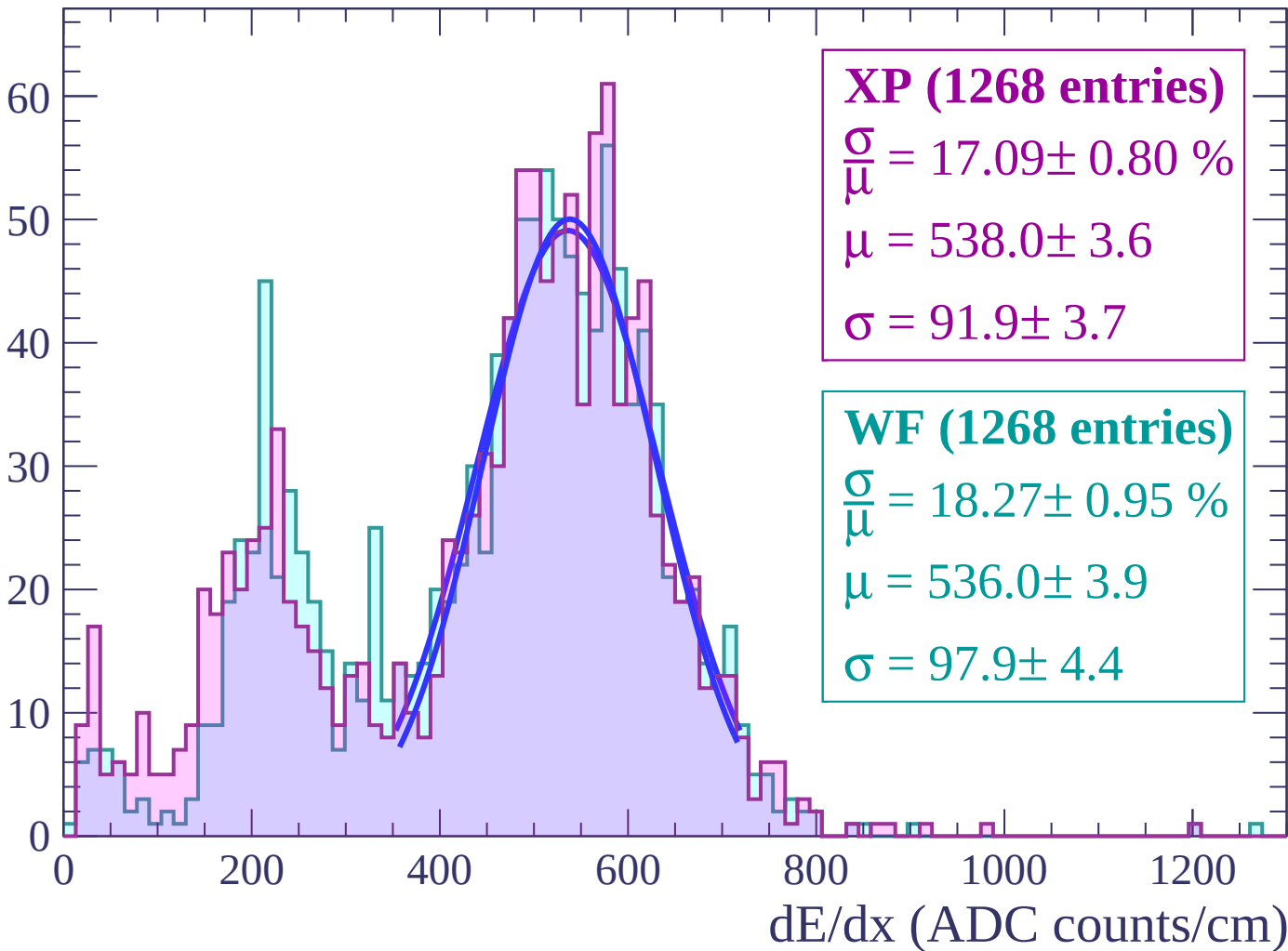
Energy loss | $-64 < \theta < -62$

Count



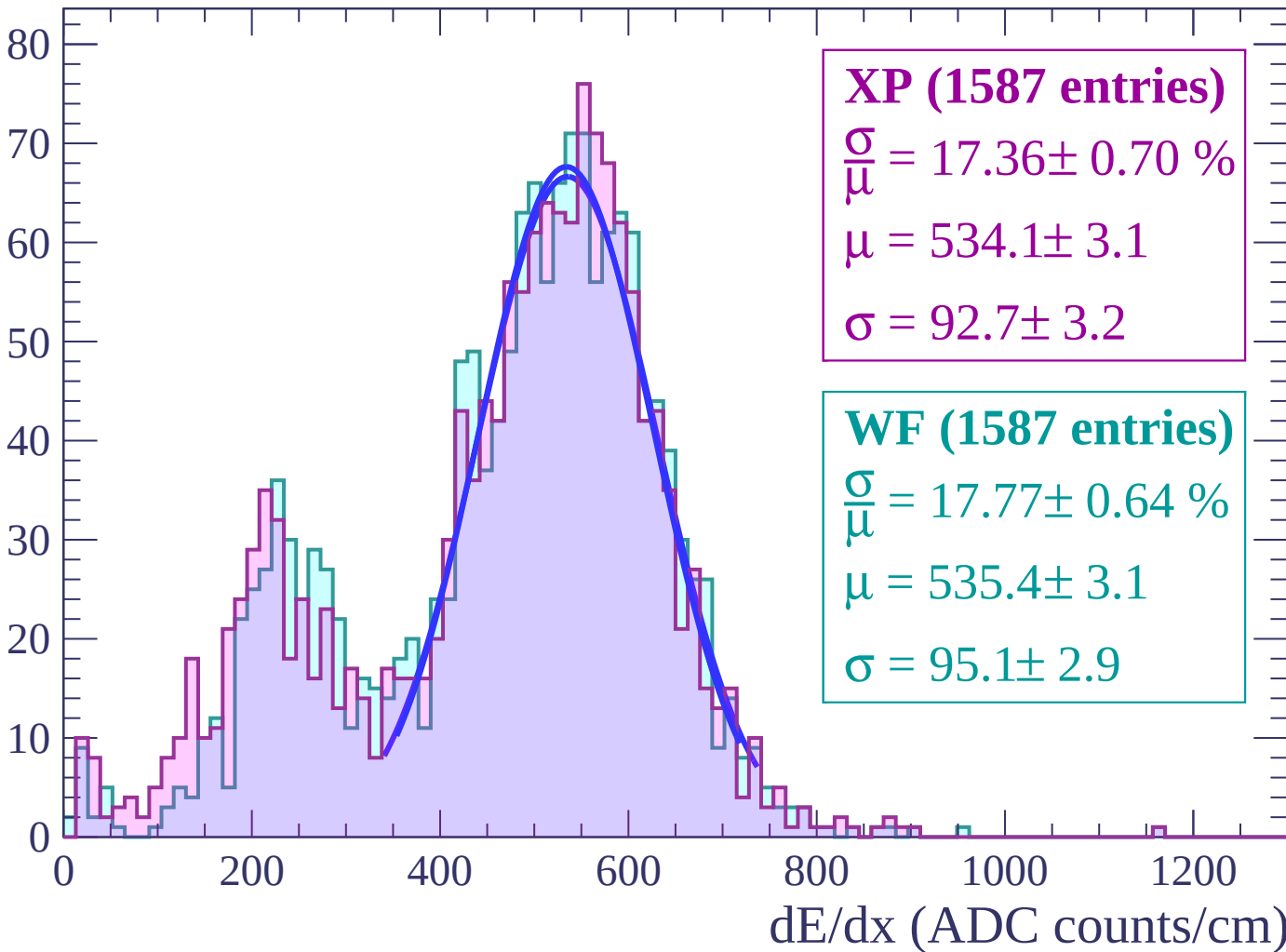
Energy loss | $-62 < \theta < -60$

Count



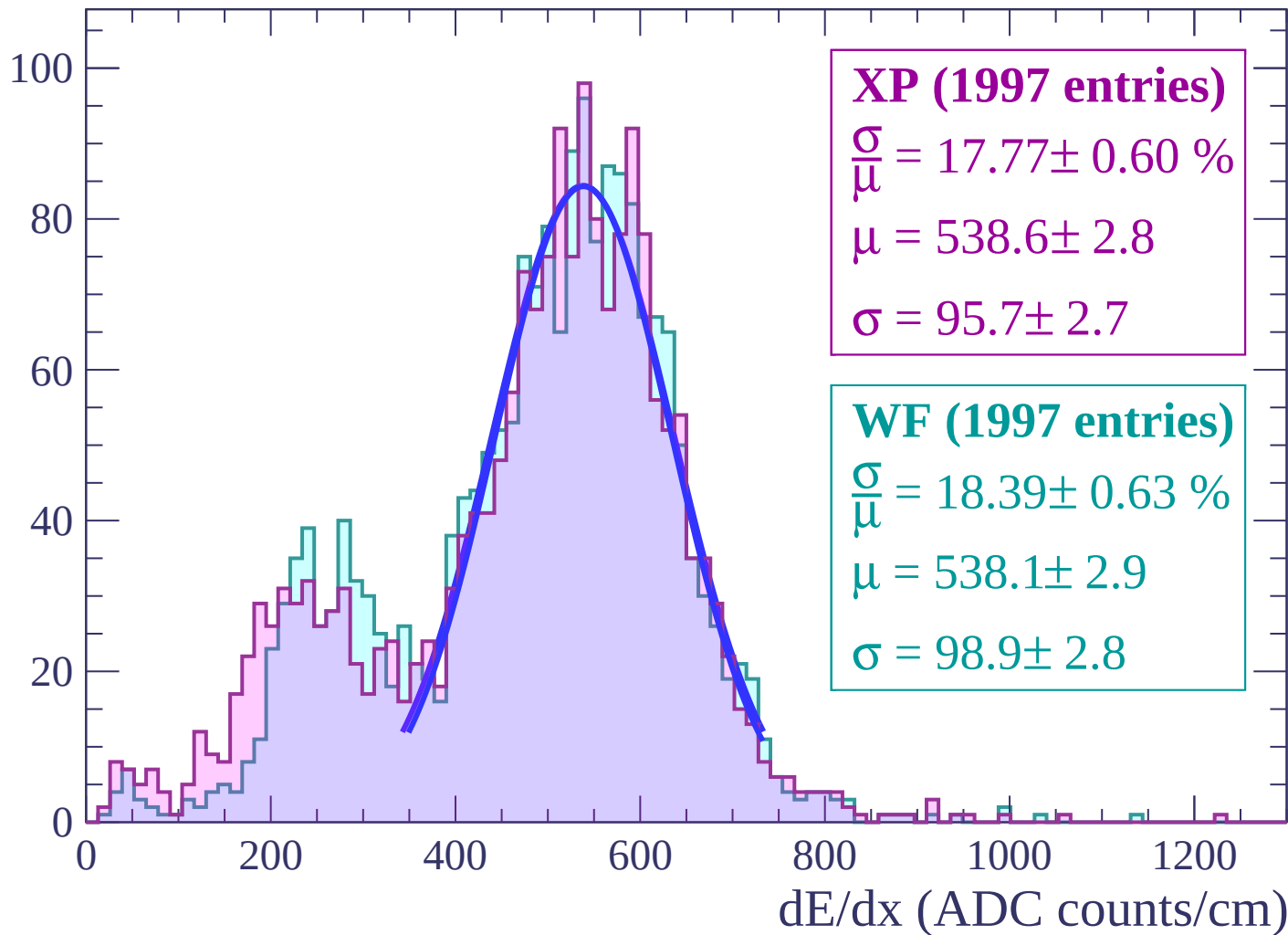
Energy loss | $-60 < \theta < -58$

Count



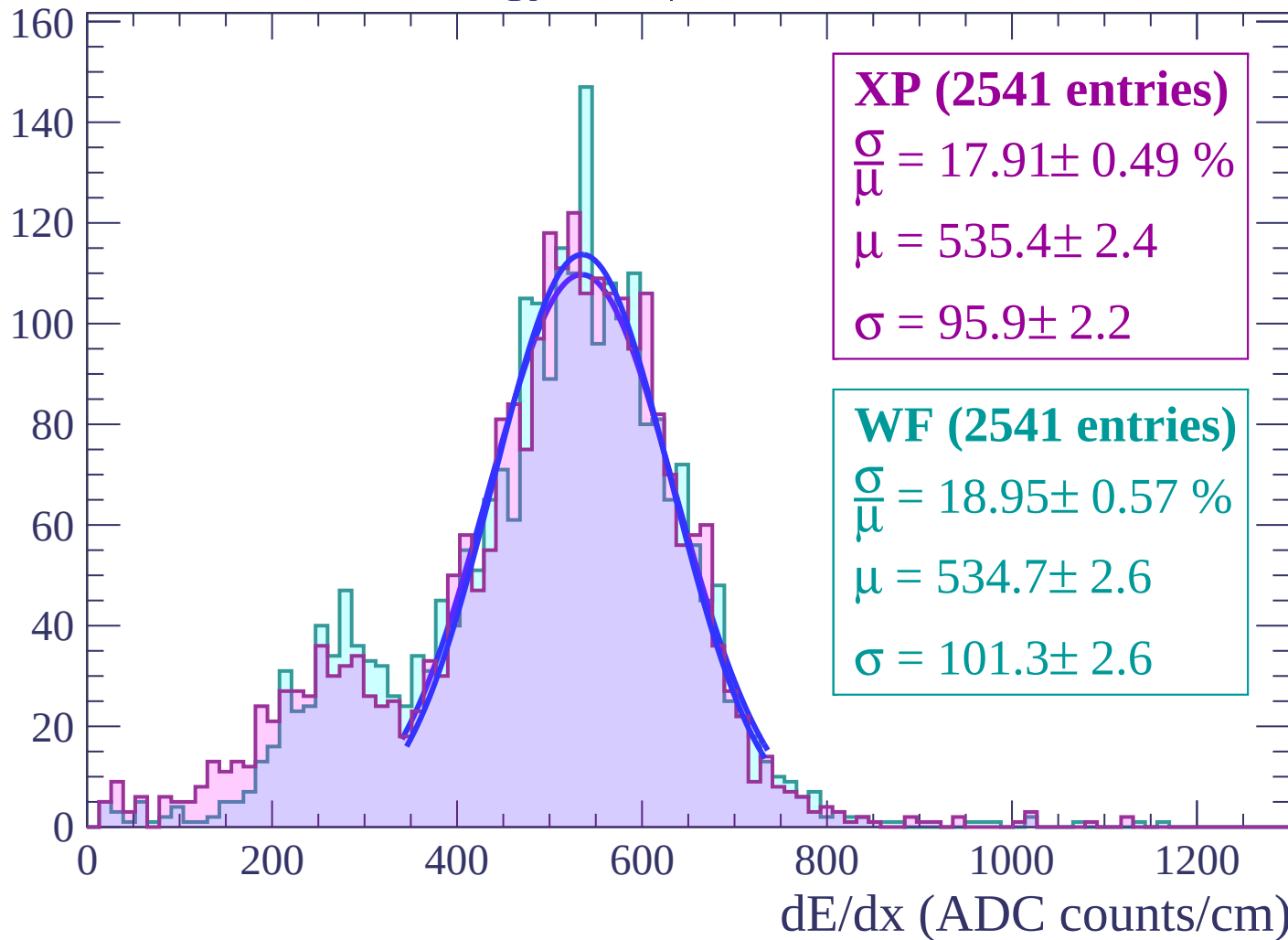
Energy loss | $-58 < \theta < -56$

Count



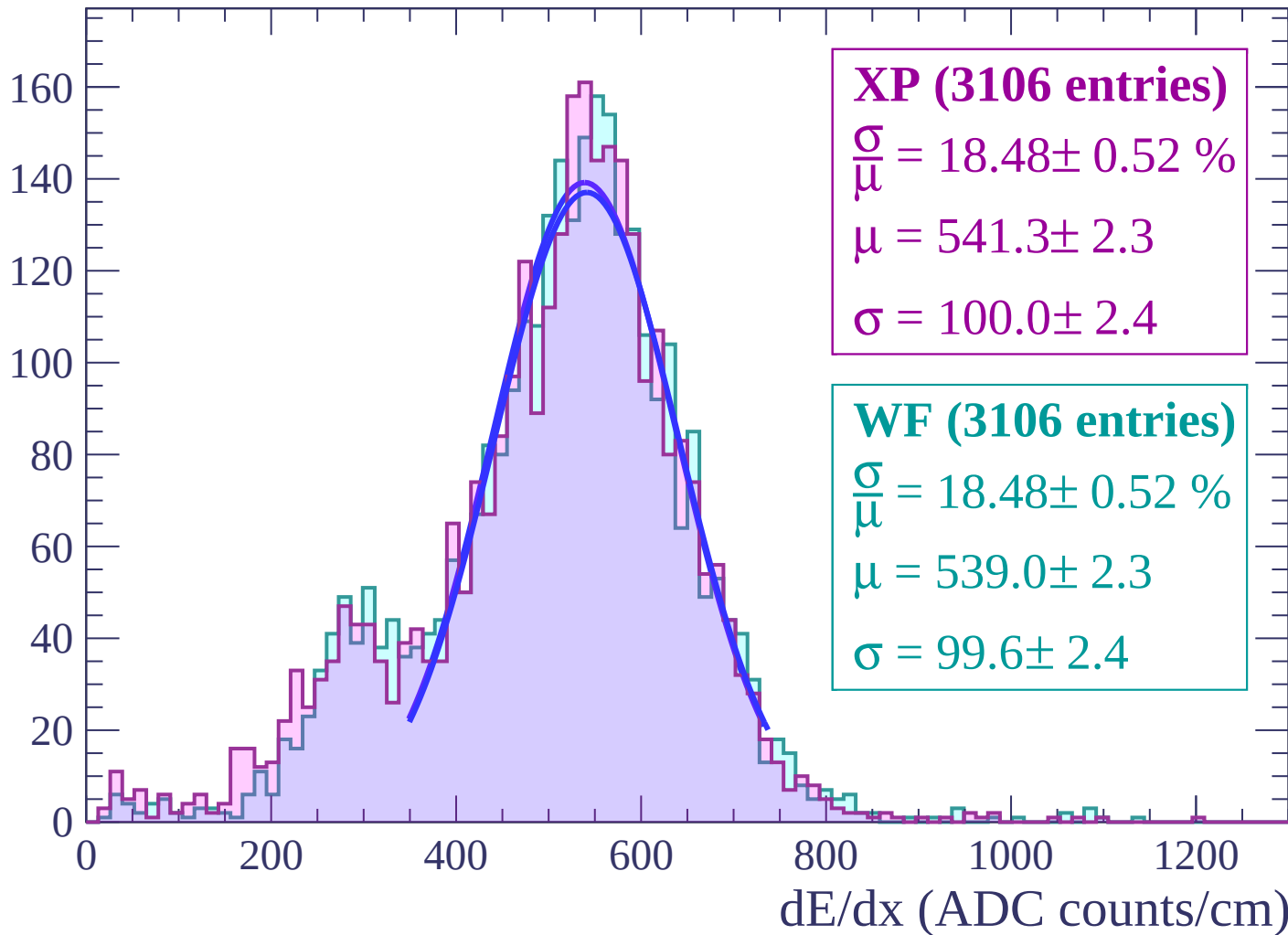
Energy loss | $-56 < \theta < -54$

Count



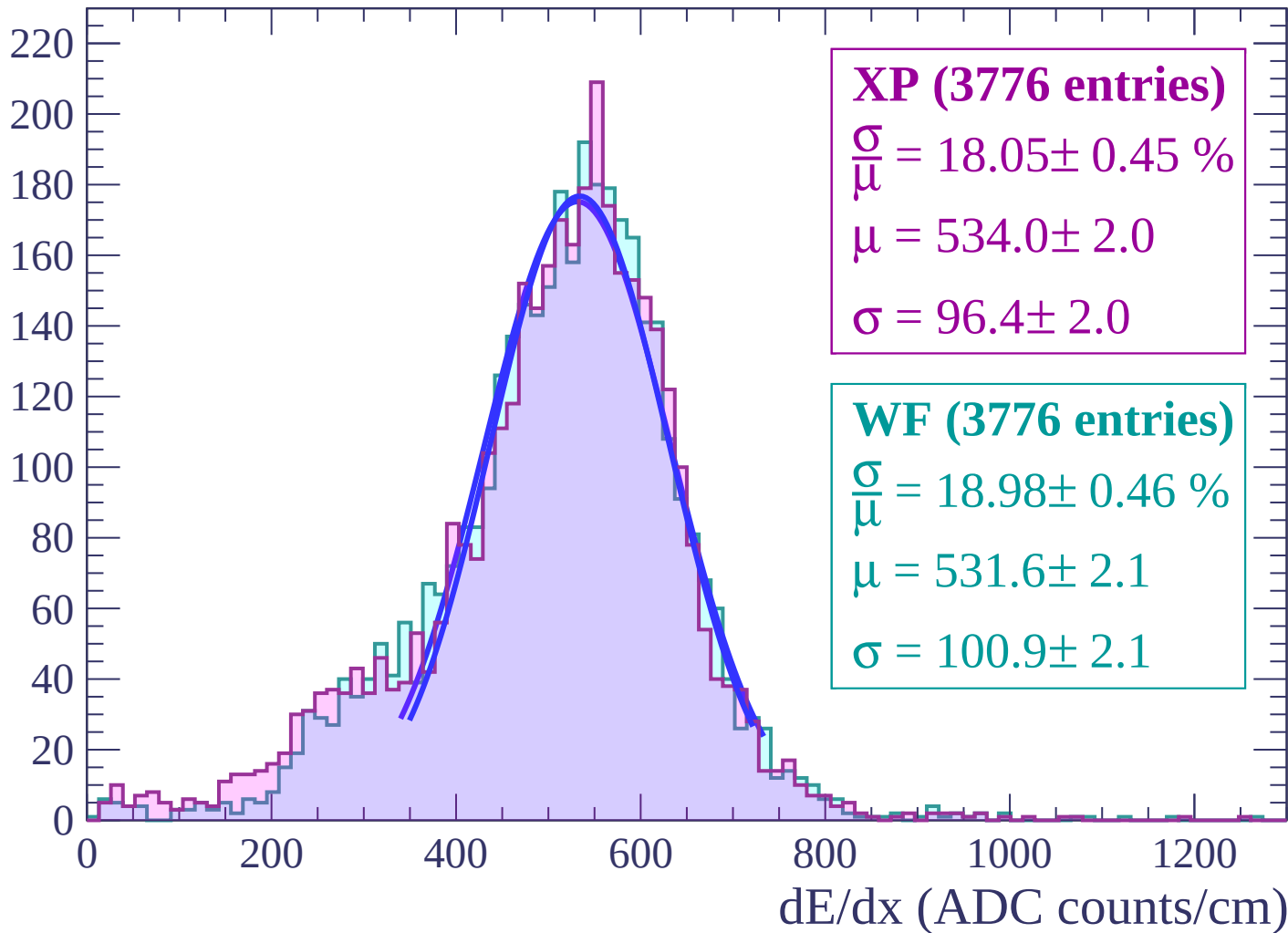
Energy loss | $-54 < \theta < -52$

Count



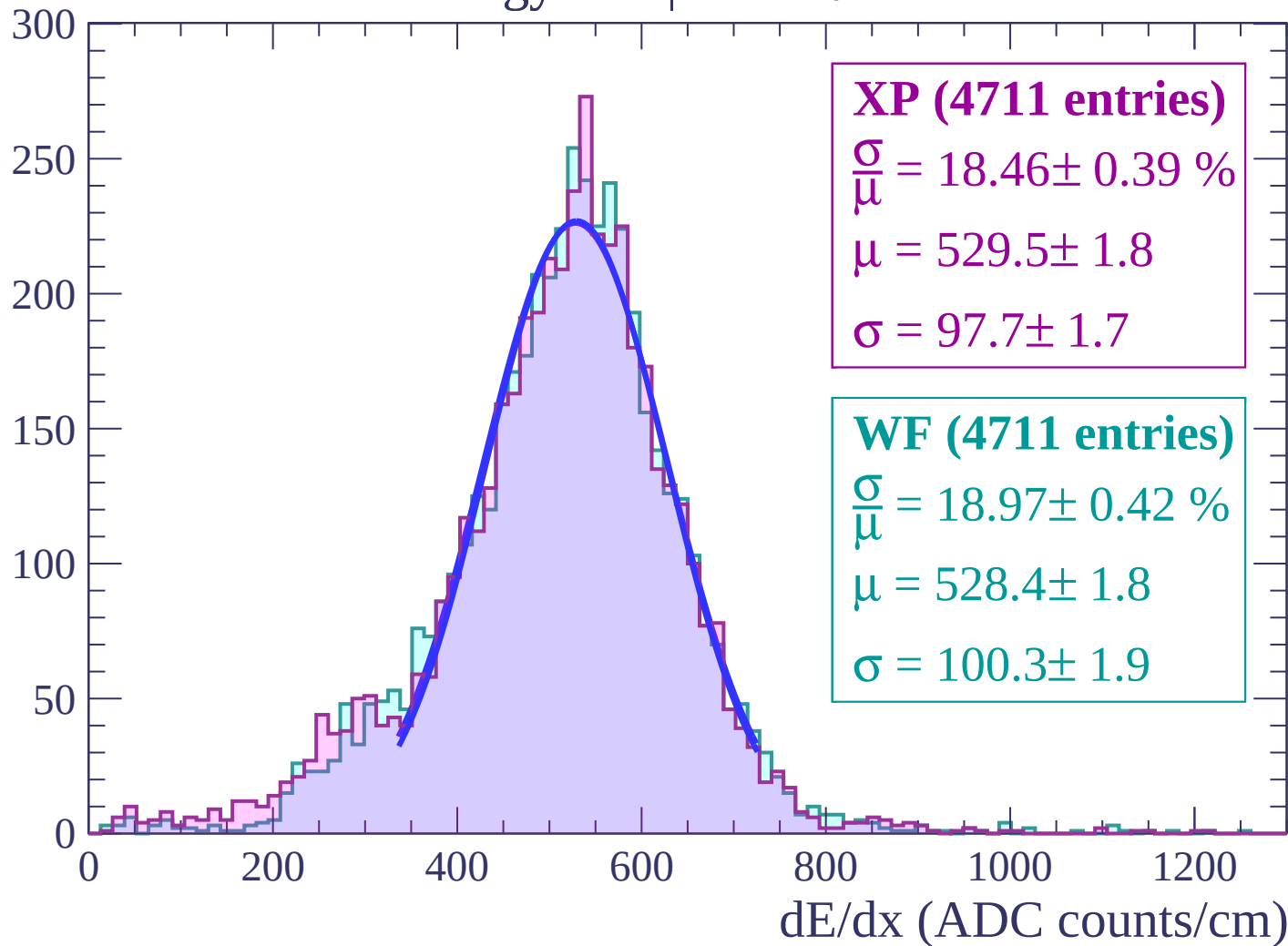
Energy loss | $-52 < \theta < -50$

Count



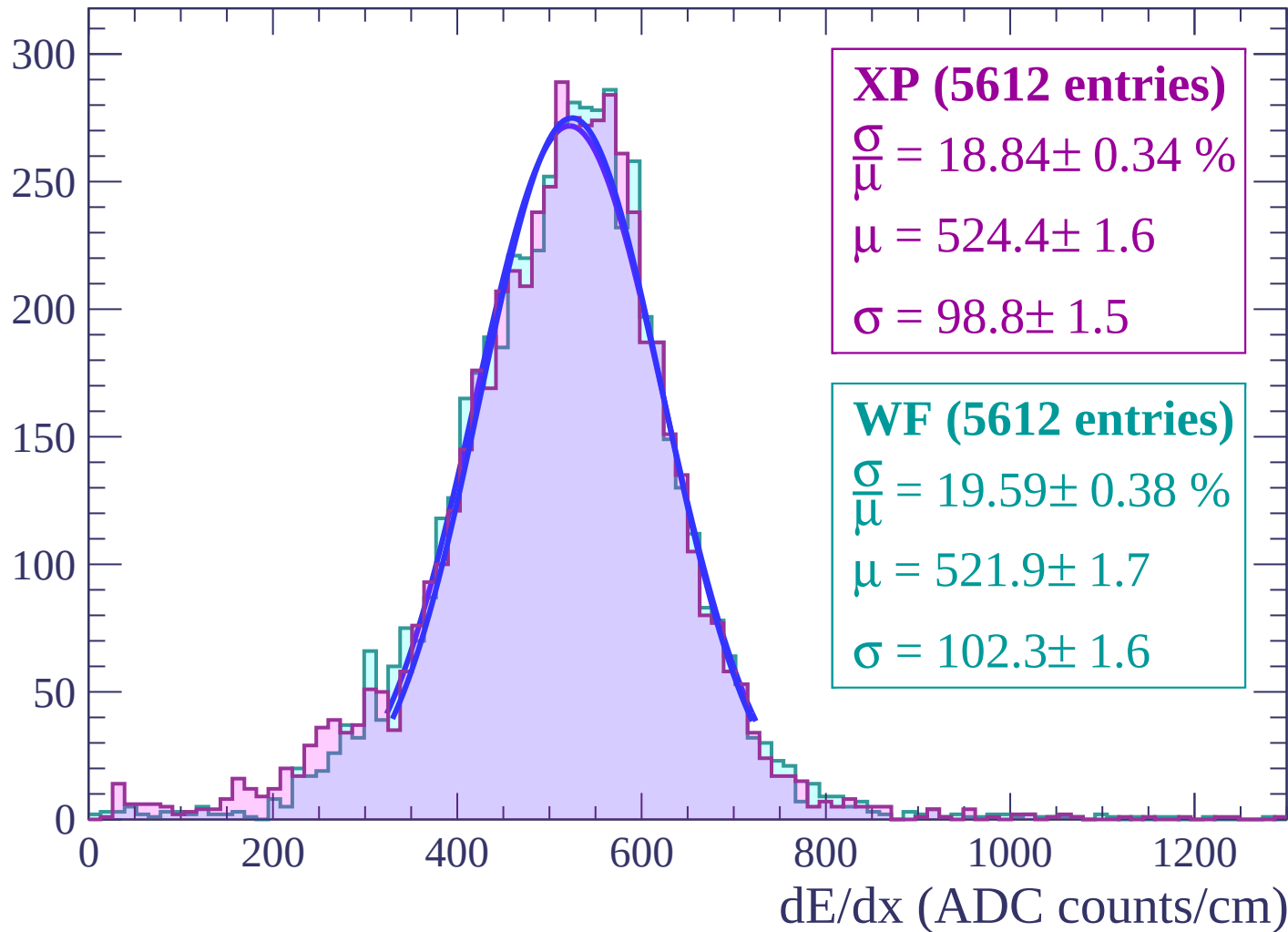
Energy loss | $-50 < \theta < -48$

Count



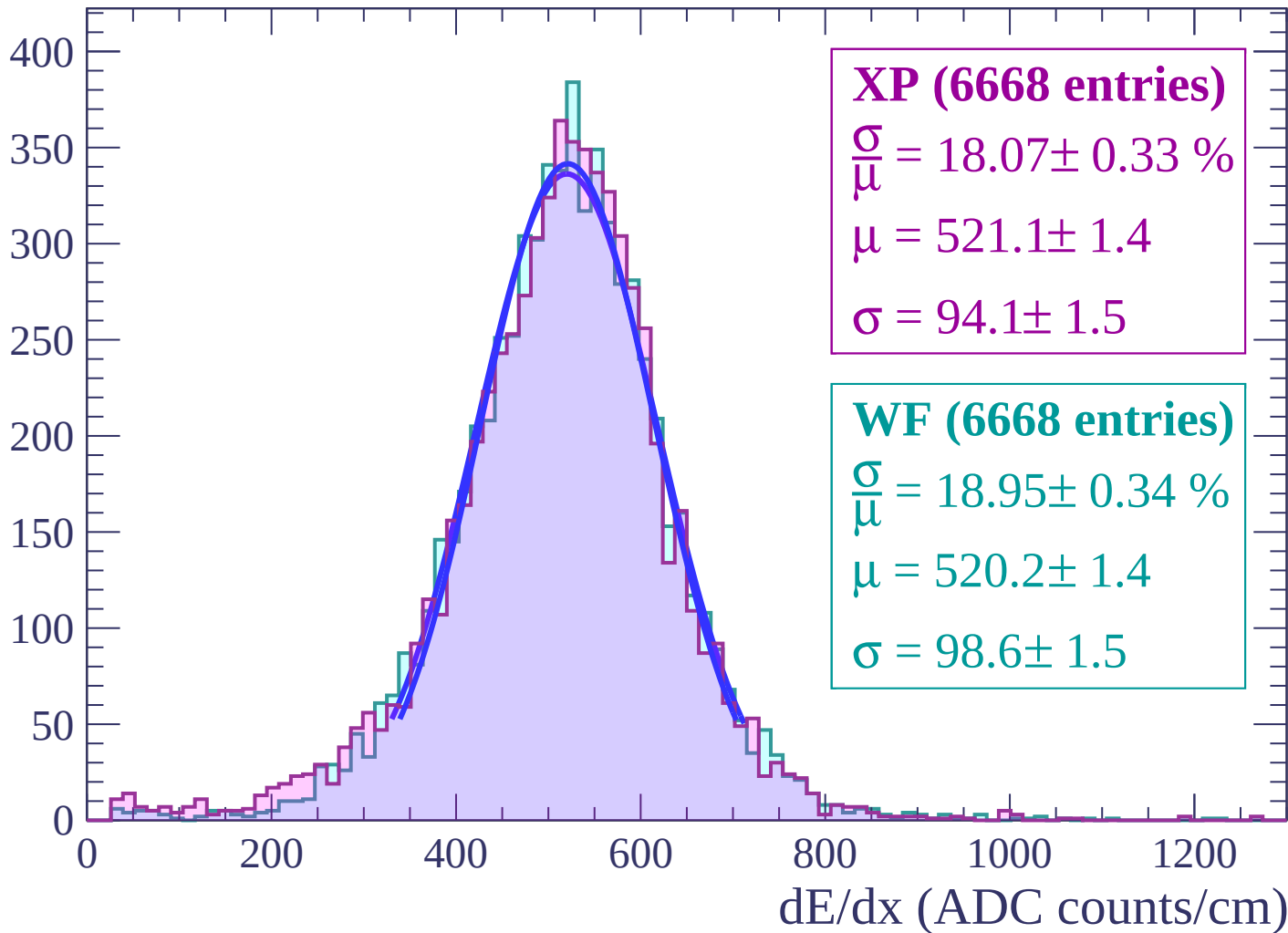
Energy loss | $-48 < \theta < -46$

Count



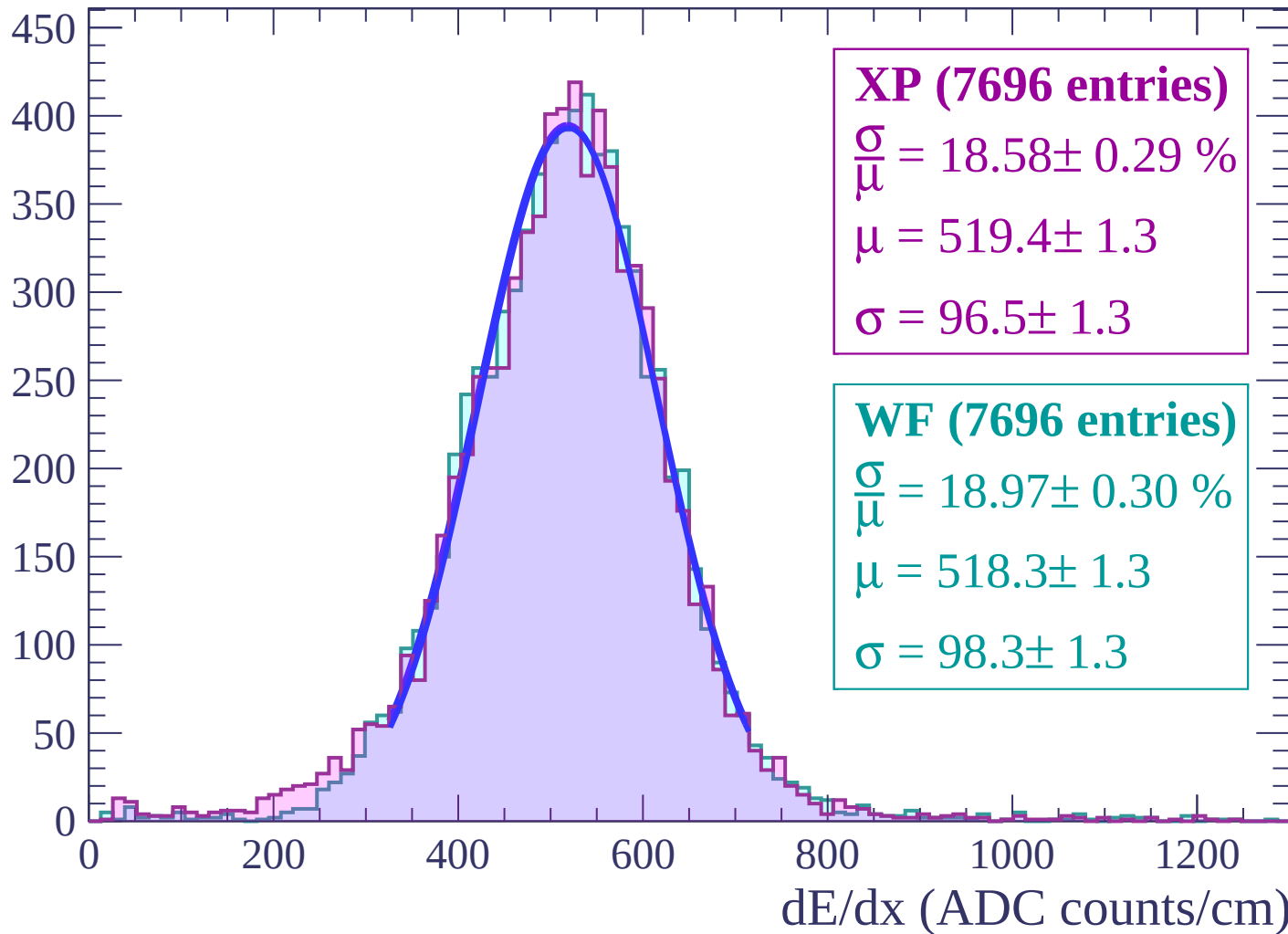
Energy loss | $-46 < \theta < -44$

Count



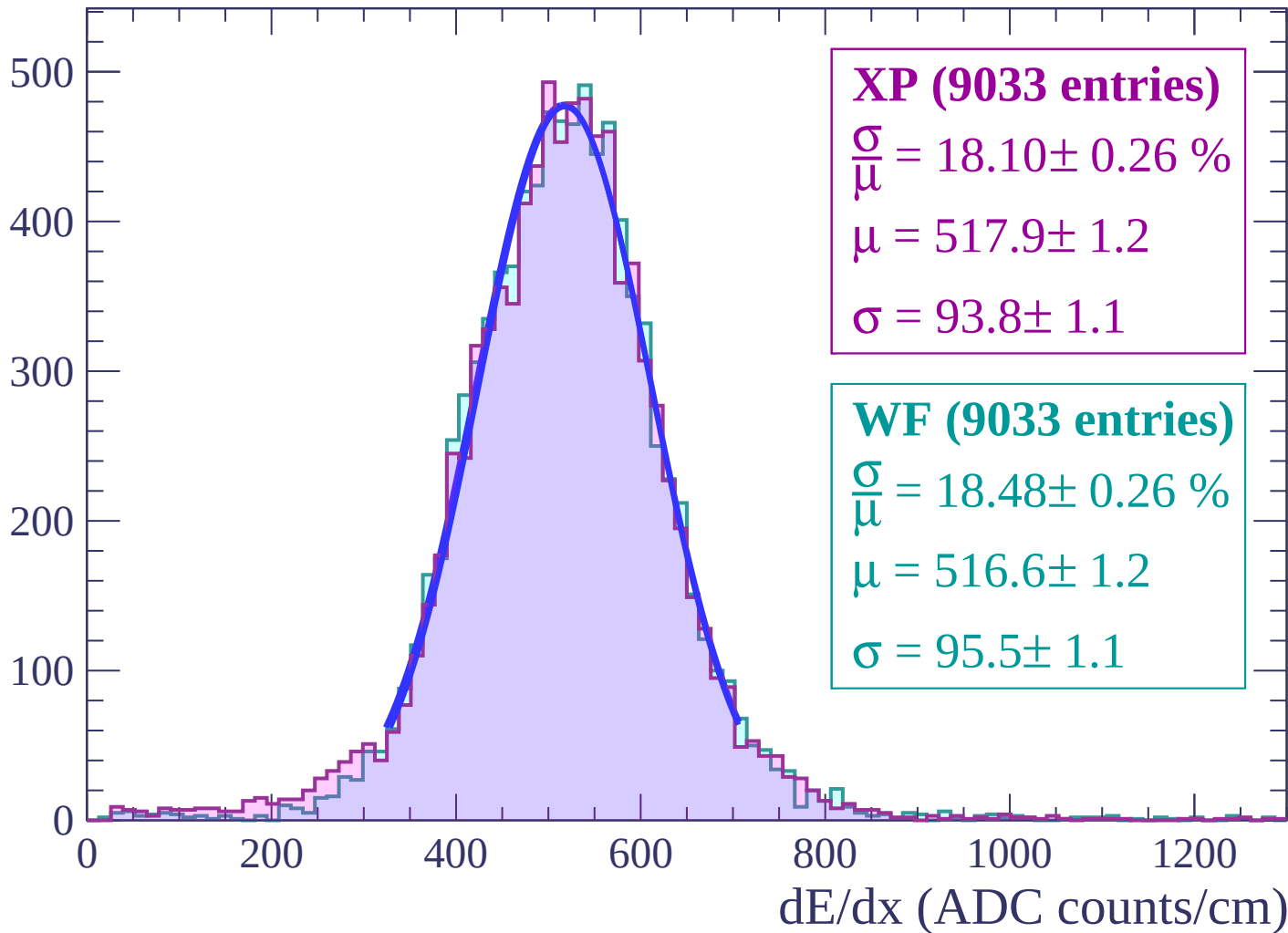
Energy loss | $-44 < \theta < -42$

Count



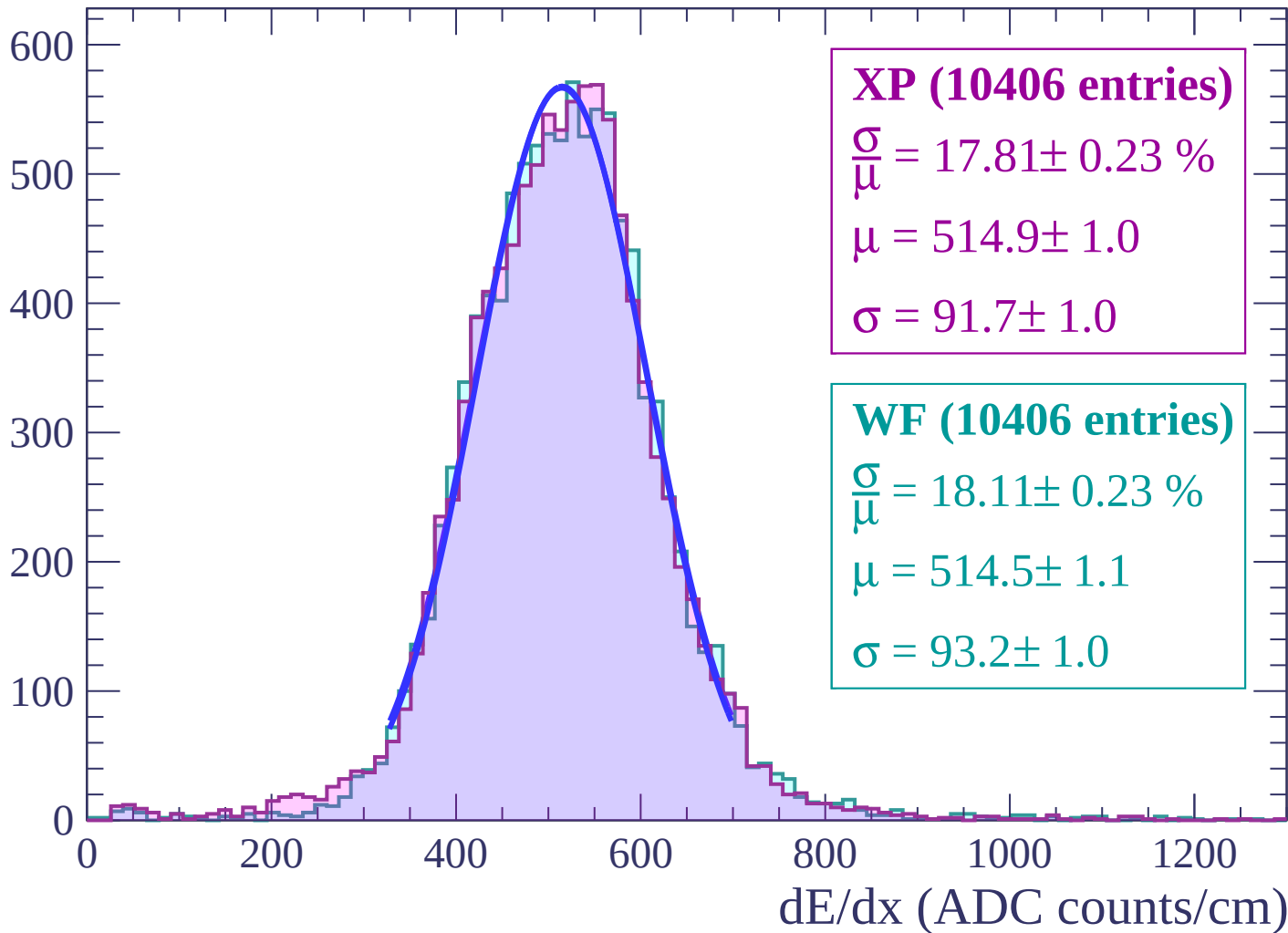
Energy loss | $-42 < \theta < -40$

Count



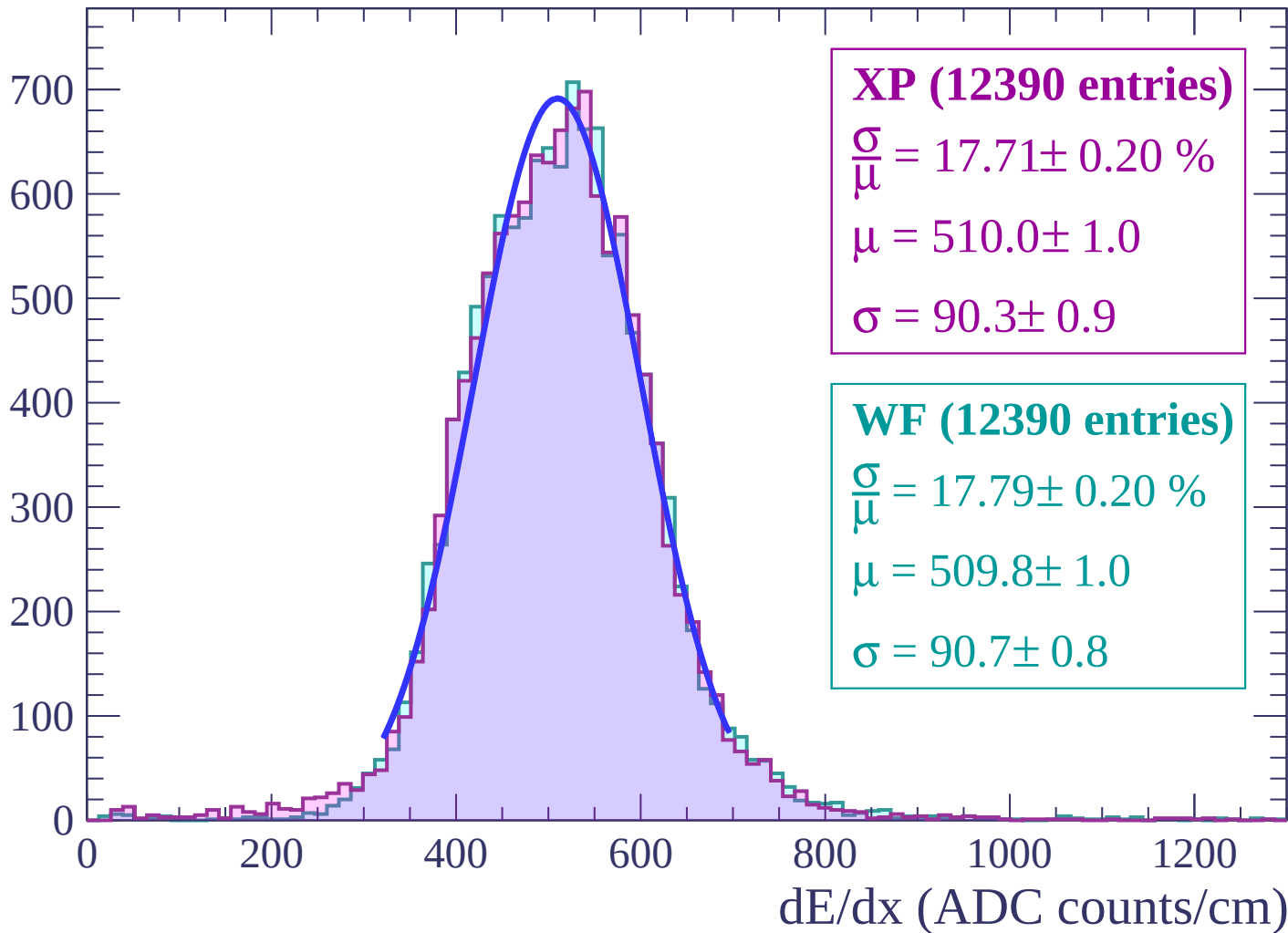
Energy loss | $-40 < \theta < -38$

Count



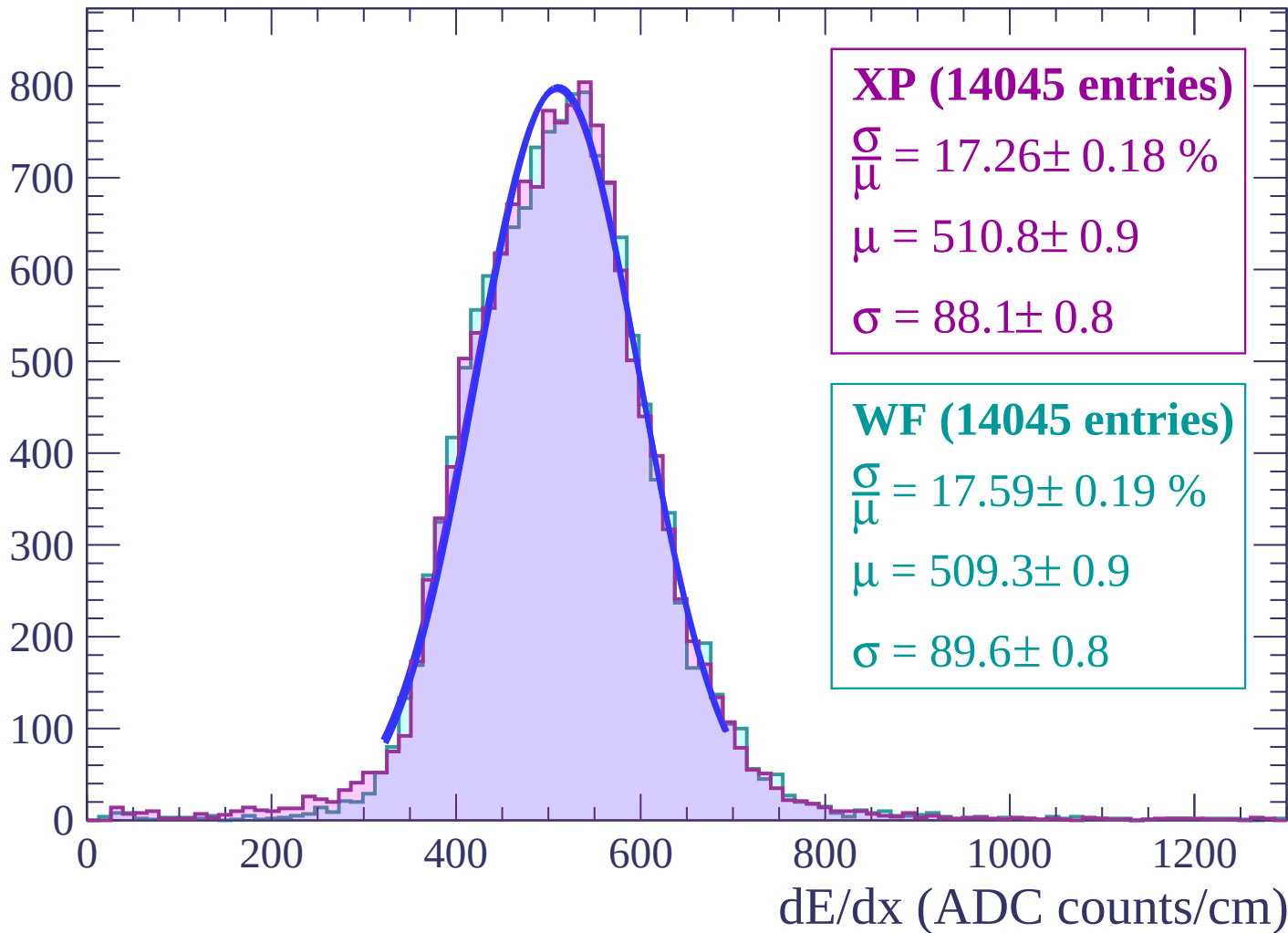
Energy loss | $-38 < \theta < -36$

Count



Energy loss | $-36 < \theta < -34$

Count



Energy loss | $-34 < \theta < -32$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (15986 entries)

$\frac{\sigma}{\mu} = 17.06 \pm 0.17 \%$

$\mu = 507.3 \pm 0.8$

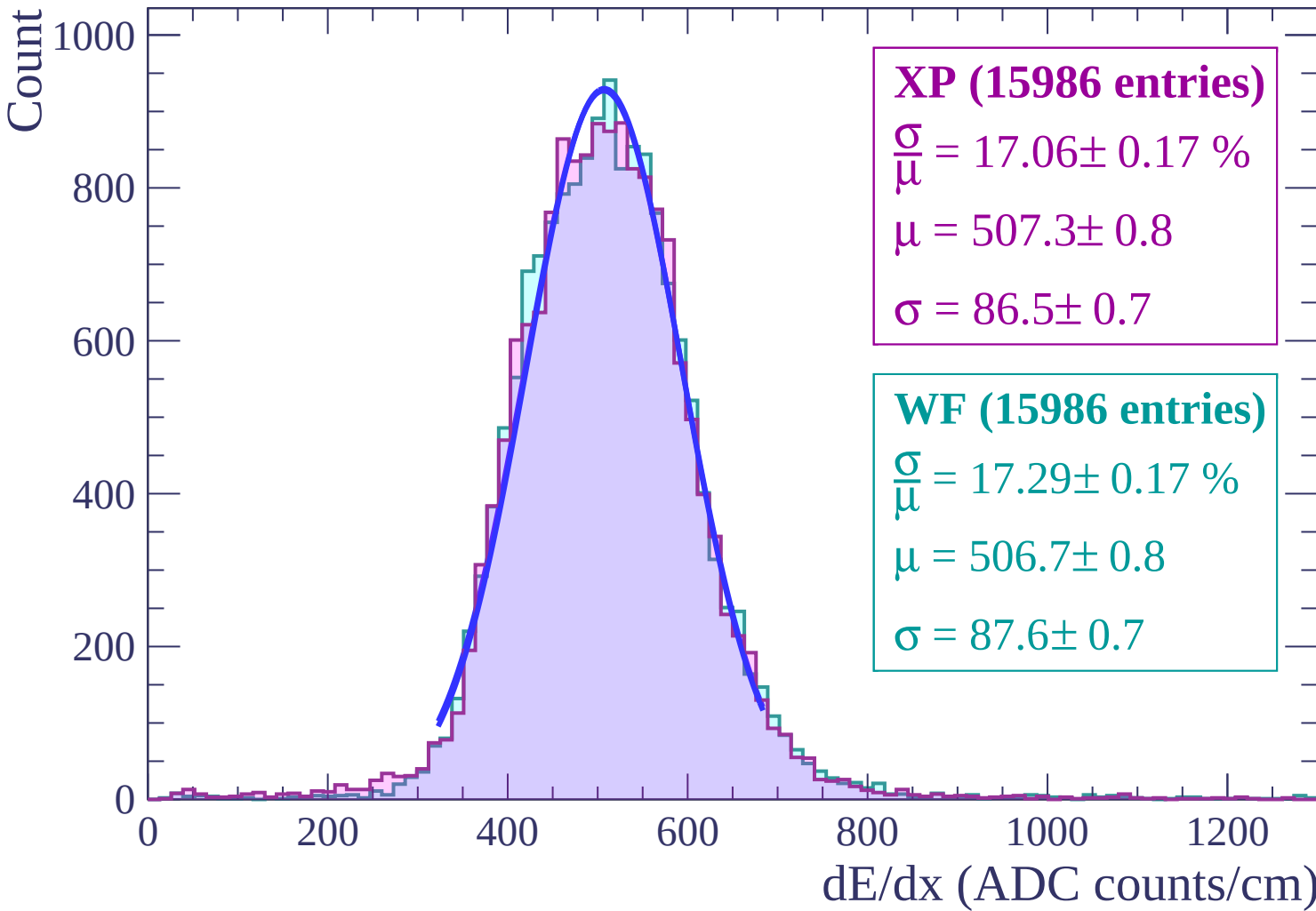
$\sigma = 86.5 \pm 0.7$

WF (15986 entries)

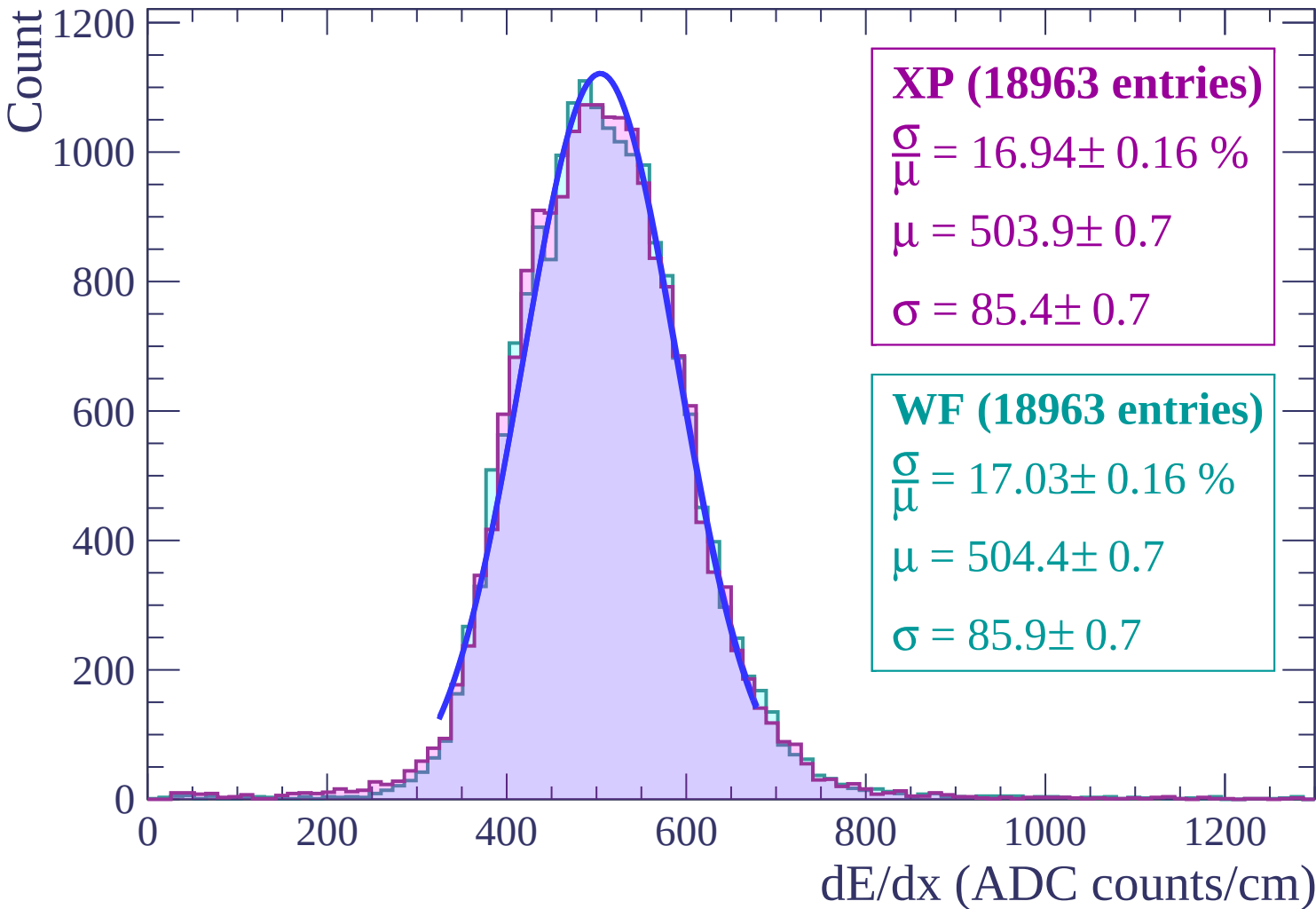
$\frac{\sigma}{\mu} = 17.29 \pm 0.17 \%$

$\mu = 506.7 \pm 0.8$

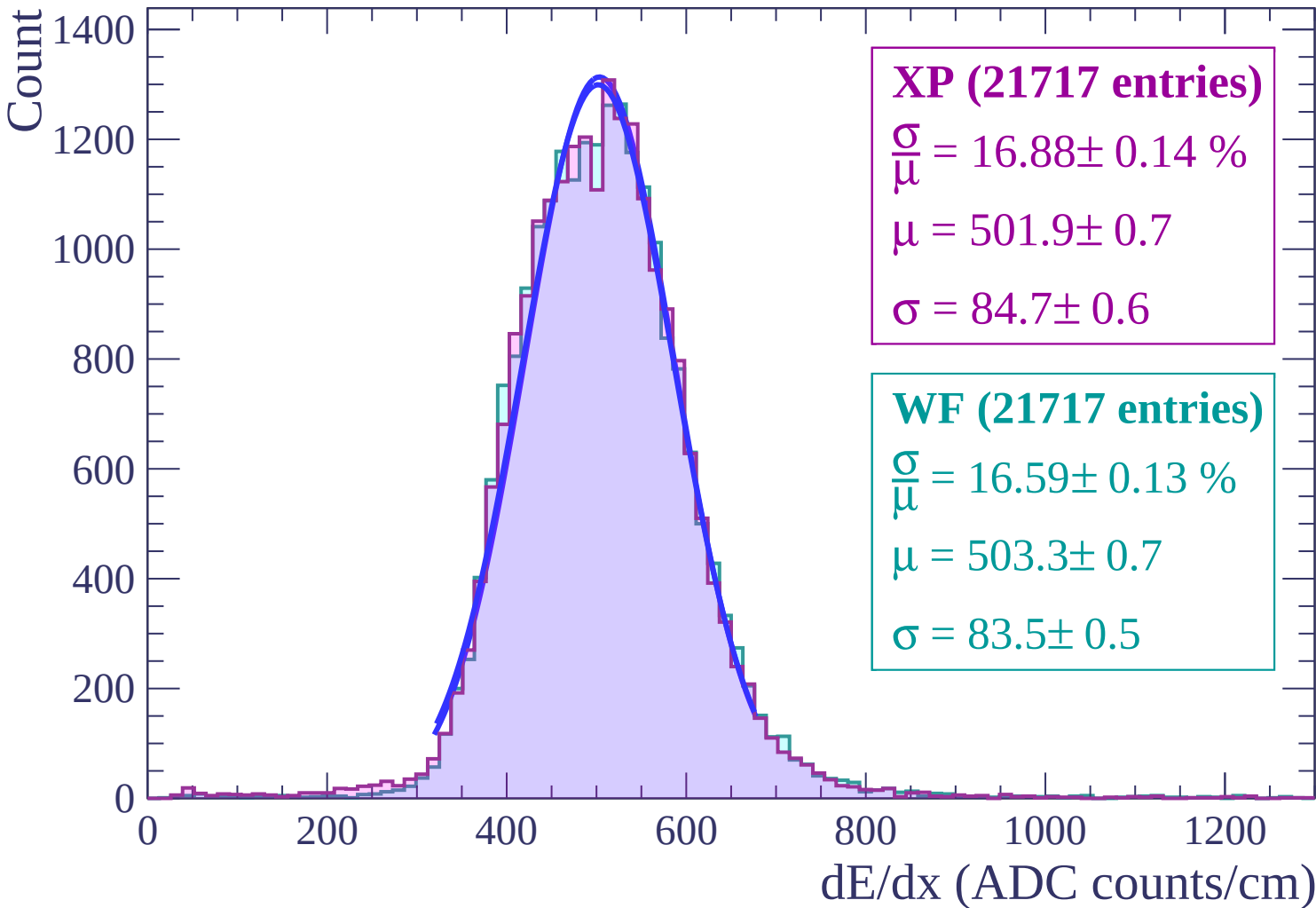
$\sigma = 87.6 \pm 0.7$



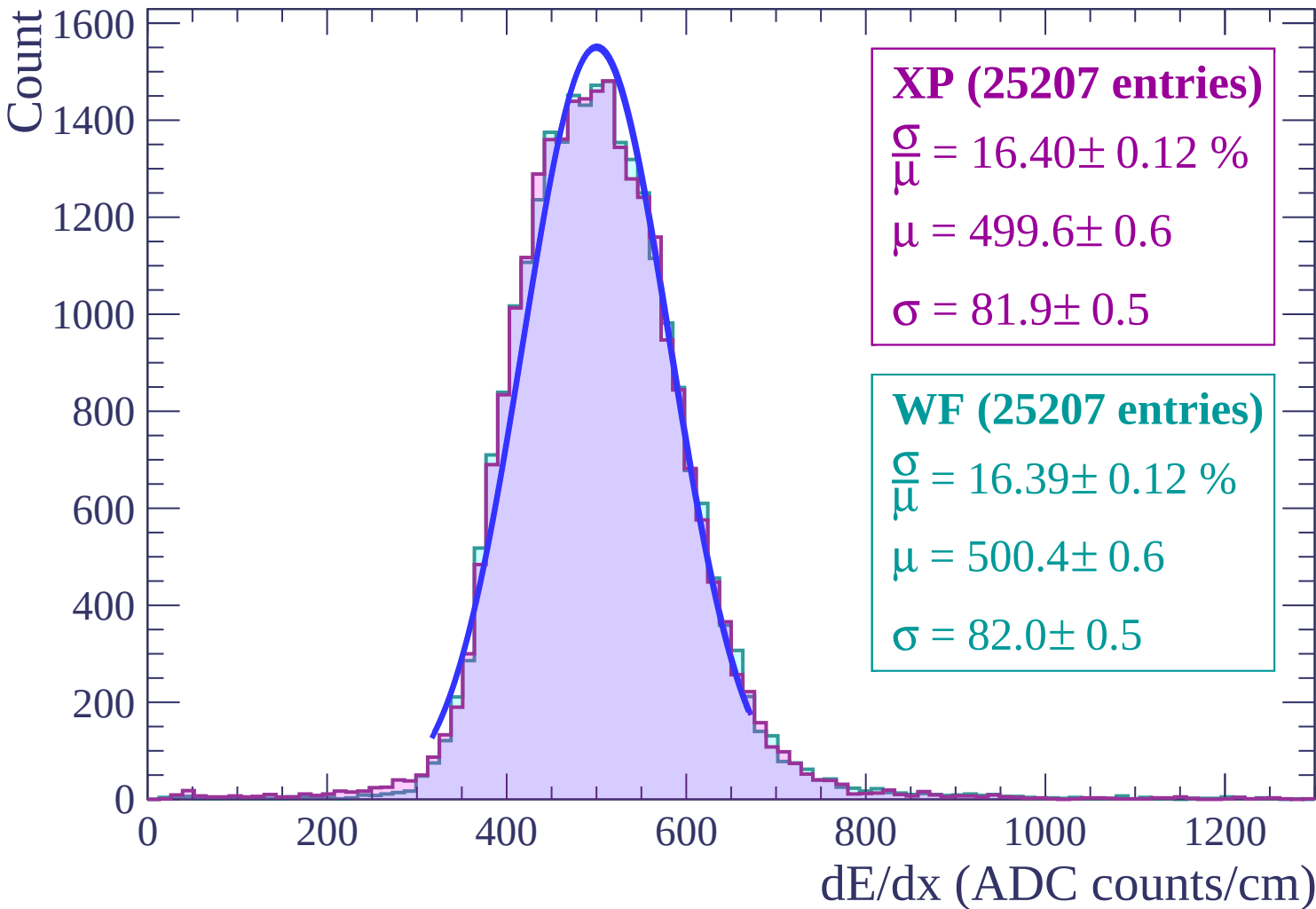
Energy loss | $-32 < \theta < -30$



Energy loss | $-30 < \theta < -28$



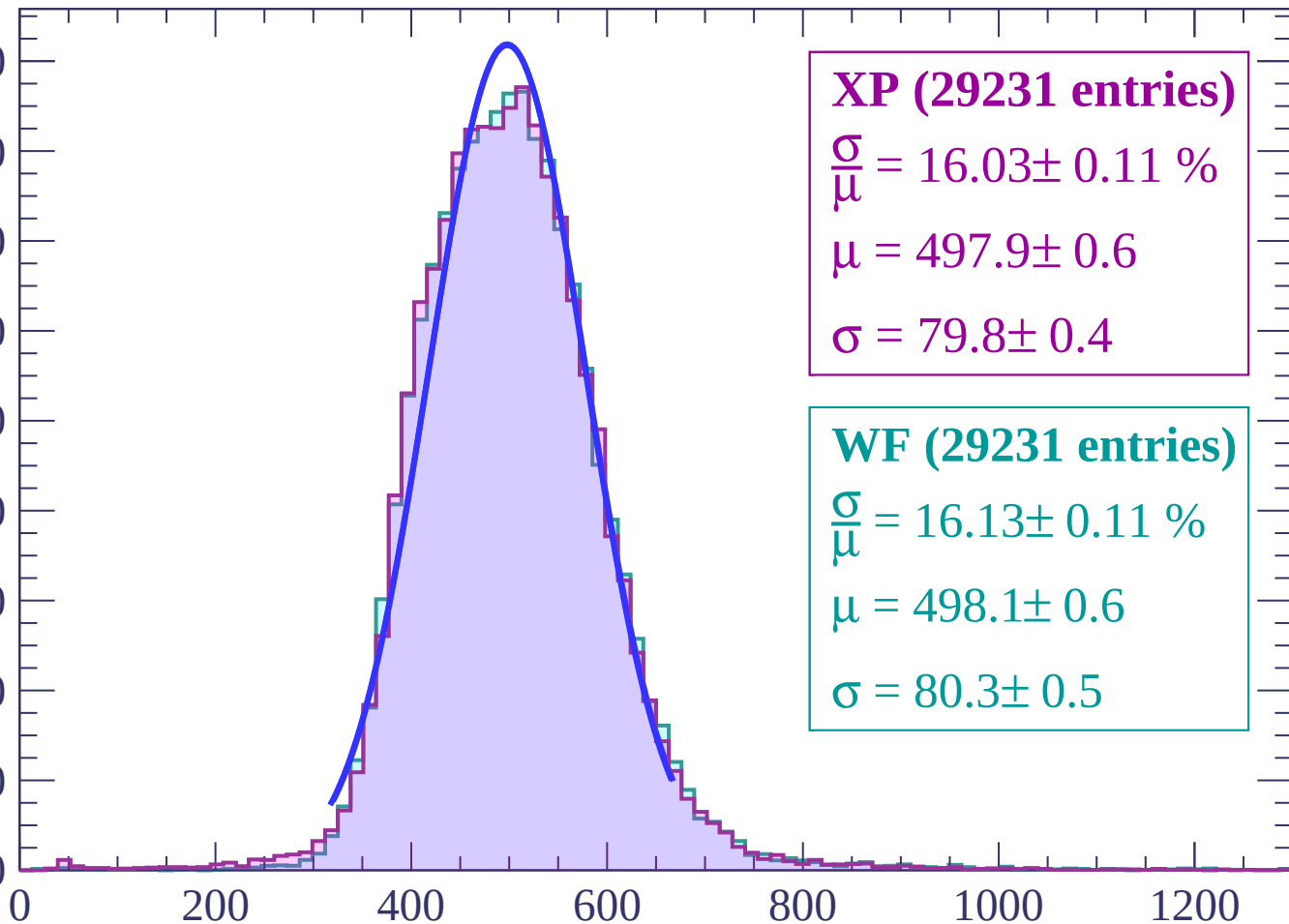
Energy loss | $-28 < \theta < -26$



Energy loss | $-26 < \theta < -24$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP (29231 entries)

$\frac{\sigma}{\mu} = 16.03 \pm 0.11 \%$

$\mu = 497.9 \pm 0.6$

$\sigma = 79.8 \pm 0.4$

WF (29231 entries)

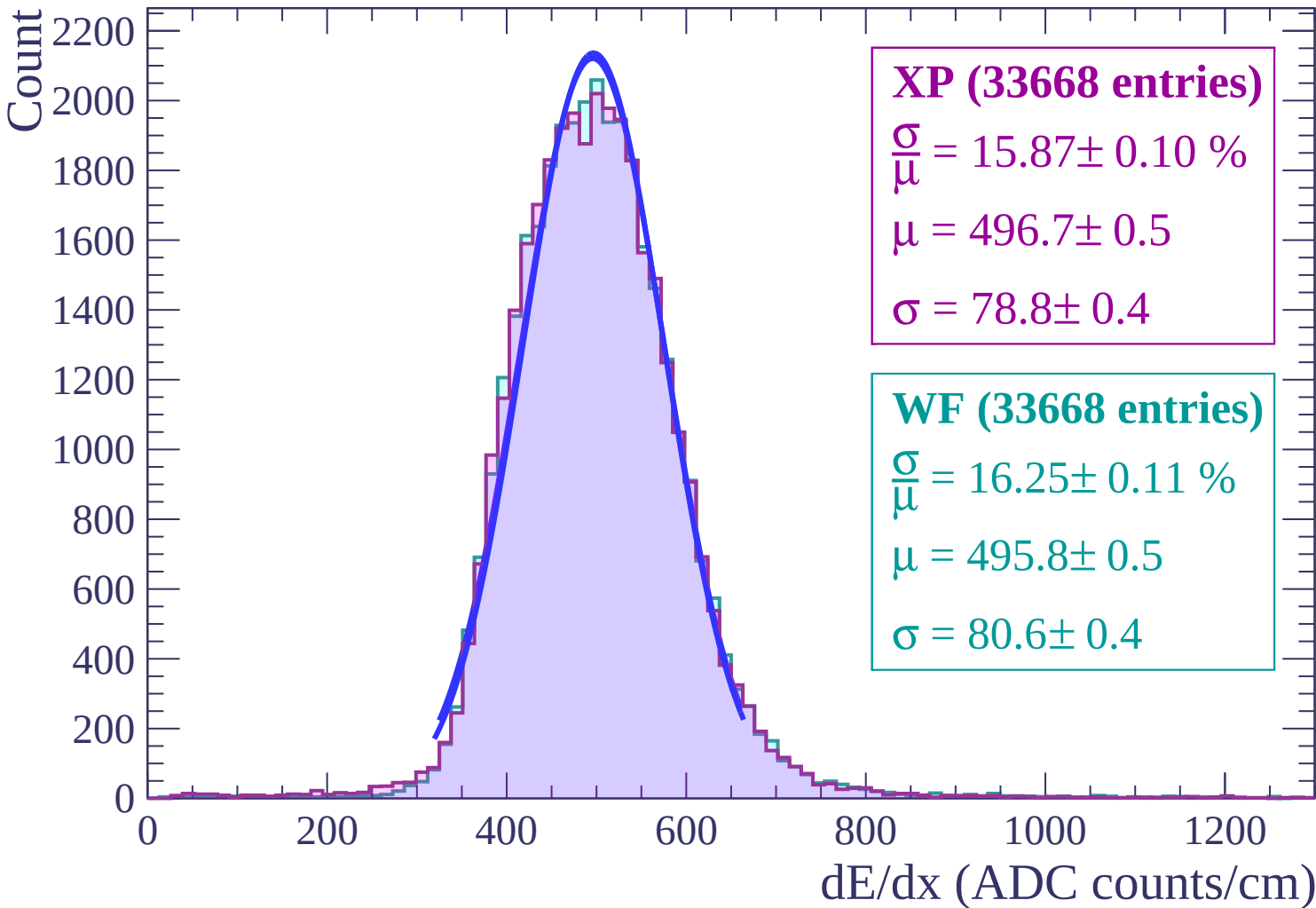
$\frac{\sigma}{\mu} = 16.13 \pm 0.11 \%$

$\mu = 498.1 \pm 0.6$

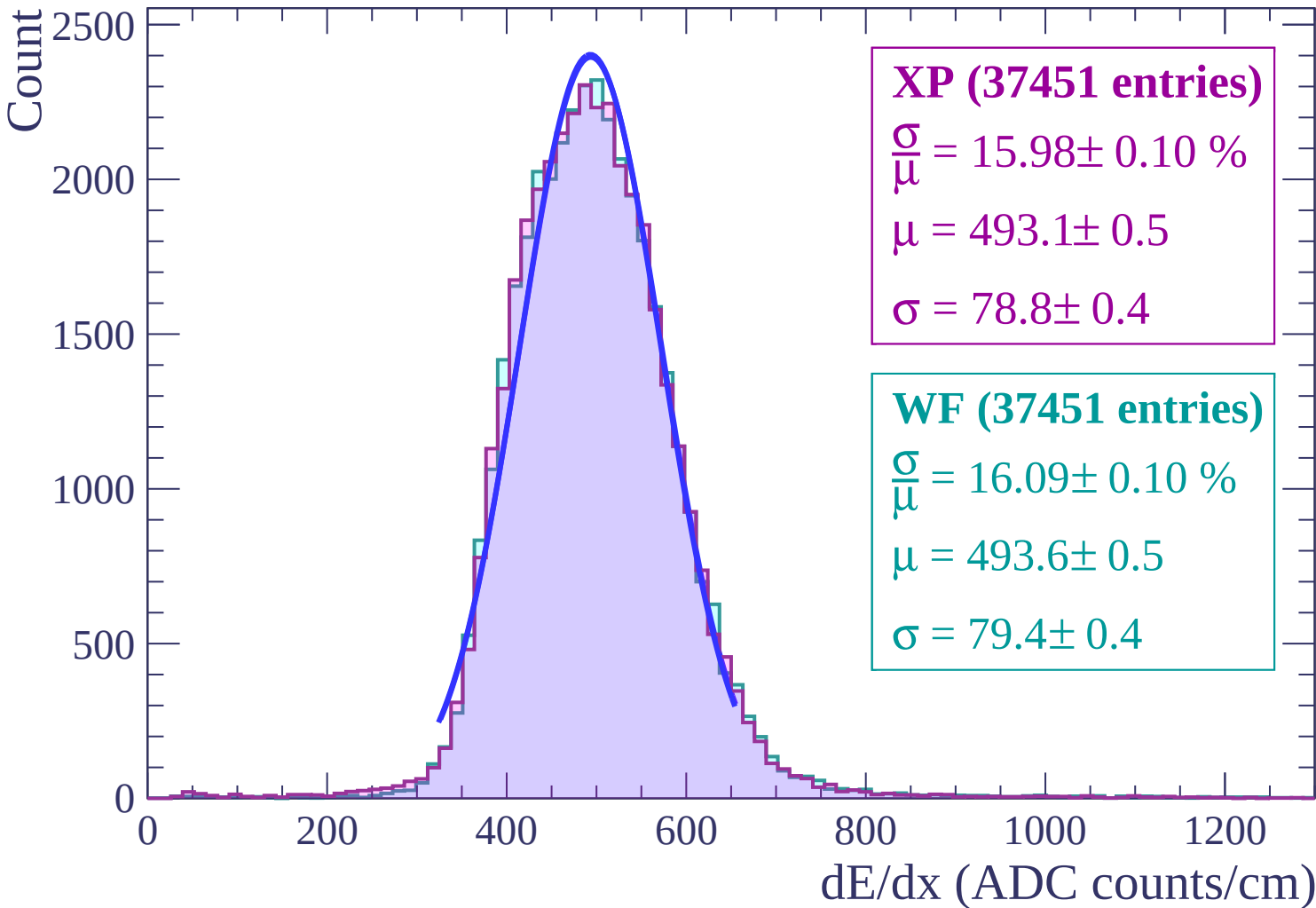
$\sigma = 80.3 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-24 < \theta < -22$

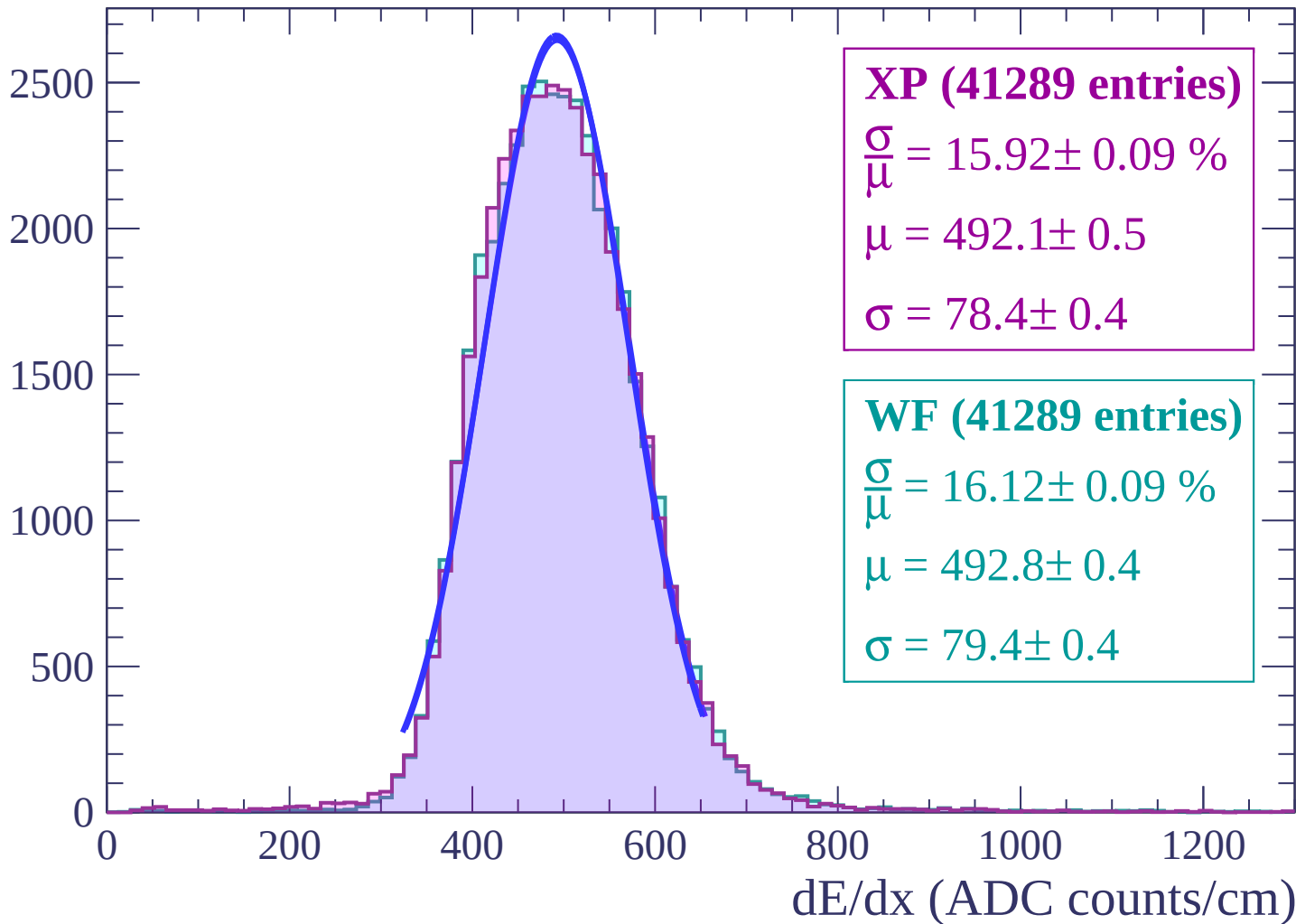


Energy loss | $-22 < \theta < -20$

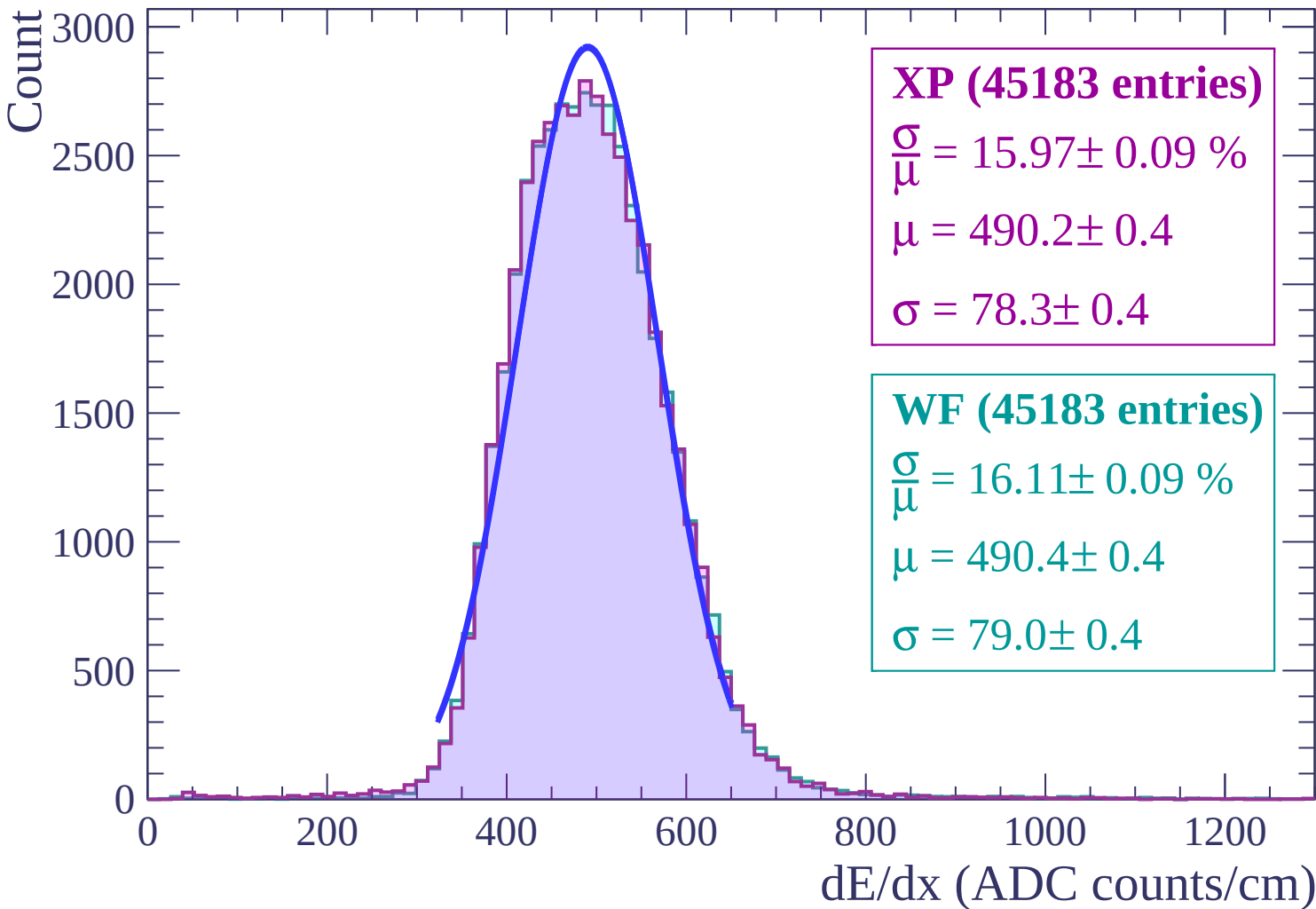


Energy loss | $-20 < \theta < -18$

Count

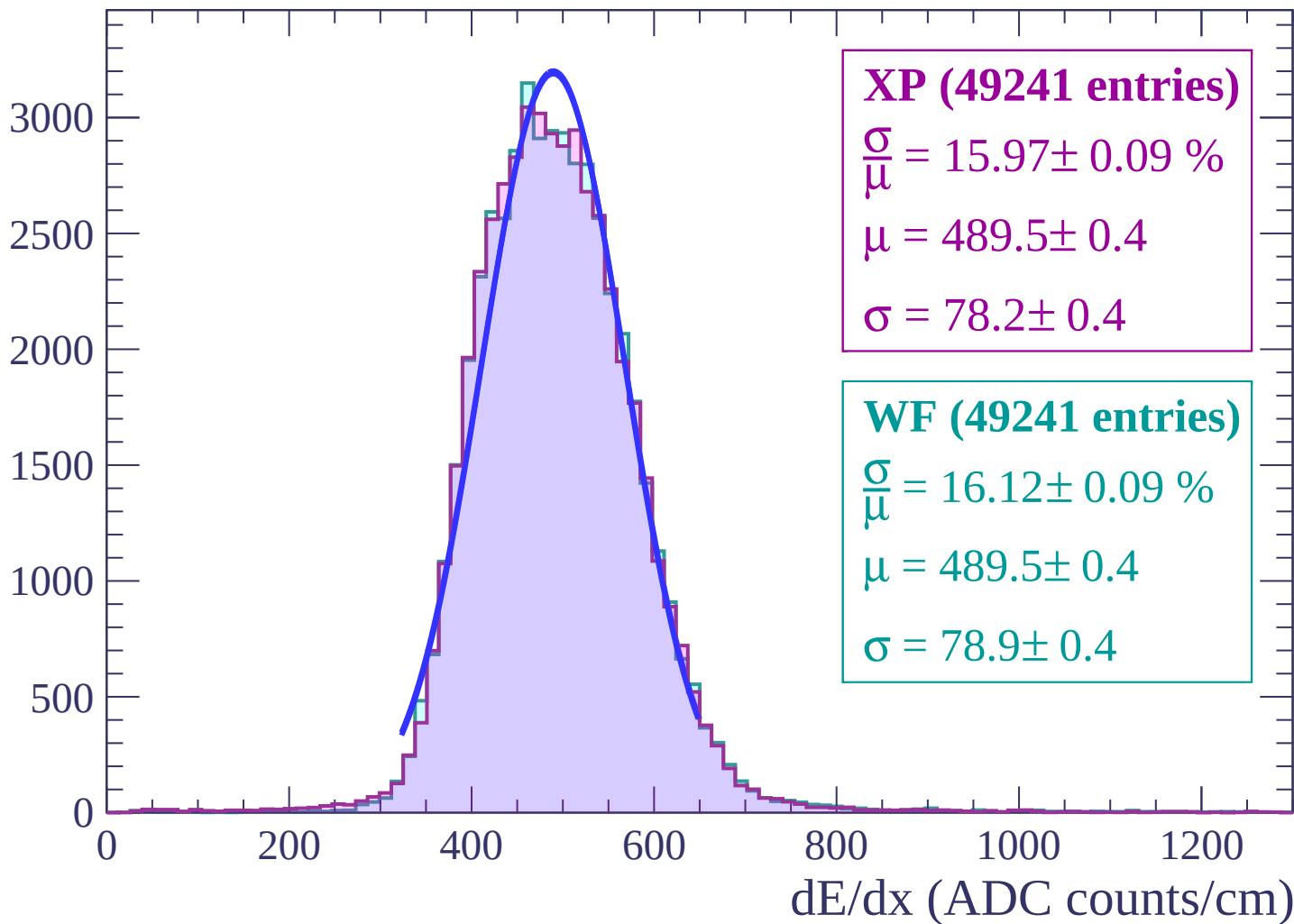


Energy loss | $-18 < \theta < -16$

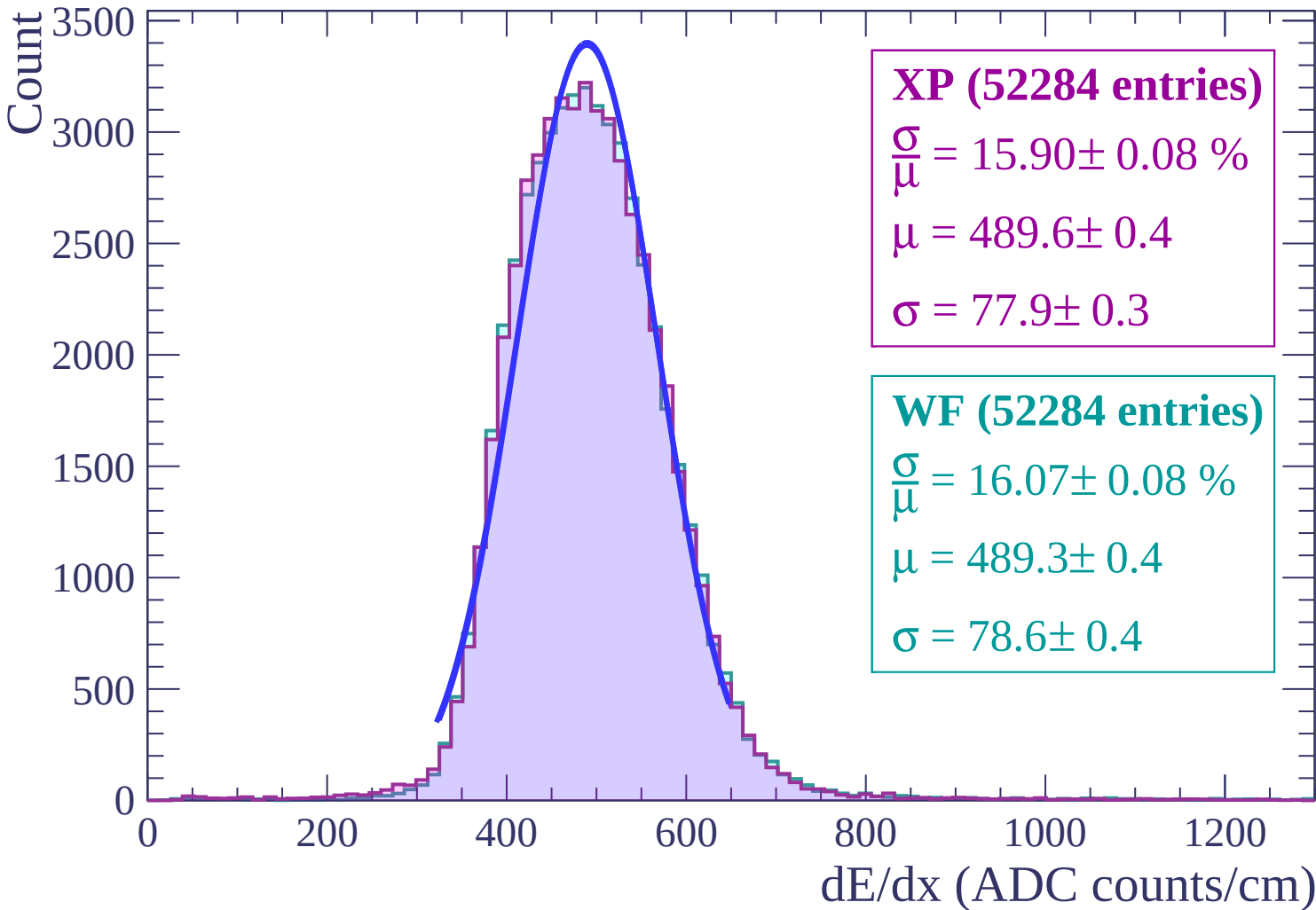


Energy loss | $-16 < \theta < -14$

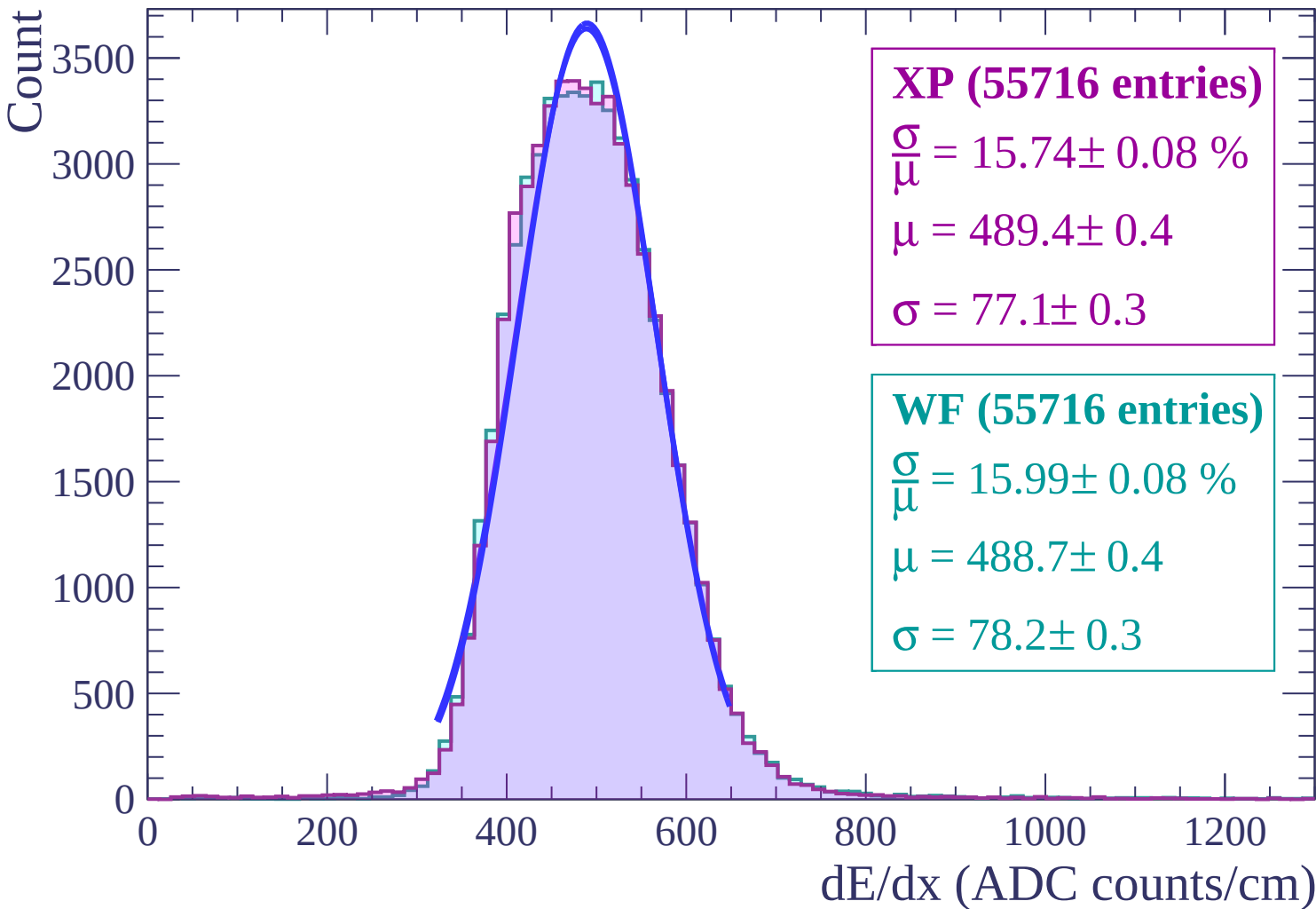
Count



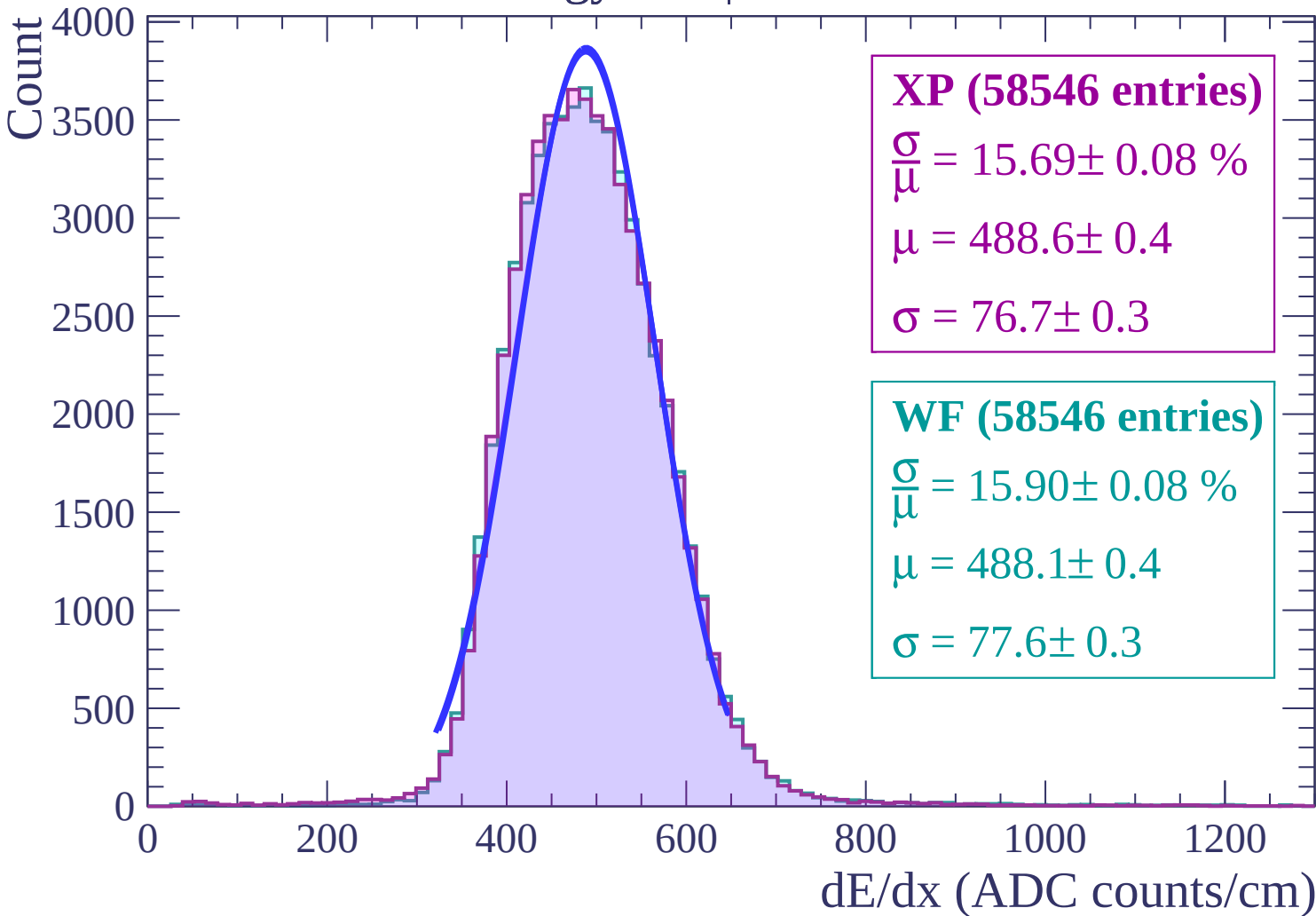
Energy loss | $-14 < \theta < -12$



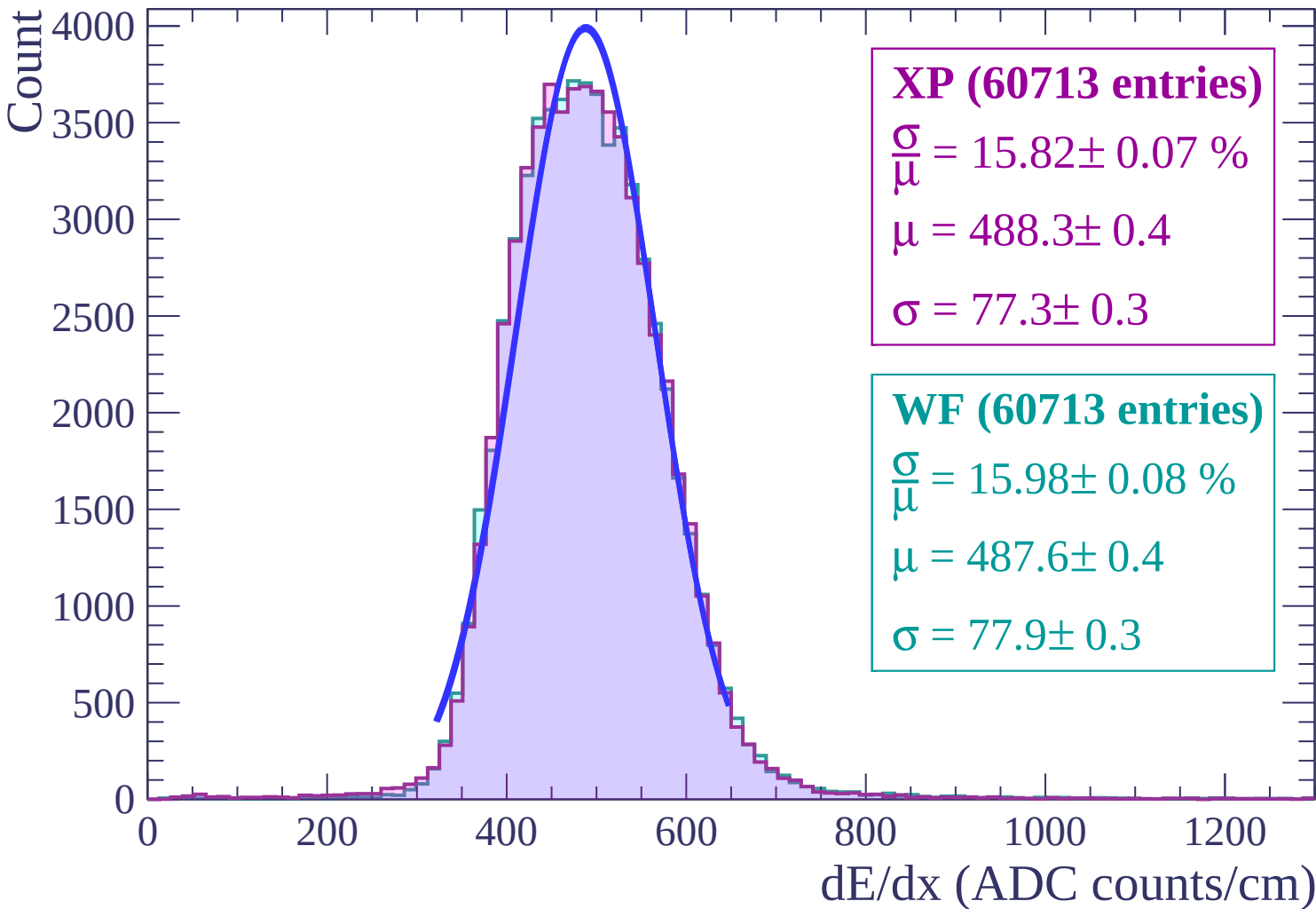
Energy loss | $-12 < \theta < -10$



Energy loss | $-10 < \theta < -8$

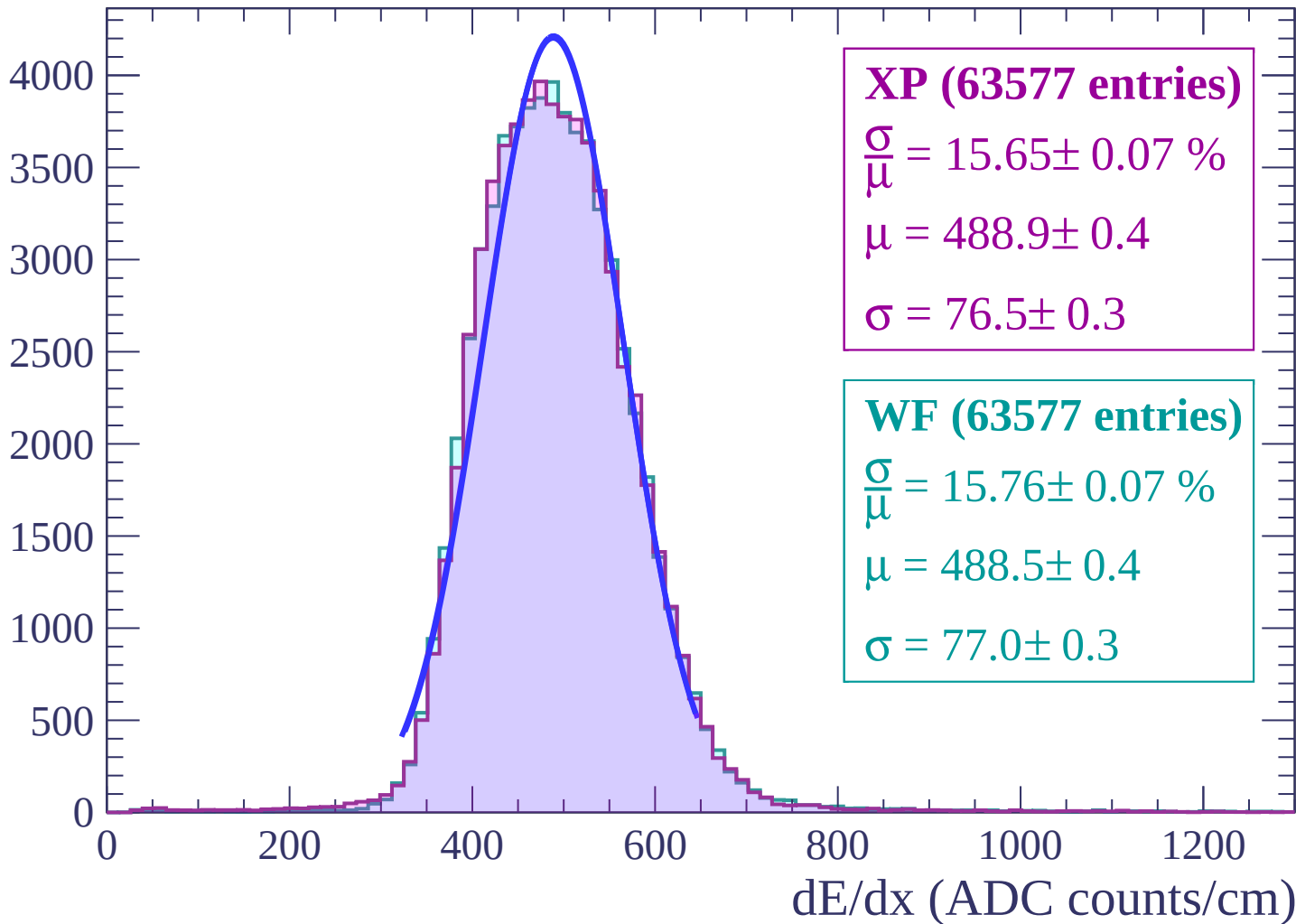


Energy loss | $-8 < \theta < -6$



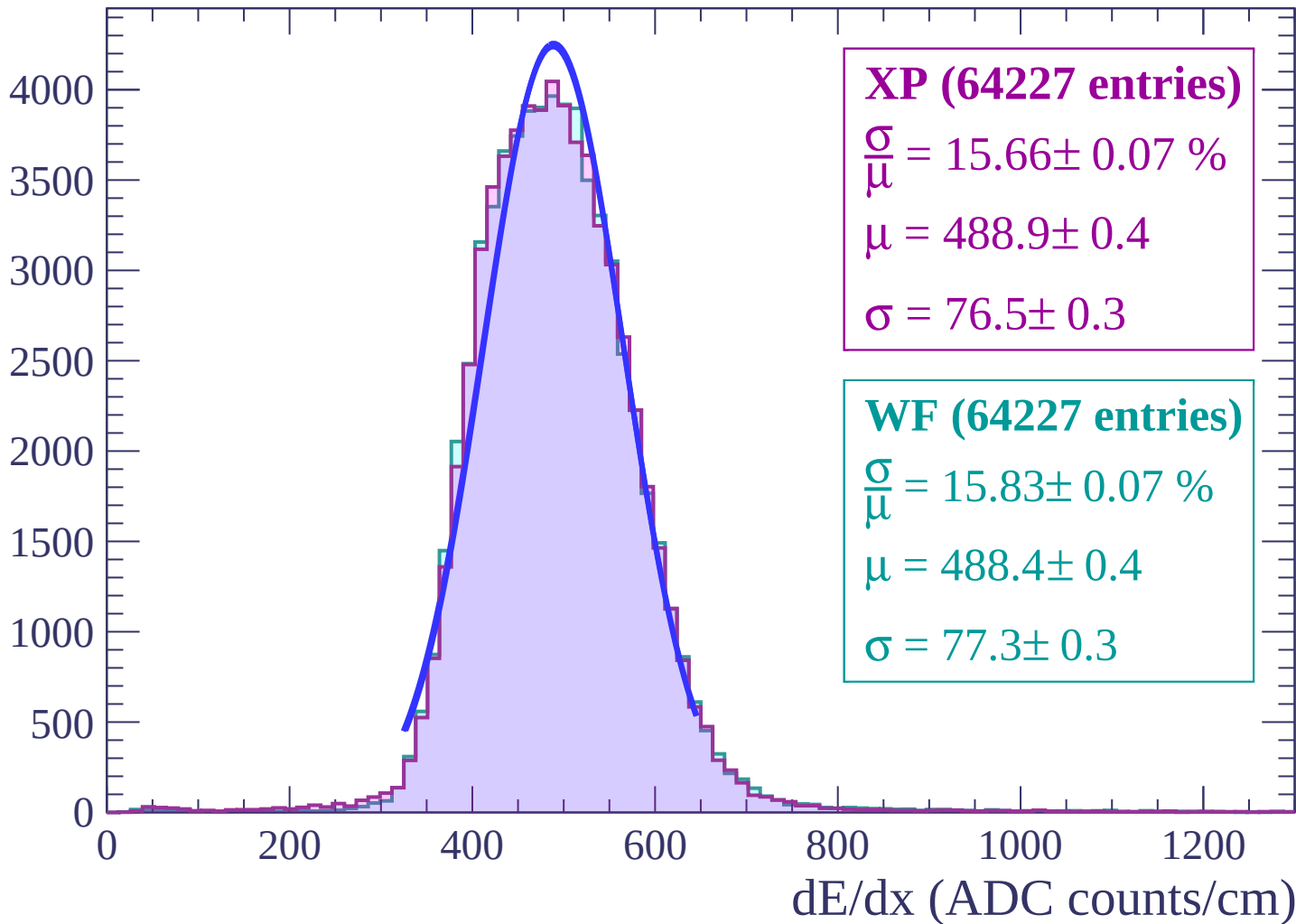
Energy loss | $-6 < \theta < -4$

Count

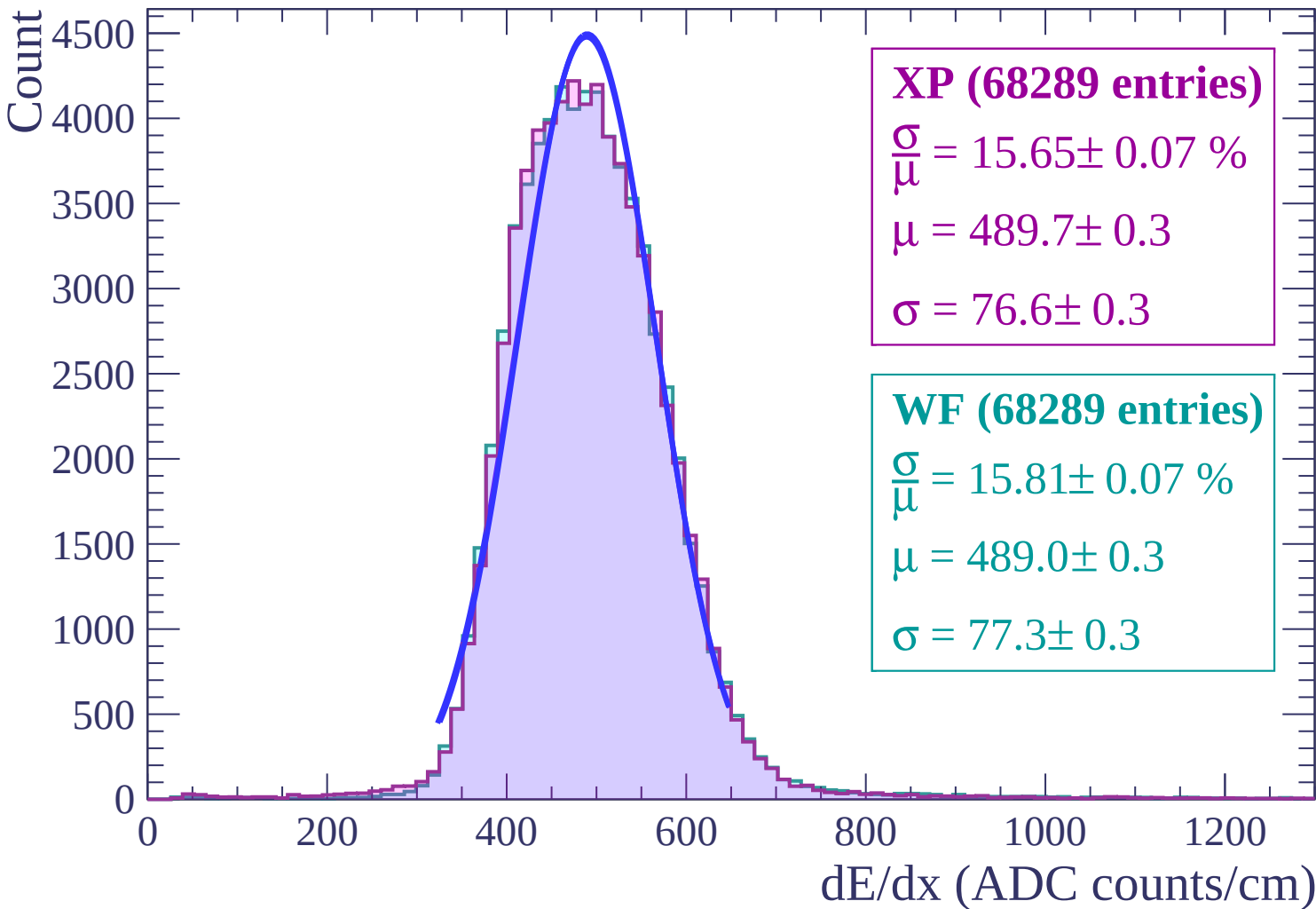


Energy loss | $-4 < \theta < -2$

Count

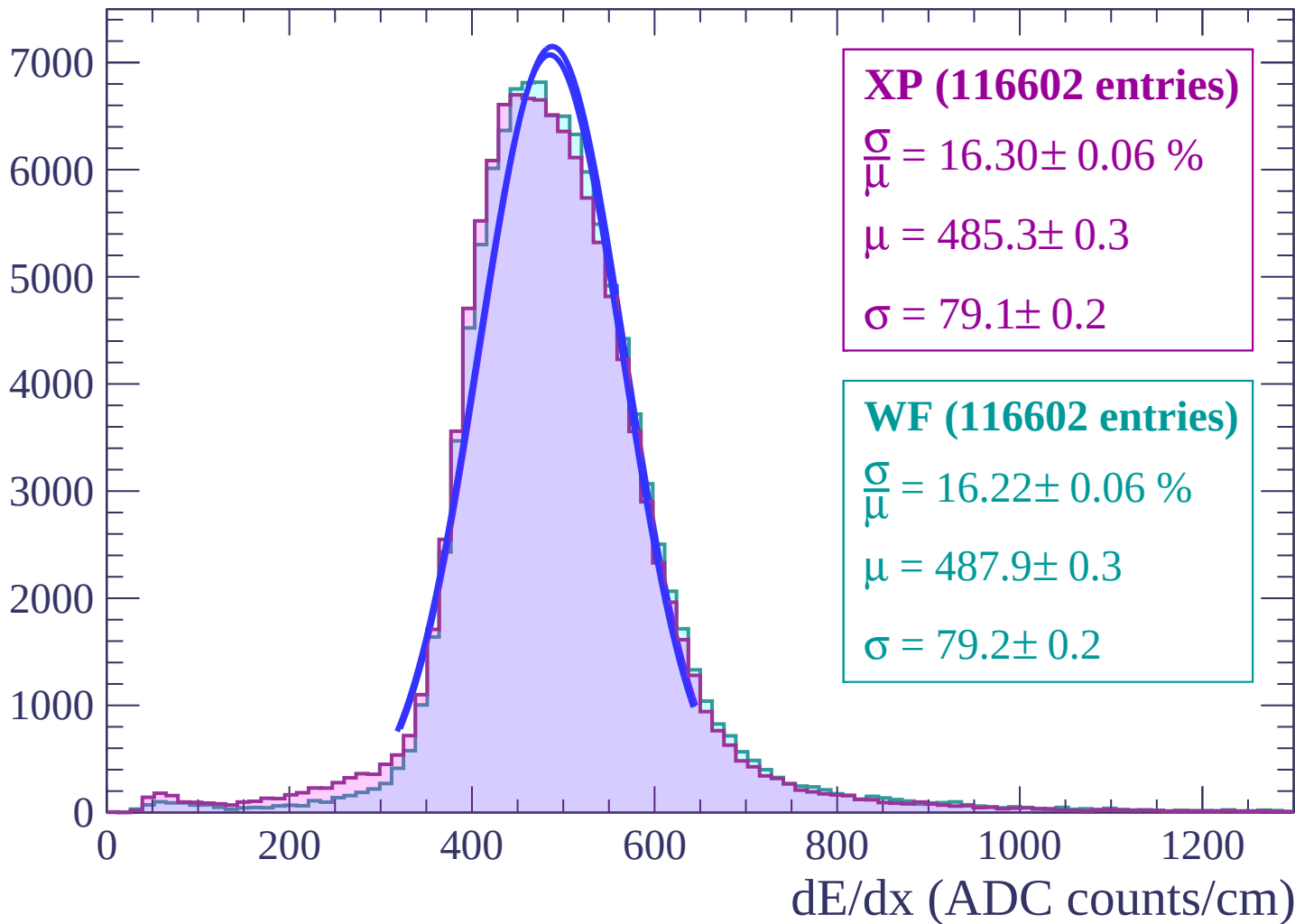


Energy loss $-2 < \theta < 0$

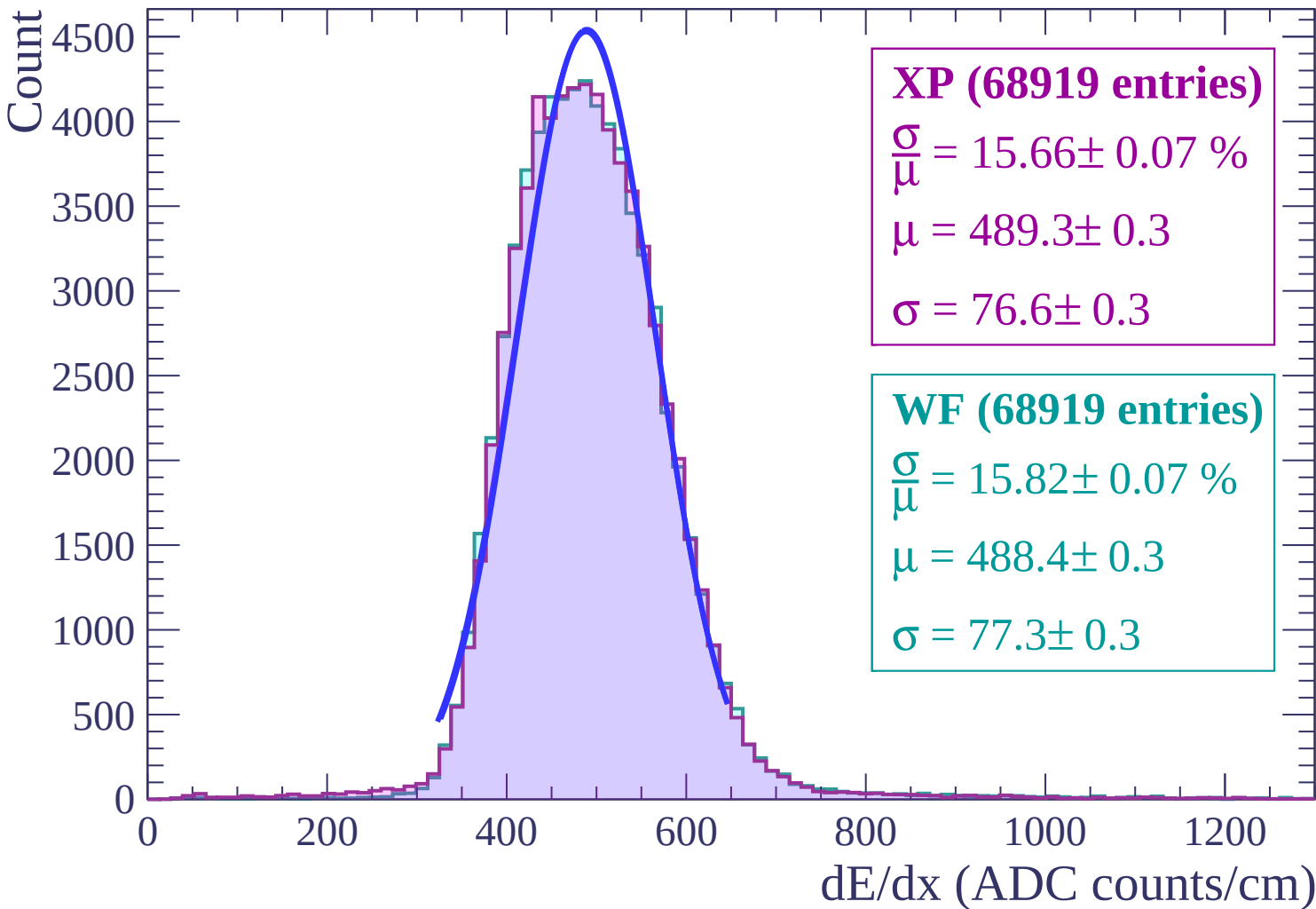


Energy loss | $0 < \theta < 2$

Count

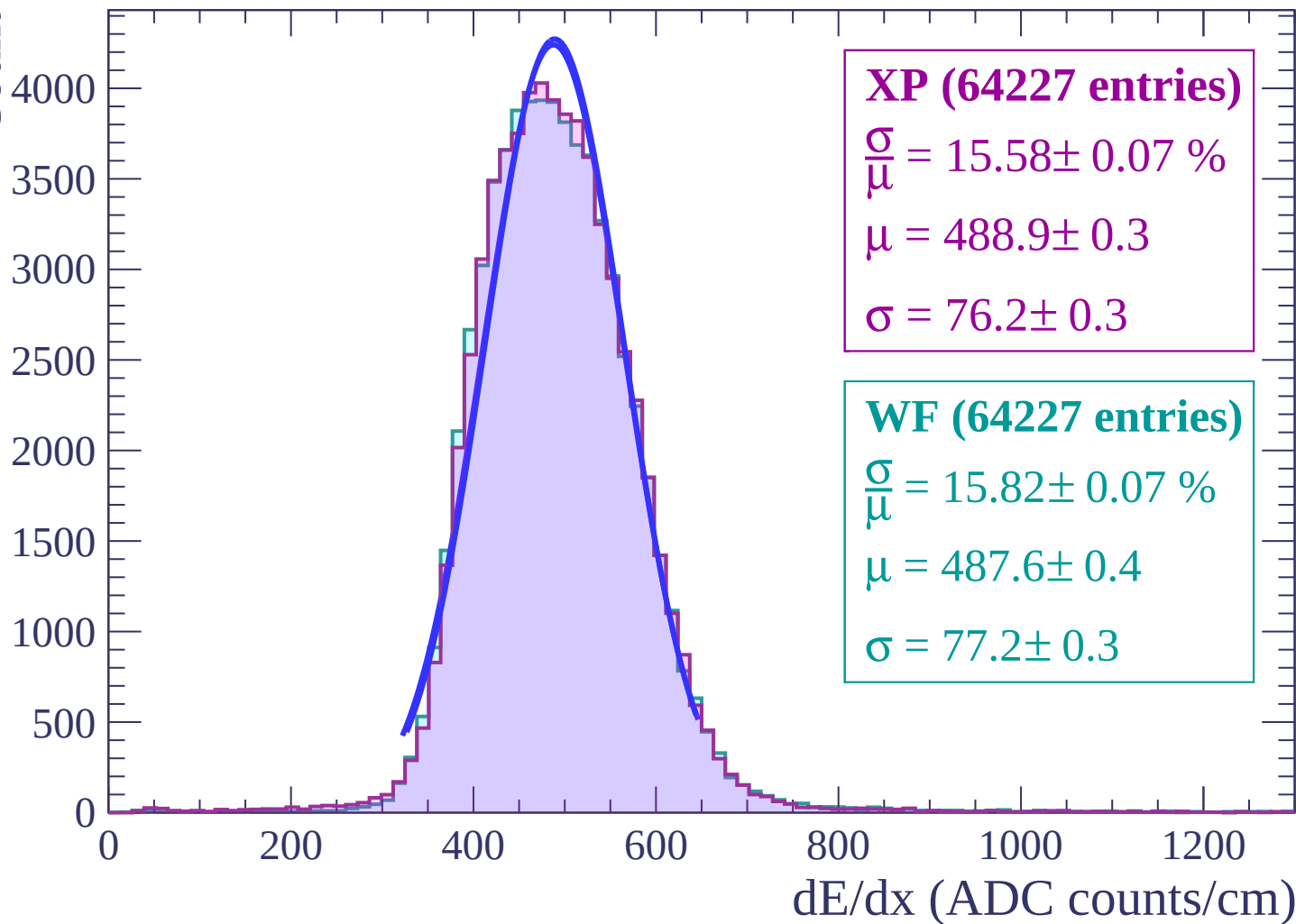


Energy loss | $2 < \theta < 4$



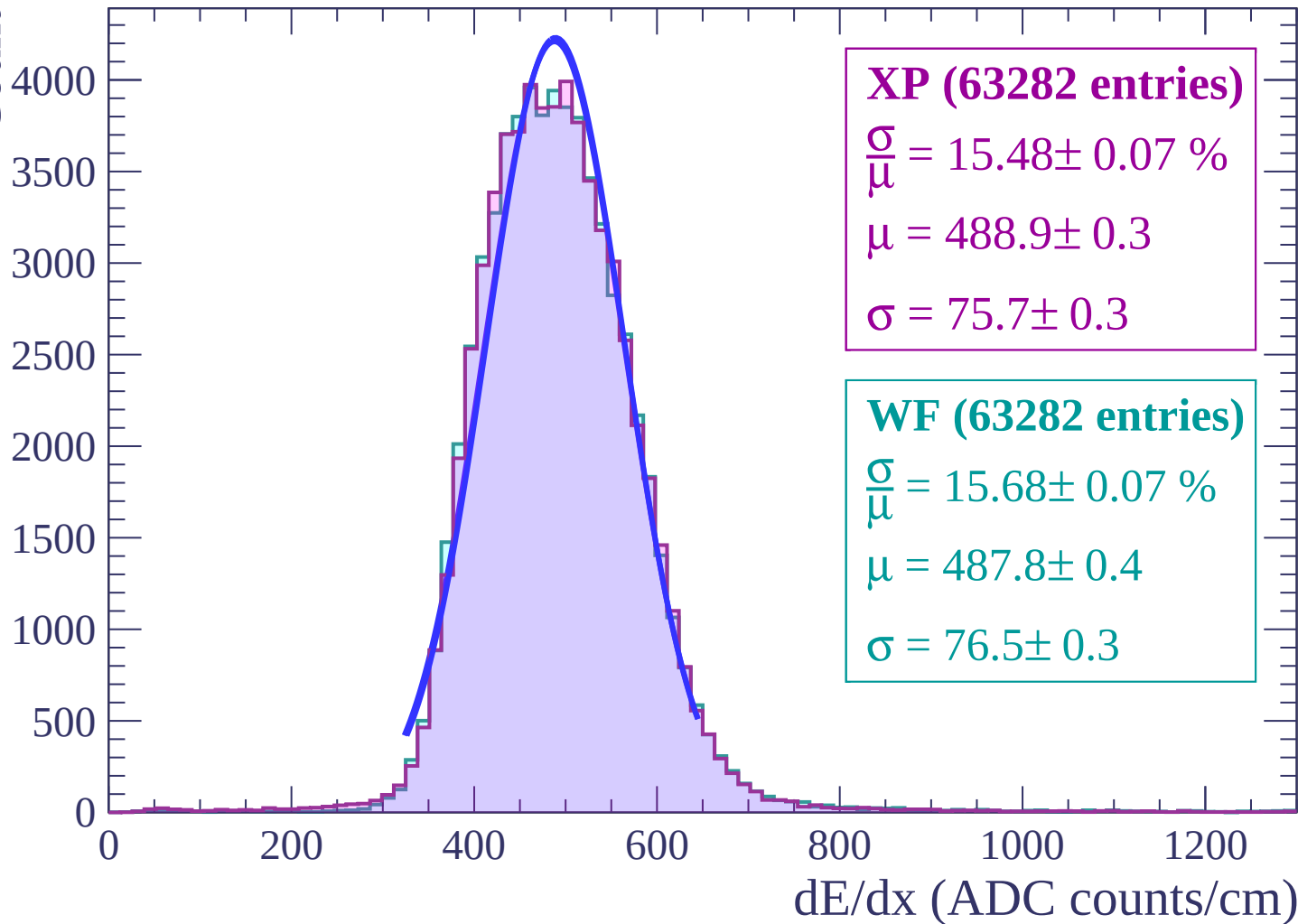
Energy loss | $4 < \theta < 6$

Count

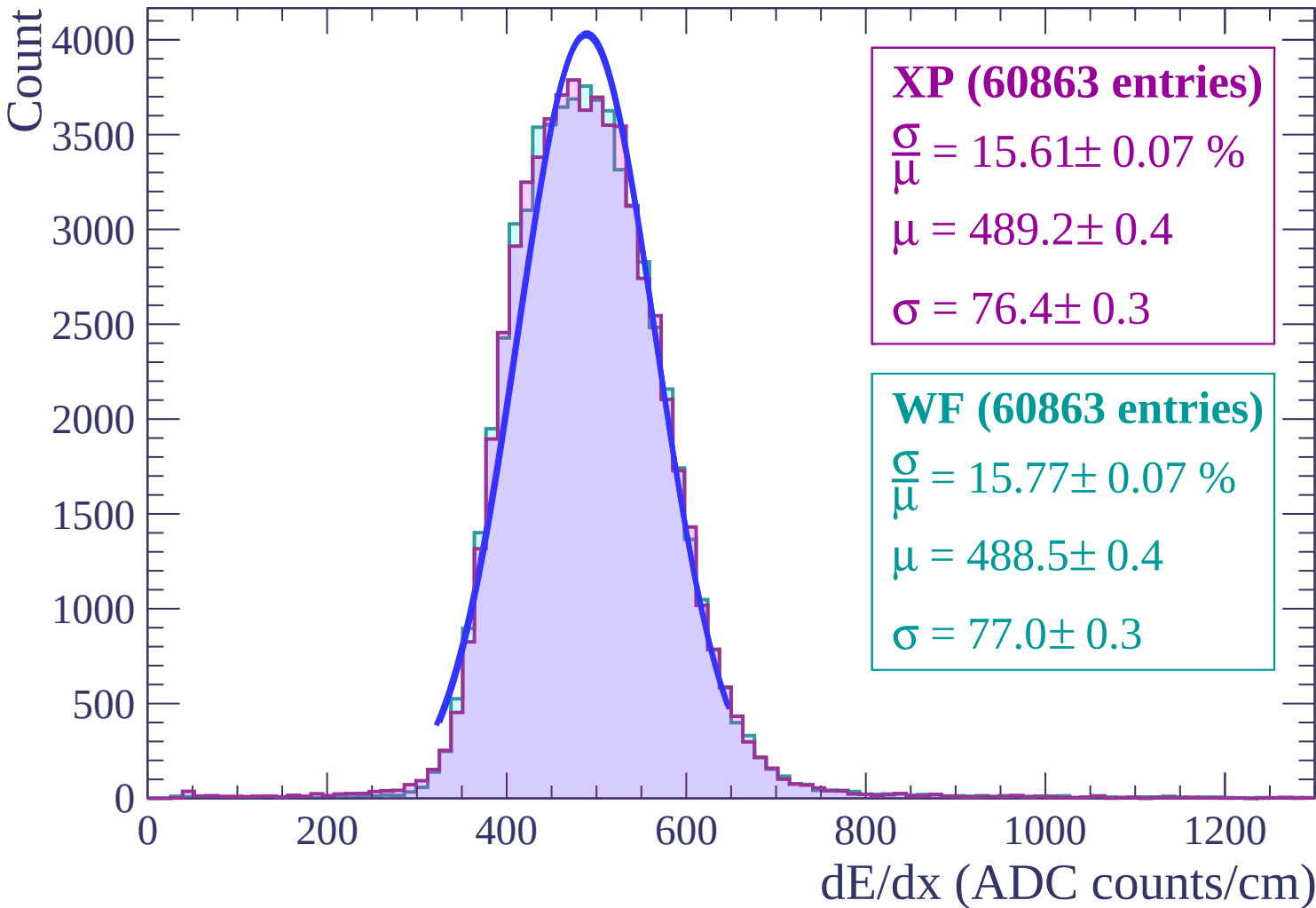


Energy loss | $6 < \theta < 8$

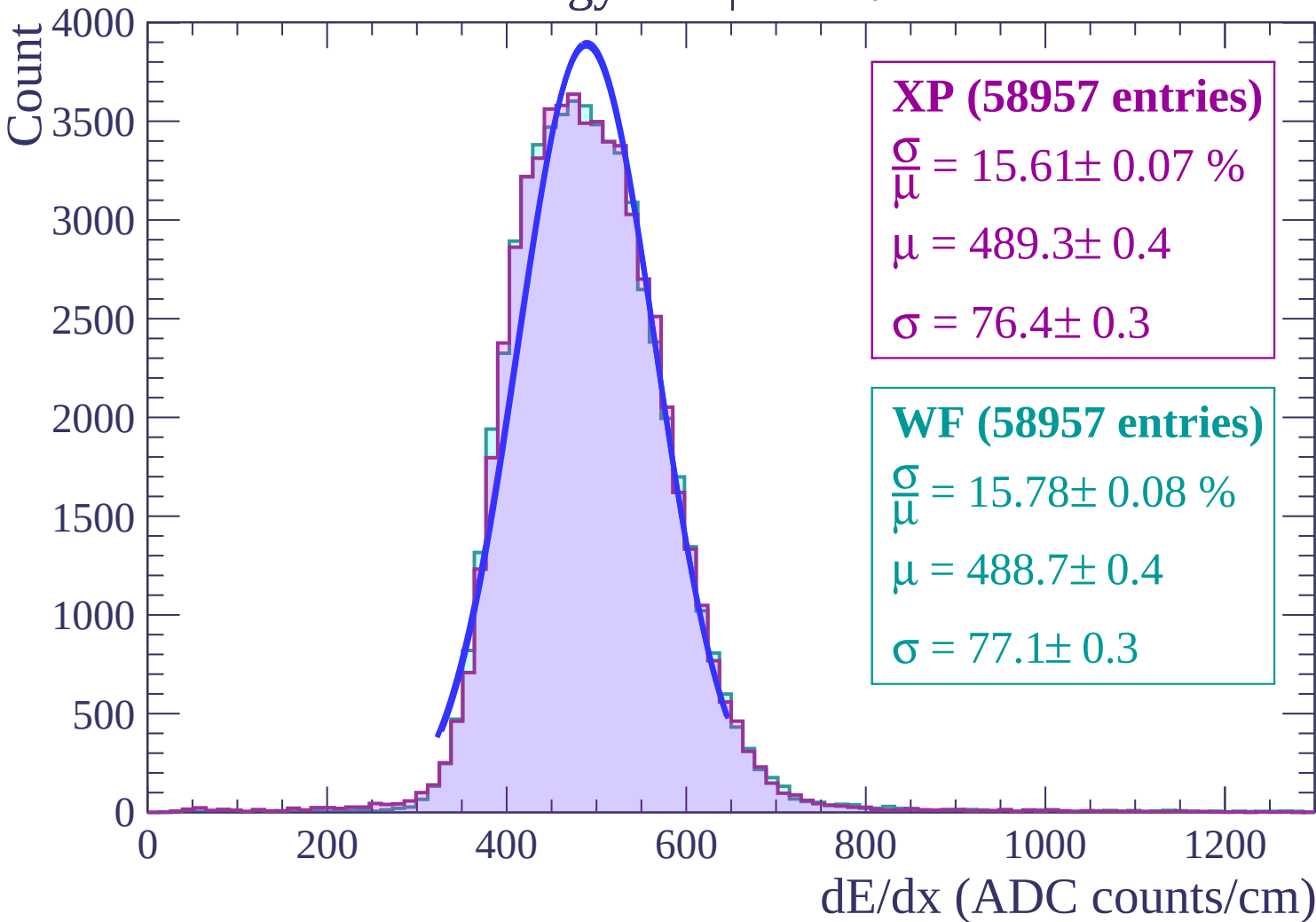
Count



Energy loss | $8 < \theta < 10$



Energy loss | $10 < \theta < 12$



Energy loss | $12 < \theta < 14$

Count

3500

3000

2500

2000

1500

1000

500

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (55865 entries)

$\frac{\sigma}{\mu} = 15.77 \pm 0.08 \%$

$\mu = 489.2 \pm 0.4$

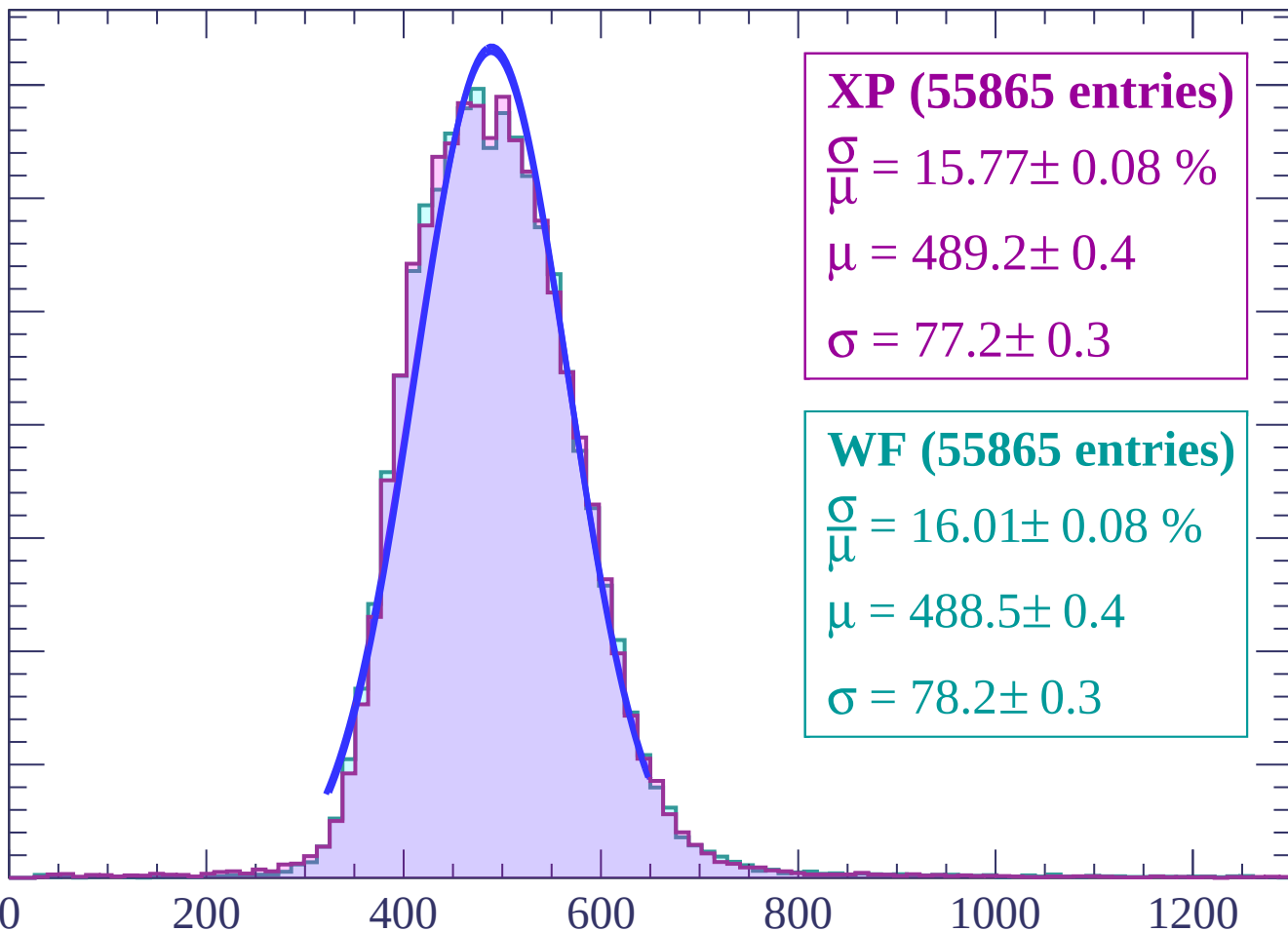
$\sigma = 77.2 \pm 0.3$

WF (55865 entries)

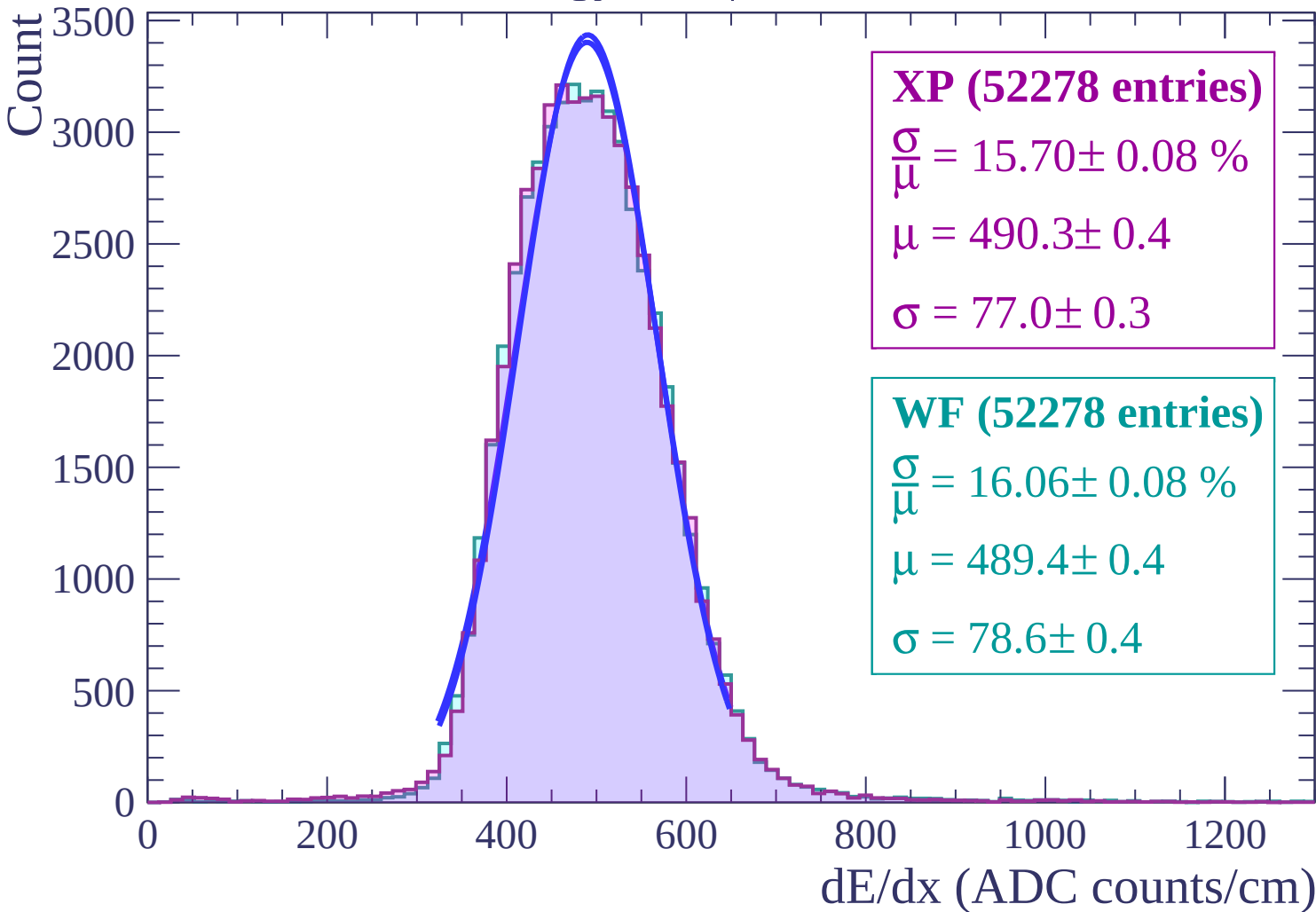
$\frac{\sigma}{\mu} = 16.01 \pm 0.08 \%$

$\mu = 488.5 \pm 0.4$

$\sigma = 78.2 \pm 0.3$



Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$

Count

3000

2500

2000

1500

1000

500

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (49291 entries)

$\frac{\sigma}{\mu} = 15.75 \pm 0.08 \%$

$\mu = 490.5 \pm 0.4$

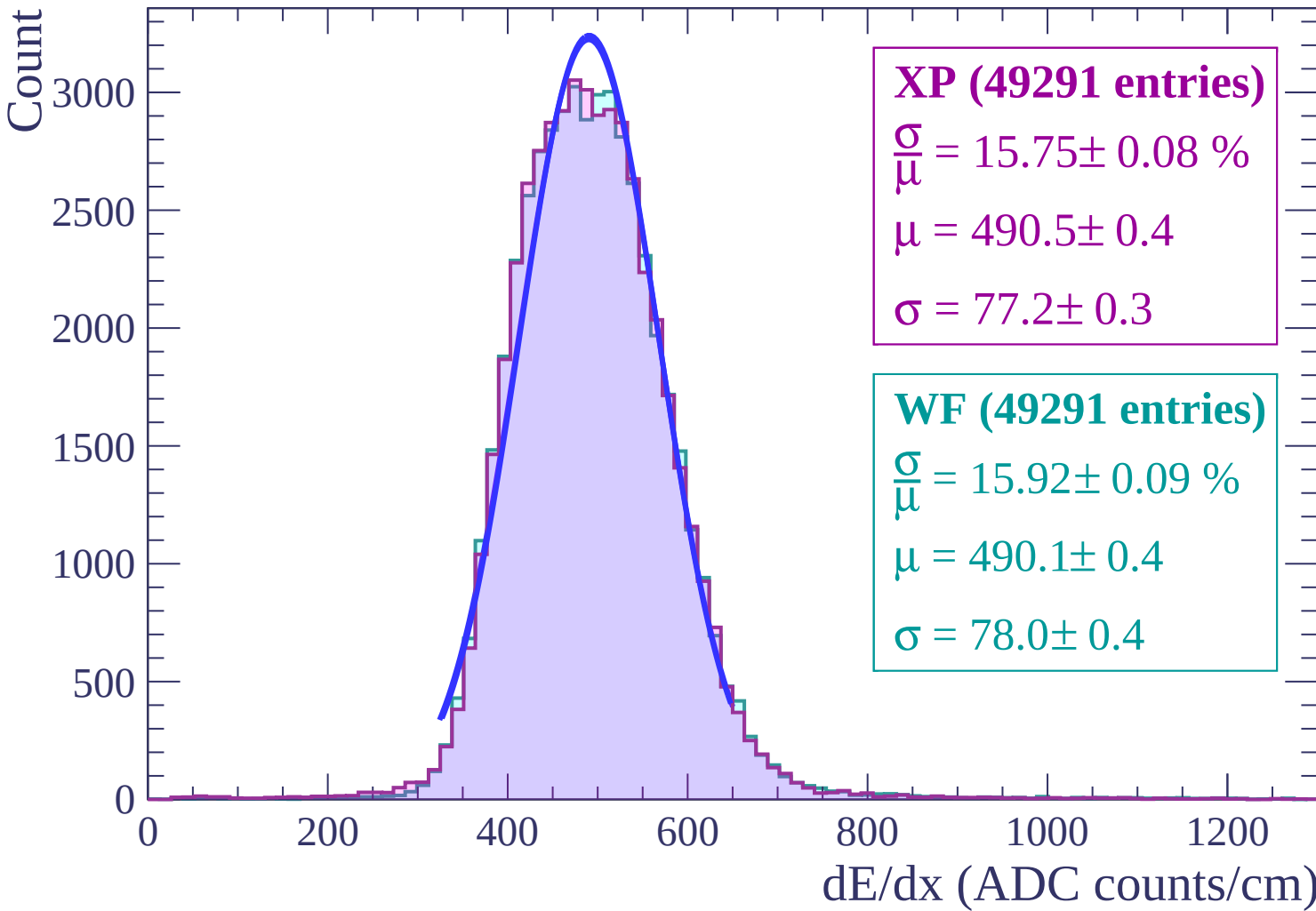
$\sigma = 77.2 \pm 0.3$

WF (49291 entries)

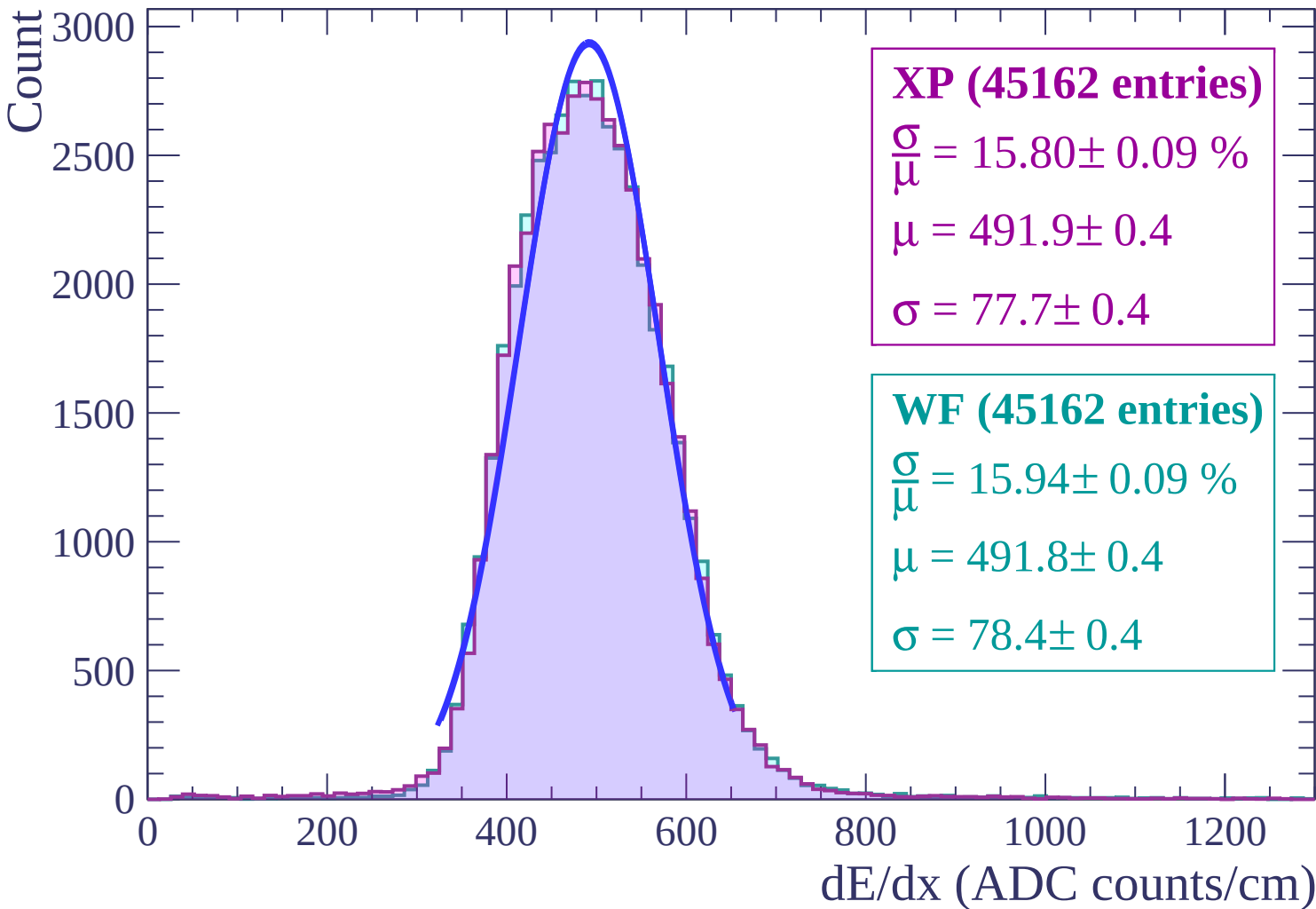
$\frac{\sigma}{\mu} = 15.92 \pm 0.09 \%$

$\mu = 490.1 \pm 0.4$

$\sigma = 78.0 \pm 0.4$



Energy loss | $18 < \theta < 20$



Energy loss | $20 < \theta < 22$

Count

2500

2000

1500

1000

500

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (41412 entries)

$\frac{\sigma}{\mu} = 15.63 \pm 0.08 \%$

$\mu = 494.0 \pm 0.4$

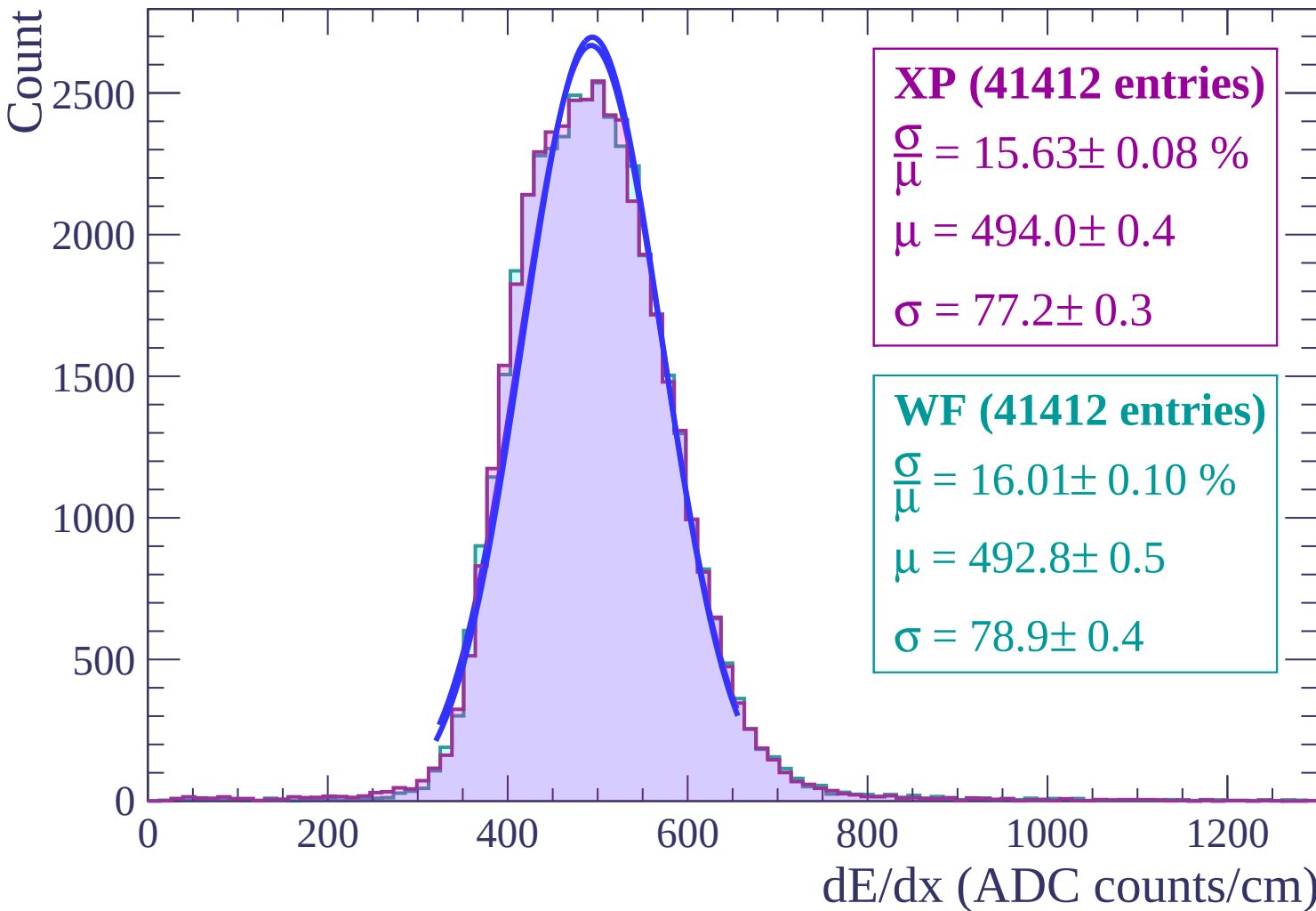
$\sigma = 77.2 \pm 0.3$

WF (41412 entries)

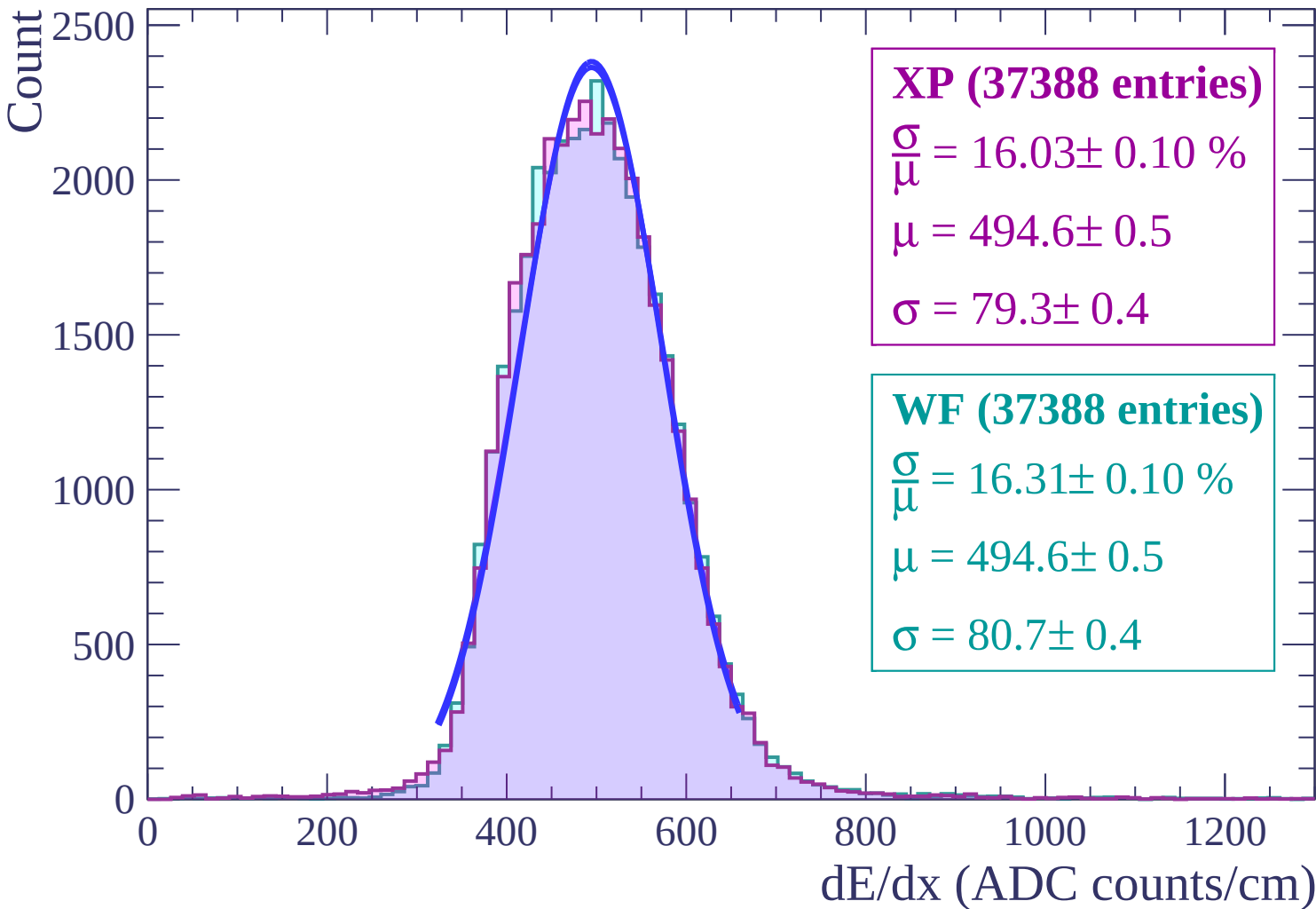
$\frac{\sigma}{\mu} = 16.01 \pm 0.10 \%$

$\mu = 492.8 \pm 0.5$

$\sigma = 78.9 \pm 0.4$



Energy loss | $22 < \theta < 24$



Energy loss | $24 < \theta < 26$

Count

2000
1800
1600
1400
1200
1000
800
600
400
200
0

XP (33510 entries)

$\frac{\sigma}{\mu} = 16.06 \pm 0.10 \%$

$\mu = 496.1 \pm 0.5$

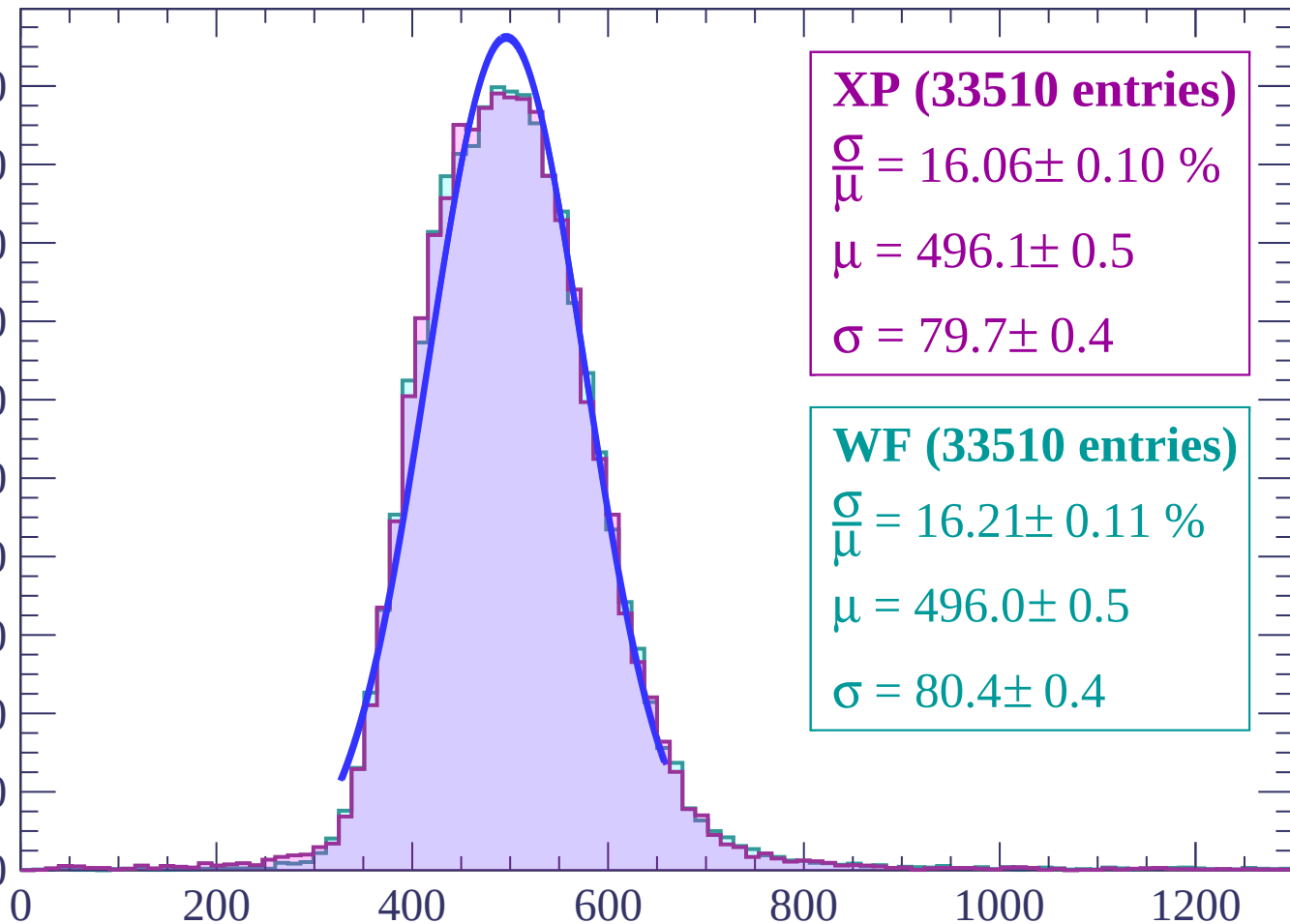
$\sigma = 79.7 \pm 0.4$

WF (33510 entries)

$\frac{\sigma}{\mu} = 16.21 \pm 0.11 \%$

$\mu = 496.0 \pm 0.5$

$\sigma = 80.4 \pm 0.4$

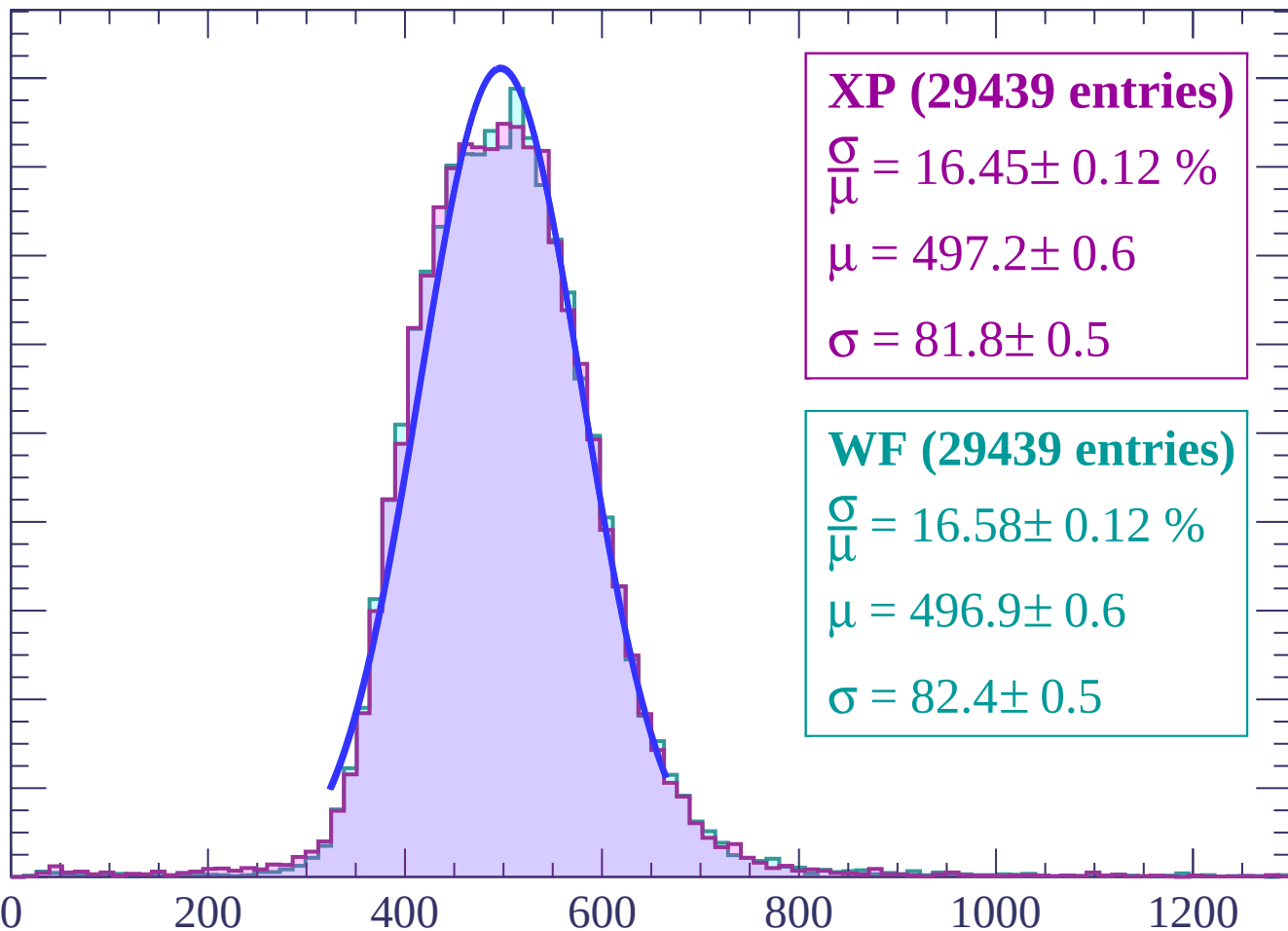


dE/dx (ADC counts/cm)

Energy loss | $26 < \theta < 28$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP (29439 entries)

$\frac{\sigma}{\mu} = 16.45 \pm 0.12 \%$

$\mu = 497.2 \pm 0.6$

$\sigma = 81.8 \pm 0.5$

WF (29439 entries)

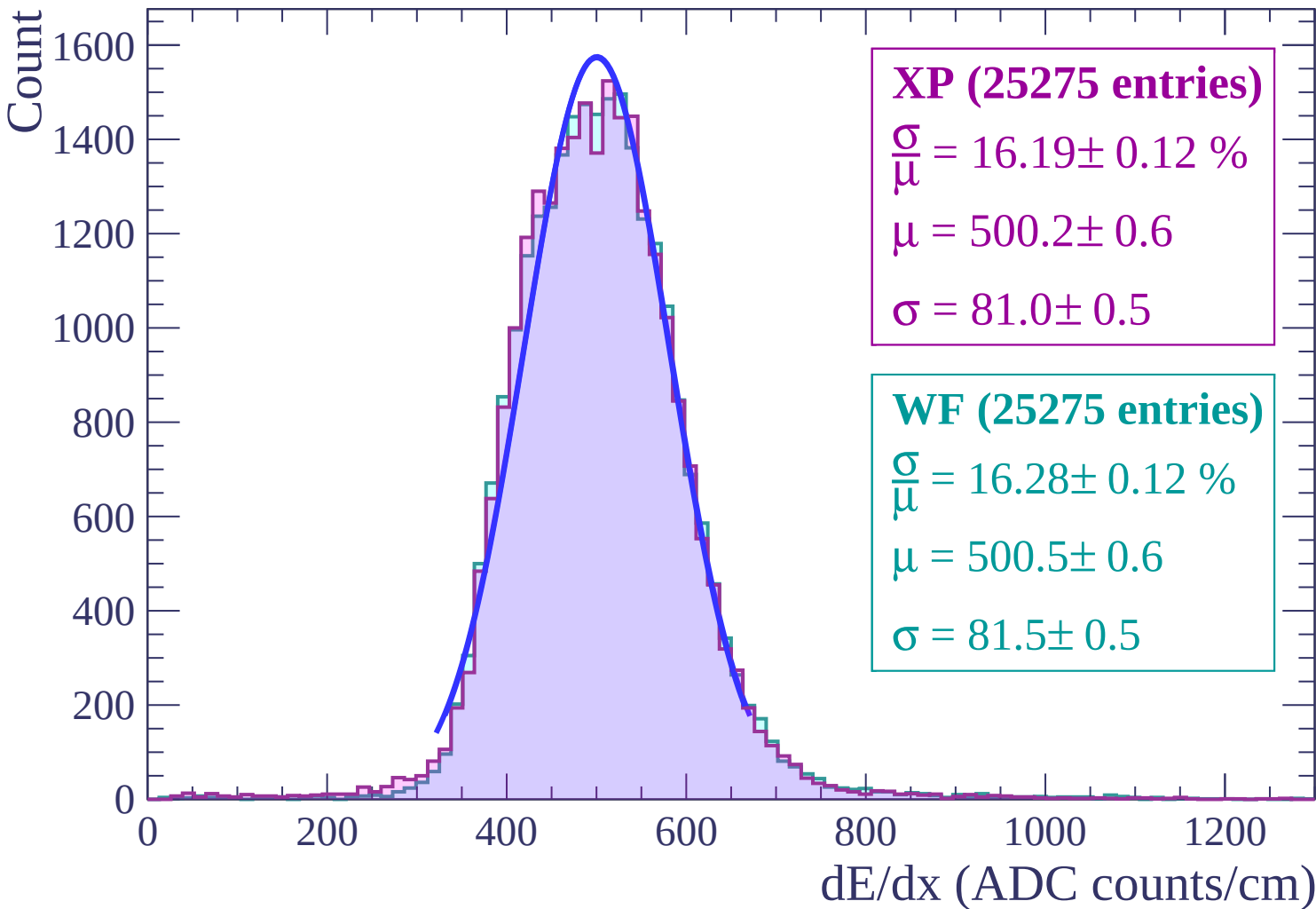
$\frac{\sigma}{\mu} = 16.58 \pm 0.12 \%$

$\mu = 496.9 \pm 0.6$

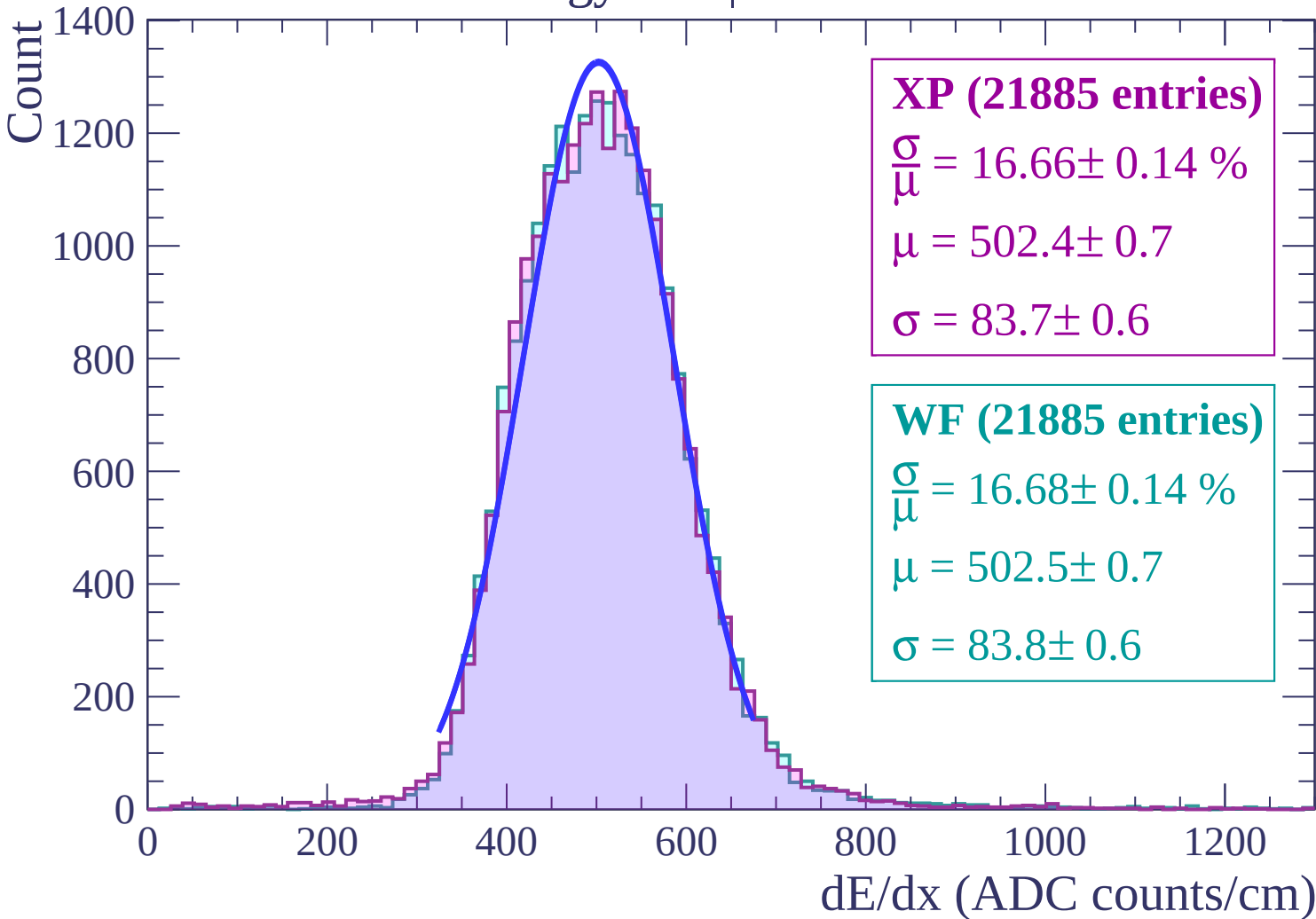
$\sigma = 82.4 \pm 0.5$

dE/dx (ADC counts/cm)

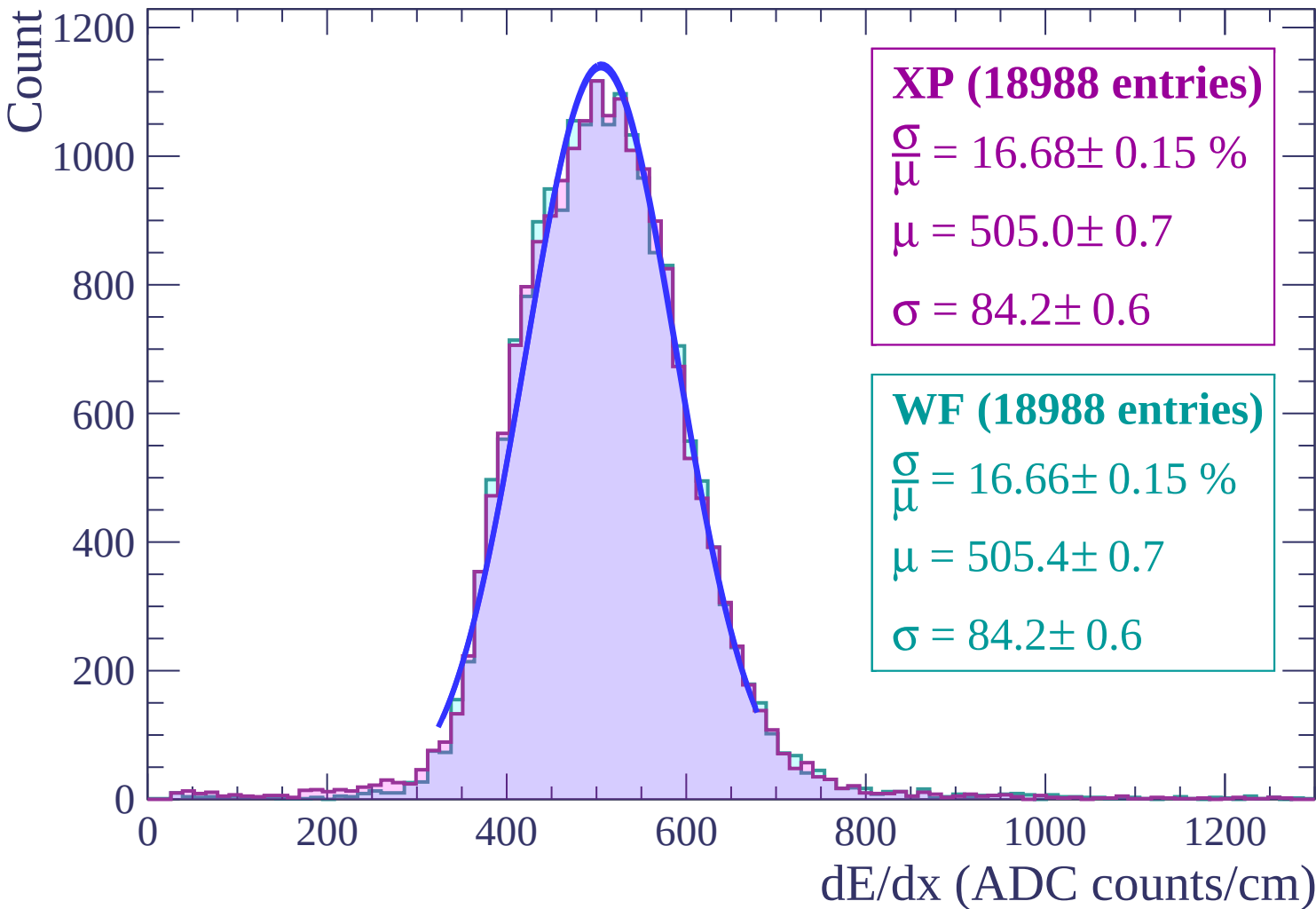
Energy loss | $28 < \theta < 30$



Energy loss | $30 < \theta < 32$



Energy loss | $32 < \theta < 34$



Energy loss | $34 < \theta < 36$

Count

1000

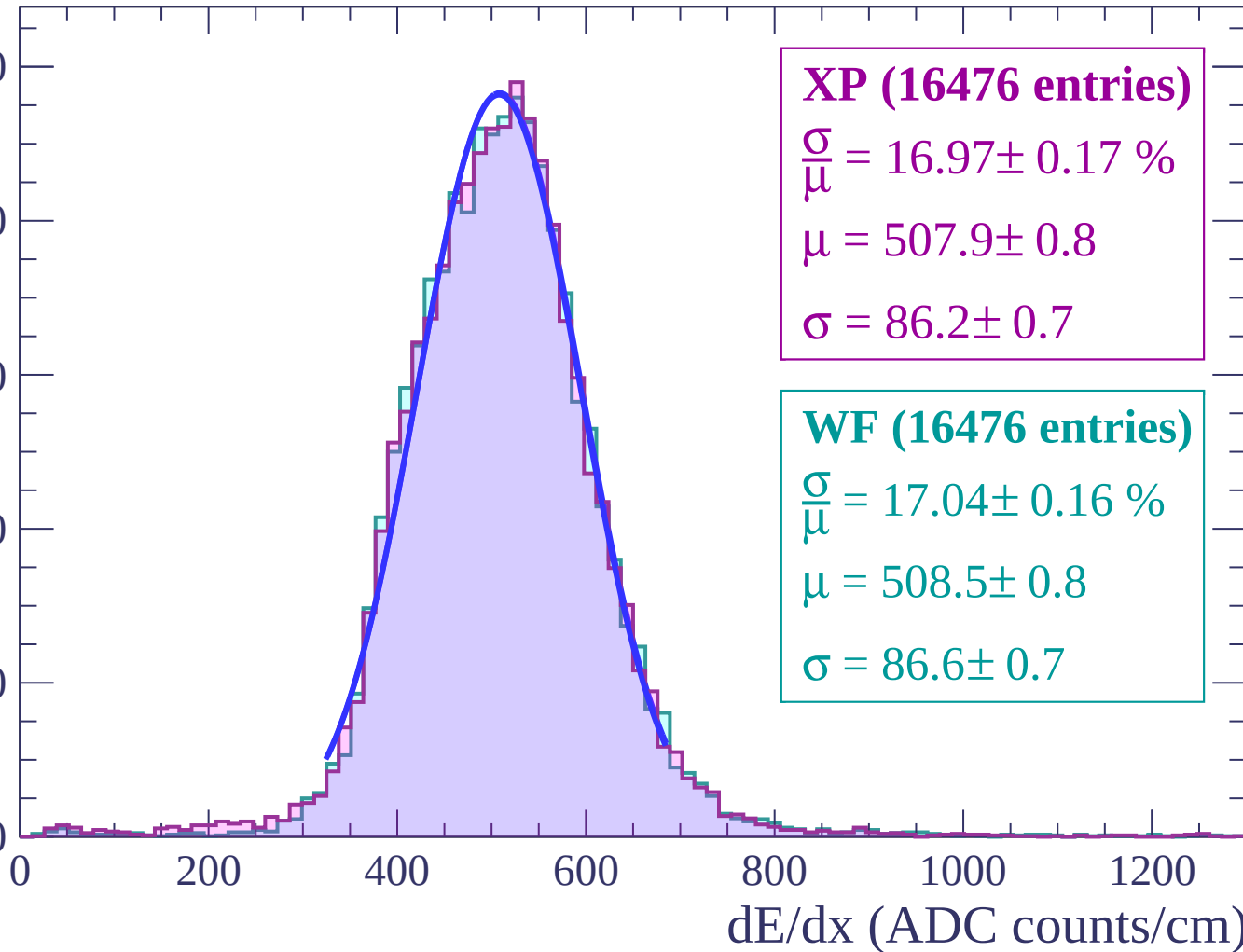
800

600

400

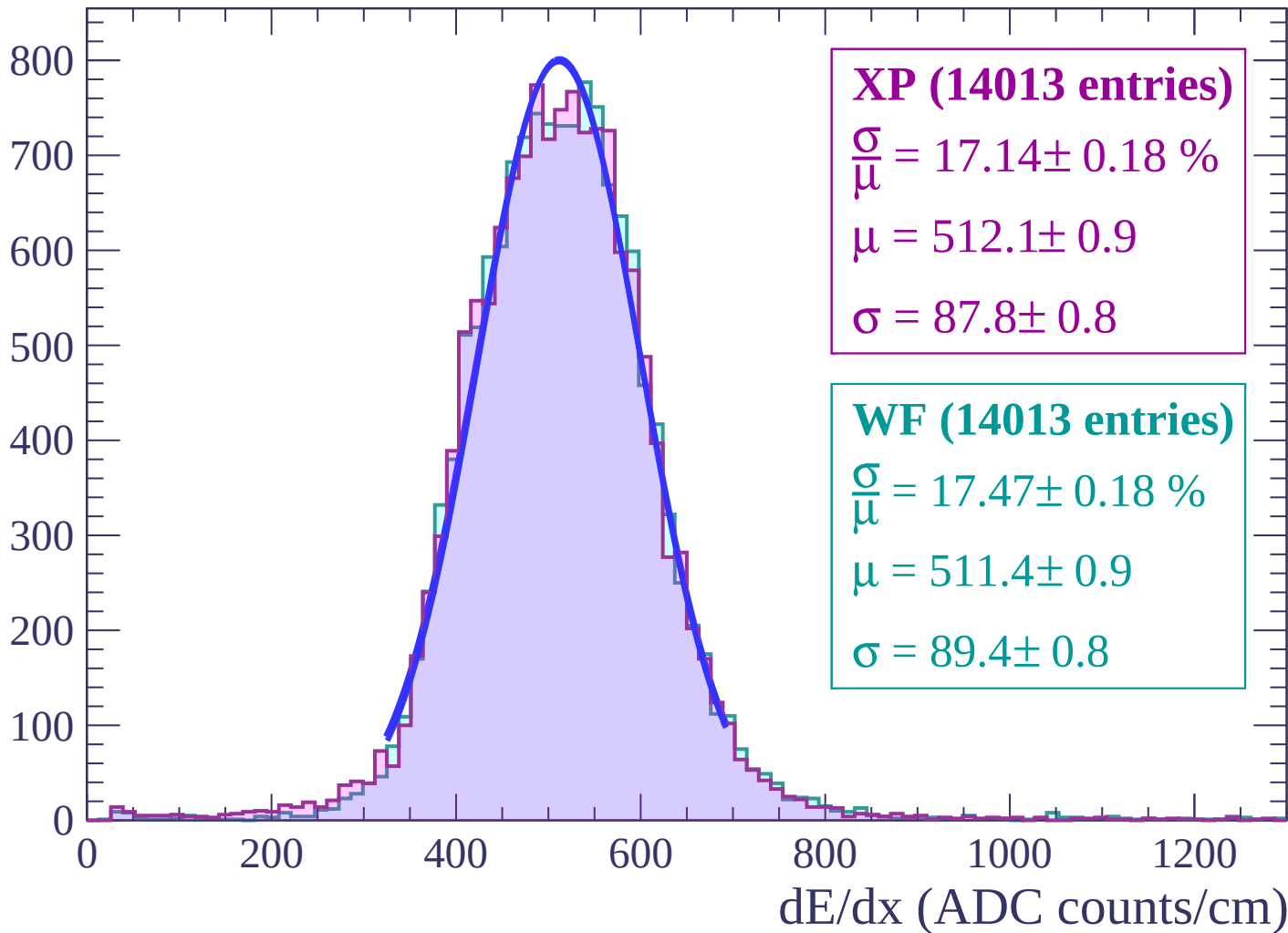
200

0



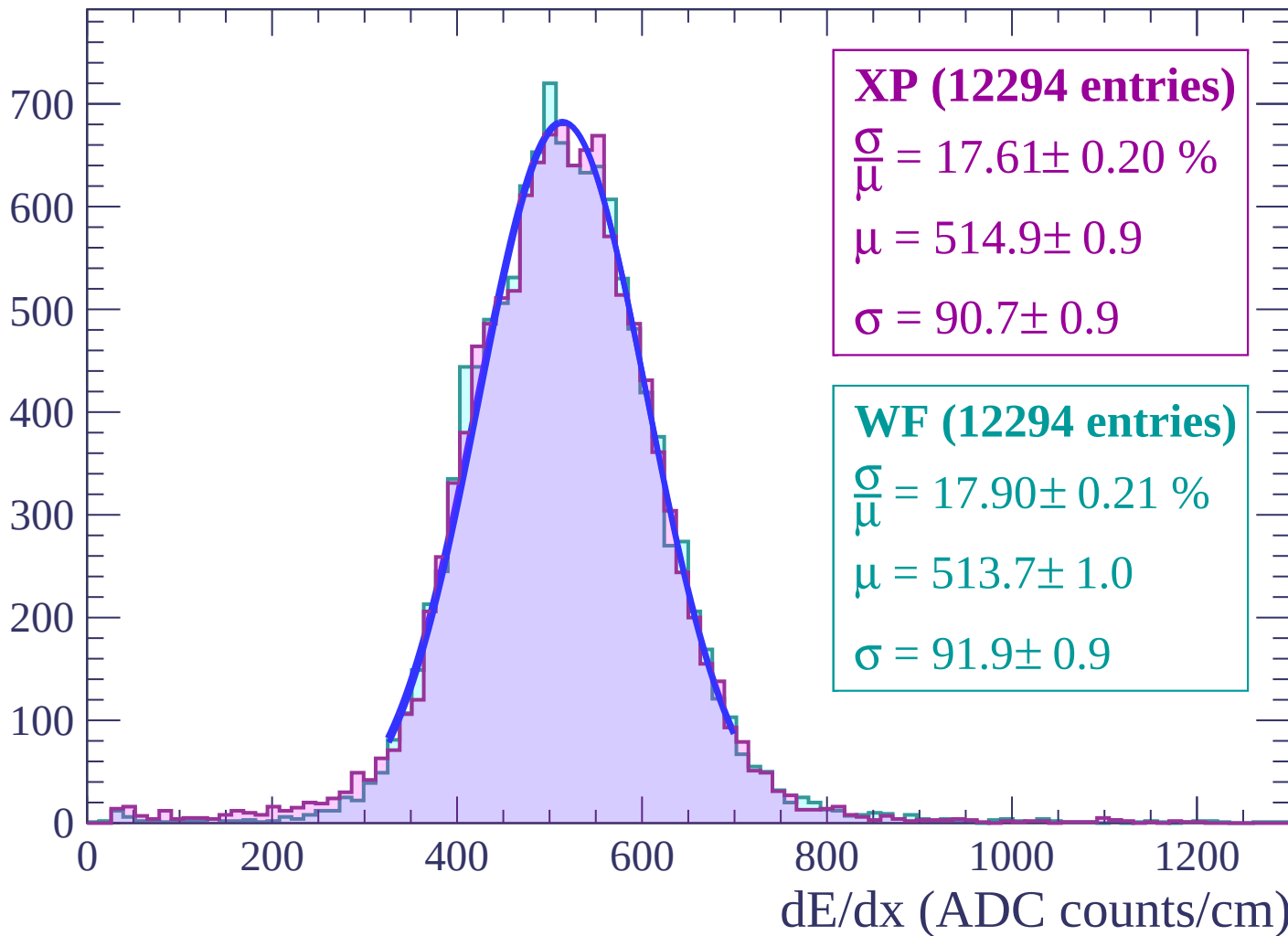
Energy loss | $36 < \theta < 38$

Count



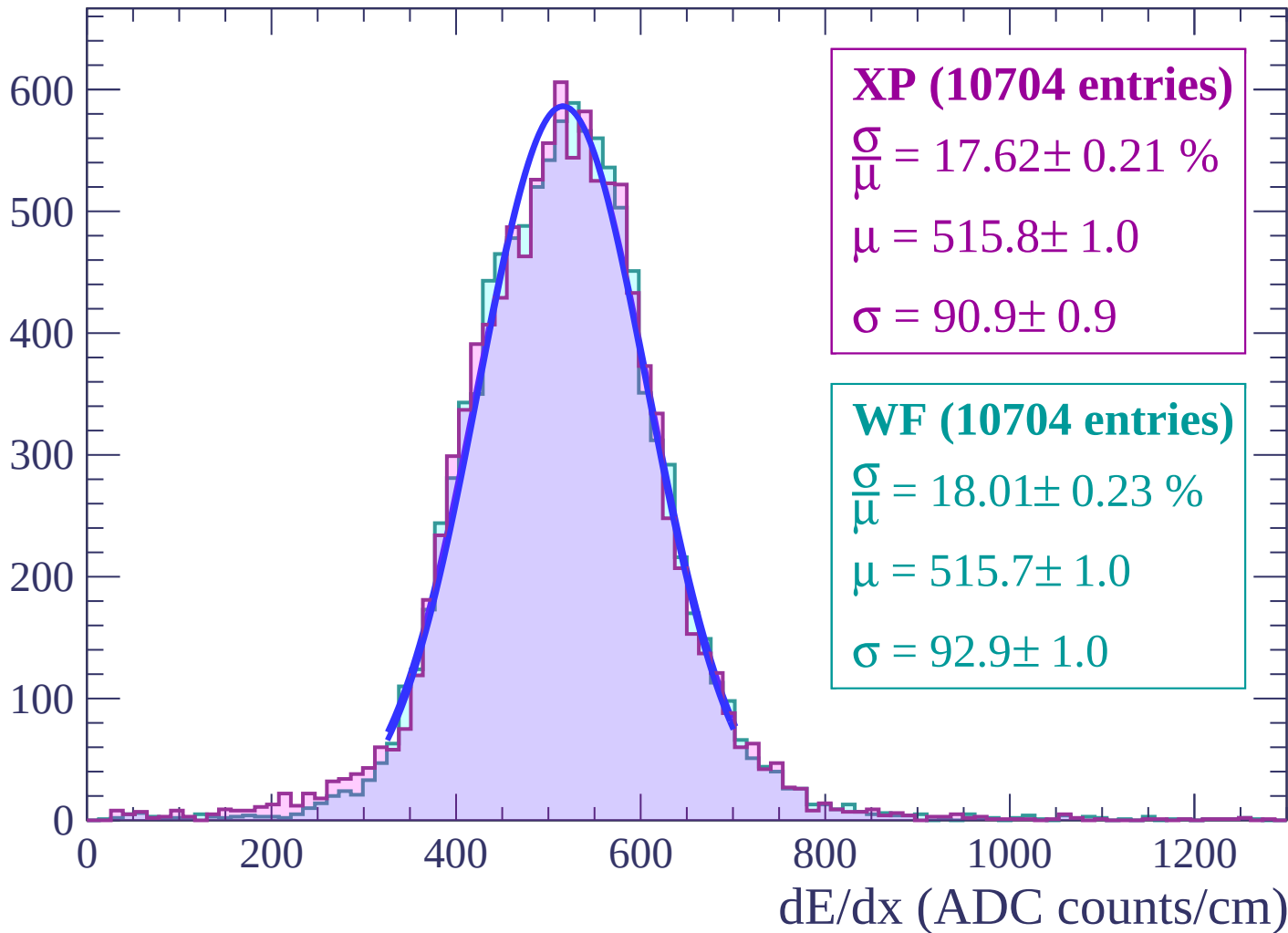
Energy loss | $38 < \theta < 40$

Count



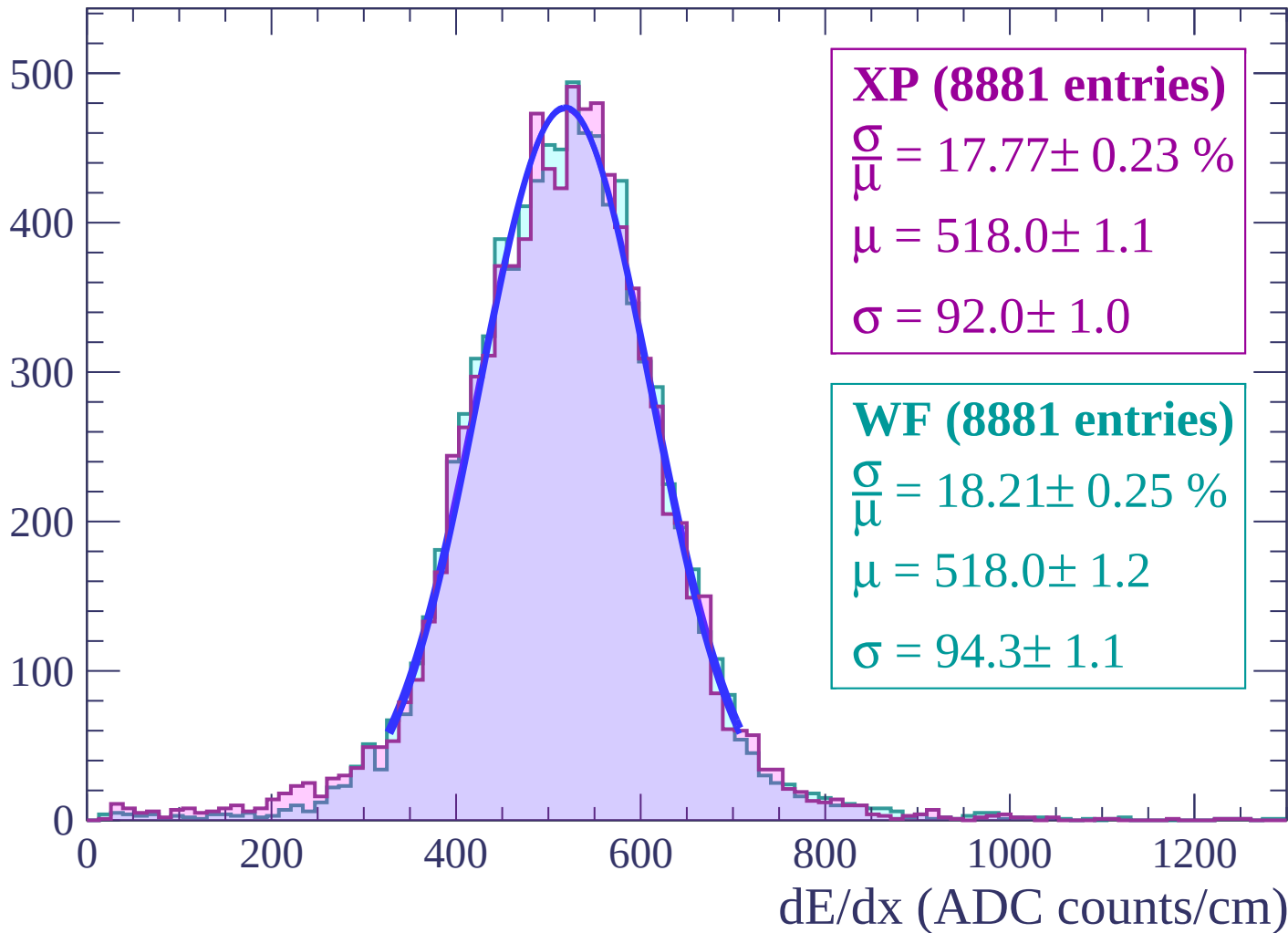
Energy loss | $40 < \theta < 42$

Count



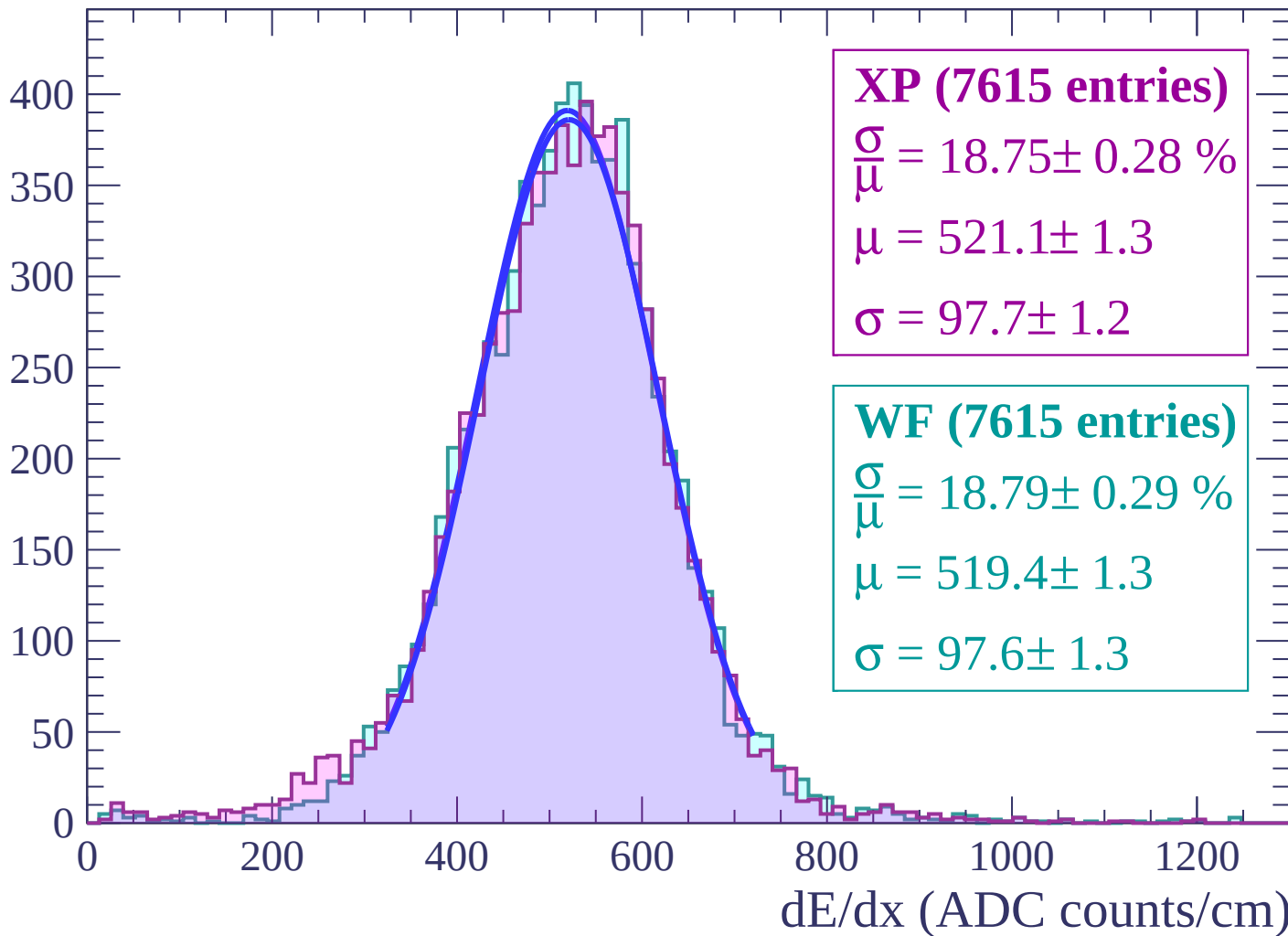
Energy loss | $42 < \theta < 44$

Count



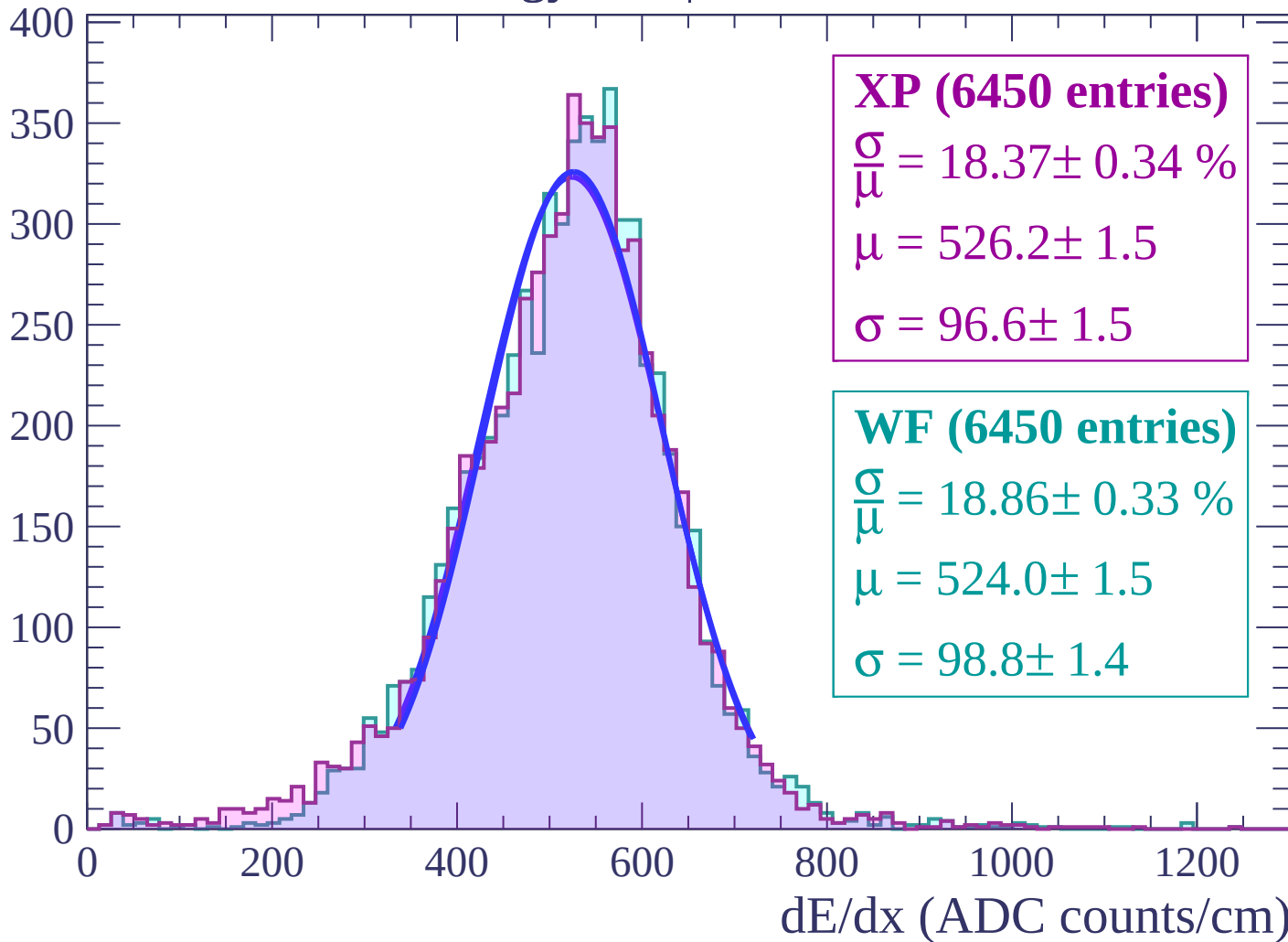
Energy loss | $44 < \theta < 46$

Count



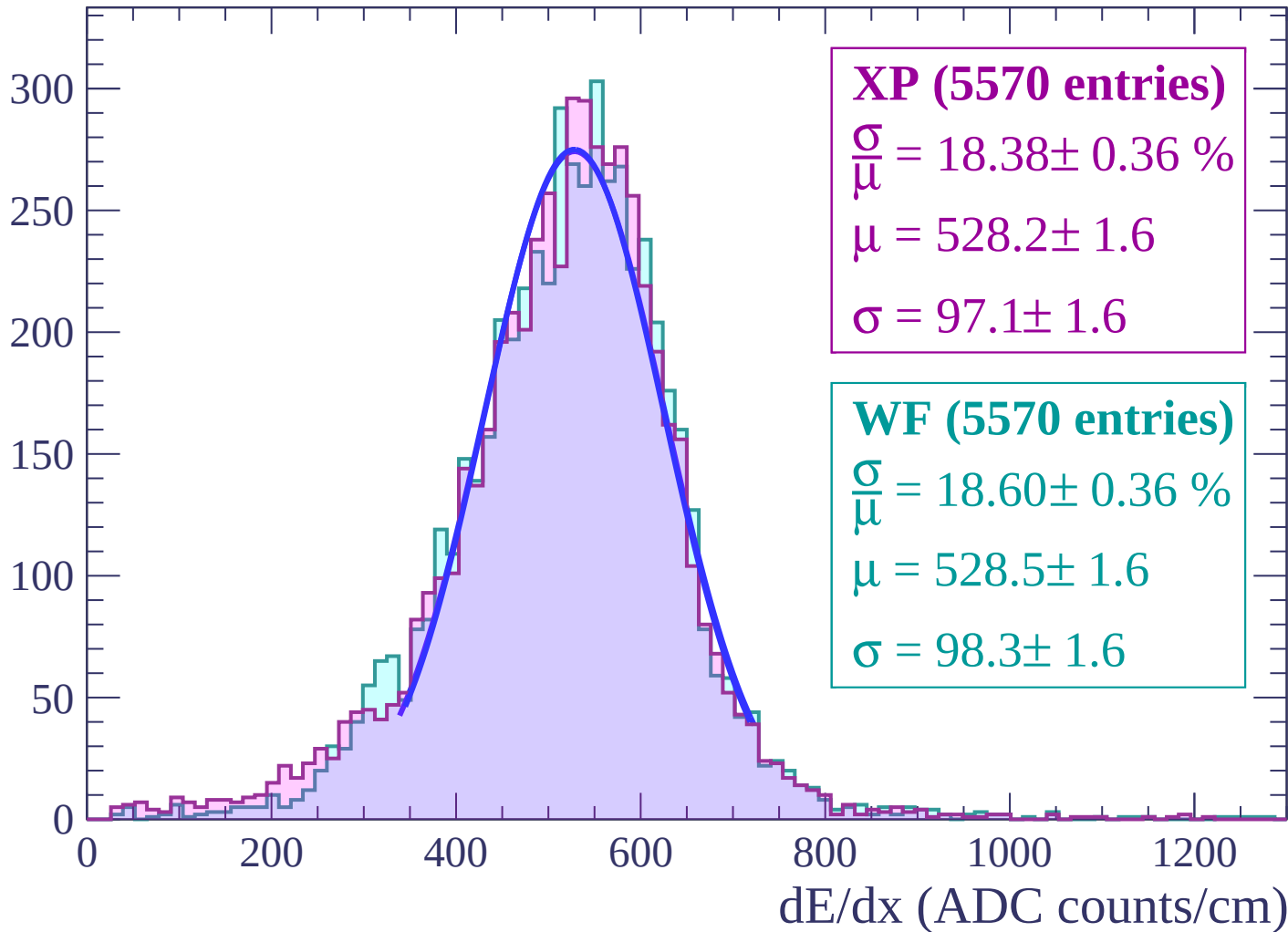
Energy loss | $46 < \theta < 48$

Count



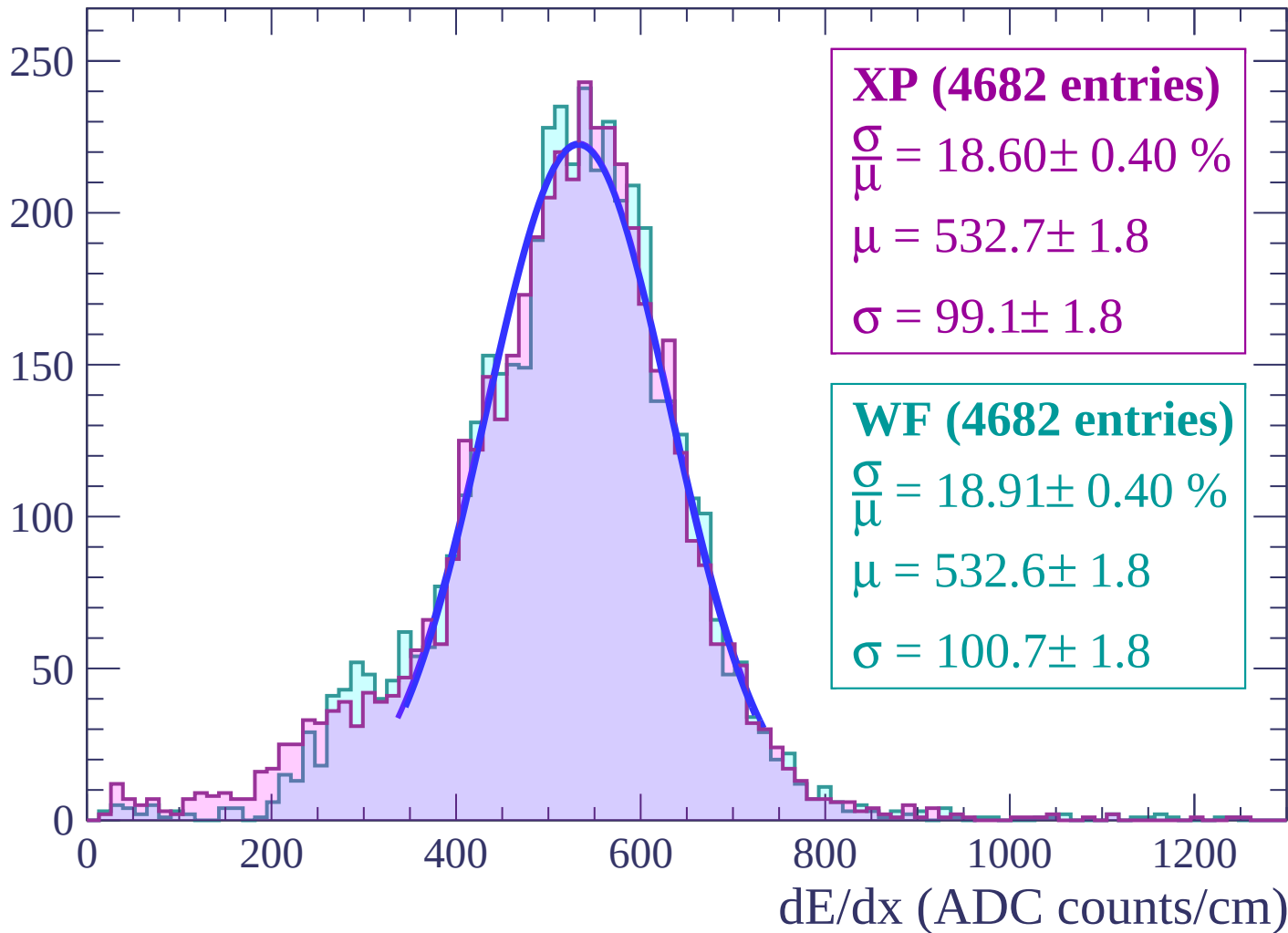
Energy loss | $48 < \theta < 50$

Count



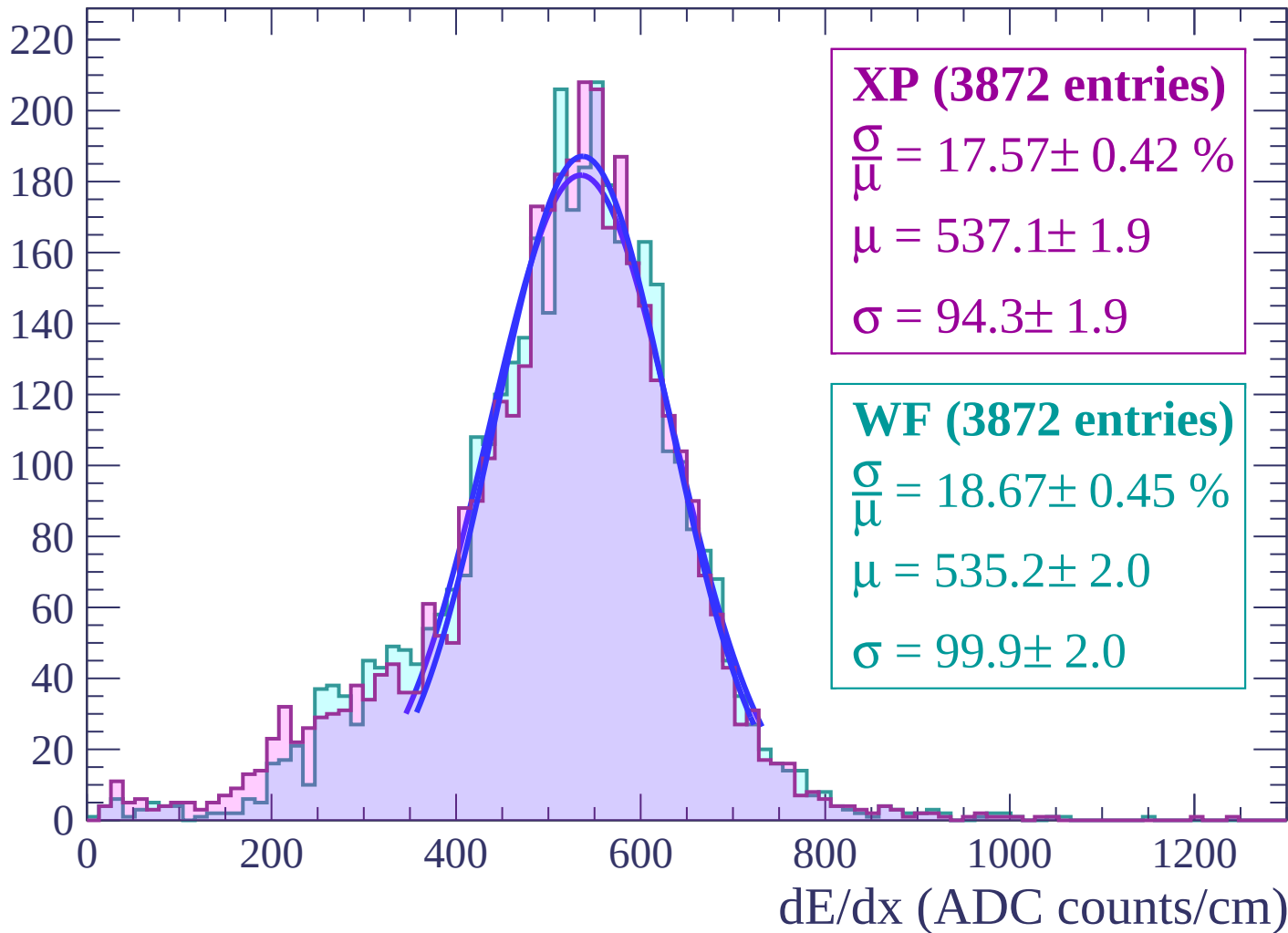
Energy loss | $50 < \theta < 52$

Count



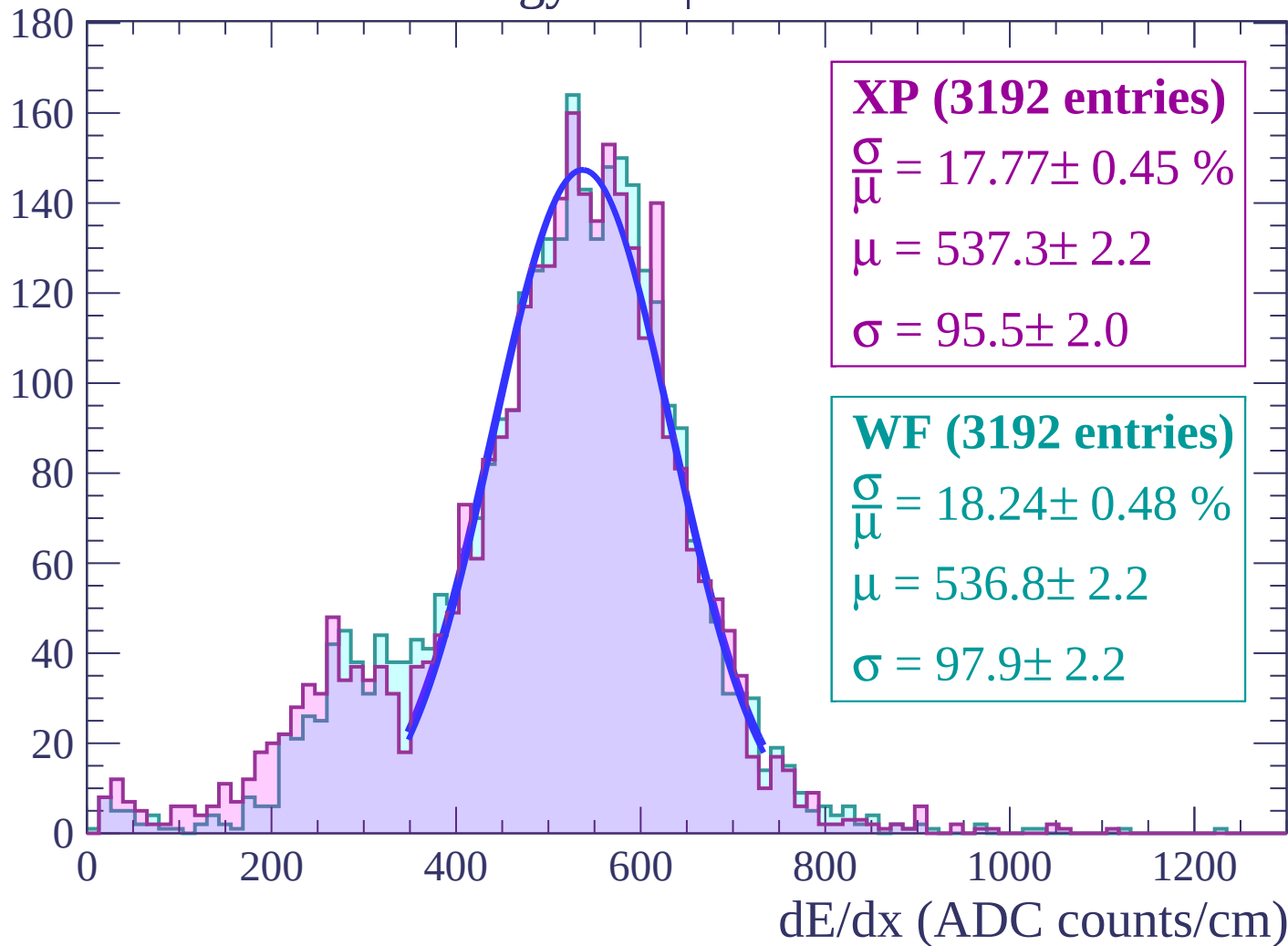
Energy loss | $52 < \theta < 54$

Count



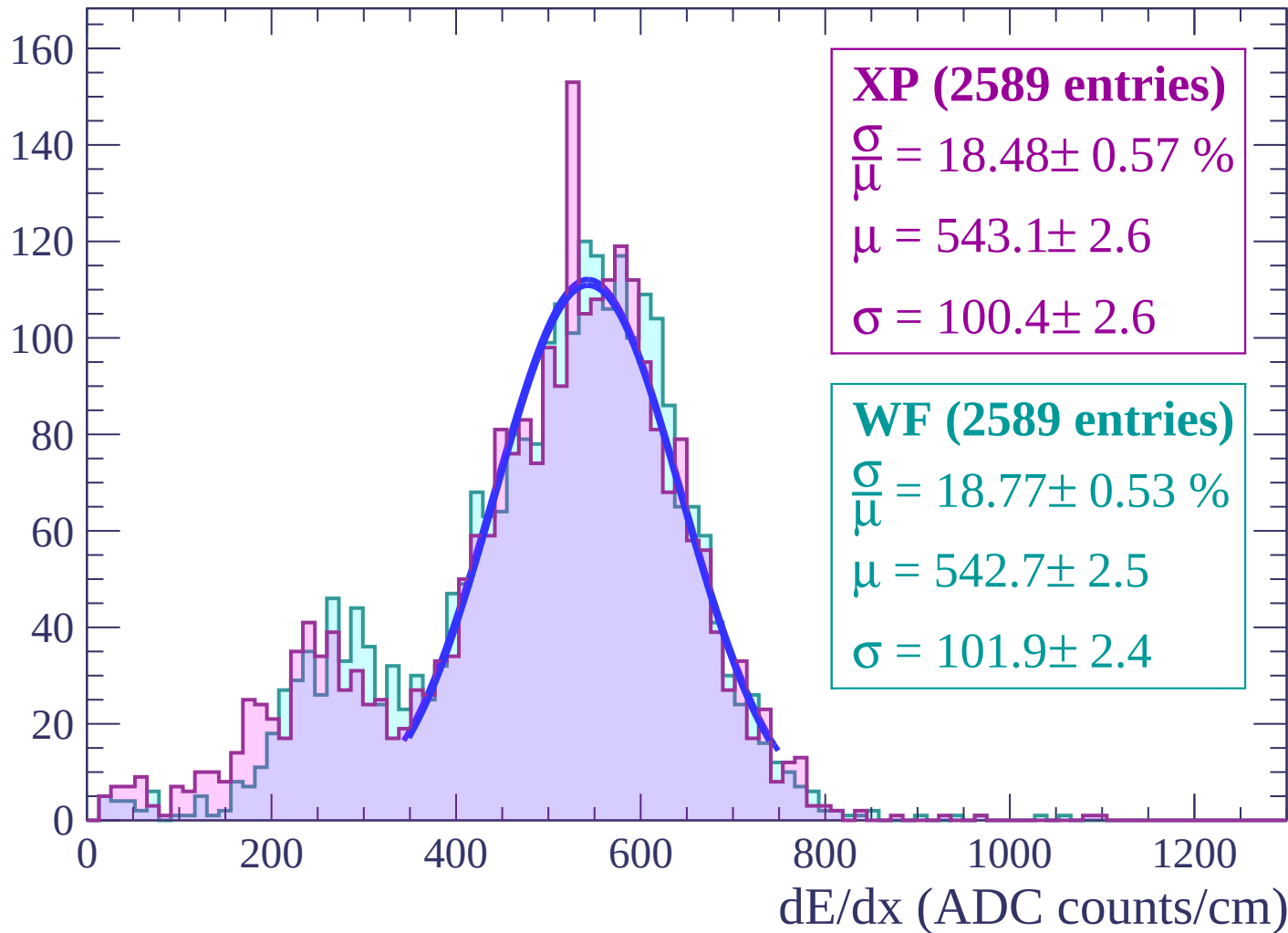
Energy loss | $54 < \theta < 56$

Count



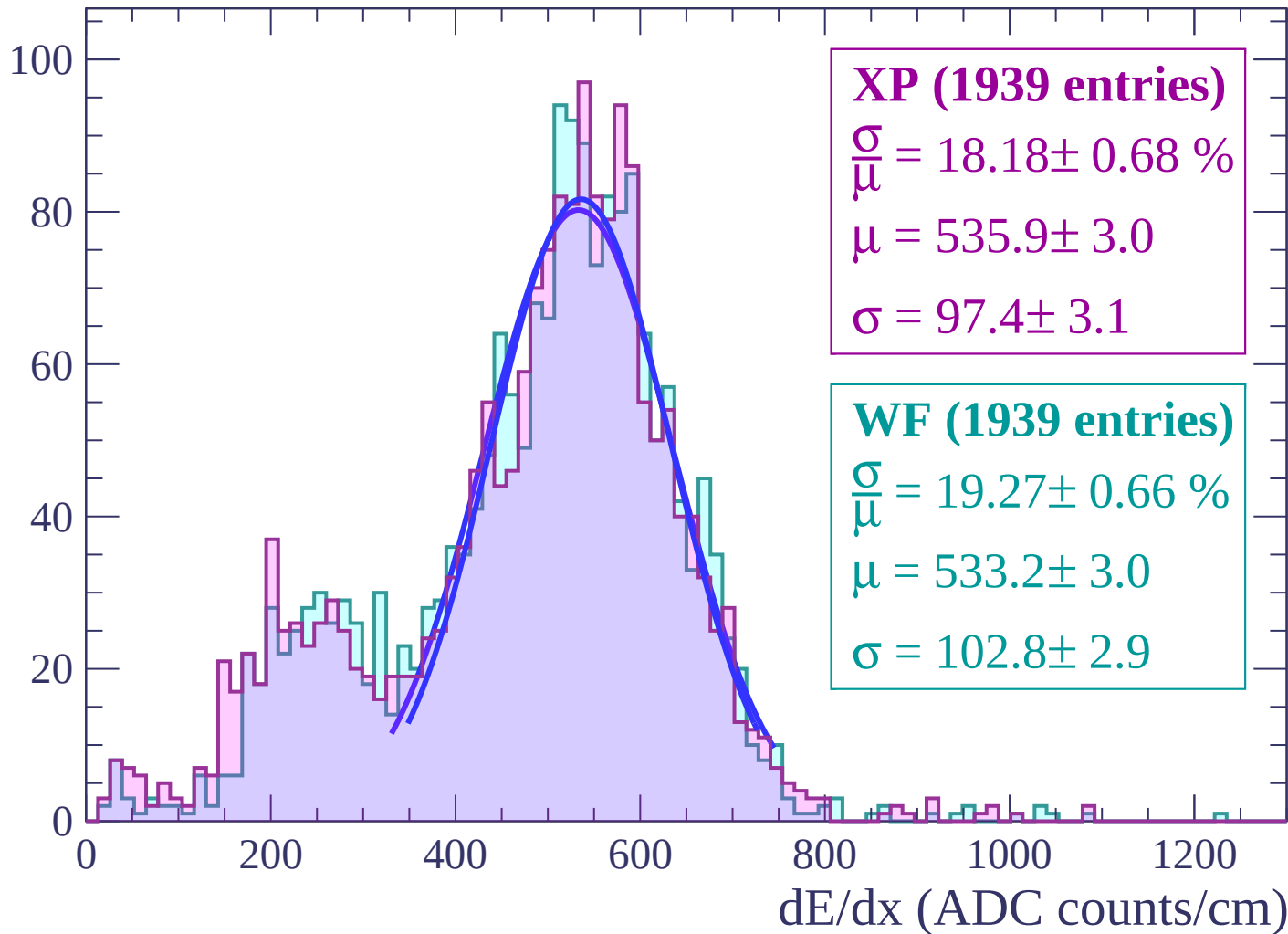
Energy loss | $56 < \theta < 58$

Count



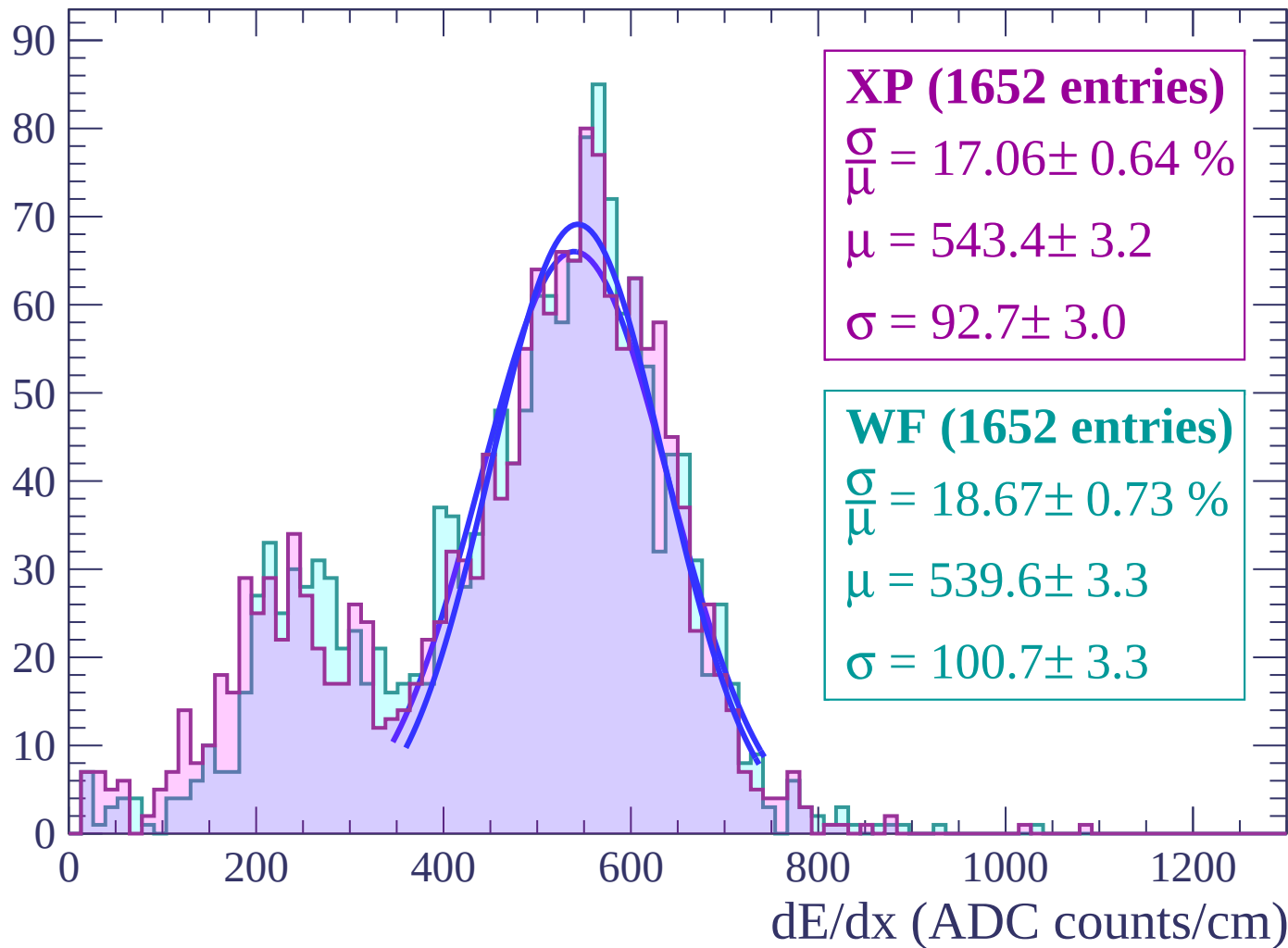
Energy loss | $58 < \theta < 60$

Count



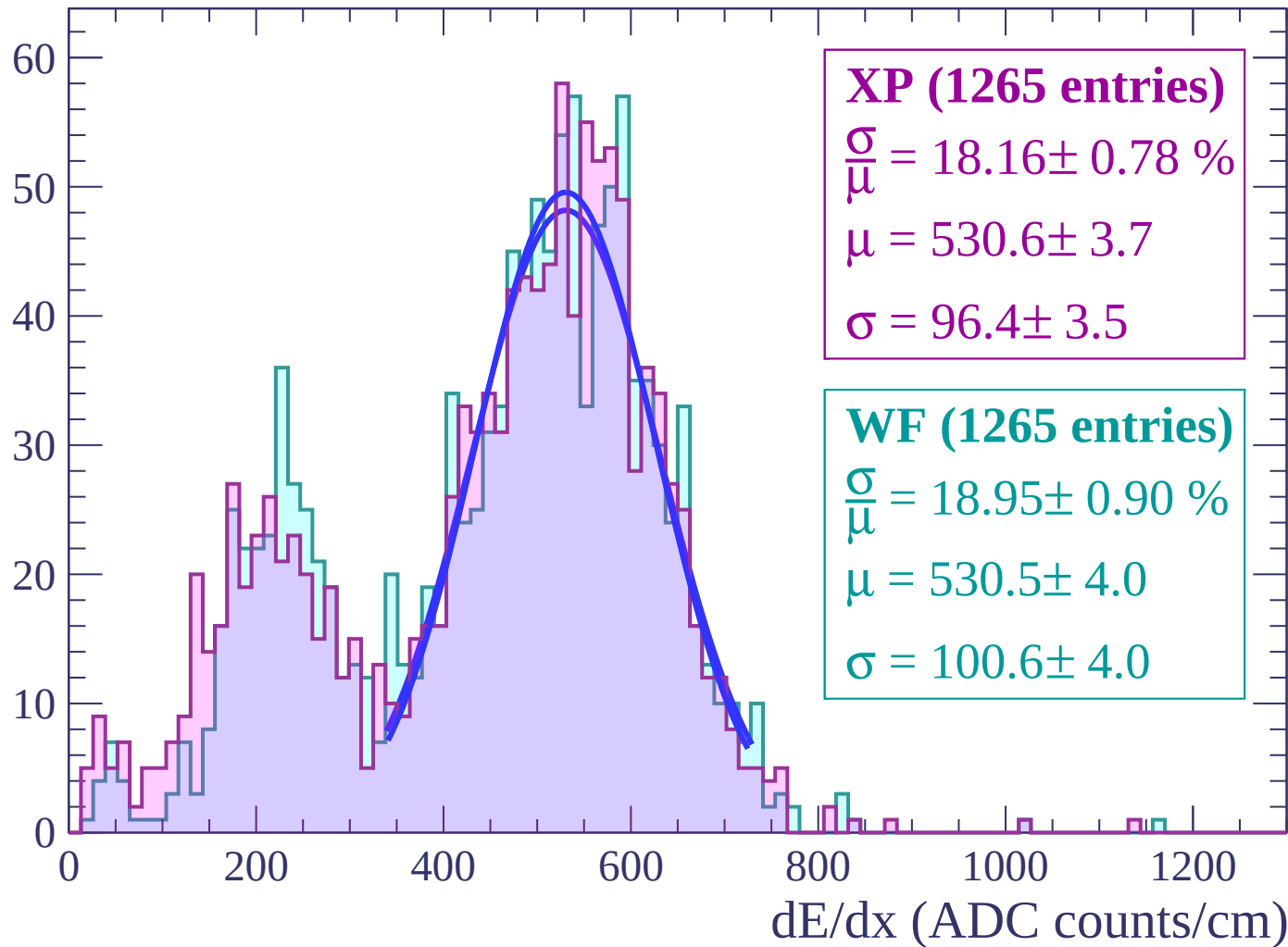
Energy loss | $60 < \theta < 62$

Count



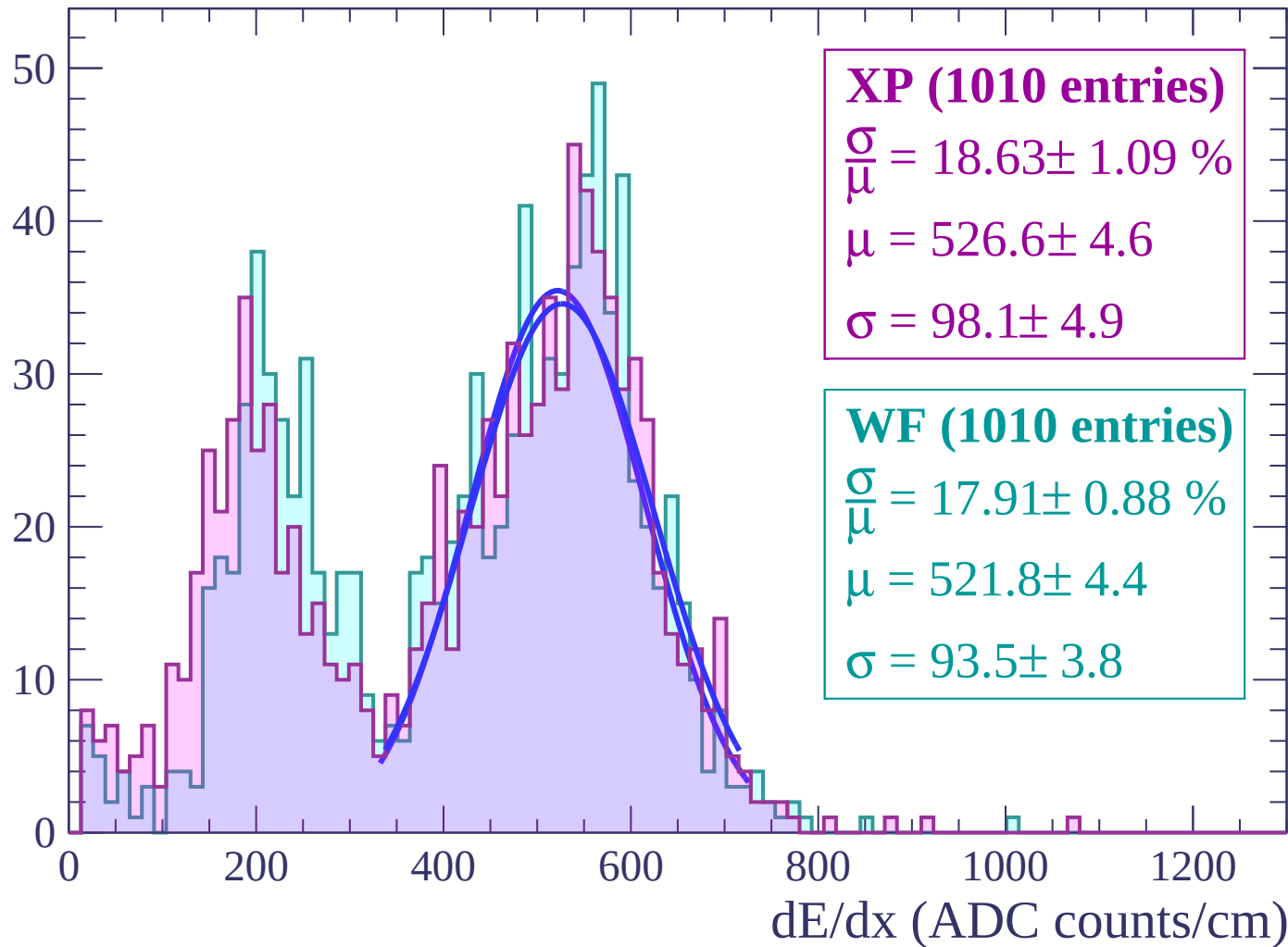
Energy loss | $62 < \theta < 64$

Count



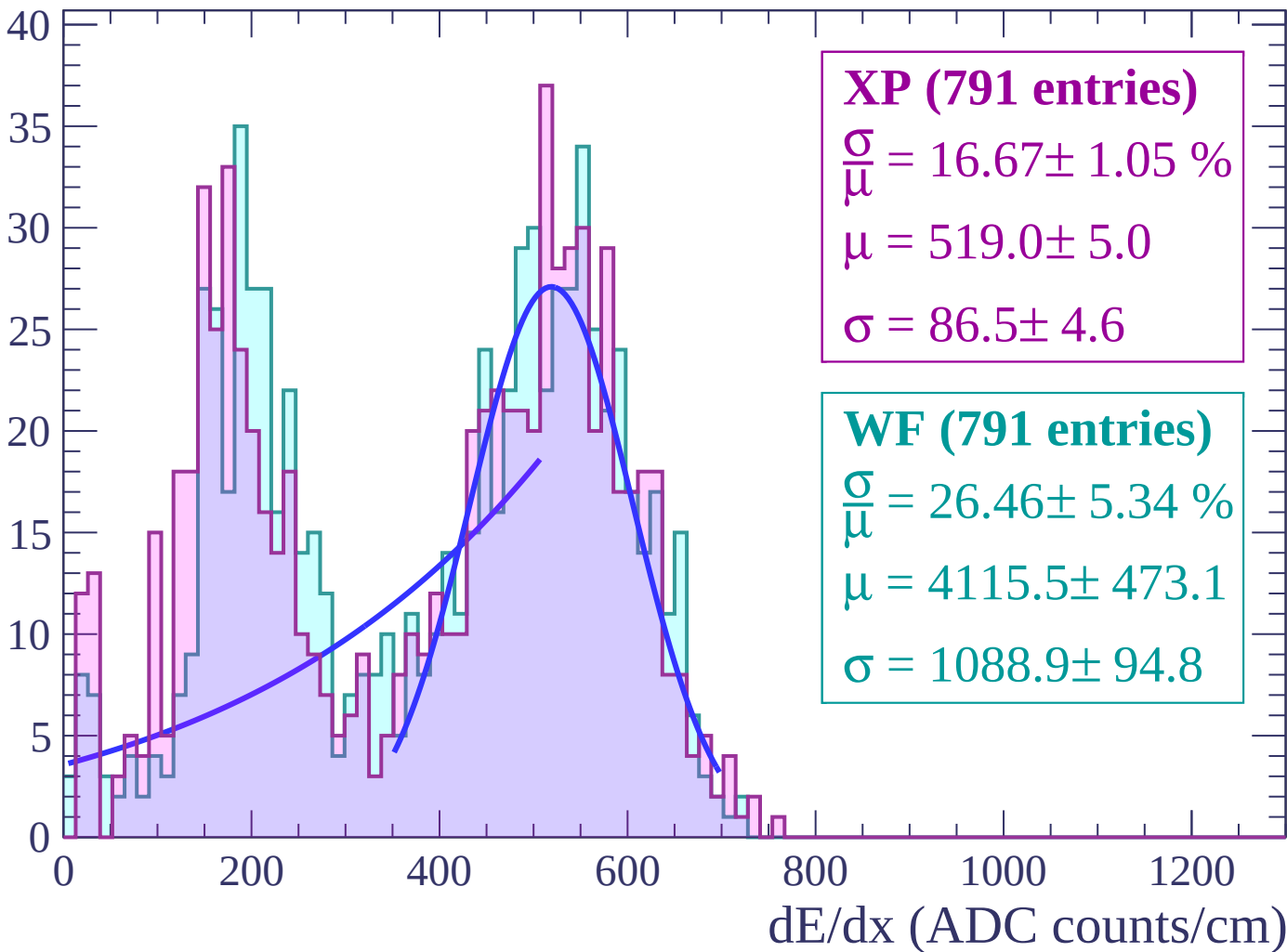
Energy loss | $64 < \theta < 66$

Count



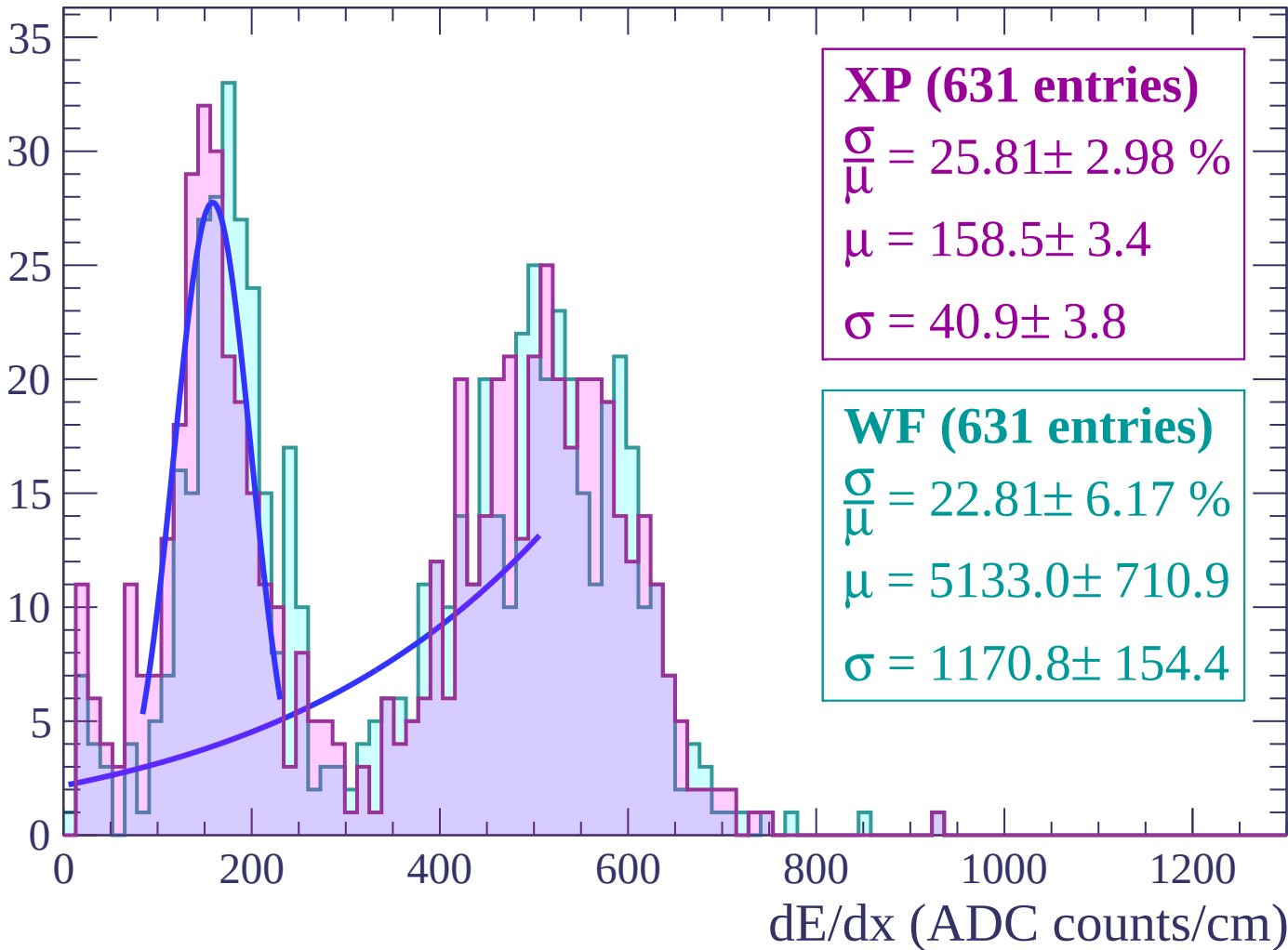
Energy loss | $66 < \theta < 68$

Count



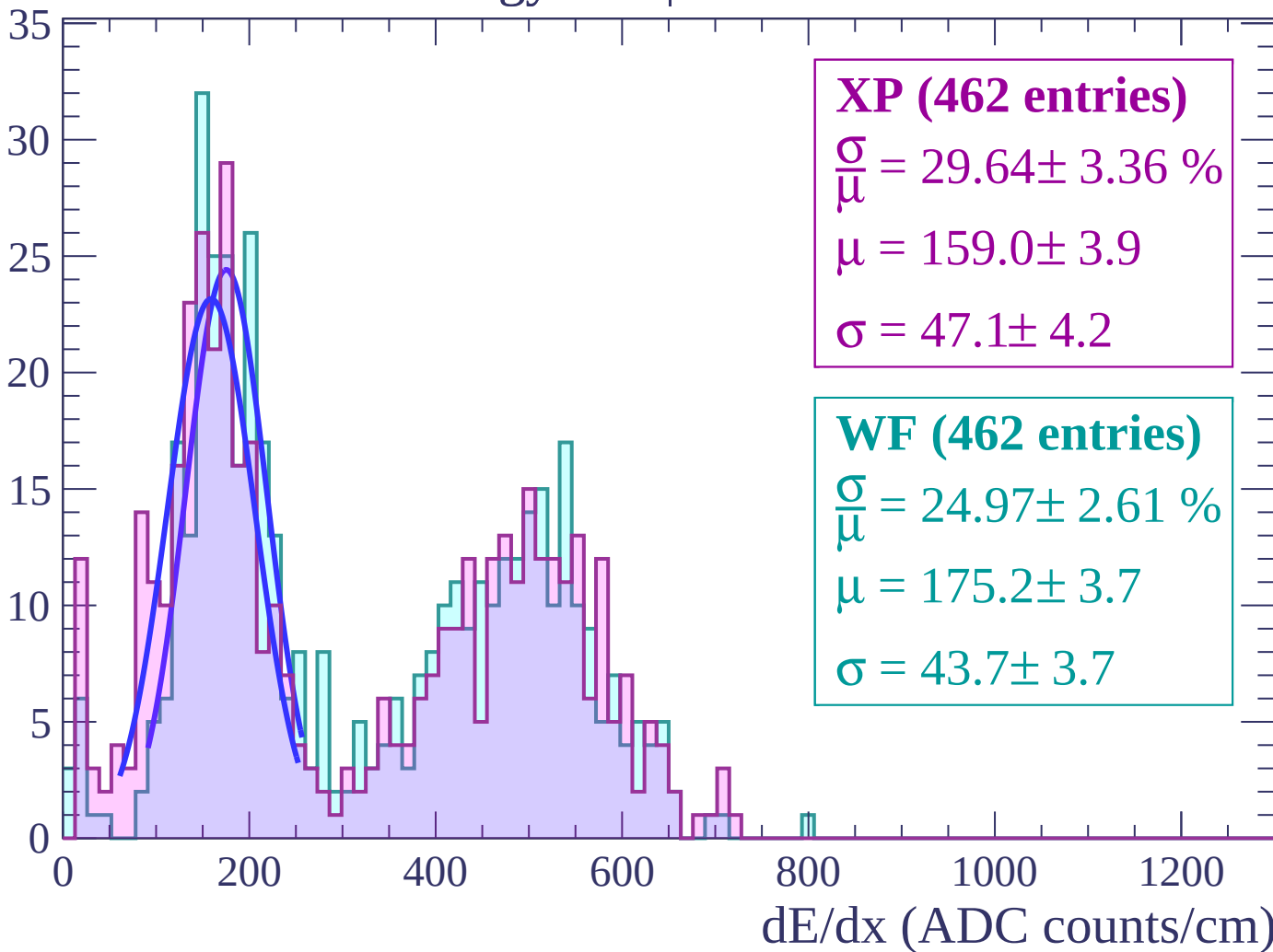
Energy loss | $68 < \theta < 70$

Count



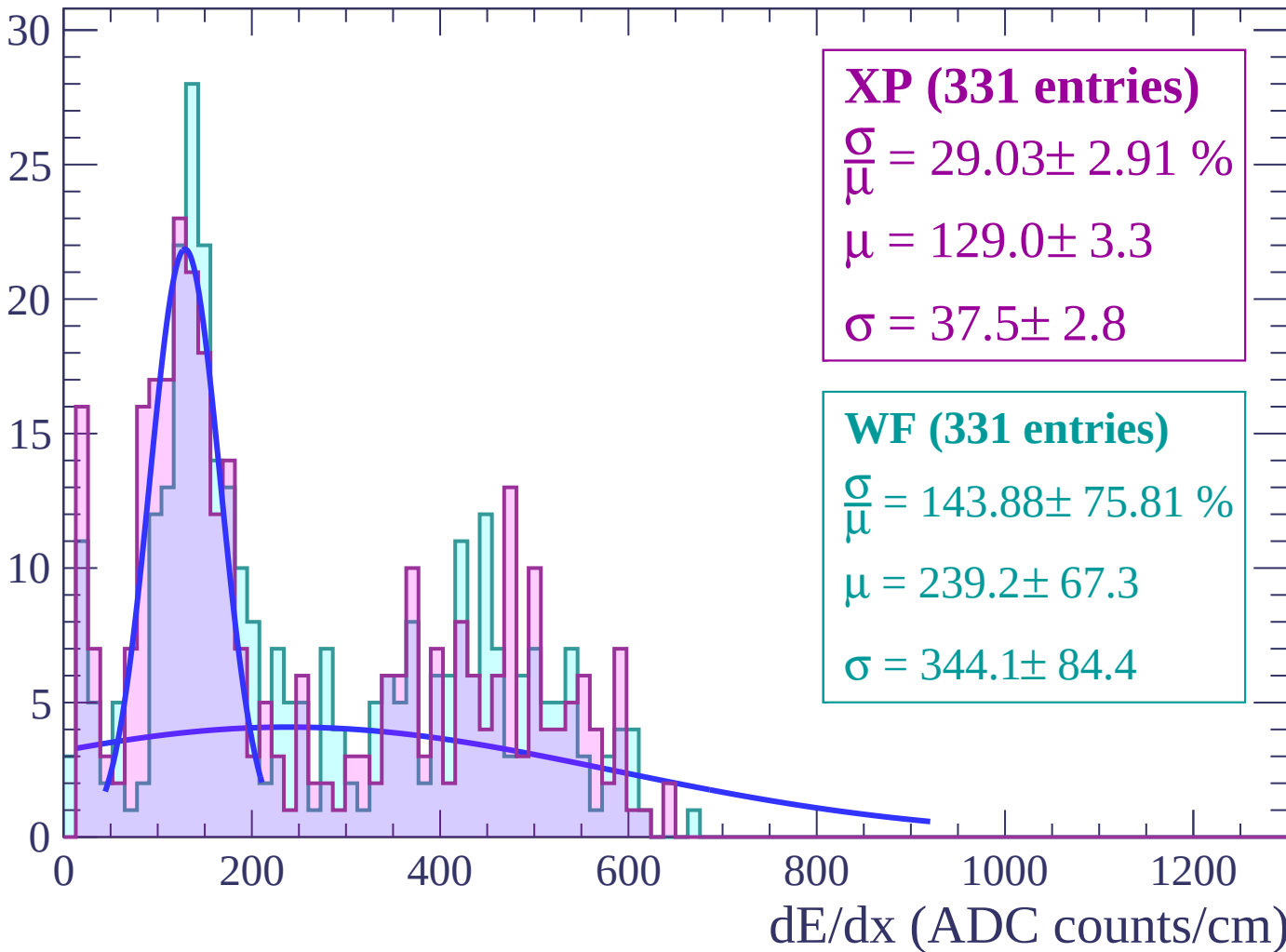
Energy loss | $70 < \theta < 72$

Count



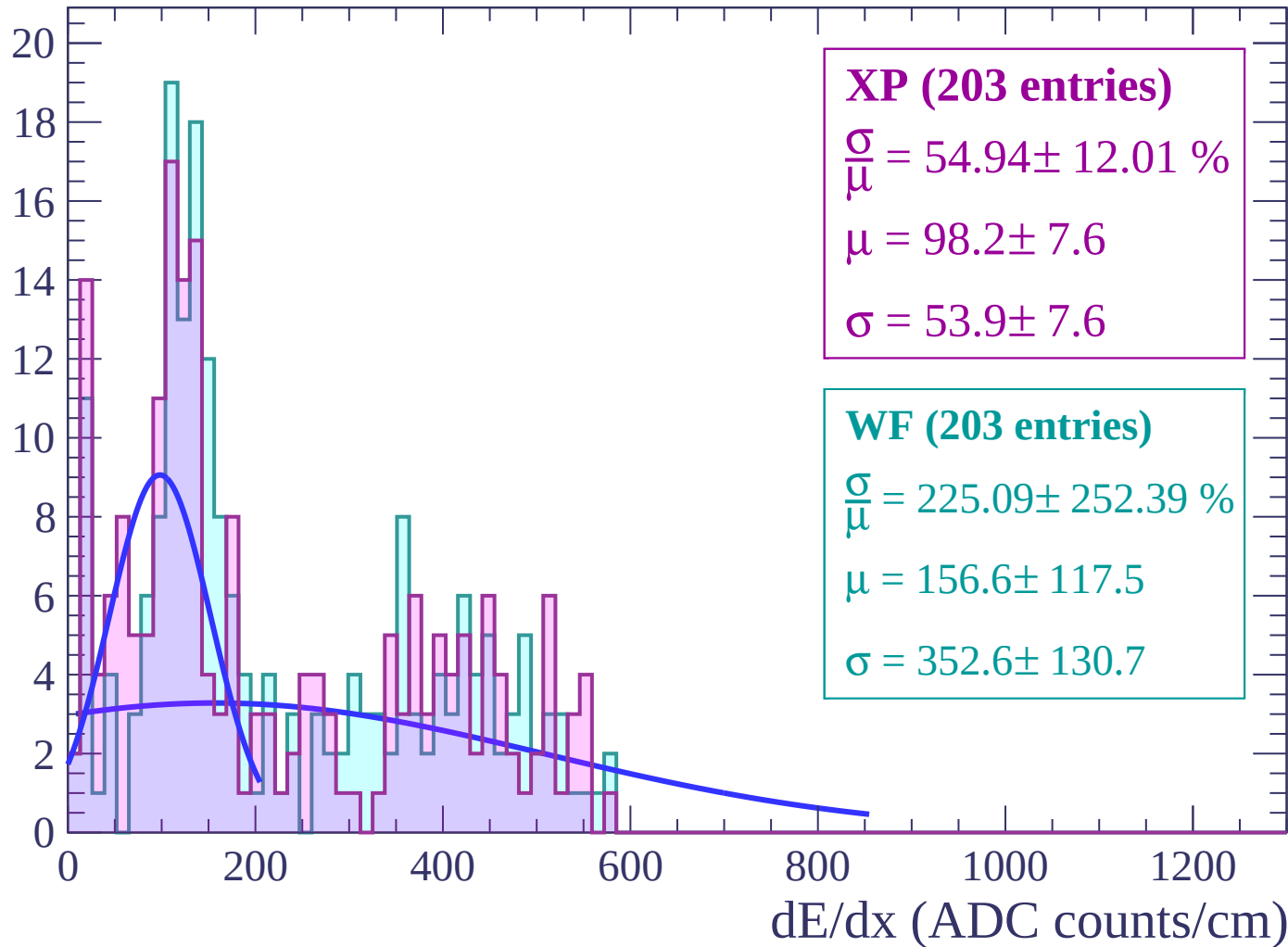
Energy loss | $72 < \theta < 74$

Count



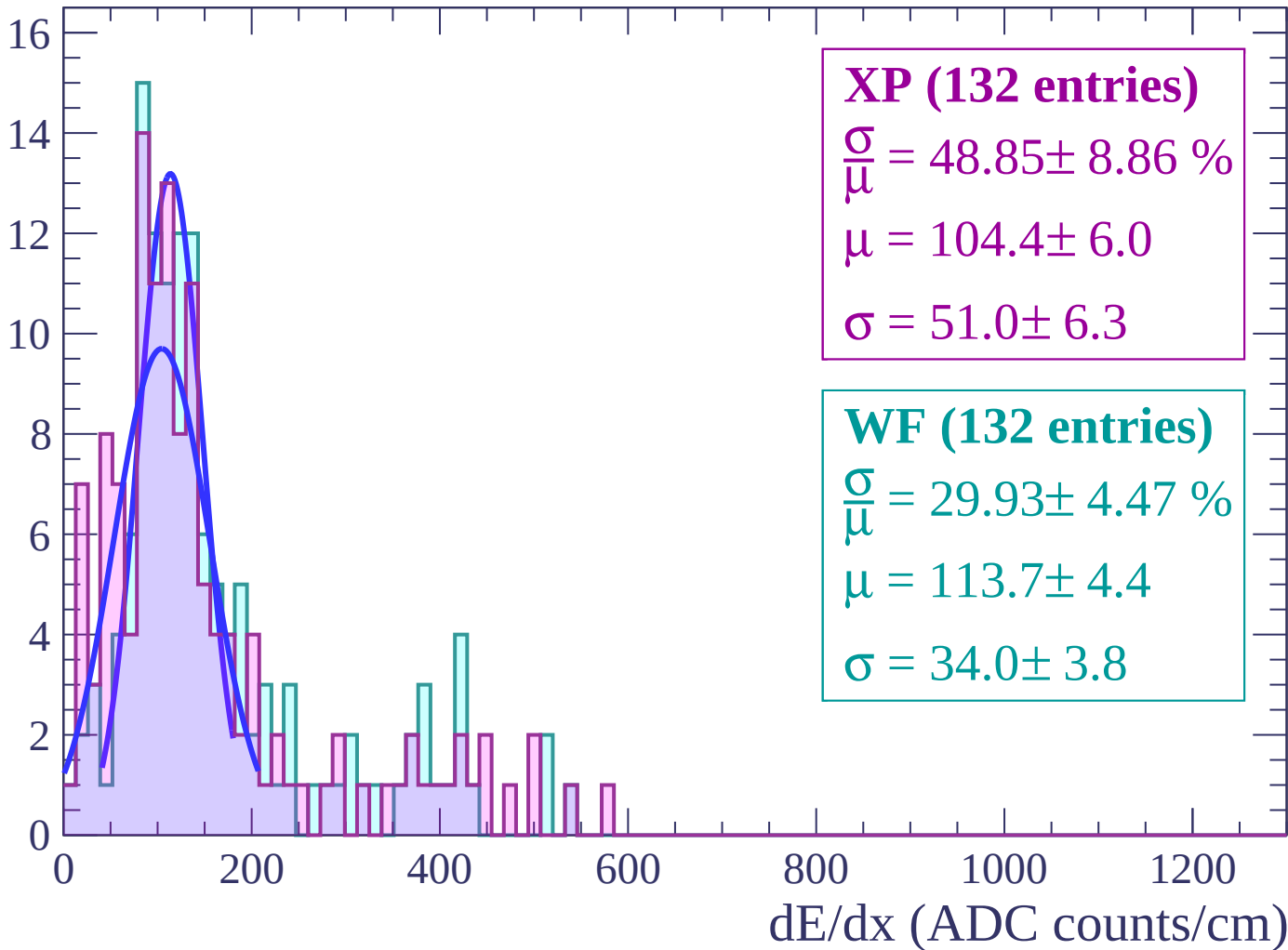
Energy loss | $74 < \theta < 76$

Count



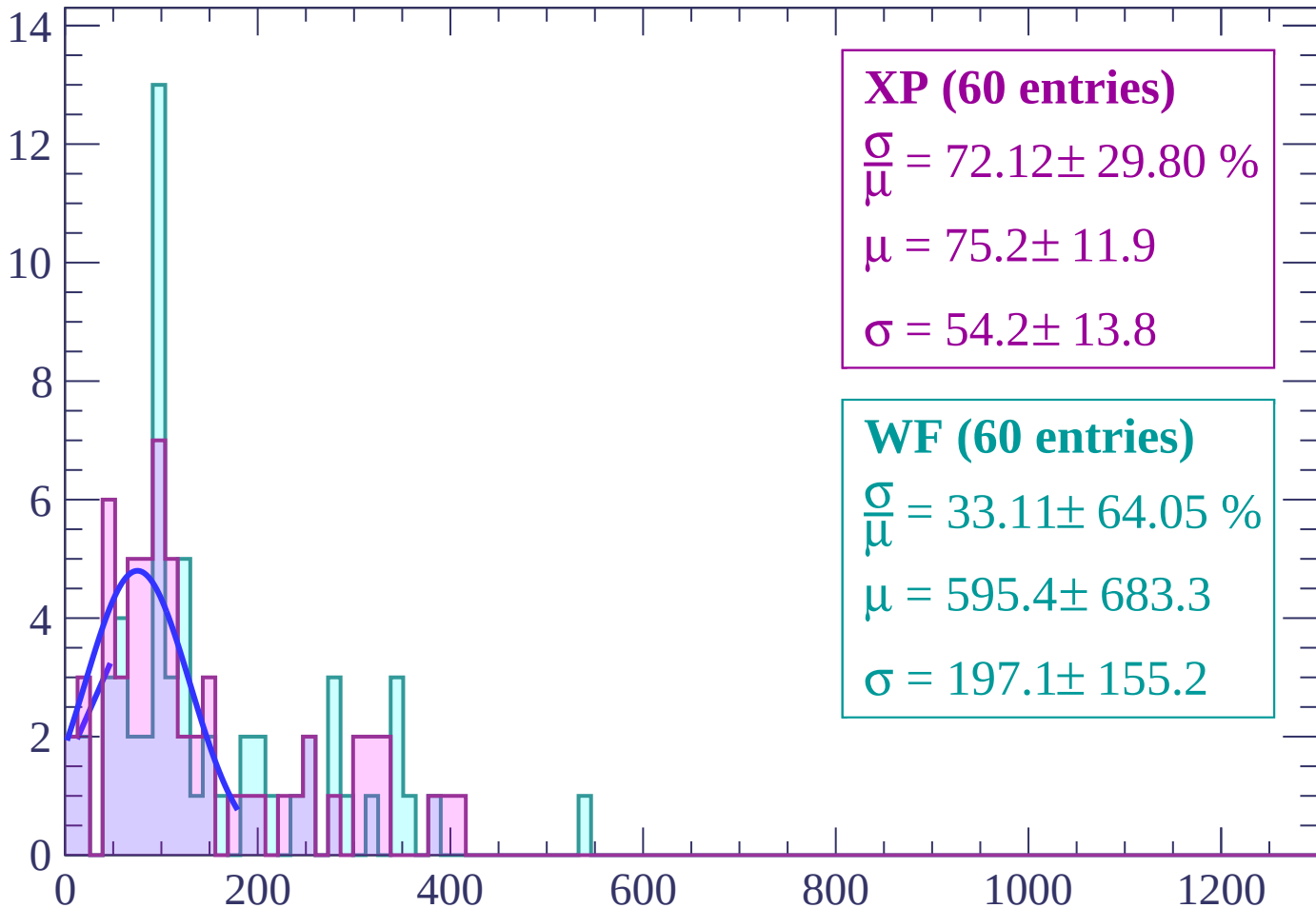
Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

Count



XP (60 entries)

$\frac{\sigma}{\mu} = 72.12 \pm 29.80 \%$

$\mu = 75.2 \pm 11.9$

$\sigma = 54.2 \pm 13.8$

WF (60 entries)

$\frac{\sigma}{\mu} = 33.11 \pm 64.05 \%$

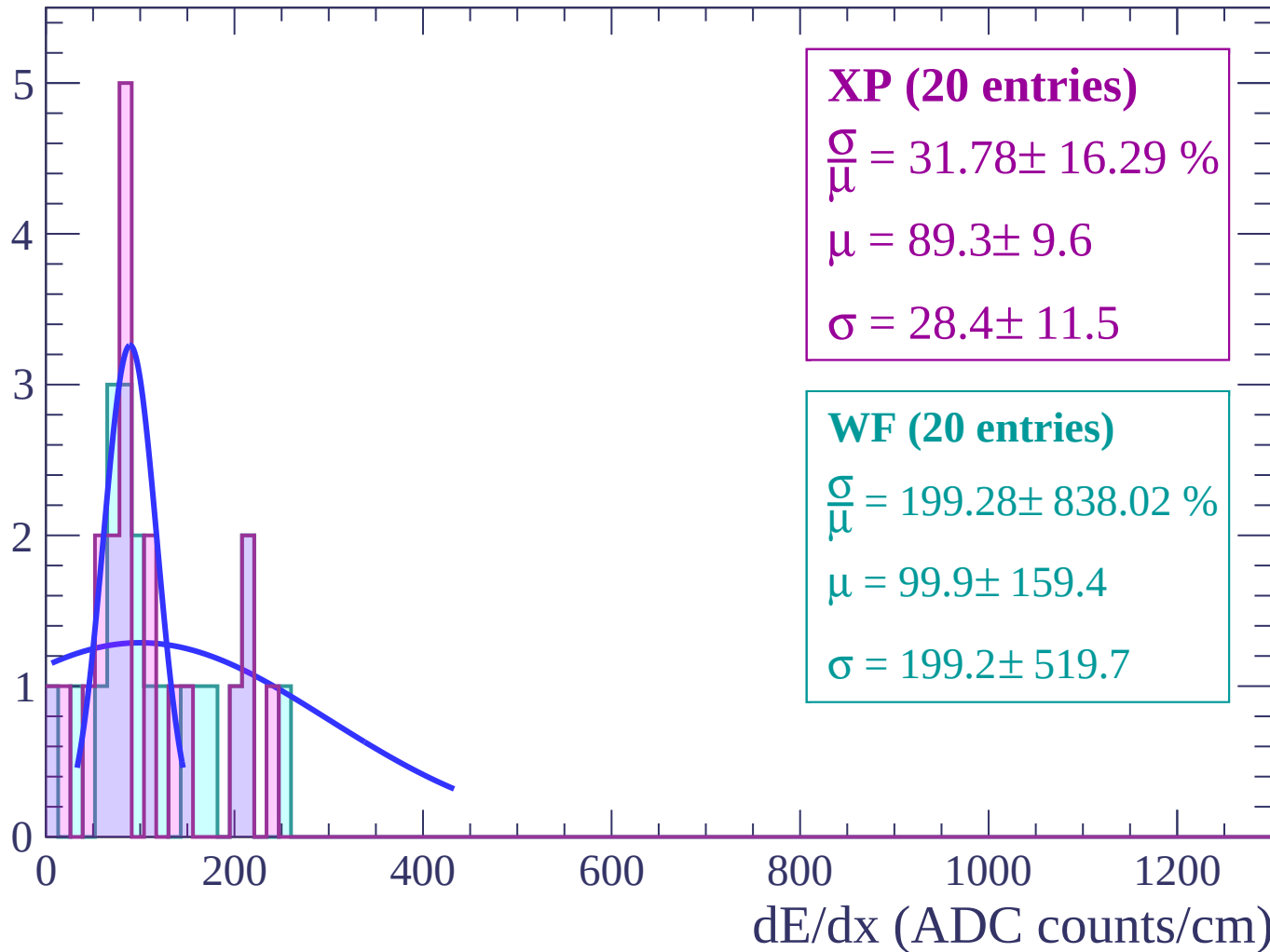
$\mu = 595.4 \pm 683.3$

$\sigma = 197.1 \pm 155.2$

dE/dx (ADC counts/cm)

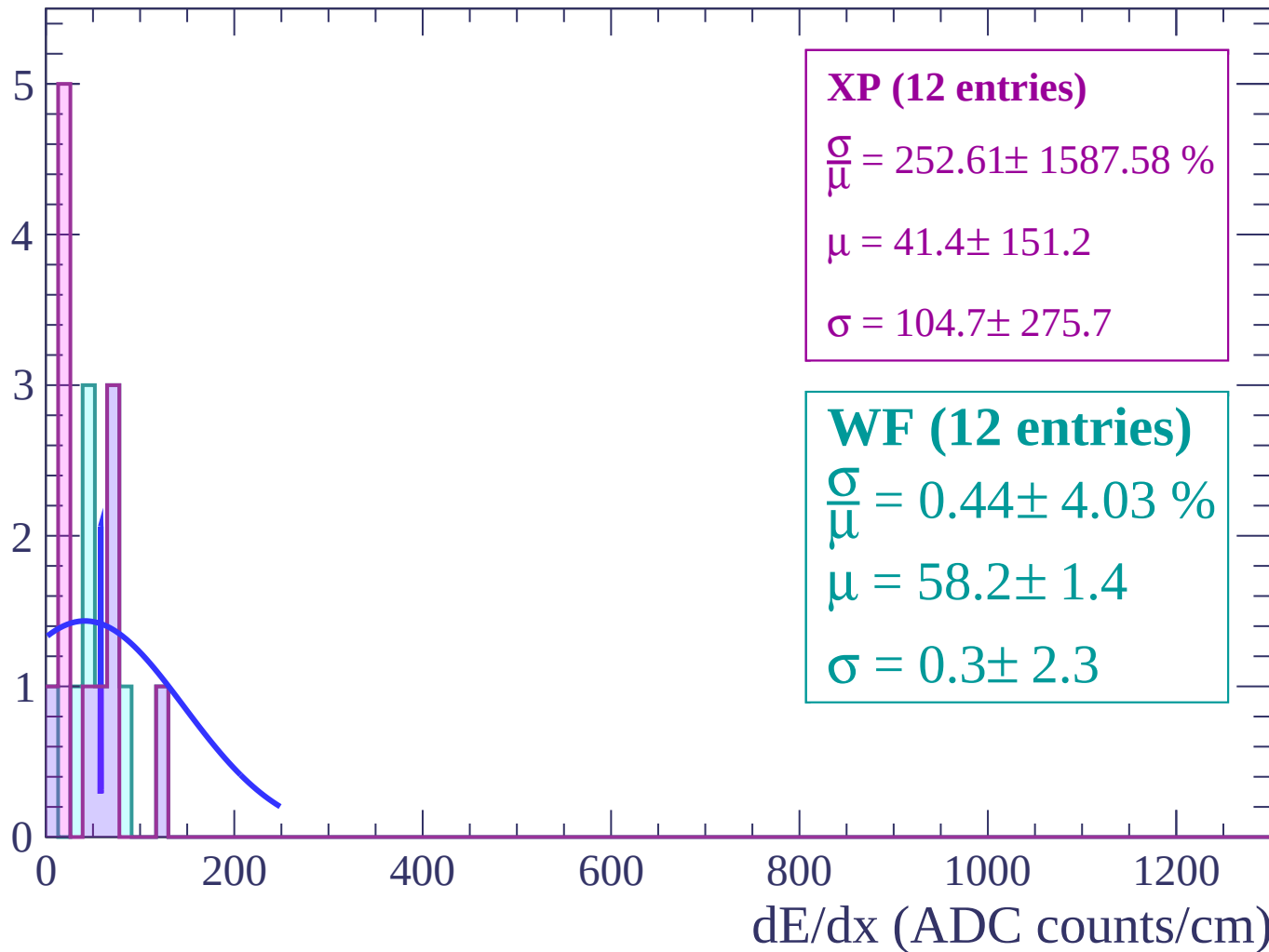
Energy loss | $80 < \theta < 82$

Count



Energy loss | $82 < \theta < 84$

Count



Energy loss | $84 < \theta < 86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $86 < \theta < 88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

